

OpenFOAM

The OpenFOAM Foundation

试用水印

User Guide

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Chapter 1

Introduction

This guide accompanies the release of version 11 of the Open Source Field Operation and Manipulation (OpenFOAM) C++ libraries. It provides a description of the basic operation of OpenFOAM, first through a set of tutorial exercises in chapter 2 and later by more detailed descriptions of different components of OpenFOAM.

OpenFOAM is software for computational fluid dynamics (CFD). It includes a collection of *applications* which perform a range of tasks in CFD. The applications use packaged functionality contained within over 150 *libraries*. As well as performing calculations of the fluid dynamics, there are applications which configure and initialise simulations, manipulate case geometry, generate computational meshes, and process and visualise results.

Applications primarily fall into two categories: *solvers*, which perform the calculations in fluid (or other continuum) mechanics; and *utilities*, that perform the other tasks described above. Prior to version 11 of OpenFOAM, individual solvers were written for numerous specific types of flow. With so many combinations of flow type and additional physics, OpenFOAM included almost 100 solvers at one time. Solvers with names like simpleFoam and pimpleFoam have been the mainstay of OpenFOAM since the early 1990s.

OpenFOAM version 11, however, introduces *modular solvers* as a major improvement to the original application solvers. The application solvers are replaced by a single foamRun solver which describes the steps of a general algorithm for fluid dynamics calculations. foamRun loads a solver module, which defines each step to characterise a particular type of flow.

Modular solvers are simpler to use and maintain than application solvers. Their source code is easier to navigate, promoting better understanding. They are more flexible; in particular, there is also a foamMultiRun solver which can take two or more domain regions and apply a different solver module to each region. In particular, modules for one or more fluids and solids can be coupled for conjugate heat transfer (CHT) for different flow types, *e.g.* multiphase.

Further details of applications, including modular solvers, are described in chapter 3. General configuration and running of OpenFOAM cases are described in chapter 4. Chapter 5 covers details of the generation of meshes using the mesh generator supplied with OpenFOAM and conversion of mesh data generated by third-party products. Post-processing of results, including visualisation, is in chapter 7. Finally, some aspects of physical modelling, *e.g.* transport and thermophysical modelling, are described in chapter 8.

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Chapter 2

Tutorials

This chapter we describes the process of setup, simulation and post-processing for some OpenFOAM test cases, with the principal aim of introducing a user to the basic procedures of running OpenFOAM. The test cases are taken from the *tutorials* directory which contains numerous example cases in OpenFOAM. The directory location is represented by the *$FOAM_TUTORIALS* variable in the OpenFOAM "environment".

The directory contains numerous cases that demonstrate the use of all the solver modules, other solvers and many utilities supplied with OpenFOAM. Most examples are stored in sub-directories corresponding to each of the modular solvers. For example, the cases that use the incompressibleFluid module are stored in *$FOAM_TUTORIALS/incompressibleFluid*. The user can explore these example cases, starting by listing the top-level of the *$FOAM_TUTORIALS* directory, by typing in a terminal

```
ls $FOAM_TUTORIALS
```

The OpenFOAM environment includes a $FOAM_RUN variable which represents a directory in the user's file system at *$HOME/OpenFOAM/<USER>-11/run* where <*USER*> is the account login name and "11" is the OpenFOAM version number. The directory provides a recommended location to store and run simulation cases. The examples presented in this chapter will be copied into the *run* directory. The user should check whether the directory exists by typing

```
ls $FOAM_RUN
```

If a message is returned saying no such directory exists, the user should create the directory by typing

```
mkdir -p $FOAM_RUN
```

Any example case from *$FOAM_TUTORIALS* can then be copied into the *run* directory. For example to try the motorBike example for the incompressibleFluid solver module, the user can copy it to the *run* directory by typing:

```
cd $FOAM_RUN
cp -r $FOAM_TUTORIALS/incompressibleFluid/motorBike .
```

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2.1 Backward-facing step

This tutorial will describe how to pre-process, run and post-process a case involving isothermal, incompressible flow across a backward-facing step. The problem is treated as two dimensional with the geometry shown in Figure 2.1. The domain consists of:

- an inlet opening (left);
- an outlet opening (right);
- an upper wall, which is horizontal before tapering gently toward the outlet;
- a lower wall, which also tapers towards the outlet, but includes an abrupt step within a short distance from the inlet;

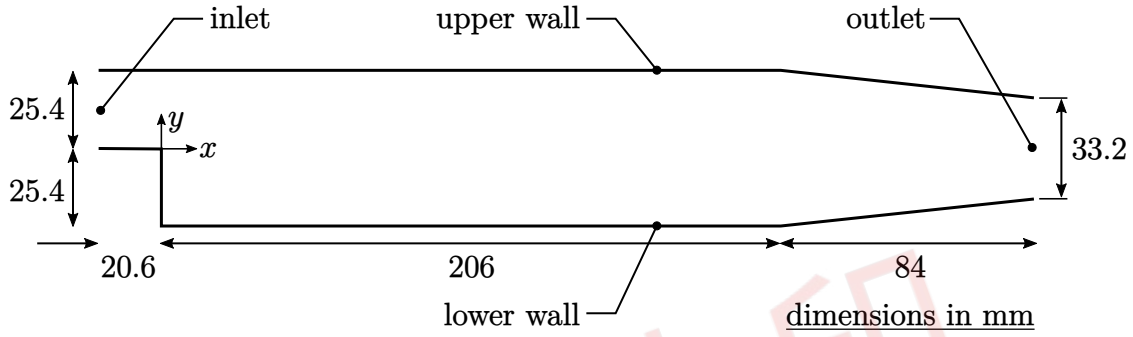


Figure 2.1: Geometry of the backward-facing step.

Flow enters the inlet in the x-direction with a speed of 10 m/s. The flow will be assumed isothermal and incompressible and will be solved using the incompressibleFluid modular solver.

2.1.1 Pre-processing

Cases are configured in OpenFOAM by editing input data files. Users should select a suitable file editor to do this, *e.g.* emacs, vi, gedit, nedit, *etc.* A case involves multiple data files, corresponding to different parts of the configuration, *e.g.* mesh, fields, properties, control parameters, etc. As described in section 4.1, the set of files is stored within a case directory, which is given a suitably descriptive name. This tutorial uses the case *$FOAM_TUTORIALS/incompressibleFluid/pitzDailySteady*, which the user should copy to their *run* directory as follows.

```
cd $FOAM_RUN
cp -r $FOAM_TUTORIALS/modules/incompressibleFluid/pitzDailySteady .
cd pitzDailySteady
```

2.1.2 Mesh generation

OpenFOAM always operates in a three dimensional Cartesian coordinate system and all geometries are generated in three dimensions (3D). It solves the case in two dimensions (2D) by specifying a special empty boundary condition on boundaries normal to the (3rd) dimension (for which no solution is required).

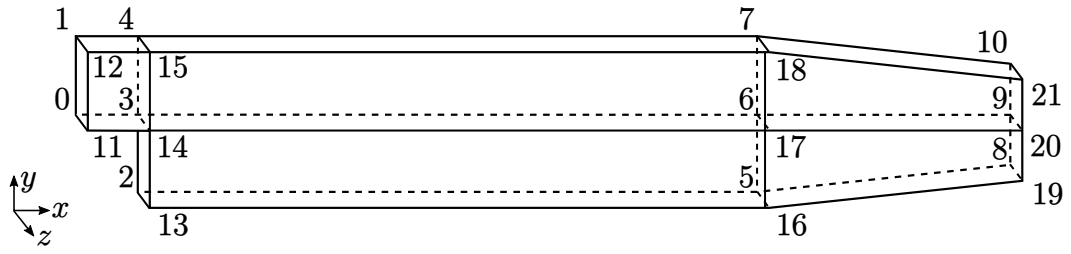


Figure 2.2: Block structure of the mesh for the backward step.

OpenFOAM includes a simple mesh generator, blockMesh, which generates meshes from a *blockMeshDict* file, located in the *system* directory for a given case. The domain is defined using blocks whose vertex locations are specified in the file. The structure of the blocks and respective vertices are shown in Figure 2.2.

The backwardStep domain consists of five blocks shown in the figure. The domain depth in the z direction (exaggerated in the figure) is 1 mm. The *blockMeshDict* file for this example is reproduced below:

```
/*--------------------------------*- C++ -*----------------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  11
     \\/     M anipulation  |
\*---------------------------------------------------------------------------*/
FoamFile
{
    format      ascii;
    class       dictionary;
    object      blockMeshDict;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

// Note: this file is a Copy of $FOAM_TUTORIALS/resources/blockMesh/pitzDaily

convertToMeters 0.001;

vertices
(
    (-20.6 0 -0.5)
    (-20.6 25.4 -0.5)
    (0 -25.4 -0.5)
    (0 0 -0.5)
    (0 25.4 -0.5)
    (206 -25.4 -0.5)
    (206 0 -0.5)
    (206 25.4 -0.5)
    (290 -16.6 -0.5)
    (290 0 -0.5)
    (290 16.6 -0.5)

    (-20.6 0 0.5)
    (-20.6 25.4 0.5)
    (0 -25.4 0.5)
    (0 0 0.5)
    (0 25.4 0.5)
    (206 -25.4 0.5)
    (206 0 0.5)
    (206 25.4 0.5)
    (290 -16.6 0.5)
    (290 0 0.5)
    (290 16.6 0.5)
);

negY
(
    (2 4 1)
    (1 3 0.3)
);

posY
```

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```
(
    (1 4 2)
    (2 3 4)
    (2 4 0.25)
);

posYR
(
    (2 1 1)
    (1 1 0.25)
);


blocks
(
    hex (0 3 4 1 11 14 15 12)
    (18 30 1)
    simpleGrading (0.5 $posY 1)

    hex (2 5 6 3 13 16 17 14)
    (180 27 1)
    edgeGrading (4 4 4 4 $negY 1 1 $negY 1 1 1 1)

    hex (3 6 7 4 14 17 18 15)
    (180 30 1)
    edgeGrading (4 4 4 4 $posY $posYR $posYR $posY 1 1 1 1)

    hex (5 8 9 6 16 19 20 17)
    (25 27 1)
    simpleGrading (2.5 1 1)

    hex (6 9 10 7 17 20 21 18)
    (25 30 1)
    simpleGrading (2.5 $posYR 1)
);

boundary
(
    inlet
    {
        type patch;
        faces
        (
            (0 1 12 11)
        );
    }
    outlet
    {
        type patch;
        faces
        (
            (8 9 20 19)
            (9 10 21 20)
        );
    }
    upperWall
    {
        type wall;
        faces
        (
            (1 4 15 12)
            (4 7 18 15)
            (7 10 21 18)
        );
    }
    lowerWall
    {
        type wall;
        faces
        (
            (0 3 14 11)
            (3 2 13 14)
            (2 5 16 13)
            (5 8 19 16)
        );
    }
    frontAndBack
    {
        type empty;
        faces
        (
            (0 3 4 1)
            (2 5 6 3)
            (3 6 7 4)
```

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```
138             (5 8 9 6)
139             (6 9 10 7)
140             (11 14 15 12)
141             (13 16 17 14)
142             (14 17 18 15)
143             (16 19 20 17)
144             (17 20 21 18)
145         );
146     }
147 );
148
149 // ************************************************************************* //
```

The file first contains header information in the form of a banner (lines 1-7), then file information contained in a *FoamFile* sub-dictionary, delimited by curly braces (`{...}`).

For the remainder of the manual:
To save space, file headers, including the banner and *FoamFile* sub-dictionary, will be removed from further verbatim quoting of case files.

The body of the *blockMeshDict* file will be briefly reviewed here, but for further details see section 5.4. The file begins with the coordinates of the block `vertices`. All vertices are scaled by the factor specified by `convertToMeters`. The file then defines the `blocks` (here, 5 of them). Each block is a hexahedral shape, given by the `hex` entry. The eight vertex labels are listed following the `hex` entry.

The number of cells is specified in each direction for each block by a vector of three integers. For example the first block specifies `(18 30 1)`, which produces 18 cells through the block in the x-direction, 30 in the x-direction and 1 in the z-direction (the unused direction).

The blocks includes mesh grading, which enables the cell lengths to vary across the block. It is includes multi-grading which is described in section 5.4.5 using parameters `negY`, `posY` and `posYR`.

Finally, the mesh splits the boundary into inlet, outlet and wall regions, included in the following patches: `upperWall` and `lowerWall` for the wall boundaries of the domain; `inlet` and `outlet` for the open boundaries. The boundary in the z-normal direction is included in a single patch named `frontAndBack`.

The mesh is generated by running blockMesh on this *blockMeshDict* file. From within the case directory, this is done, simply by typing in the terminal:

```
blockMesh
```

The running status of blockMesh is reported in the terminal window. Any mistakes in the *blockMeshDict* file are picked up by blockMesh and the resulting error message directs the user to the source of the error.

2.1.3 Viewing the mesh

It is sensible to verify the mesh is generated correctly before running the simulation. The mesh can be viewed in ParaView, the post-processing tool supplied with OpenFOAM. The ParaView post-processing is conveniently launched on OpenFOAM case data by executing the paraFoam script from within the case directory.

Any UNIX/Linux executable (application, script, *etc.*) can be run in two ways: as a foreground process, *i.e.* one in which the shell waits until the executable has finished before returning the command prompt; or, as a background process, which allows the shell to accept additional commands while the executable is still running. Since it is

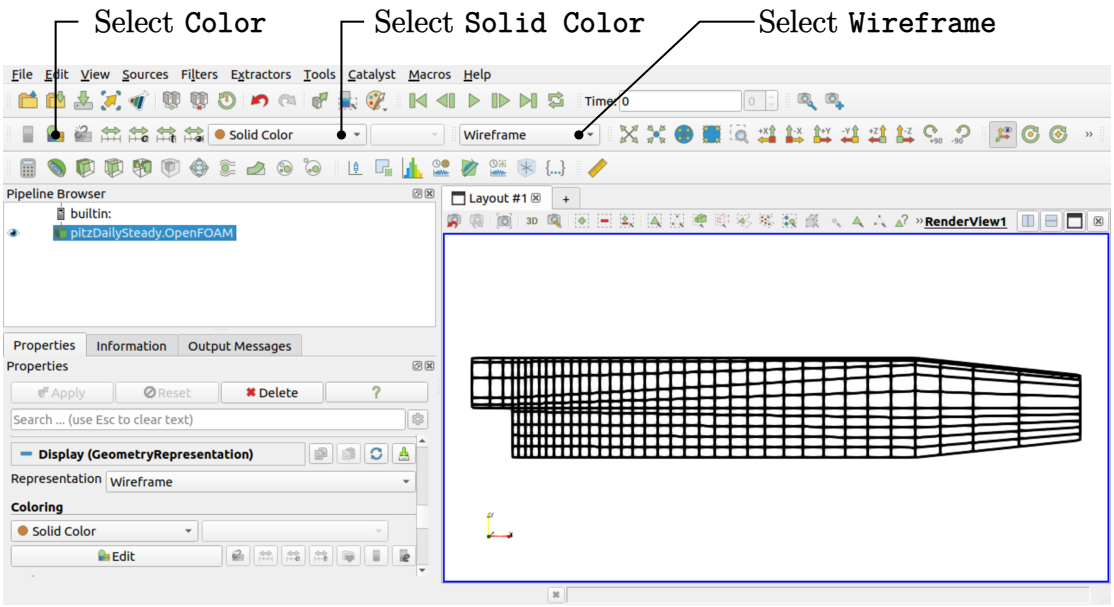


Figure 2.3: Viewing the mesh in ParaView (cell density reduced).

convenient to keep ParaView open while running other commands from the terminal, we will launch it in the background using the `&` operator by typing

```
paraFoam &
```

在工程文件夹内添加一个para.foam后缀文件

This launches the ParaView window as shown in Figure 7.1. In the Pipeline Browser, ParaView registers `pitzDailySteady.OpenFOAM`, representing the pitzDailySteady case.

For the remainder of the manual:

The first time ParaView is launched, users are faced with a splash screen which can be permanently deactivated by clicking the relevant checkbox before closing.

Before clicking the Apply button, the user can select some geometry from the Mesh Parts panel in the Properties window (may require scrolling to find). Because the case is small, it is easiest to select all the data by checking the box adjacent to the Mesh Parts panel title, which automatically checks all individual components within the respective panel. The user should then click the Apply button to load the geometry into ParaView.

The user can control the visual representation of the selected module *either* using the second row of controls at the top of ParaView *or* by scrolling down further to the Display panel that. The user should make the selections using the second row of controls as shown in Figure 2.3, or as described below from within the Display panel.

1. in the `Coloring` section, select `Solid Color`;
2. click Edit (in `Coloring`) and select an appropriate colour *e.g.* black (for a white background);
3. select `Wireframe` from the `Representation` menu.

Especially the first time the user starts ParaView, **it is recommended** that they manipulate the view as described in section 7.1.5. In particular, since this is a 2D case, it is recommended that Camera Parallel Projection is selected at the bottom of the View

(Render View) panel. The selection can be saved as a user default by clicking the Save current view settings button to the right of the View (Render View) heading (the furthest right of the four buttons). The background colour can be also set in the View (Render View) panel at the bottom of the Properties window.

Note that, **many parameters in the** Properties **window are only visible by clicking the** Advanced Properties **gearwheel button** (⚙) at the top of the Properties window, next to the search box.

2.1.4 Boundary and initial conditions

Once the mesh generation is complete, the user can look at the configuration of the initial fields for this case. The case starts at time $t = 0$ s, so the initial field data is stored in a *0* sub-directory of the *cavity* directory. The *0* sub-directory contains several files including *p* and *U*, which represent the pressure (p) and velocity ($\mathbf{U}$) fields, respectfully. Within these files the initial values and boundary conditions must be set. Let us examine the *p* file below.

```
16  dimensions      [0 2 -2 0 0 0 0];
17
18  internalField   uniform 0;
19
20  boundaryField
21  {
22      inlet
23      {
24          type            zeroGradient;
25      }
26
27      outlet
28      {
29          type            fixedValue;
30          value           uniform 0;
31      }
32
33      upperWall
34      {
35          type            zeroGradient;
36      }
37
38      lowerWall
39      {
40          type            zeroGradient;
41      }
42
43      frontAndBack
44      {
45          type            empty;
46      }
47  }
48
49  // ************************************************************************* //
```

There are three principal entries in field data files:

`dimensions` specifies the dimensions of the field, here *kinematic* pressure, *i.e.* $\mathrm{m}^2\,\mathrm{s}^{-2}$ (see section 4.2.6 for more information);

`internalField` the internal field data which can be `uniform`, described by a single value; or `nonuniform`, where the values of the field must be specified for all cells (see section 4.2.8 for more information);

`boundaryField` the boundary field data that includes boundary conditions and data for all the boundary patches (see section 4.2.8 for more information).

For this pitzDailySteady case, the initial fields are set to be uniform. Here the pressure is kinematic, and since the solution does not involve energy and thermodynamics, its absolute value is not relevant, so is set to `uniform 0` for convenience.

The boundary includes the `upperWall`, `lowerWall`, `inlet` and `outlet` patches. The walls and inlet are both assigned the zeroGradient boundary condition for `p`, meaning "the normal gradient of pressure is zero". The outlet uses the fixedValue boundary condition for `p` with `value` of `uniform 0`. The `frontAndBack` patch, describing the front and back planes of the 2D case, is specified as empty.

The user can similarly examine the velocity field in the *0/U* file. The `dimensions` are those expected for velocity, the internal field is initialised as uniform zero, which in the case of velocity must be expressed by 3 vector components, *i.e.* `uniform (0 0 0)` (see section 4.2.5 for more information).

A no-slip condition is assumed on the walls, specified by a noSlip condition. The inlet flow speed is 10 m/s in the x-direction so represented by a fixedValue condition with `value` of `uniform (10 0 0)`. The outlet reverts to the zeroGradient condition. The `frontAndBack` patch must be set to empty.

2.1.5 Physical properties

Physical properties and model configurations for the case are stored in dictionary files in the *constant* directory. Properties for this example are specified in the following *physicalProperties* file.

```
16
17  viscosityModel  constant;
18
19  nu              1e-05;
20
21  // ************************************************************************* //
```

First it includes the `viscosityModel` entry which is set to `constant`. With that model, a single kinematic viscosity is then specified by the keyword `nu`, representing the Greek symbol ν phonetically for the kinematic viscosity. The value is set to $\nu = 1 \times 10^{-5}$ m^2 s^{-1}.

2.1.6 Momentum transport

An estimate of the Reynolds number is required to determine whether the flow is expected to be turbulent. The Reynolds number is defined as:

$$Re = \frac{|\mathbf{U}|L}{\nu} \tag{2.1}$$

where $|\mathbf{U}|$ and L are the characteristic speed and length respectively and ν is the kinematic viscosity. Using the inlet (or step) height $L = 25.4$ mm and $|\mathbf{U}| = 10$ m/s, $Re = 25400$. For flow in a pipe, transition typically occurs when $Re \approx> 2000$, so this case can be assumed to be turbulent.

The *momentumTransport* file characterises the viscous stress in the fluid by a variety of models, *e.g.* Newtonian and non-Newtonian fluids, turbulence, visco-elasticity and more. For this example, the file includes the configuration of turbulence modelling as shown below.

```
16
17  simulationType RAS;
18
19  RAS
20  {
```

```
21      // Tested with kEpsilon, realizableKE, kOmega, kOmega2006, kOmegaSST, v2f,
22      // ShihQuadraticKE, LienCubicKE.
23      model           kEpsilon;
24
25      turbulence      on;
26
27      printCoeffs     on;
28
29      viscosityModel  Newtonian;
30  }
31
32
33  // ************************************************************************* //
```

The type of simulation is first specified by the `simulationType` keyword. The `RAS` entry indicates a Reynolds-averaged simulation, the standard form of turbulence modelling. The `RAS` sub-dictionary includes the `model` entry which is set to the well-known k–ε model by the `kEpsilon` entry. The `turbulence` keyword provides a switch to turn the modelling on and off. The model coefficients have default values which can be overridden with additional entries in the *momentumTransport* file. When the `printCoeffs` switch in on, the coefficients are printed to the terminal when the case is run. The `viscosityModel` entry confirms the fluid is modelled as Newtonian (which is the default model, so the entry could be omitted).

The k–ε model solves transport equations for: k, the turbulent kinetic energy; and, ε, the turbulent dissipation rate. The initial and boundary conditions for those fields are configured in the *0/k* and *0/epsilon* files, respectfully.

In particular, the turbulent fields must be initialised with suitable internal and inlet values. Turbulent kinetic energy k can be calculated by

$$k = \frac{3}{2}\left(|\mathbf{U}|I\right)^2 \tag{2.2}$$

from an estimate of *turbulent intensity* $I = \mathbf{U}'_{\mathrm{RMS}}/|\mathbf{U}|$, the ratio of the root-mean-square (RMS) of turbulent fluctuations $\mathbf{U}'_{\mathrm{RMS}}$ to the mean flow speed $|\mathbf{U}|$. This example uses an estimate $I = 5\%$, such that $k = 1.5 \times (10 \times 0.05)^2 = 0.375\ \mathrm{m^2\,s^{-2}}$. In the *0/k* file, `0.375` is used both for the initial `internalField` and the inlet `value`.

The turbulent dissipation rate ε can be calculated by

$$\varepsilon = C_\mu^{0.75}\frac{k^{1.5}}{l_{\mathrm{m}}}. \tag{2.3}$$

from $C_\mu = 0.09$ and an estimate of Prandtl mixing length l_{m}. This example uses an estimate $l_{\mathrm{m}} = 10\% \times$ step height $= 2.54$ mm, such that $\varepsilon = 0.09^{0.75} \times 0.375^{1.5}/0.00254 = 14.855\ \mathrm{m^2 s^{-3}}$. In the *0/epsilon* file, `14.855` is used both for the initial `internalField` and the inlet `value`.

The turbulence model is deployed with wall functions to model the behaviour at wall boundaries. Wall functions are applied as boundary conditions on the individual wall patches which enables different wall function models to be applied to different wall regions. The choice of wall function models are specified through the turbulent viscosity field, ν_t in the *0/nut* file.

```
16
17  dimensions      [0 2 -1 0 0 0 0];
18
19  internalField   uniform 0;
20
21  boundaryField
22  {
23      inlet
24      {
```

```
25          type            calculated;
26          value           uniform 0;
27      }
28      outlet
29      {
30          type            calculated;
31          value           uniform 0;
32      }
33      upperWall
34      {
35          type            nutkWallFunction;
36          value           uniform 0;
37      }
38      lowerWall
39      {
40          type            nutkWallFunction;
41          value           uniform 0;
42      }
43      frontAndBack
44      {
45          type            empty;
46      }
47  }
48
49
50  // ************************************************************************* //
```

This case uses standard wall functions, specified by the `nutkWallFunction` type on the `upperWall` and `lowerWall` patches. Alternative wall function models include the rough wall functions, specified through the `nutRoughWallFunction` keyword.

When wall functions are specified through boundary conditions in the *0/nut* file, corresponding conditions must be applied to the wall patches for the turbulence fields. The *0/eps* and *0/k* files show that ε is assigned the epsilonWallFunction condition and k is assigned the kqRWallFunction condition at the wall patches. The latter is a generic boundary condition that can be applied to any field that are of a turbulent kinetic energy type, *e.g.* k, q or Reynolds Stress R.

2.1.7 Control

Input data relating to the control of time and reading and writing of the solution data are read in from the *controlDict* file. The user should view this file; as a case control file, it is located in the *system* directory.

```
16
17  application     foamRun;
18
19  solver          incompressibleFluid;
20
21  startFrom       startTime;
22
23  startTime       0;
24
25  stopAt          endTime;
26
27  endTime         2000;
28
29  deltaT          1;
30
31  writeControl    timeStep;
32
33  writeInterval   100;
34
35  purgeWrite      0;
36
37  writeFormat     ascii;
38
39  writePrecision  6;
40
41  writeCompression off;
42
43  timeFormat      general;
44
45  timePrecision   6;
```

```
46
47  runTimeModifiable true;
48
49  cacheTemporaryObjects
50  (
51      kEpsilon:G
52  );
53
54  functions
55  {
56      #includeFunc streamlinesLine
57      (
58          name=streamlines,
59          start=(-0.0205 0.001 0.00001),
60          end=(-0.0205 0.0251 0.00001),
61          nPoints=10,
62          fields=(p k U)
63      )
64
65      #includeFunc writeObjects(kEpsilon:G)
66  }
67
68  // ************************************************************************* //
```

The file first includes an `application` entry which is *not* used by OpenFOAM. It is instead used by run scripts, named Allrun, which are accompany many of the example cases. The following `solver` entry describes the solver module used for the simulation. This example uses the incompressibleFluid module for steady or transient turbulent flow of incompressible isothermal fluids with optional mesh motion and change.

The start/stop times and the time step for the run must be set. OpenFOAM provides flexible options for time controls which are described in section 4.4. Like most cases, this example starts the simulation at time $t = 0$ which instructs the solver to read its field data from a directory named *0*. Therefore we set the `startFrom` keyword to `startTime` and then specify the `startTime` keyword to be `0`.

The aim of the simulation is to reach the steady solution where the recirculation region is fully developed adjacent to the step. The incompressibleFluid module can run as a steady-state solver by setting the time derivatives $\partial/\partial t$ to zero in all the equations. This is achieved by setting the `ddtSchemes` to `steadyState` in the *fvSchemes* file, discussed later.

In this mode, the time step, represented by the keyword `deltaT`, is only used as a time increment (since it is no longer used to discretise $\partial/\partial t$). Its value does not affect the solution, so for steady solutions it is set to `1` so that time simply represents the number of solution steps.

The `endTime` keyword sets a time at which the solver application stops running. The value of `2000` provides an adequate number of solution steps to enable the steady solution to converge to a reasonable level of accuracy.

As the simulation progresses we wish to write results at intervals of time of interest for visualisation and other post-processing. The `writeControl` keyword presents several options for setting the time at which the results are written; here the `timeStep` option specifies that results are written every nth time step where the value n is specified under the `writeInterval` keyword. The interval of `100` means results are written at 100, 200, *etc.*

OpenFOAM creates a new directory *named after the current time*, *e.g.* 100, on each occasion that it writes a set of data, as discussed in full in section 4.1. It writes out the results for each solution field, *e.g.* `U`, `p`, `k`, into the time directories.

2.1.8 Discretisation and linear-solver settings

The user specifies the choice of finite volume discretisation schemes in the *fvSchemes* file in the *system* directory. The specification of the linear equation solvers and tolerances and other algorithm controls is made in the *fvSolution* file, also in the *system* directory.

The details of those two files are described in Sections 4.5 and 4.6, respectively. For this example, the following points are important:

- the `ddtSchemes` defaults to `steadyState` in *fvSchemes*, to invoke a steady-state calculation;
- steady-state solution uses an algorithm based on SIMPLE whose controls are configured in a `SIMPLE` sub-dictionary in the *fvSolution* file;
- the `SIMPLE` sub-dictionary contains the `consistent` switch which is set to `yes`, applying the "consistent" form of the SIMPLE algorithm (SIMPLEC);
- the convergence of SIMPLEC is very sensitive to the `relaxationFactors` in the *fvSolution* file; the values of `0.9` are carefully tuned and are not suitable for the standard SIMPLE algorithm;
- SIMPLEC's sensitive convergence generally makes it only reliable for cases with simple geometries.

2.1.9 Running an application

To run a simulation with a single domain region, the foamRun application is run. It loads the relevant solver module, from the `solver` entry in the *controlDict* file, to perform the calculation. On this occasion, we will run foamRun in the terminal foreground by typing the following from within the case directory.

```
foamRun
```

The progress of the simulation is reported in the terminal window. It describes successive solutions steps, giving the initial and final residuals for each equation, and conservation errors.

The `SIMPLE` algorithm controls in *fvSolution* include a `residualControl` sub-dictionary with a set of tolerances for `p`, `U` and turbulence fields. The incompressibleFluid solver terminates When the initial residuals for **all** fields fall below their respective tolerances. In this example, the solver terminates at 287 iterations with the following statement.

```
SIMPLE solution converged in 287 iterations
```

Results from the simulation are written into time directories within the *pitzDailySteady* case directory. The user can list the directory contents with the "`ls`" command to see the time directories (`100`, `200` and `287`) containing the results.

2.1.10 Time selection in ParaView

Once the results are written to time directories, they can be viewed using ParaView. The first step is to activate the time selector, including the Time text box on right hand side of the top row of buttons, as shown in Figure 7.4.

If ParaView is opened *before there are any solution time directories* (*i.e.* only a *0* directory), the time selector must later be re-activated to recognise the solution time directories by:

- selecting the **top** of the Properties window (scroll up the panel if necessary) in ParaView;
- toggling the Cache Mesh button at the top of the panel (under the Refresh Times button;
- clicking the Apply button.

The time selector is then updated with Time becoming a drop down menu with the time directories from the case (0, 100, 200 and 287). In order to view the solution at 287, the user can use the `VCR Controls` or `Current Time Controls` to change the current time to 287.

For the remainder of the manual:

In ParaView, if the window panels, *e.g.* Properties, do not contain the expected entries, ensure that the relevant module is highlighted in blue in the Pipeline Browser. For example, Refresh Times only appears when the top module (here, `pitzDailySteady.OpenFOAM`) is selected.

2.1.11 Colouring surfaces

To view pressure, the user can *either* make selections from the second row of buttons at the top of ParaView as shown in Figure 2.4 *or* scroll down to the Display panel in Properties and make the following selections:

1. select `Surface` from the `Representation` menu;
2. select p in `Coloring`
3. click the Rescale button to set the colour scale to the data range, if necessary.

The pressure field should appear as shown above, with the pressure increasing to a maximum at the contraction of the channel towards the outlet. With the point icon (p), the pressure field is interpolated across each cell to give a continuous appearance. Instead if the user selects the cell icon, (p), from the `Coloring` menu, a single value for pressure will be attributed to each cell, represented by a single colour with no grading.

A colour legend is included which can be disabled by clicking the Toggle Color Legend Visibility button at the left of the second row of buttons at the top of ParaView. These buttons are part of the `Active Variable Controls` toolbar, shown in Figure 7.4). The Edit Color Map button, second on the left in `Active Variable Controls` toolbar, opens the Color Map Editor window, as shown in Figure 2.5, where the user can set a range of attributes of the colour scale and the color bar.

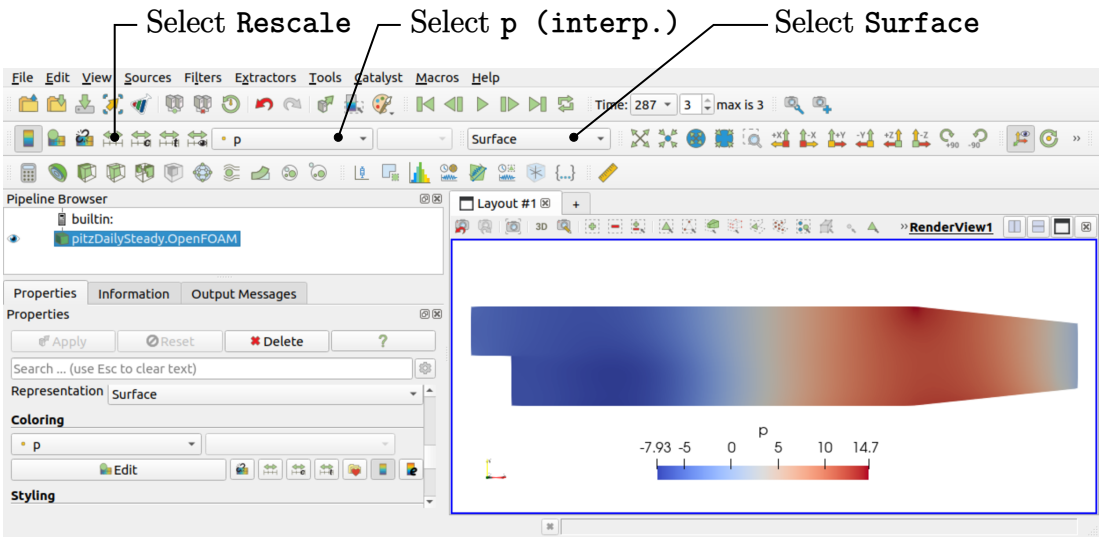


Figure 2.4: Displaying pressure contours for the backward step case.

In particular, ParaView defaults to using a colour scale of blue to white to red rather than the more common blue to green to red (rainbow). Therefore *the first time* that the user executes ParaView, they may wish to change the colour scale. This can be done by selecting the Choose Preset button (with the heart icon) in the Color Scale Editor. The conventional color scale for CFD is Blue to Red Rainbow which is only listed if the user types the name in the Search bar or checks the gearwheel to the right of that bar.

After selecting Blue to Red Rainbow and clicking Apply and Close, the user can click the Save as Default button at the absolute bottom of the panel (file save symbol) so that ParaView will always adopt this type of colour bar.

The user can also edit the color legend properties, such as text size, font selection and numbering format for the scale, by clicking the Edit Color Legend Properties to the far right of the search bar, as shown in Figure 2.5.

2.1.12 Cutting plane (slice)

If the user rotates the image, by holding down the left mouse button in the image window and moving the cursor, they can see that they have now coloured the complete geometry surface by the pressure. In order to produce a 2D contour plot the user should first create a cutting plane, or 'slice'. With the `pitzDailySteady.OpenFOAM` module highlighted in the Pipeline Browser, the user should select the `Slice` filter from the `Filters` menu in the top menu of ParaView (accessible at the top of the screen on some systems). The `Slice` filter can be found in the `Common` sub-menu or among the `Common and Data Analysis` buttons, the third row of buttons at the top of ParaView (see Figure 7.4).

Selecting the `Slice` filter creates a new item in the Pipeline Browser. In the Properties window, the cutting plane should have an origin at a point on the z-axis, *e.g.* $(0, 0, 0)$ and its normal should be set to $(0, 0, 1)$ (click the Z Normal button).

When Apply is clicked, the slice appears in the RenderView window, while the original `pitzDailySteady.OpenFOAM` module disappears. The visibility of each module is enabled and disabled by the eye button to the left of each module in the pipeline browser.

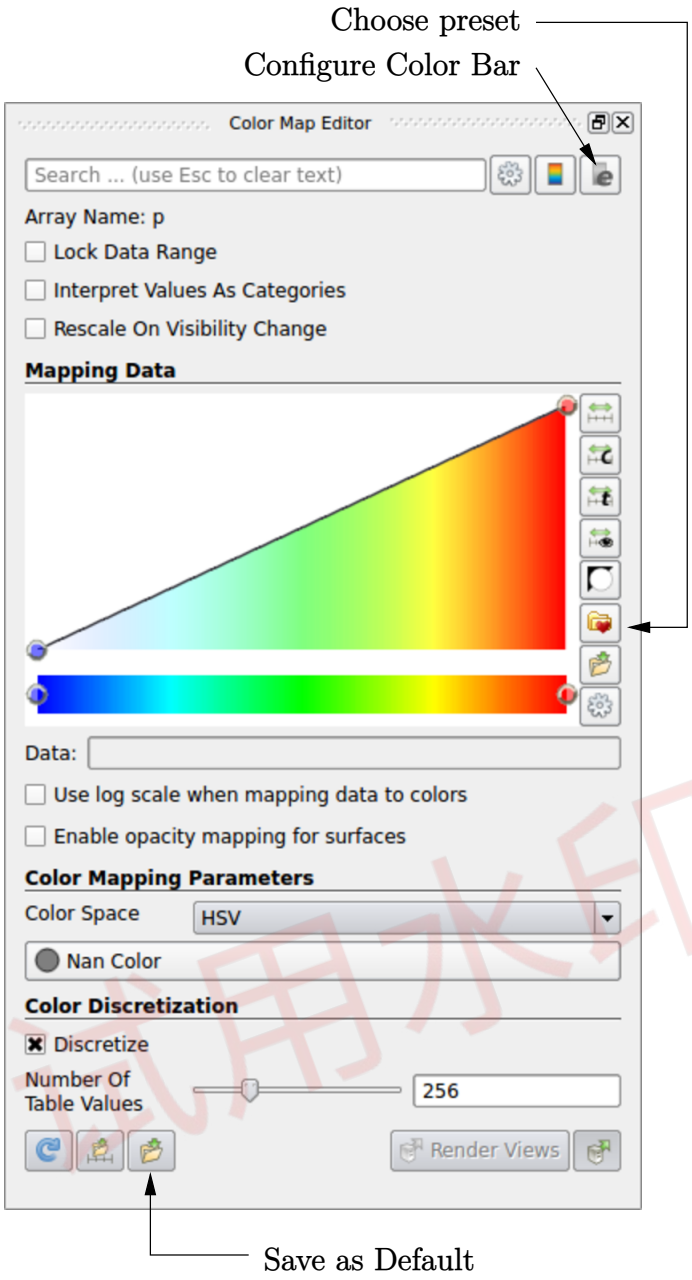


Figure 2.5: Color Map Editor.

For the remainder of the manual:

In ParaView, if items do not appear to be displayed in the RenderView window, ensure the relevant module is visible by switching on the eye button in the Pipeline Browser.

2.1.13 Vector plots

Before drawing vectors of the flow velocity, turn off the display of the `Slice` module by highlighting it in the Pipeline Browser and clicking the eye button to the left of it. The aim is to generate a vector glyph for velocity at the *centre of each cell*. We therefore first need to filter the cell centres from the mesh geometry as described in section 7.1.7. With the `pitzDailySteady.OpenFOAM` module highlighted in the Pipeline Browser, the user should select `Cell Centers` from the `Filters->Alphabetical` menu and then click Apply.

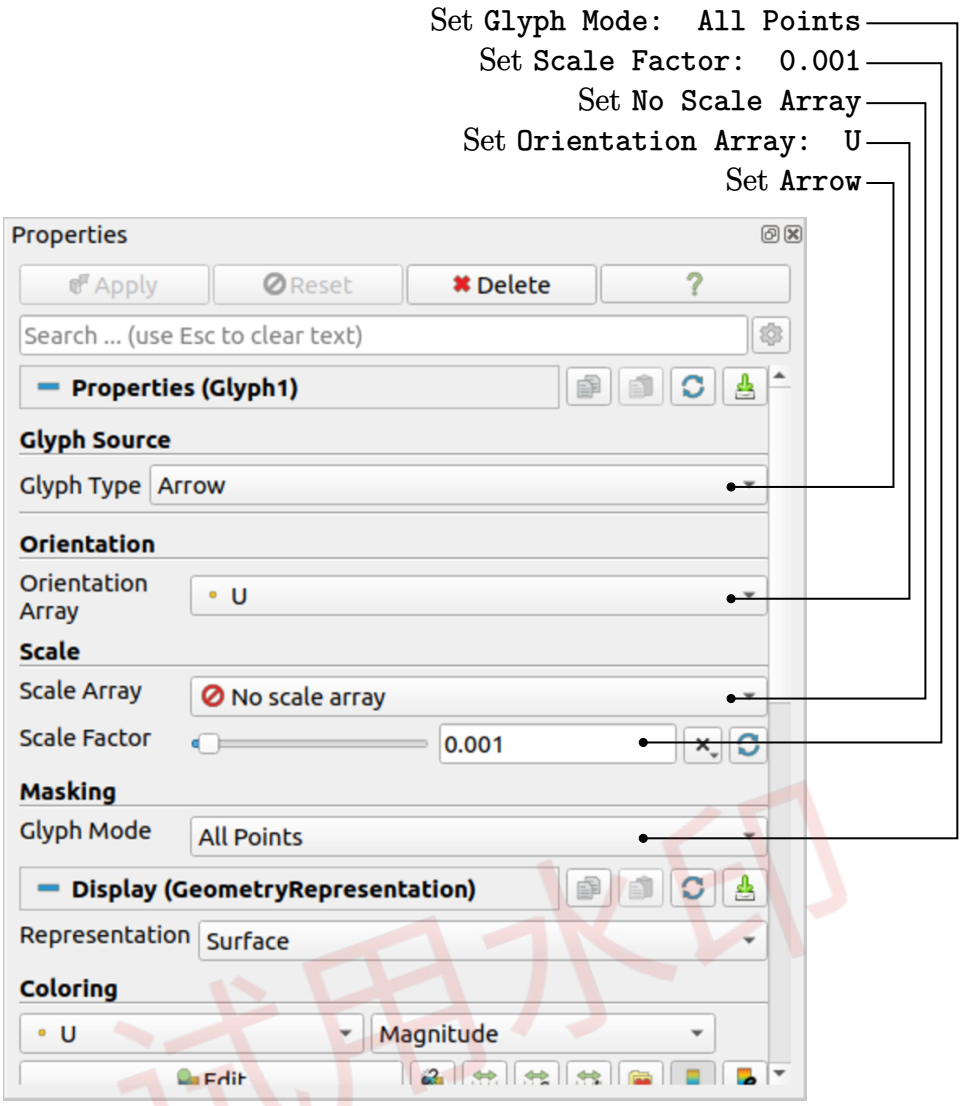


Figure 2.6: Properties panel for the `Glyph` filter.

With the `Centers` highlighted in the Pipeline Browser, the user should then select `Glyph` from the `Filters->Common` menu (or the third row of buttons). The Properties window panel should appear as shown in Figure 2.6.

When displaying velocity vectors, there are four principal settings required for the glyphs:

- the glyph type, set to `Arrow`;
- the arrow direction, set by `Orientation Array`;
- the arrow lengths, set by `Scale Array` and Scale Factor;
- the number of arrows, set by `Glyph Mode`.

On clicking Apply, the glyphs appear as a single colour, *e.g.* white. The user should colour the glyphs by velocity magnitude which, as usual, is controlled by setting `U` in the drop down menu towards the left of the second row of buttons.

The Legend is also displayed. The user can configure the legend by clicking `Edit Color Map` (second button from left, second row). This opens the the Color Map Editor

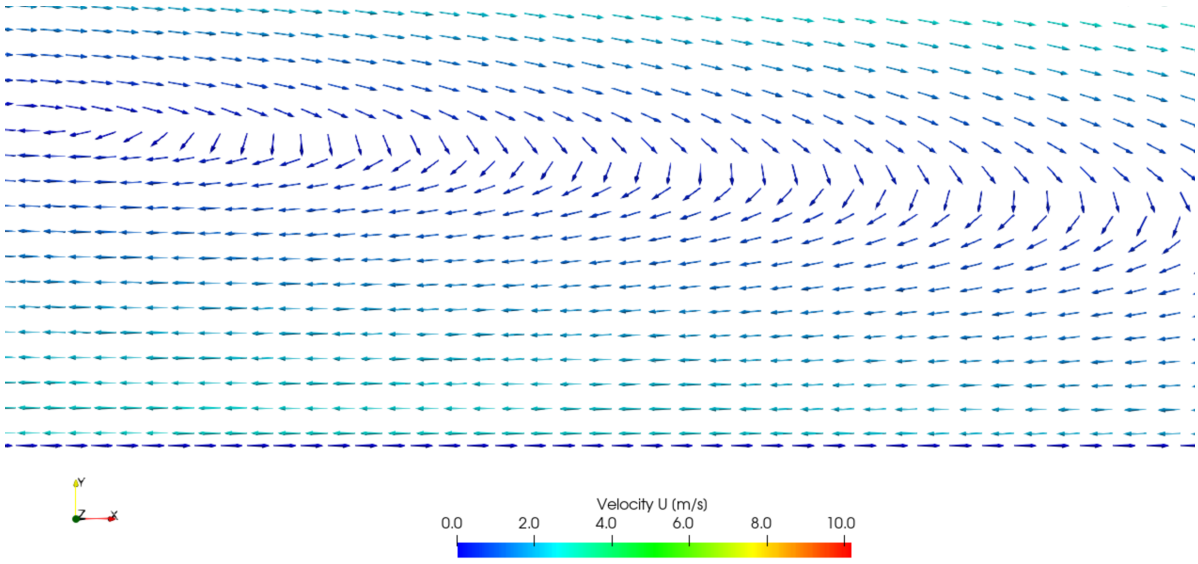


Figure 2.7: Velocities in the backward facing step.

window. The legend can be configured by clicking by the button furthest to the right of the search box. This button opens the Edit Color Legend Properties window. The Advanced Properties gearwheel button should be checked to the right of the search box.

Titles and labels can be fully configured. For Figure 2.7, the legend title is set to `Velocity U [m/s]`. The labels are specified to 1 fixed decimal place by unchecking Automatic Label Format and entering `%-#6.1f` in the Label Format text box. Add Range Labels is also unchecked.

Sample output is shown in Figure 2.7, zooming in on part of the recirculation region downstream of the step. At the lower wall, glyphs point in a direction opposing that of the flow adjacent to the wall. This glitch is caused by these glyphs being drawn at the face centres of the wall boundary patch, where the the velocity magnitude is 0 due to the no-slip condition. Without any direction to the vectors, ParaView orientates the arrows in a default x-direction. A quick way to remove these vectors is:

- go back to the `pitzDailySteady.OpenFOAM` module at the top of the Pipeline Browser;
- in the Mesh Parts panel, uncheck the lowerWall patch;
- click Apply.

2.1.14 Popular filters in ParaView

ParaView includes over 200 filters which can be listed by the `Filters->Alphabetical` menu. Only a small fraction, *e.g.* 10-15, of these filters are relevant for CFD, which we suggest adding to the `Filters->Favourites` menu. To do this, select `Manage Favourites` from the `Filters->Favourites` menu. Search for the following important filters and click `Add>>` to add them to the `Favourites` menu:

- `Extract Block`, to select components of the domain, *e.g.* boundary patches and internal cells;
- `Slice`, to insert a plane through the geometry;

- Cell Centers and Glyph, principally to draw velocity vectors;
- Stream Tracer and Tube, to draw streamlines;
- Contour, to draw contour lines (on surfaces) and iso-surfaces;
- Feature Edges, to capture features on a surface for better image definition.

2.1.15 Contours

Before drawing contour lines of the flow speed, turn off the display of the Glyph module by highlighting it in the Pipeline Browser and clicking the eye button to its left. The user should then highlight the Slice module and colour by velocity U. We will then aim to draw contour lines on the slice at intervals of U of 1, 2, ..., 10.

The contour lines can be drawn by applying the Contour filter to the slice. The filter draws lines in 2D, or surfaces in 3D, along constant values of *scalar* quantities. Contours cannot be drawn directly from U, since it is a *vector*, so we first need to generate a scalar field of the magnitude of U.

The mag(U) field can be generated using post-processing with function objects, described in section 7.3. The user should run the foamPostProcess utility, calling the mag function object using the -func option as follows:

```
foamPostProcess -func "mag(U)"
```

The utility loops over all time directories. For each time directory, it reads in U, calculates mag(U) and writes it out as a field file back into the time directory. The mag(U) field must then be loaded into ParaView by: first, from the pitzDailySteady-.OpenFOAM in the Pipeline Browser, clicking Refresh Times; then, scrolling down to the Fields panel, selecting mag(U) and clicking Apply.

The user should re-select the Slice module in the Pipeline Browser, then apply the Contour filter. In the Properties panel, the user should select mag(U) from the Contour By menu. Under Isosurfaces, the user should first delete the default value by clicking the – button, then add a range of 10 values as shown in Figure 2.8. The contours can be displayed with a Wireframe representation with solid black Coloring.

2.1.16 Streamline plots

Before drawing streamlines, the user should turn off the display of the Contour module. To display streamlines for this 2D example, the user should first highlight the Slice module in the Pipeline Browser and then apply the Stream Tracer filter.

A new StreamTracer module opens which is configured through its Properties window. Tracer is created by tracking lines in the direction of flow, starting from *seed points*. With Integration Direction BOTH, lines are tracked both upstream and downstream of the seed points. The user should scroll down the Properties window to configure the Seeds. The default Seed Type in Line which seeds points along a line drawn between specified points. In the Line Parameters, the user should can set the two points to $(0, -0.025, 0)$ to $(0.2, 0.0.25, 0)$. The Resolution specifies the number of seed points distributed along the line, which should be reduced to 25. On clicking Apply the tracer is generated as shown in Figure 2.9. The user can experiment with the line points and resolution to produce different stream tracer output.

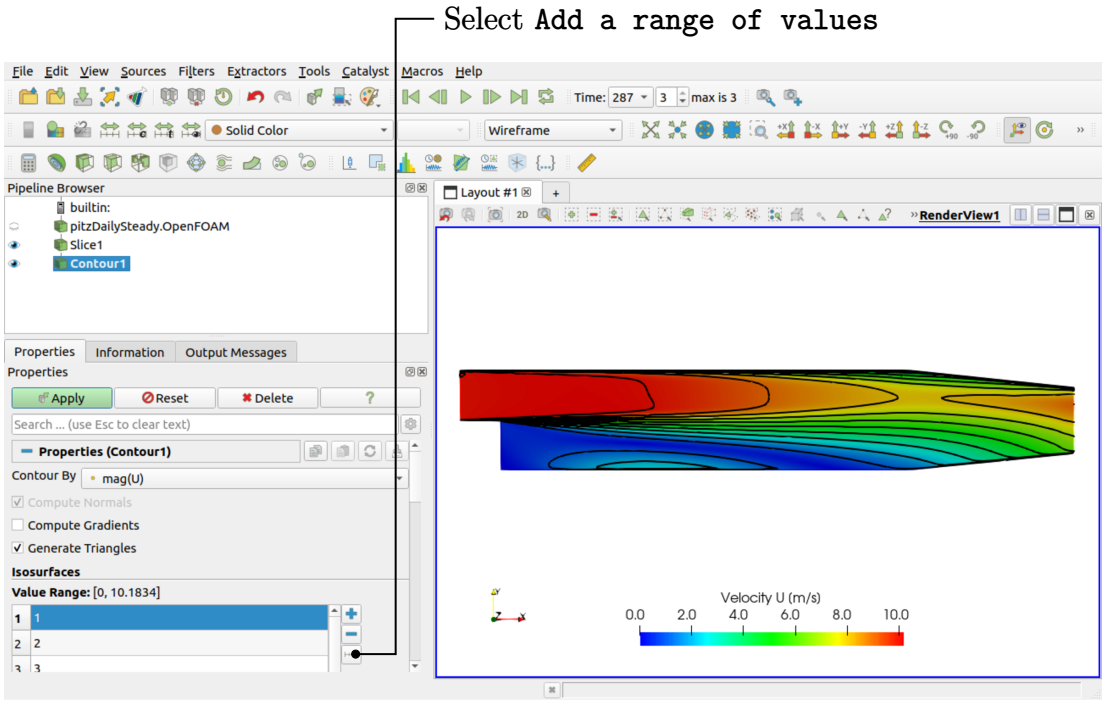


Figure 2.8: Contours in the backward facing step.

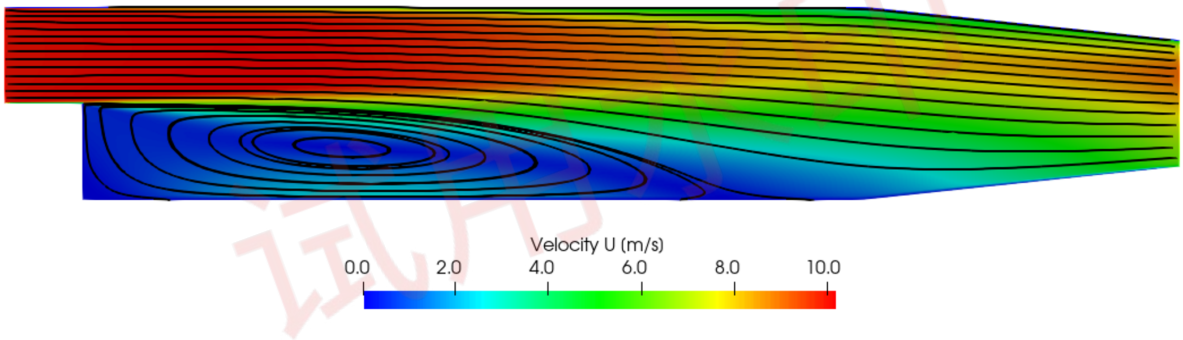


Figure 2.9: Streamlines in the backward facing step.

2.1.17 Inlet boundary condition

The user should examine the flow at the inlet boundary of the domain. The velocity condition is specified in *0/U* as `fixedValue` with a `value` of `(10 0 0)`. This value is applied to *all* faces of the inlet boundary patch. The inlet is adjacent to wall boundaries where the `noSlip` condition is applied, giving rise to a sudden change in `U`, between adjacent boundary faces.

The user should zoom in around the inlet region of the geometry using the right button of the mouse. The `Slice` module should be activated in the Pipeline Browser and coloured by *cell values* of `p` (p selection). In order to highlight the variation in pressure in cells close to the inlet, the user should apply a custom range of $-5 < p < -1$ as shown in Figure 2.10. A custom range is applied by clicking the Rescale to Custom Data Range button ( ), located fifth from the left of the second row of buttons. A panel opens, in which the minimum and maximum values of the range should be entered before clicking the Rescale button. The velocity profile can be shown by returning to the `Glyph` module in the Pipeline Browser and making it visible. The profile can be illustrated by scaling

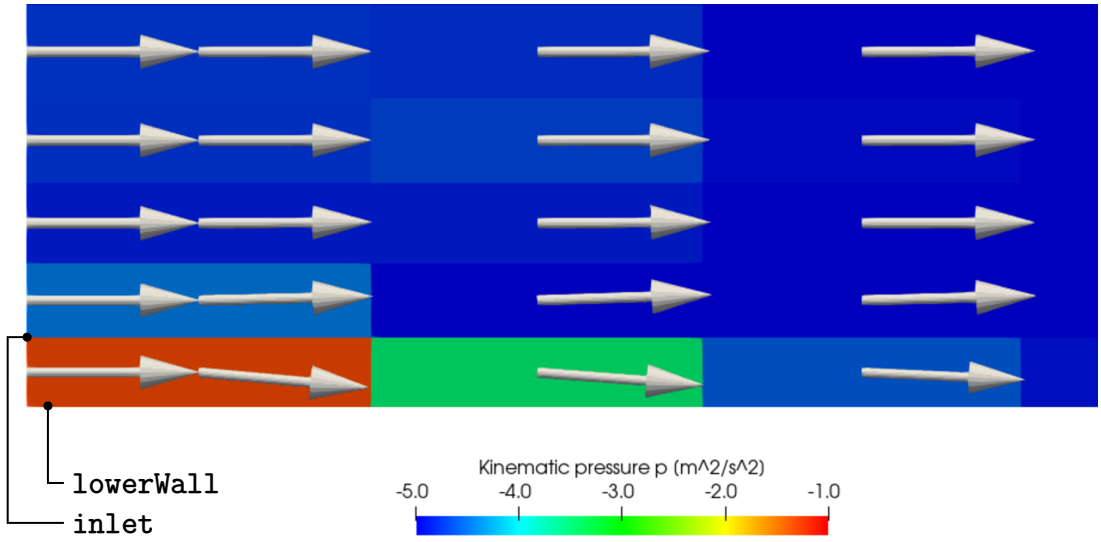


Figure 2.10: Uniform flow at the inlet in the backward facing step.

the arrows by the the flow speed. In the Properties of the `Glyph`, scroll down to the Scale panel and set `Scale Array` to `U`, with `Scale By Magnitude` and a `Scale Factor` of `8e-05`. The vectors are assigned a `Solid Color` of white.

The figure shows the inlet region adjacent to the lower wall boundary. At the left of the image, the vectors show a uniform profile. Shear at the wall causes the flow to decelerate, starting in the near-wall cell. The deceleration causes an increase in pressure, which produces a driving gradient that redirects the flow slightly away from the wall, to obey mass conservation.

In order to reduce the pressure increase, the boundary condition can be modified so that the inlet velocity is no longer uniform. A flowRateInletVelocity boundary condition is a general boundary condition for `U`, specifying flow at an inlet. Documentation for this boundary condition can be viewed using the foamInfo script, a general tool which provides documentation for applications, models and tools in OpenFOAM. It is run for the flowRateInletVelocity boundary condition as follows (noting that the name can sometimes be abbreviated for simplicity, here `flowRateInlet`, rather than `flowRateInletVelocity`).

```
foamInfo flowRateInlet
```

It locates and prints the header file of the related code and extracts the `Description` and `Usage` information from the file. It then identifies associated models, *i.e.* other boundary conditions in this case, and lists example cases that use the model. The foamInfo script is not perfect, but provides useful information quickly in at least nine times out of ten.

The documentation explains that the flowRateInletVelocity condition can specify the flow by a `massFlowRate`, `volumetricFlowRate` or `meanVelocity`. The user should open the *0/U* in their editor and locate the boundary field entry for the `inlet` patch. The condition with a `meanVelocity` can then be applied by changing the `inlet` sub-dictionary as follows:

```
inlet
{
    type            flowRateInletVelocity; // modify
```

```
    meanVelocity  10;                    // insert
    value         uniform (10 0 0);      // leave
}
```

The condition evaluates the velocity on the boundary, setting the `value` for all faces on the boundary patch. The `value` entry is therefore redundant for OpenFOAM, but it can be needed by ParaView to display initial values of initialised fields at patches. Therefore, **the `value` entry is retained** for ParaView's benefit.

After saving the *0/U* file, the user can re-run the simulation to check first that the flowRateInletVelocity emulates the original fixedValue condition. The *controlDict* file specifies that case starts from time `0` and it will overwrite previous results in time directories. The user can test the new condition simply by re-running the foamRun solver.

```
foamRun
```

The solver runs as before, terminating at 287 iterations. The user can return to ParaView and click the Refresh Times button in the `pitzDailySteady.OpenFOAM` module of the Pipeline Browser. There is no change to the results, demonstrating that the flowRateInletVelocity is setting a uniform `value` of `(10 0 0)` at this stage.

The user should now modify the flowRateInletVelocity condition in the *0/U* file by including a `profile` for the velocity. The documentation highlights two customised profile function, `turbulentBL` and `laminarBL`, which provide *power-law* and *quadtratic* profiles for fully-developed turbulent and laminar boundary layers, respectively. In this example, add the `turbulentBL` profile to the boundary condition as follows:

```
inlet
{
    type          flowRateInletVelocity;
    meanVelocity  10;
    profile       turbulentBL; // add
    value         uniform (10 0 0);
}
```

Before running the simulation again, it it recommended to delete the previous solution time directories. The results should be deleted now because the solver will likely terminate at a different time, so the results from the final time directory from the old case will not be overwritten, potentially causing confusion.

The foamListTimes utility provides a quick, simple way to delete solution time directories, *i.e.* retaining the `0` directory. First run foamListTimes in the terminal as follows:

```
foamListTimes
```

This returns the list of the solution time directories, `100`, `200` and `287`, but not `0`. The listed directories can be deleted by including the `-rm` remove option to foamListTimes, *i.e.* running the following command.

```
foamListTimes -rm
```

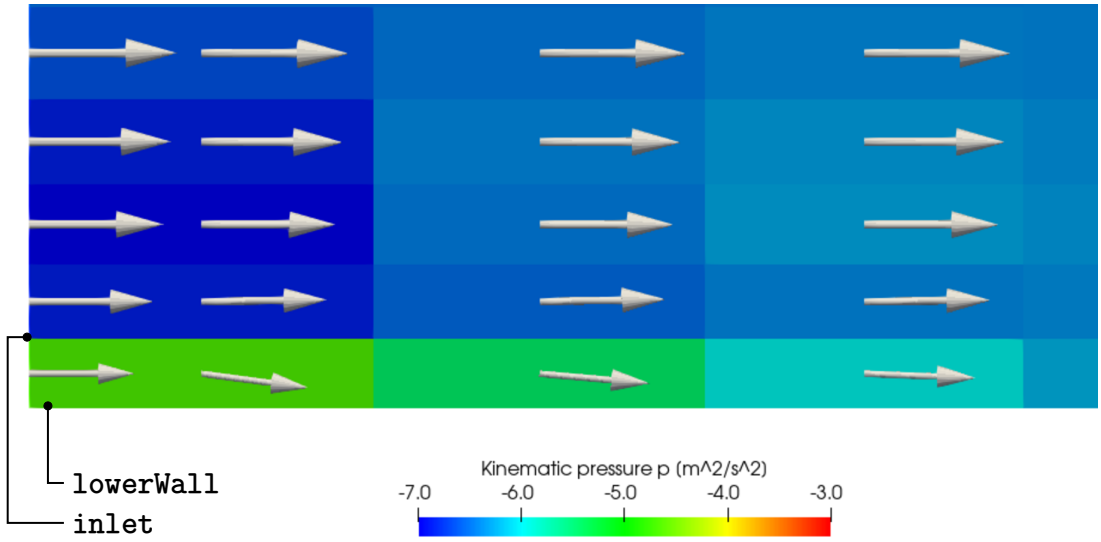


Figure 2.11: Turbulent boundary layer at the inlet in the backward facing step.

With the time directories containing the results now deleted, the foamRun solver should be run as before. This time foamRun terminates at 277 iterations. The user should return to ParaView and click the Refresh Times button in the `pitzDailySteady.OpenFOAM` module of the Pipeline Browser.

The change in output times may create confusion for ParaView, causing it to query times in the time selector with `(?)` symbols. The selector entries can be rebuilt by reverting to time `0` or clicking Cache Mesh and Apply. The user can then select the last time in the sequence (277). The velocity profile and pressure are updated as shown in Figure 2.11. For pressure, the custom range is moved to $-7 < p < -3$. The velocity is no longer uniform at the inlet, but forms a profile according to the `turbulentBL` function. The velocity magnitude at the face adjacent to the lower wall is significantly reduced from previously so the deceleration due to shear is also reduced in the near-wall cell. The pressure difference between the left corner and surrounding cells is consequently not as high, approximately $+2$ $m^2\,s^{-2}$ (from -7 to -5) compared to $+4$ $m^2\,s^{-2}$ (from -5 to -1) previously.

2.1.18 Turbulence model

The backward step case is set up to allow users to try out different turbulence models quickly. There is comment in the *momentumTransport* file in the *constant* directory, listing different turbulence models tested on this case. Some of the models solve equations for fields other than k and ε, *e.g.* ω (`omega`). To minimise the work when changing models, files for these other fields are already included in the *0* directory. Similarly, entries for schemes and solvers for these fields are included in the *fvSchemes* and *fvSolution* files, respectively (in the *system* directory).

The user should open the *momentumTransport* file in their editor to change the turbulence model. They can change the model to realizable k–ε by the following setting.

```
model      realizableKE;
```

The simulation can now be re-run using this model by deleting the solution time directories and executing foamRun as follows.

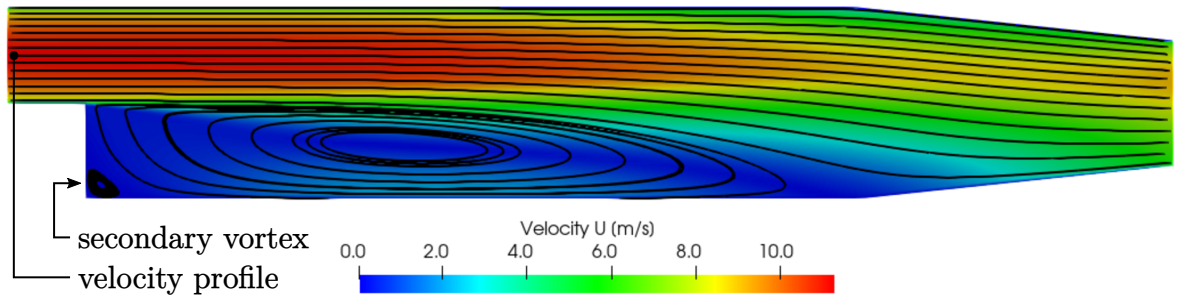


Figure 2.12: Streamlines with the realizable k–ε model.

```
foamListTimes -rm
foamRun
```

The solver terminates this time at 255 iterations. The user can now return to ParaView and click Refresh Times to view the results at time 255. The results are shown in Figure 2.12 using the slice with the velocity field and the streamlines filters configured earlier.

The realizable k–ε model is less diffusive that the standard k–ε model. It captures a secondary vortex at the base of the step which is approximately half the step height. The recirculation region is also longer, with reattachment occurring at the point the lower wall begins to taper towards the outlet.

The user can also test other turbulence models. The k–ω SST model is a popular choice in industrial CFD which can be selected by the following setting in the *momentumTransport* file.

```
model       kOmegaSST;
```

It requires initialisation of specific turbulent dissipation rate ω. Item can be calculated similarly to ε in Equation 2.3 using l_m as follows:

$$\omega = C_\mu^{-0.25} \frac{k^{0.5}}{l_\mathrm{m}}. \tag{2.4}$$

Using the estimate $l_\mathrm{m} = 2.54$ mm as before, $\omega = 0.09^{-0.25} \times 0.375^{0.5}/0.00254 = 440.2\ \mathrm{s}^{--1}$. In the *0/omega* file, `440.2` is used both for the initial `internalField` and the inlet `value`.

The simulation can now be re-run using this model by deleting the solution time directories and executing foamRun as before. The solver does not terminate early by converging to within the tolerances specified in the `residualControl` sub-dictionary of the *fvSolution* file. Instead, it terminates at 2000 iterations, the `endTime` specified in the *controlDict* file.

The results with the k–ω SST model show a larger secondary vortex at the base of the step. The vortex does not stabilise to a steady-state, but instead oscillates a small amount over successive solution steps. The oscillations can be seen by examining the velocity vectors at different solution steps, *e.g.* `1000`, `1100`, `1200`, *etc.* Figure 2.13 shows the extent of the secondary vortex and indicates where vectors oscillate around the reattachment point of the vortex.

The secondary vortex can be stabilised by a change to the numerical scheme for momentum advection. The user should open the *fvSchemes* file from the *system* directory. The discretisation of the advection terms is specified by the keyword entries of the

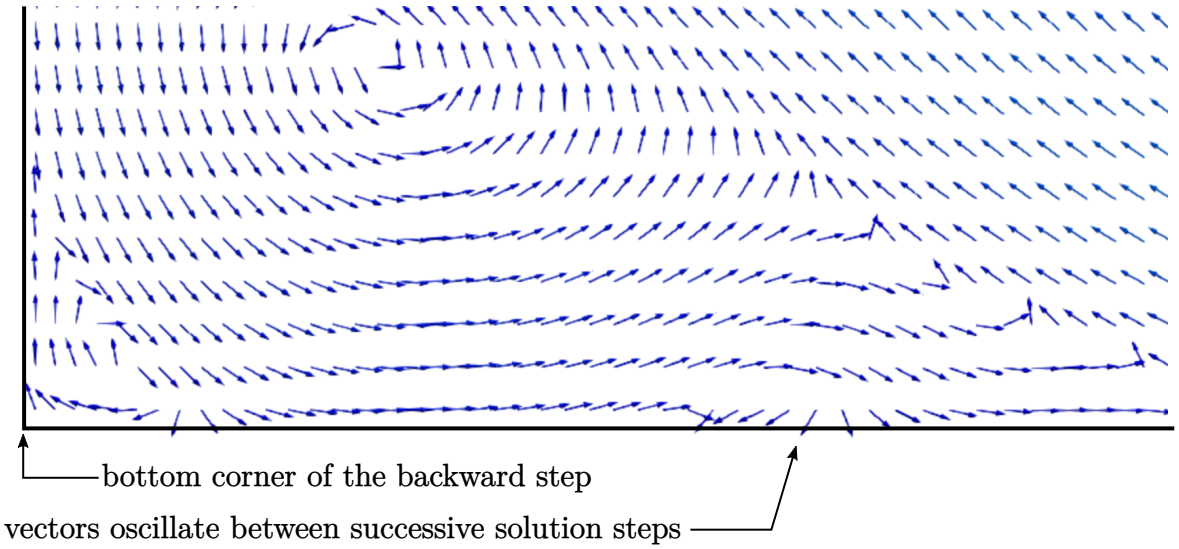


Figure 2.13: Secondary vortex with k–ω SST model.

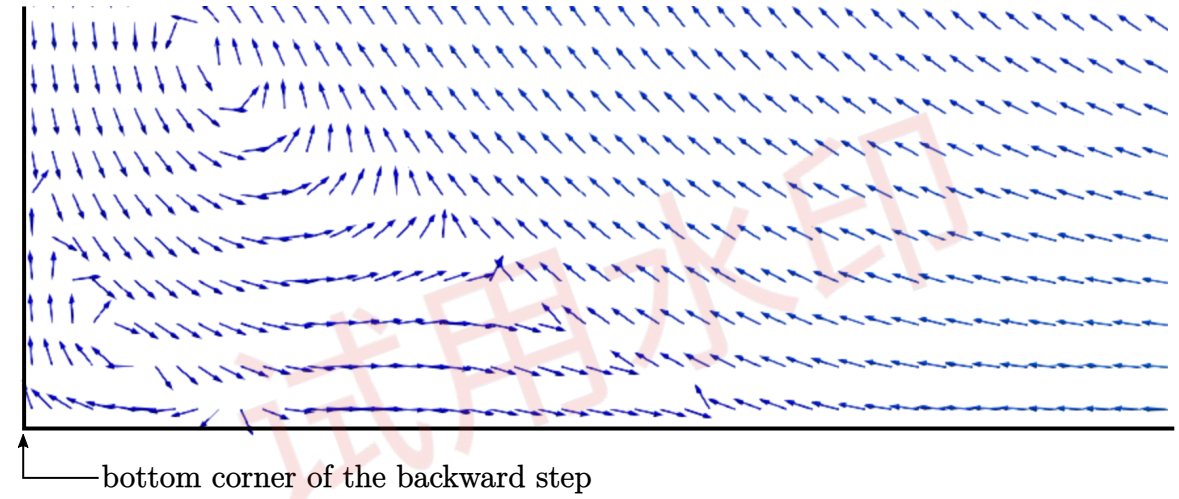


Figure 2.14: Secondary vortex with k–ω SST model with limiting on `grad(U)`.

form "`div(phi,...)`" in the `divSchemes` sub-dictionary. Momentum advection uses the `linearUpwind` scheme as shown below.

```
div(phi,U)      bounded Gauss linearUpwind grad(U);
```

The `linearUpwind` scheme interpolates fields from cell centres to faces by extrapolation using the cell gradient. The `grad(U)` entry specifies the form of the gradient calculation, using the scheme specified in the `gradSchemes` sub-dictionary. In the example case, it includes only a `default` scheme for calculating all gradient terms in all equations.

In order to stabilise the secondary vortex, the `cellLimited` scheme can be applied specifically to the discretisation of the velocity gradient `grad(U)`. The syntax is shown below.

```
gradSchemes
{
    default         Gauss linear;
```

```
        grad(U)         cellLimited Gauss linear 1;
    }
```

After saving the *fvSchemes* file, the simulation can be re-run by deleting the solution time directories and executing foamRun as before. With the `cellLimited` scheme applied, the solution converges normally, with the solver terminating at 285 iterations. The secondary vortex stabilises as shown in Figure 2.14.

2.2 Breaking of a dam

In this example we shall solve a problem of a simplified dam break in 2 dimensions using the incompressibleVoF modular solver. The feature of the problem is a transient flow of two fluids separated by a sharp interface, or free surface. The two-phase algorithm in incompressibleVoF is based on the volume of fluid (VoF) method in which a phase transport equation is used to determine the relative volume fraction of the two phases, or phase fraction α, in each computational cell. Physical properties are calculated as weighted averages based on this fraction. The nature of the VoF method means that an interface between the phases is not explicitly computed, but rather emerges as a property of the phase fraction field. Since the phase fraction can have any value between 0 and 1, the interface is never precisely defined, but occupies a volume around the region where a sharp interface should exist.

The test setup consists of a column of water at rest located behind a membrane on the left side of a tank. At time $t = 0$ s, the membrane is removed and the column of water collapses. During the collapse, the water impacts an obstacle at the bottom of the tank and creates a complicated flow structure, including several captured pockets of air. The geometry and the initial setup is shown in Figure 2.15.

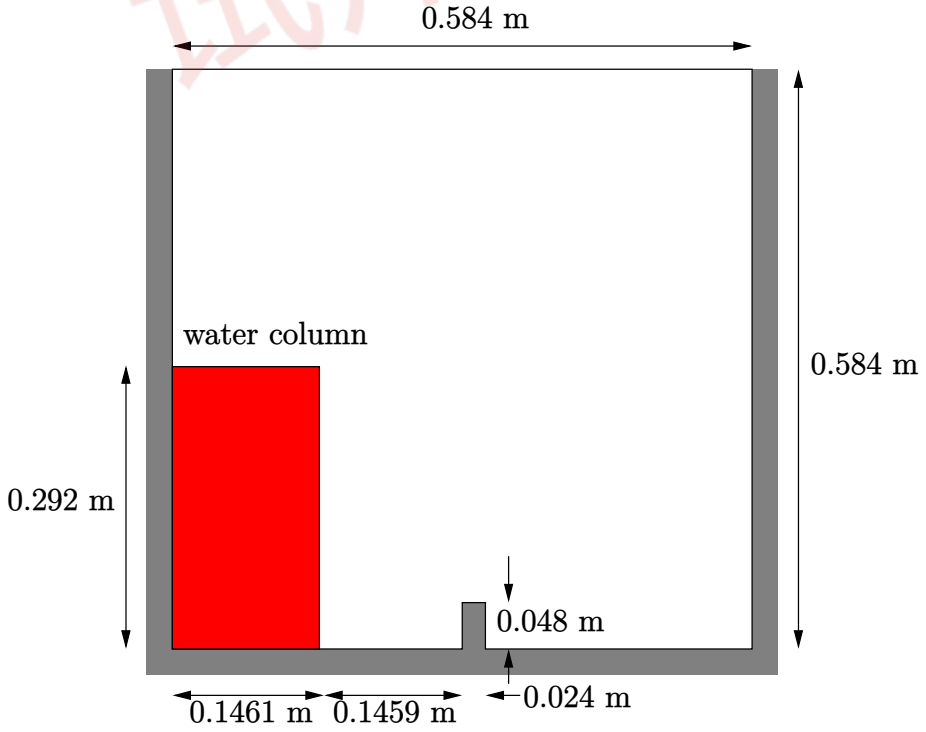


Figure 2.15: Geometry of the dam break.

2.2.1 Mesh generation

The user should go to their *run* directory and copy the *damBreak* case from the *$FOAM_TUTORIALS/incompressibleVoF/damBreakLaminar* directory, *i.e.*

```
run
cp -r $FOAM_TUTORIALS/incompressibleVoF/damBreakLaminar/damBreak .
```

Go into the damBreak case directory and generate the mesh running blockMesh as described previously. The damBreak mesh consist of five blocks; the *blockMeshDict* entries are given below.

```
16  convertToMeters 0.146;
17
18  vertices
19  (
20      (0 0 0)
21      (2 0 0)
22      (2.16438 0 0)
23      (4 0 0)
24      (0 0.32876 0)
25      (2 0.32876 0)
26      (2.16438 0.32876 0)
27      (4 0.32876 0)
28      (0 4 0)
29      (2 4 0)
30      (2.16438 4 0)
31      (4 4 0)
32      (0 0 0.1)
33      (2 0 0.1)
34      (2.16438 0 0.1)
35      (4 0 0.1)
36      (0 0.32876 0.1)
37      (2 0.32876 0.1)
38      (2.16438 0.32876 0.1)
39      (4 0.32876 0.1)
40      (0 4 0.1)
41      (2 4 0.1)
42      (2.16438 4 0.1)
43      (4 4 0.1)
44  );
45
46  blocks
47  (
48      hex (0 1 5 4 12 13 17 16) (23 8 1) simpleGrading (1 1 1)
49      hex (2 3 7 6 14 15 19 18) (19 8 1) simpleGrading (1 1 1)
50      hex (4 5 9 8 16 17 21 20) (23 42 1) simpleGrading (1 1 1)
51      hex (5 6 10 9 17 18 22 21) (4 42 1) simpleGrading (1 1 1)
52      hex (6 7 11 10 18 19 23 22) (19 42 1) simpleGrading (1 1 1)
53  );
54
55  defaultPatch
56  {
57      type empty;
58  }
59
60  boundary
61  (
62      leftWall
63      {
64          type wall;
65          faces
66          (
67              (0 12 16 4)
68              (4 16 20 8)
69          );
70      }
71      rightWall
72      {
73          type wall;
74          faces
75          (
76              (7 19 15 3)
77              (11 23 19 7)
78          );
79      }
```

试用水印

```
80      lowerWall
81      {
82          type wall;
83          faces
84          (
85              (0 1 13 12)
86              (1 5 17 13)
87              (5 6 18 17)
88              (2 14 18 6)
89              (2 3 15 14)
90          );
91      }
92      atmosphere
93      {
94          type patch;
95          faces
96          (
97              (8 20 21 9)
98              (9 21 22 10)
99              (10 22 23 11)
100         );
101     }
102 );
103
104
105 // ************************************************************************* //
```

The mesh is written into a set of files in a *polyMesh* directory in the *constant* directory. The use can list the contents of the directory to reveal the set of files/

```
ls constant/polyMesh
```

The files include: *points*, a list of the cell vertices; *faces*, a list of the cell faces; *owner* and *neighbour*, containing the indices of cells connected to a given face; *boundary*, a description of the boundary patches.

2.2.2 Boundary conditions

The *boundary* file can be read and understood by the user. The user should take a look at its contents, either by opening it in a file editor or printing out in the terminal window using the cat utility.

```
cat constant/polyMesh/boundary
```

The file contains a list of five boundary patches: `leftWall`, `rightWall`, `lowerWall`, `atmosphere` and `defaultFaces`. The user should notice the `type` of the patches. Firstly, the `atmosphere` is a standard patch, *i.e.* has no special attributes, merely an entity on which boundary conditions can be specified. Then, the `defaultFaces` patch is formed of **block faces that are omitted** from the `boundary` sub-dictionary in the *blockMeshDict* file. Those block faces form a patch whose properties are specified in a `defaultPatch` sub-dictionary in the *blockMeshDict* file. In this case, the default `type` is set to empty since the patch normal is in the direction we will not solve in this 2D case.

The `leftWall`, `rightWall` and `lowerWall` patches are each a wall. Like the generic patch, the wall type contains no geometric or topological information about the mesh and only differs from the plain patch in that it identifies the patch as a wall. This is required by some modelling, *e.g.* turbulent wall functions and some turbulence models which include the distance to the nearest wall in their calculations.

With VoF specifically, surface tension models can include wall adhesion at the contact point between the interface and wall surface. Wall adhesion models can be applied through a special boundary condition on the `alpha` (α) field, *e.g.* the constantAlphaContactAngle boundary condition, which requires the user to specify a static contact angle, `theta0`.

This example ignores surface tension effects between the wall and interface. This can be done can do this by setting the static contact angle, $\theta_0 = 90°$, but a simpler approach is to apply the zeroGradient condition to `alpha` on the walls.

The `top` boundary is free to the atmosphere so needs to permit both outflow and inflow according to the internal flow. We therefore use a combination of boundary conditions for pressure and velocity that does this while maintaining stability. They are:

- prghTotalPressure, applied to the pressure field, minus the hydrostatic component, $p_{\rho gh}$, given by Equation 6.3;
- pressureInletOutletVelocity, applied to velocity $\mathbf{U}$, which sets zeroGradient on all components of $\mathbf{U}$, except where there is inflow, in which case a fixedValue condition is applied to the *tangential* component;
- inletOutlet applied to other fields, which is a zeroGradient condition when flow outwards, fixedValue when flow is inwards.

At all wall boundaries, the fixedFluxPressure boundary condition is applied to the pressure field, which adjusts the pressure gradient so that the boundary flux matches the velocity boundary condition for solvers that include body forces such as gravity and surface tension.

The `defaultFaces` patch representing the front and back planes of the 2D problem, is, as usual, an empty type.

2.2.3 Phases

The fluid phases are specified in the *phaseProperties* file in the *constant* directory as follows:

```
16
17  phases          (water air);
18
19  sigma           0.07;
20
21
22  // ************************************************************************* //
```

It lists two phases, `water` and `air`. Equations for phase fraction are solved for the phases in the list, **except the last phase** listed, *i.e.* `air` in this case. Since there are only two phases, only one phase fraction equation is solved in this case, for the water phase fraction α_{water}, specified in the file *alpha.water* in the *0* directory.

The *phaseProperties* file also contains an entry for the surface tension between the two phases, specified by the keyword `sigma` in units N m^{-1}.

2.2.4 Setting initial fields

Unlike the previous cases, we shall now specify a non-uniform initial condition for the phase fraction α_{water} where

$$\alpha_{\text{water}} = \begin{cases} 1 & \text{for the water phase} \\ 0 & \text{for the air phase} \end{cases} \tag{2.5}$$

This is done by running the setFields utility. It requires a *setFieldsDict* dictionary, located in the *system* directory, whose entries for this case are shown below.

```
16
17  defaultFieldValues
18  (
19      volScalarFieldValue alpha.water 0
20  );
21
22  regions
23  (
24      boxToCell
25      {
26          box (0 0 -1) (0.1461 0.292 1);
27          fieldValues
28          (
29              volScalarFieldValue alpha.water 1
30          );
31      }
32  );
33
34
35  // ************************************************************************* //
```

The `defaultFieldValues` sets the default value of the fields, *i.e.* the value the field takes unless specified otherwise in the `regions` list. That list contains sub-dictionaries with `fieldValues` that override the defaults in a specified region. Regions are defined by functions that define a set of points, faces, or cells. Here `boxToCell` creates a box defined by minimum and maximum bounds that defines the set of cells of the water region. The phase fraction α_{water} is specified as 1 in this region.

The setFields utility reads fields from file and, after re-calculating those fields, will write them back to file. In the damBreak case, the `alpha.water` field is initially stored in its original form with the name `alpha.water.orig`. A field file with the `.orig` extension is read in when the actual file does not exist, so setFields will read `alpha.water.orig` but write the resulting output to `alpha.water` (or `alpha.water.gz` if compression is switched on). This way the original file is not overwritten, so can be reused.

The user should execute setFields like any other utility by:

```
setFields
```

Using paraFoam, check that the initial `alpha.water` field corresponds to the desired distribution as in Figure 2.16.

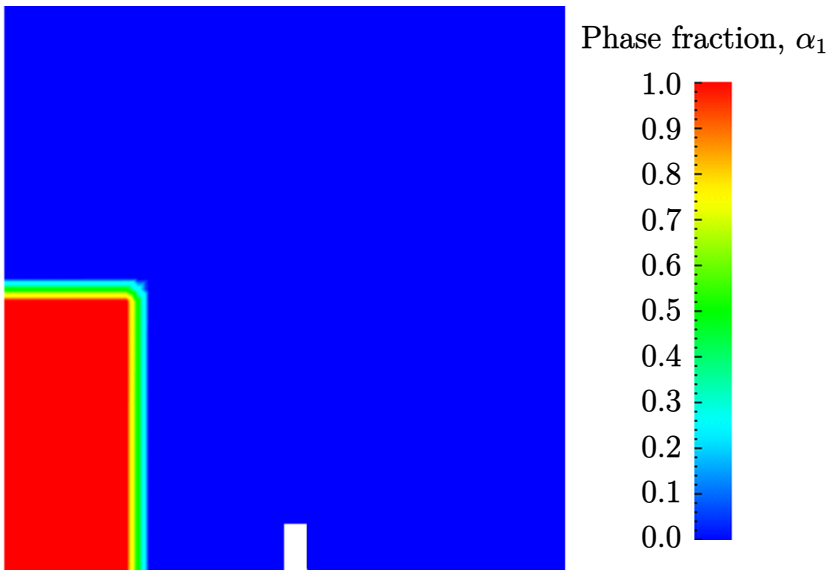


Figure 2.16: Initial conditions for phase fraction `alpha.water`.

2.2.5 Fluid properties

The physical properties for the air and water phases are specified in *physicalProperties.air* and *physicalProperties.water* files, respectively, in the *constant* directory. Physical properties describe characteristics of the fluid in a absence of flow. Each file specifies the viscosity model through the `viscosityModel` keyword, which is set to `constant` to indicate the value is unchanging. The viscosity is then specified by the `nu` keyword in units $m^2\,s^{-1}$. The density of each fluid is also specified by the keyword `rho` in units $kg\,m^{-3}$. The *physicalProperties.air* file is shown below as an example:

```
16
17  viscosityModel  constant;
18
19  nu              1.48e-05;
20
21  rho             1;
22
23
24  // ************************************************************************* //
```

If the viscosity does change according to the flow, *e.g.* as in non-Newtonian or viscoelastic fluids, then those models are specified through the *momentumProperties* file, as described in section 8.3.

2.2.6 Gravity

Gravitational acceleration is uniform across the domain and is specified in a file named *g* in the *constant* directory. Unlike a normal field file, *e.g.* *U* and *p*, *g* is a uniformDimensionedVectorField and so simply contains a set of `dimensions` and a `value` that represents $(0, 9.81, 0)\ m\,s^{-2}$ for this case:

```
16
17  dimensions      [0 1 -2 0 0 0 0];
18  value           (0 -9.81 0);
19
20
21  // ************************************************************************* //
```

2.2.7 Turbulence modelling

As in the cavity example, the choice of turbulence modelling method is selectable at runtime through the `simulationType` keyword in *momentumTransport* dictionary. In this example, we wish to run without turbulence modelling so we set `laminar`:

```
16
17  simulationType  laminar;
18
19
20  // ************************************************************************* //
```

2.2.8 Time step control

The simulation of the damBreak case is fully transient so the time step requires attention. The Courant number Co is an important consideration relating to time step. It is dimensionless parameter which can be defined for each cell as:

$$Co = \frac{\delta t |\mathbf{U}|}{\delta x} \tag{2.6}$$

where δt is the time step, $|\mathbf{U}|$ is the magnitude of the velocity through that cell and δx is the cell size in the direction of the velocity. With *explicit* solution, stability requires

the maximum $Co < 1$ at least; stricter limits exist depending on the choice of advection scheme. *Implicit* solutions do not have the same stability limit of the maximum Co, but temporal accuracy becomes more relevant as Co increased beyond 1.

Time step control is particularly important with interface-capturing. The incompressibleVoF solver module uses the multidimensional universal limiter for explicit solution (MULES), created by Henry Weller, to maintain boundedness of the phase fraction. Co needs to be limited depending on the choice of MULES algorithm. With the original *explicit* MULES algorithm, an upper limit of $Co \approx 0.25$ for stability is typically required. However, there is also the *semi-implicit* version of MULES, specified by the `MULESCorr` switch in the *fvSolution* file. For semi-implicit MULES, there is really no upper limit in Co for stability, but instead the level is determined by requirements of temporal accuracy.

In general it is difficult to specify a fixed time-step to satisfy the Co criterion since $|\mathbf{U}|$ is changing from cell to cell during the simulation. Instead, automatic adjustment of the time step is specified in the *controlDict* by switching `adjustTimeStep` to `on` and specifying the maximum Co for the phase fields, `maxAlphaCo`, and other fields, `maxCo`. In this example, the `maxAlphaCo` and `maxCo` are set to 1.0. The upper limit on time step `maxDeltaT` can be set to a value that will not be exceeded in this simulation, *e.g.* 1.0.

By using automatic time step control, the steps themselves are never rounded to a convenient value. Consequently if we request that OpenFOAM saves results at a fixed number of time step intervals, the times at which results are saved are somewhat arbitrary. However with automatic time step adjustment, results can be written at fixed times using the `adjustableRunTime` option for `writeControl` in the *controlDict* dictionary. With this option, the automatic time stepping procedure further adjusts their time steps so that it 'hits' on the exact times specified by the `writeInterval`, set to `0.05` in this example. The *controlDict* dictionary entries are shown below.

```
16
17  application     foamRun;
18
19  solver          incompressibleVoF;
20
21  startFrom       startTime;
22
23  startTime       0;
24
25  stopAt          endTime;
26
27  endTime         1;
28
29  deltaT          0.001;
30
31  writeControl    adjustableRunTime;
32
33  writeInterval   0.05;
34
35  purgeWrite      0;
36
37  writeFormat     binary;
38
39  writePrecision  6;
40
41  writeCompression off;
42
43  timeFormat      general;
44
45  timePrecision   6;
46
47  runTimeModifiable yes;
48
49  adjustTimeStep  yes;
50
51  maxCo           1;
52  maxAlphaCo      1;
53
54  maxDeltaT       1;
55
56
57  // ************************************************************************* //
```

2.2.9 Discretisation schemes

The MULES method, used by the incompressibleVoF modular solver, maintains boundedness of the phase fraction independently of the underlying numerical scheme, mesh structure, *etc.* The choice of schemes for convection are therefore not restricted to those that are strongly stable or bounded, such as upwind differencing.

The convection schemes settings are made in the *divSchemes* sub-dictionary of the *fvSchemes* dictionary. In this example, the convection term in the momentum equation, $\nabla \bullet (\rho \mathbf{U}\mathbf{U})$, denoted by the `div(rhoPhi,U)` keyword, uses `Gauss linearUpwind grad(U)` to produce good accuracy.

The $\nabla \bullet (\mathbf{U}\alpha)$ term, represented by the `div(phi,alpha)` keyword uses a bespoke `interfaceCompression` scheme where the specified coefficient is a factor that controls the compression of the interface where: 0 corresponds to no compression; 1 corresponds to conservative compression; and, anything larger than 1, relates to enhanced compression of the interface. We generally use a value of 1.0, as in this example.

The other discretised terms use commonly employed schemes so that the *fvSchemes* dictionary entries is as follows.

```
16
17  ddtSchemes
18  {
19      default         Euler;
20  }
21
22  gradSchemes
23  {
24      default         Gauss linear;
25  }
26
27  divSchemes
28  {
29      div(rhoPhi,U)  Gauss linearUpwind grad(U);
30      div(phi,alpha)  Gauss interfaceCompression vanLeer 1;
31      div(((rho*nuEff)*dev2(T(grad(U))))) Gauss linear;
32  }
33
34  laplacianSchemes
35  {
36      default         Gauss linear corrected;
37  }
38
39  interpolationSchemes
40  {
41      default         linear;
42  }
43
44  snGradSchemes
45  {
46      default         corrected;
47  }
48
49
50  // ************************************************************************* //
```

2.2.10 Linear-solver control

In the *fvSolution* file, the sub-dictionary in *solvers* for *alpha.water* contains elements that are specific to the MULES algorithm as shown below.

```
"alpha.water.*"
{
   nAlphaCorr      2;
   nAlphaSubCycles 1;

   MULESCorr       yes;
   nLimiterIter    5;

   solver          smoothSolver;
```

```
    smoother        symGaussSeidel;
    tolerance       1e-8;
    relTol          0;
}
```

MULES calculates two limiters to keep the phase fraction within the lower and upper bounds of 0 and 1. The limiter calculation is iterative, with the number of iterations specified by `nLimiterIter`. As a rule of thumb, `nLimiterIter` can be set to $3 + Co$ to maintain boundedness.

The semi-implicit version of MULES is activated by the `MULESCorr` switch. It first calculates an implicit, upwind solution before applying MULES as a higher-order correction. The linear solver must be configured for the implicit, upwind solution, through the `solver`, `smoother`, `tolerance` and `relTol` parameters.

The `nAlphaCorr` keyword which controls the number of iterations of the phase fraction equation within a solution step. The iteration is used to overcome nonlinearities in the advection which are present in this case due to the `interfaceCompression` scheme.

2.2.11 Running the code

Running of the code has been described in the previous tutorial. Try the following, that uses tee, a command that enables output to be written to both standard output and files:

```
cd $FOAM_RUN/damBreak
foamRun | tee log
```

The code will now be run interactively, with a copy of output stored in the *log* file.

2.2.12 Post-processing

Post-processing of the results can now be done in the usual way. The user can monitor the development of the phase fraction `alpha.water` in time, *e.g.* see Figure 2.17.

2.2.13 Running in parallel

The results from the previous example are generated using a fairly coarse mesh. We now wish to increase the mesh resolution and re-run the case. Using a finer mesh, we can then demonstrate the parallel processing capability of OpenFOAM.

The user should first clone the damBreak case, *e.g.* by

```
run
foamCloneCase damBreak damBreakFine
```

Enter the new case directory and change the `blocks` description in the *blockMeshDict* dictionary to

```
blocks
(
    hex (0 1 5 4 12 13 17 16) (46 10 1) simpleGrading (1 1 1)
    hex (2 3 7 6 14 15 19 18) (40 10 1) simpleGrading (1 1 1)
    hex (4 5 9 8 16 17 21 20) (46 76 1) simpleGrading (1 2 1)
    hex (5 6 10 9 17 18 22 21) (4 76 1) simpleGrading (1 2 1)
```

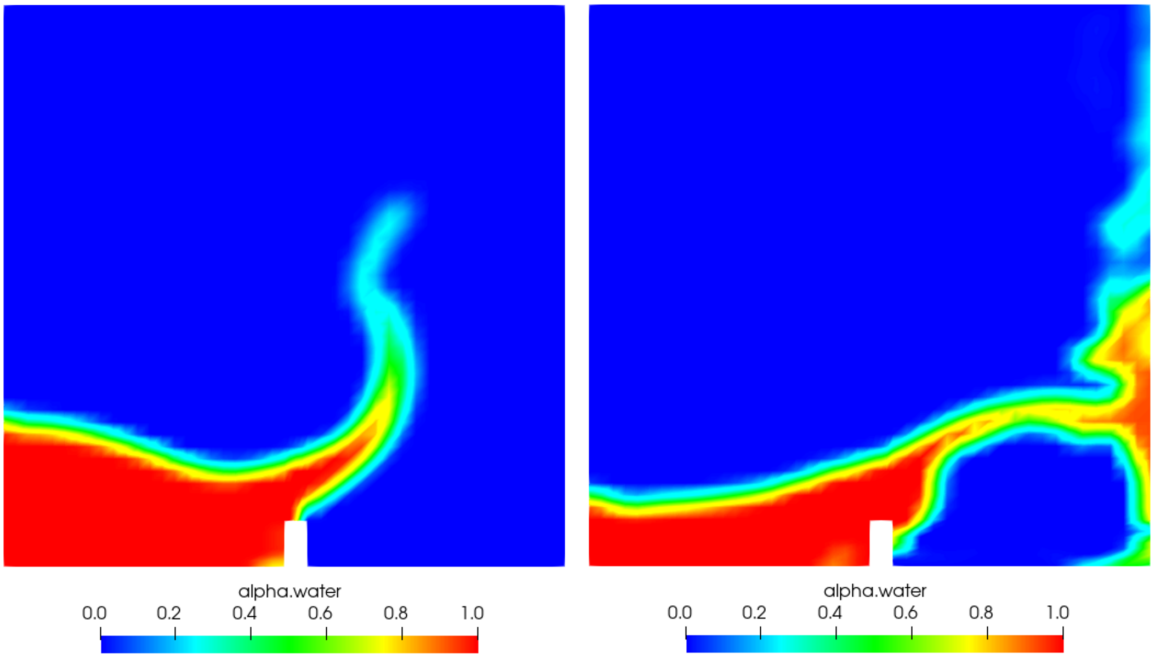


Figure 2.17: Phase fraction α at $t = 0.25$ s (left) and 0.50 s (right).

```
    hex (6 7 11 10 18 19 23 22) (40 76 1) simpleGrading (1 2 1)
);
```

Here, the entry is presented as printed from the *blockMeshDict* file; in short the user must change the mesh densities, *e.g.* the `46 10 1` entry, and some of the mesh grading entries to `1 2 1`. Once the dictionary is correct, generate the mesh by running blockMesh.

As the mesh has now changed from the damBreak example, the user must re-initialise the phase field `alpha.water` in the *0* time directory since it contains a number of elements that is inconsistent with the new mesh. Note that there is no need to change the `U` and `p_rgh` fields since they are specified as `uniform` which is independent of the number of elements in the field.

The user should then rerun the setFields utility. However, the mesh size is now inconsistent with the number of elements in the *alpha.water* file in the *0* directory, so the user must delete that file so that setFields uses the original *alpha.water.orig* file.

```
rm 0/alpha.water
setFields
```

Parallel computing uses domain decomposition, in which the geometry and associated fields are broken into pieces and allocated to separate processors for solution. The first step required to run a parallel case is therefore to decompose the domain using the decomposePar utility. There is a dictionary associated with decomposePar named *decomposeParDict* which is located in the *system* directory. Also, sample dictionaries can be found within the *etc* directory in the OpenFOAM installation, which can be copied to the case directory by running the foamGet script, *e.g.* (you do not need to do this):

```
foamGet decomposeParDict
```

The first entry is `numberOfSubdomains` which specifies the number of subdomains into which the case will be decomposed, usually corresponding to the number of processors available for the case.

This example uses the `simple method` of decomposition. It requires the `simpleCoeffs` to be configured according to the following criteria. The domain is split into pieces, or subdomains, in the x, y and z directions, the number of subdomains in each direction being given by the vector $\mathbf{n}$. As this geometry is 2 dimensional, the 3rd direction, z, cannot be split, hence n_z must equal 1. The n_x and n_y components of $\mathbf{n}$ split the domain in the x and y directions and must be specified so that the number of subdomains specified by n_x and n_y equals the specified `numberOfSubdomains`, *i.e.* $n_x n_y =$ `numberOfSubdomains`.

It is beneficial to keep the number of cell faces adjoining the subdomains to a minimum so, for a square geometry, it is best to keep the split between the x and y directions should be fairly even. For example, let us assume we wish to run on 4 processors. We would set `numberOfSubdomains` to 4 and `n` to `(2, 2, 1)`. The user should run decomposePar by the following command.

```
decomposePar
```

The terminal output shows that the decomposition is distributed evenly between the sub-domains. The decomposition writes the mesh and fields of each sub-domain into separate sub-directories named `processor`<`N`>, where `N` is the sub-domain ID, *e.g.* `0`, `1`, `2`, *etc.* The user should list the files in the case directory to confirm that four directories `processor0`, `processor1`, `processor2` and `processor3` exist.

This example presents running in parallel with the openMPI implementation of the standard message-passing interface (MPI). The following command runs on 4 cores of a local multi-processor CPU.

```
mpirun -np 4 foamRun -parallel
```

The user can consult section 3.4 for more details of how to run a case in parallel. For example, the user may run on more nodes over a network by creating a file that lists the host names of the machines on which the case is to be run as described in section 3.4.3.

2.2.14 Post-processing a case run in parallel

When the case runs in parallel, the results are written into time directories within the `processor`<`N`> sub-directories. The user can confirm this by listing the time directories for the *processor0* directory.

```
ls processor0
```

It is possible to post-process an individual sub-domain by treating the individual `processor` directory as a case in its own right. For example, to view the `processor1` sub-domain in ParaView, the user can launch paraFoam by running the following command.

```
paraFoam -case processor1
```

Figure 2.18 shows the mesh from this sub-domain, following the decomposition of the domain using the `simple` method.

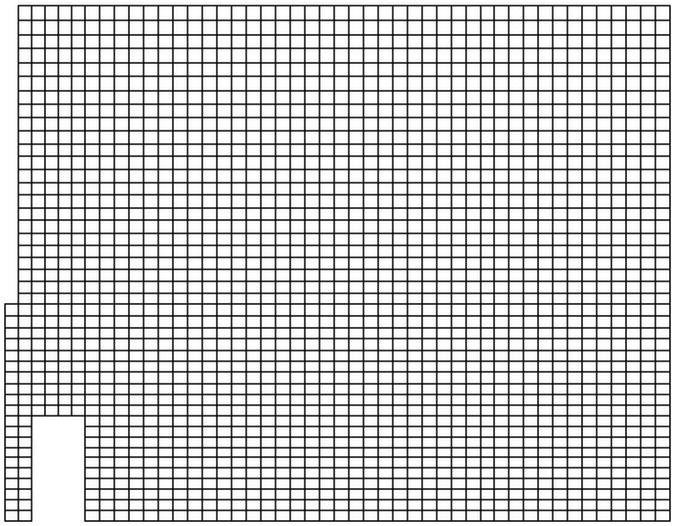


Figure 2.18: Mesh of processor 2 in parallel processed case.

Alternatively, the decomposed fields and mesh can be reassembled for post-processing in serial. The reconstructPar utility takes the field files from time directories for the `processor` sub-domain and builds equivalent field files for the complete domain. The user should run the utility as follows.

```
reconstructPar
```

The fields are reconstructed are written to solution time directories in the case directory. These fields can be visualised as normal in ParaView. The results from the fine mesh are shown in Figure 2.19. The user can see that the resolution of interface has improved significantly compared to the coarse mesh.

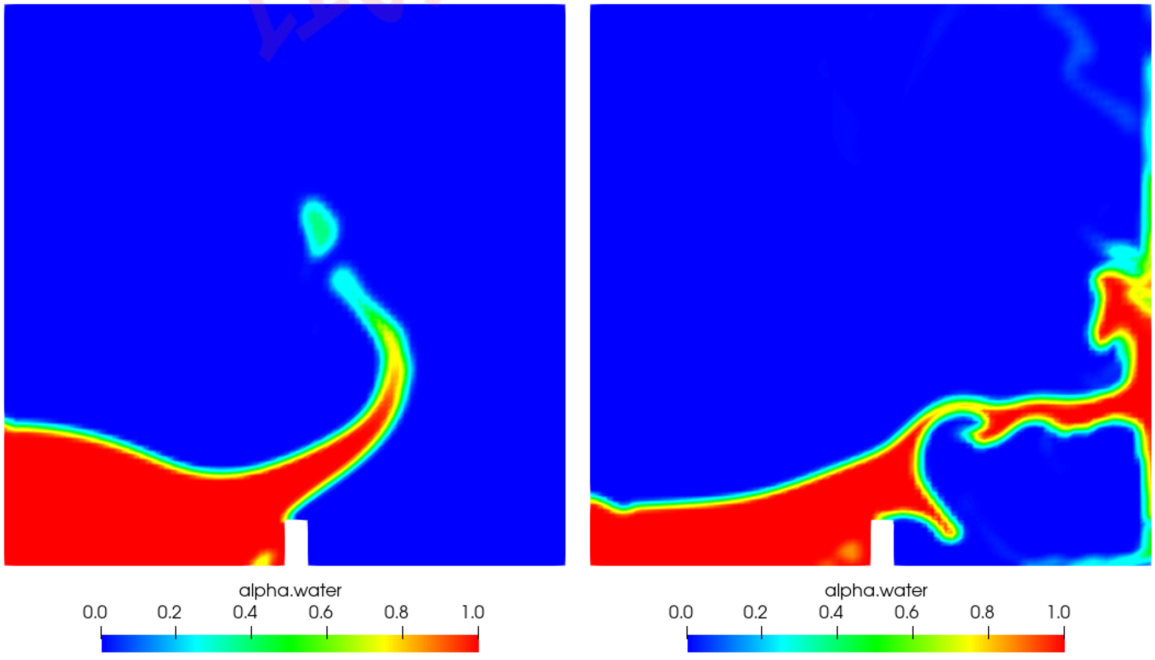


Figure 2.19: Phase fraction α at $t = 0.25$ s (left) and 0.50 s (right).

2.3 Stress analysis of a plate with a hole

This tutorial describes how to pre-process, run and post-process a case involving linear-elastic, steady-state stress analysis on a square plate with a circular hole at its centre. The plate dimensions are: side length 4 m and radius $R = 0.5$ m. It is loaded with a uniform traction of $\sigma = 10$ kPa over its left and right faces as shown in Figure 2.20. Two symmetry planes can be identified for this geometry and therefore the solution domain need only cover a quarter of the geometry, shown by the shaded area in Figure 2.20.

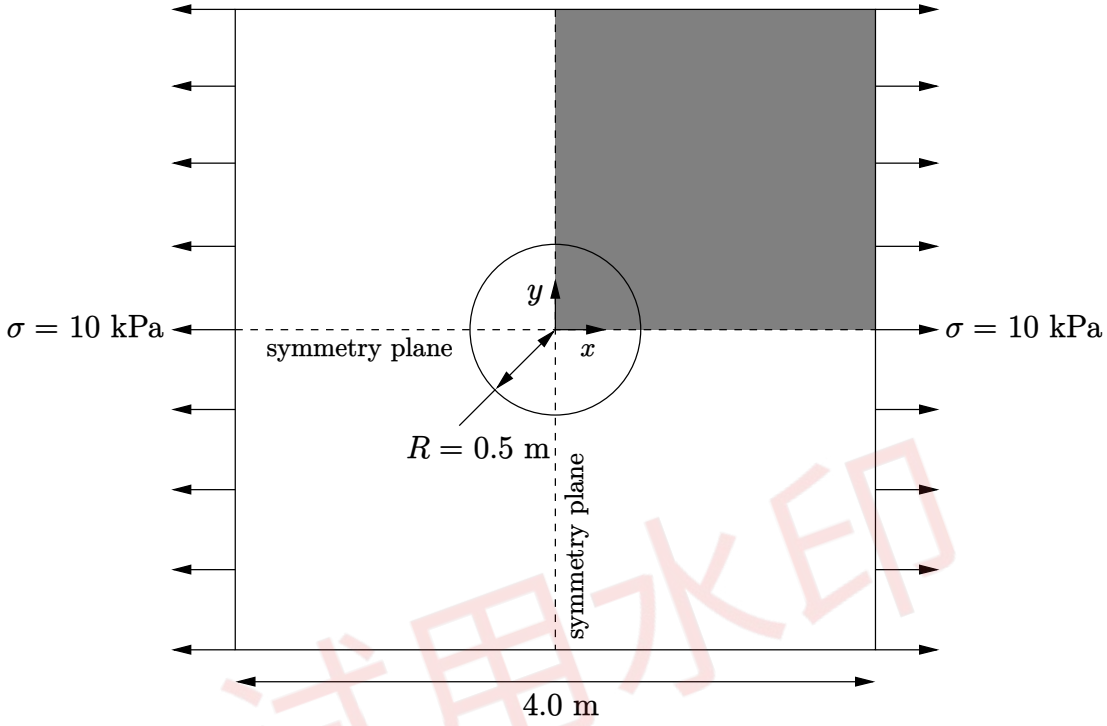


Figure 2.20: Geometry of the plate with a hole.

The problem can be approximated as 2D since the load is applied in the plane of the plate. In a Cartesian coordinate system there are two possible assumptions to take in regard to the behaviour of the structure in the third dimension: (1) the plane stress condition, in which the stress components acting out of the 2D plane are assumed to be negligible; (2) the plane strain condition, in which the strain components out of the 2D plane are assumed negligible. The plane stress condition is appropriate for solids whose third dimension is thin as in this case; the plane strain condition is applicable for solids where the third dimension is thick.

An analytical solution exists for loading of an infinitely large, thin plate with a circular hole. The solution for the stress normal to the vertical plane of symmetry is

$$(\sigma_{xx})_{x=0} = \begin{cases} \sigma\left(1 + \dfrac{R^2}{2y^2} + \dfrac{3R^4}{2y^4}\right) & \text{for } |y| \geq R \\ 0 & \text{for } |y| < R \end{cases} \tag{2.7}$$

Results from the simulation will be compared with this solution. At the end of the tutorial, the user can: investigate the sensitivity of the solution to mesh resolution and mesh grading; and, increase the size of the plate in comparison to the hole to try to estimate the error in comparing the analytical solution for an infinite plate to the solution of this problem of a finite plate.

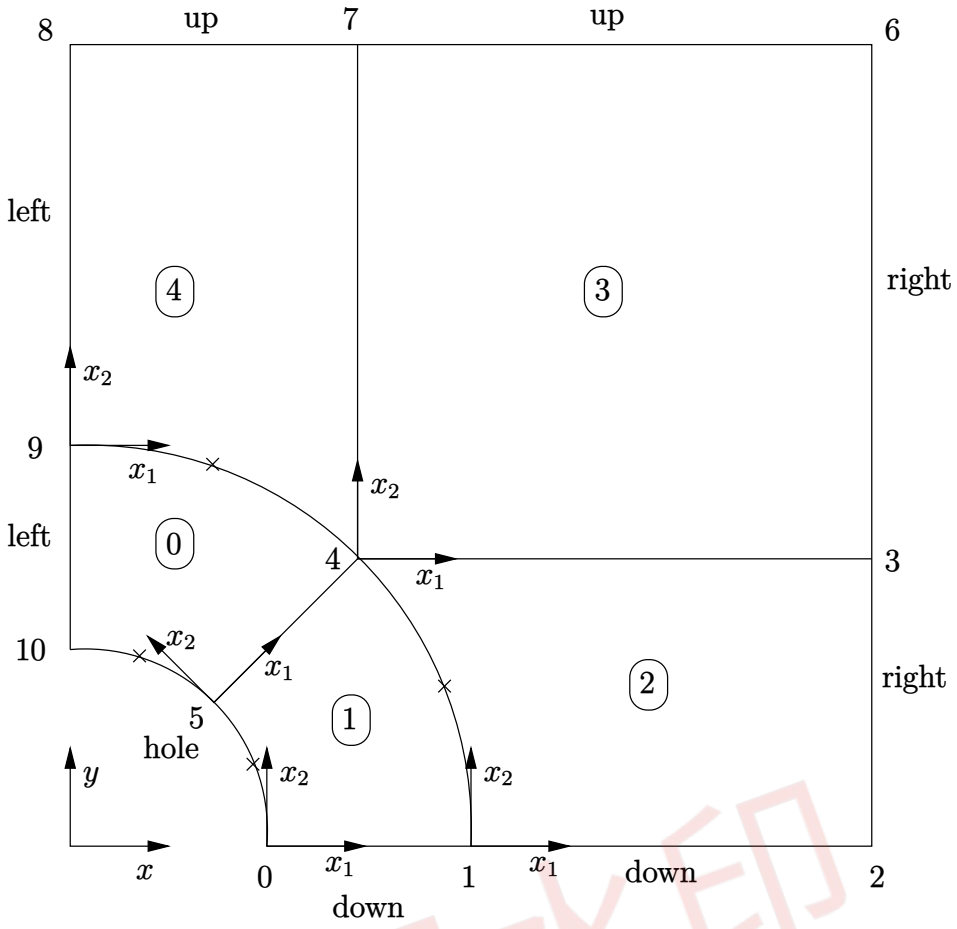


Figure 2.21: Block structure of the mesh for the plate with a hole.

The example uses the solidDisplacement modular solver. The user should go to their *run* directory, copy the plateHole case from the *$FOAM_TUTORIALS/solidDisplacement* directory and finally change into the *plateHole* case directory.

```
run
cp -r $FOAM_TUTORIALS/solidDisplacement/plateHole .
cd plateHole
```

2.3.1 Mesh generation

The domain consists of four blocks, some of which have arc-shaped edges. The block structure for the part of the mesh in the $x - y$ plane is shown in Figure 2.21. As already mentioned in section 2.1.2, all geometries are generated in 3D in OpenFOAM even if the case is to be as a 2D problem. Therefore a dimension of the block in the z direction has to be chosen; here, 0.5 m is selected. It does not affect the solution since the traction boundary condition is specified as a stress rather than a force, thereby making the solution independent of the cross-sectional area. The user should open the *blockMeshDict* file in an editor, as listed below.

```
16  convertToMeters 1;
17
18  vertices
19  (
20      (0.5 0 0)
21      (1 0 0)
22      (2 0 0)
```

```
23          (2 0.707107 0)
24          (0.707107 0.707107 0)
25          (0.353553 0.353553 0)
26          (2 2 0)
27          (0.707107 2 0)
28          (0 2 0)
29          (0 1 0)
30          (0 0.5 0)
31          (0.5 0 0.5)
32          (1 0 0.5)
33          (2 0 0.5)
34          (2 0.707107 0.5)
35          (0.707107 0.707107 0.5)
36          (0.353553 0.353553 0.5)
37          (2 2 0.5)
38          (0.707107 2 0.5)
39          (0 2 0.5)
40          (0 1 0.5)
41          (0 0.5 0.5)
42      );
43
44      blocks
45      (
46          hex (5 4 9 10 16 15 20 21) (10 10 1) simpleGrading (1 1 1)
47          hex (0 1 4 5 11 12 15 16) (10 10 1) simpleGrading (1 1 1)
48          hex (1 2 3 4 12 13 14 15) (20 10 1) simpleGrading (1 1 1)
49          hex (4 3 6 7 15 14 17 18) (20 20 1) simpleGrading (1 1 1)
50          hex (9 4 7 8 20 15 18 19) (10 20 1) simpleGrading (1 1 1)
51      );
52
53      edges
54      (
55          arc 0 5 (0.469846 0.17101 0)
56          arc 5 10 (0.17101 0.469846 0)
57          arc 1 4 (0.939693 0.34202 0)
58          arc 4 9 (0.34202 0.939693 0)
59          arc 11 16 (0.469846 0.17101 0.5)
60          arc 16 21 (0.17101 0.469846 0.5)
61          arc 12 15 (0.939693 0.34202 0.5)
62          arc 15 20 (0.34202 0.939693 0.5)
63      );
64
65      boundary
66      (
67          left
68          {
69              type symmetryPlane;
70              faces
71              (
72                  (8 9 20 19)
73                  (9 10 21 20)
74              );
75          }
76          right
77          {
78              type patch;
79              faces
80              (
81                  (2 3 14 13)
82                  (3 6 17 14)
83              );
84          }
85          down
86          {
87              type symmetryPlane;
88              faces
89              (
90                  (0 1 12 11)
91                  (1 2 13 12)
92              );
93          }
94          up
95          {
96              type patch;
97              faces
98              (
99                  (7 8 19 18)
100                 (6 7 18 17)
101             );
102         }
103         hole
104         {
105             type patch;
```

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```
106             faces
107             (
108                 (10 5 16 21)
109                 (5 0 11 16)
110             );
111         }
112         frontAndBack
113         {
114             type empty;
115             faces
116             (
117                 (10 9 4 5)
118                 (5 4 1 0)
119                 (1 4 3 2)
120                 (4 7 6 3)
121                 (4 9 8 7)
122                 (21 16 15 20)
123                 (16 11 12 15)
124                 (12 13 14 15)
125                 (15 14 17 18)
126                 (15 18 19 20)
127             );
128         }
129     );
130
131
132     // ************************************************************************* //
```

Until now, we have only specified straight edges in the geometries of previous tutorials but here we need to specify curved edges. These are specified under the `edges` keyword entry which is a list of non-straight edges. The syntax of each list entry begins with the type of curve, including `arc`, `simpleSpline`, `polyLine` *etc.*, described further in section 5.4.3. In this example, all the edges are circular and so can be specified by the `arc` keyword entry. The following entries are the labels of the start and end vertices of the arc and a point vector through which the circular arc passes.

The blocks in this *blockMeshDict* do not all have the same orientation. As can be seen in Figure 2.21 the x_2 direction of block 0 is equivalent to the $-x_1$ direction for block 4. This means care must be taken when defining the number and distribution of cells in each block so that the cells match up at the block faces.

Six patches are defined: one for each side of the plate, one for the hole and one for the front and back planes. The `left` and `down` patches are both a symmetry plane. Since this is a *geometric* constraint, it is included in the definition of the *mesh*, rather than being purely a specification on the boundary condition of the fields. Therefore they are defined as such using a special symmetryPlane type as shown in the *blockMeshDict*.

The `frontAndBack` patch represents the plane which is ignored in a 2D case. Again this is a geometric constraint so is defined within the mesh, using the empty type as shown in the *blockMeshDict*. For further details of boundary types and geometric constraints, the user should refer to section 5.3.

The remaining patches are of the regular patch type. The mesh should be generated using blockMesh and can be viewed in paraFoam as described in section 2.1.3. It should appear as in Figure 2.22.

2.3.2 Boundary and initial conditions

Once the mesh generation is complete, the initial field with boundary conditions must be set. For a stress analysis case without thermal stresses, only displacement `D` needs to be set. The *0/D* is as follows:

```
16  dimensions      [0 1 0 0 0 0 0];
17
18  internalField   uniform (0 0 0);
19
20  boundaryField
```

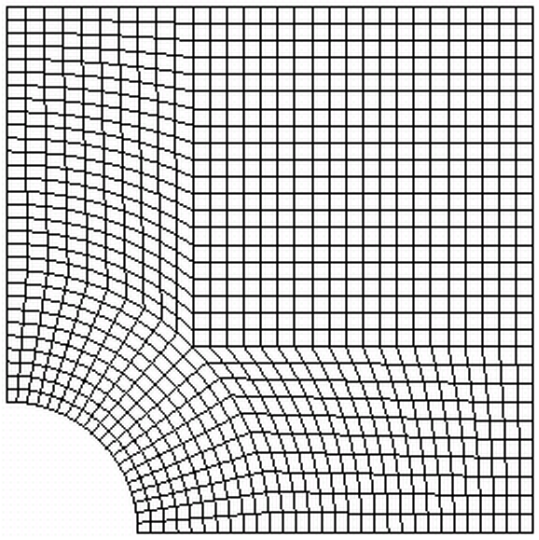


Figure 2.22: Mesh of the hole in a plate problem.

```
{
    left
    {
        type            symmetryPlane;
    }
    right
    {
        type            tractionDisplacement;
        traction        uniform (10000 0 0);
        pressure        uniform 0;
        value           uniform (0 0 0);
    }
    down
    {
        type            symmetryPlane;
    }
    up
    {
        type            tractionDisplacement;
        traction        uniform (0 0 0);
        pressure        uniform 0;
        value           uniform (0 0 0);
    }
    hole
    {
        type            tractionDisplacement;
        traction        uniform (0 0 0);
        pressure        uniform 0;
        value           uniform (0 0 0);
    }
    frontAndBack
    {
        type            empty;
    }
}

// ************************************************************************* //
```

Firstly, it can be seen that the displacement initial conditions are set to $(0, 0, 0)$ m. The `left` and `down` patches **must** be both of symmetryPlane type since they are specified as such in the mesh description in the *constant/polyMesh/boundary* file. Similarly the `frontAndBack` patch is declared empty.

The other patches are traction boundary conditions, set by a specialist tractionDisplacement boundary type. The traction boundary conditions are specified by a linear combination of: (1) a boundary traction vector under keyword `traction`; (2) a pressure that produces a traction normal to the boundary surface that is defined as negative when pointing out of the surface, under keyword `pressure`. The `up` and `hole` patches are zero

traction so the boundary traction and pressure are set to zero. For the `right` patch the traction should be $(1e4, 0, 0)$ Pa and the pressure should be 0 Pa.

2.3.3 Physical properties

The physical properties for the case are set in the *physicalProperties* dictionary in the *constant* directory, shown below:

```
16
17  rho
18  {
19      type        uniform;
20      value       7854;
21  }
22
23  nu
24  {
25      type        uniform;
26      value       0.3;
27  }
28
29  E
30  {
31      type        uniform;
32      value       2e+11;
33  }
34
35  Cv
36  {
37      type        uniform;
38      value       434;
39  }
40
41  kappa
42  {
43      type        uniform;
44      value       60.5;
45  }
46
47  alphav
48  {
49      type        uniform;
50      value       1.1e-05;
51  }
52
53  planeStress     yes;
54  thermalStress   no;
55
56
57  // ************************************************************************* //
```

The file includes mechanical properties of steel:

- Density `rho` $= 7854$ kg m^{-3}
- Young's modulus `E` $= 2 \times 10^{11}$ Pa
- Poisson's ratio `nu` $= 0.3$

The `planeStress` switch is set to `yes` to adopt the plane stress assumption in this 2D case. The solidDisplacementFoam solver may optionally solve a thermal equation that is coupled with the momentum equation through the thermal stresses that are generated. The user specifies at run time whether OpenFOAM should solve the thermal equation by the `thermalStress` switch (currently set to `no`). The thermal properties are also specified for steel for this case, *i.e.*:

- Specific heat capacity `Cp` $= 434$ Jkg^{-1}K^{-1}
- Thermal conductivity `kappa` $= 60.5$ Wm^{-1}K^{-1}
- Thermal expansion coefficient `alphav` $= 1.1 \times 10^{-5}$ K^{-1}

For thermal calculations, the temperature field variable `T` is present in the *0* directory.

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2.3.4 Control

As before, the information relating to the control of the solution procedure are read in from the *controlDict* dictionary. For this case, the `startTime` is 0 s. The time step is not important since this is a steady state case; in this situation it is best to set the time step `deltaT` to 1 so it simply acts as an iteration counter for the steady-state case. The `endTime`, set to 100, then acts as a limit on the number of iterations. The `writeInterval` can be set to 20.

The *controlDict* entries are as follows:

```
16
17  application     foamRun;
18
19  solver          solidDisplacement;
20
21  startFrom       startTime;
22
23  startTime       0;
24
25  stopAt          endTime;
26
27  endTime         100;
28
29  deltaT          1;
30
31  writeControl    timeStep;
32
33  writeInterval   20;
34
35  purgeWrite      0;
36
37  writeFormat     ascii;
38
39  writePrecision  6;
40
41  writeCompression off;
42
43  timeFormat      general;
44
45  timePrecision   6;
46
47  runTimeModifiable true;
48
49
50  // ************************************************************************* //
```

2.3.5 Discretisation schemes and linear-solver control

Let us turn our attention to the *fvSchemes* dictionary. Firstly, the problem we are analysing is steady-state so the user should select `SteadyState` for the time derivatives in `timeScheme`. This essentially switches off the time derivative terms. Not all solvers, especially in fluid dynamics, work for both steady-state and transient problems but solidDisplacementFoam does work, since the base algorithm is the same for both types of simulation.

The momentum equation in linear-elastic stress analysis includes several explicit terms containing the gradient of displacement. The calculations benefit from accurate and smooth evaluation of the gradient. Normally, in the finite volume method the discretisation is based on Gauss's theorem. The Gauss method is sufficiently accurate for most purposes but, in this case, the least squares method will be used. The user should therefore open the `fvSchemes` dictionary in the *system* directory and ensure the `leastSquares` method is selected for the `grad(U)` gradient discretisation scheme in the `gradSchemes` sub-dictionary:

```
16
17  d2dt2Schemes
18  {
19      default         steadyState;
```

```
20  }
21
22  ddtSchemes
23  {
24      default         Euler;
25  }
26
27  gradSchemes
28  {
29      default         leastSquares;
30  }
31
32  divSchemes
33  {
34      default         none;
35      div(sigmaD)     Gauss linear;
36  }
37
38  laplacianSchemes
39  {
40      default         Gauss linear corrected;
41  }
42
43  interpolationSchemes
44  {
45      default         linear;
46  }
47
48  snGradSchemes
49  {
50      default         none;
51  }
52
53  // ************************************************************************* //
```

The *fvSolution* dictionary in the *system* directory controls the linear equation solvers and algorithms used in the solution. The user should first look at the *solvers* sub-dictionary and notice that the choice of `solver` for `D` is `GAMG`. The solver `tolerance` should be set to 10^{-6} for this problem. The solver relative tolerance, denoted by `relTol`, sets the required reduction in the residuals within each iteration. It is uneconomical to set a tight (low) relative tolerance within each iteration since a lot of terms in each equation are explicit and are updated as part of the segregated iterative procedure. Therefore a reasonable value for the relative tolerance is 0.01, or possibly even higher, say 0.1, or in some cases even 0.9 (as in this case).

```
16
17  solvers
18  {
19      "(D|e)"
20      {
21          solver          GAMG;
22          tolerance       1e-06;
23          relTol          0.9;
24          smoother        GaussSeidel;
25          nCellsInCoarsestLevel 20;
26      }
27  }
28
29  PIMPLE
30  {
31      compactNormalStress yes;
32  }
33
34
35  // ************************************************************************* //
```

2.3.6 Running the code

The user can now try running foamRun in the background of the terminal. When running an OpenFOAM application in the background, the standard output (log information) should be redirected to a file. In the command below, standard output is written to a file named *log* which can be examined afterwards.

```
foamRun > log &
```

The user should check the convergence information by viewing the generated *log* file. It shows the number of iterations and the initial and final residuals of the displacement in each direction being solved. The final residual should always be less than 0.9 times the initial residual as this iteration tolerance set. By the end of the simulation, the initial residuals have reduced towards the convergence tolerance of 10^{-6}.

2.3.7 Post-processing

The solidDisplacementFoam solver outputs the stress field σ as a symmetric tensor field `sigma`. To post-process individual scalar field components, σ_{xx}, σ_{xy} *etc.*, the user can run the foamPostProcess utility, calling the components function object on the `sigma` field using the `-func` option as follows:

```
foamPostProcess -func "components(sigma)"
```

Components named `sigmaxx`, `sigmaxy` *etc.* are written to time directories of the case. The σ_{xx} stresses can be viewed in ParaView as shown in Figure 2.23.

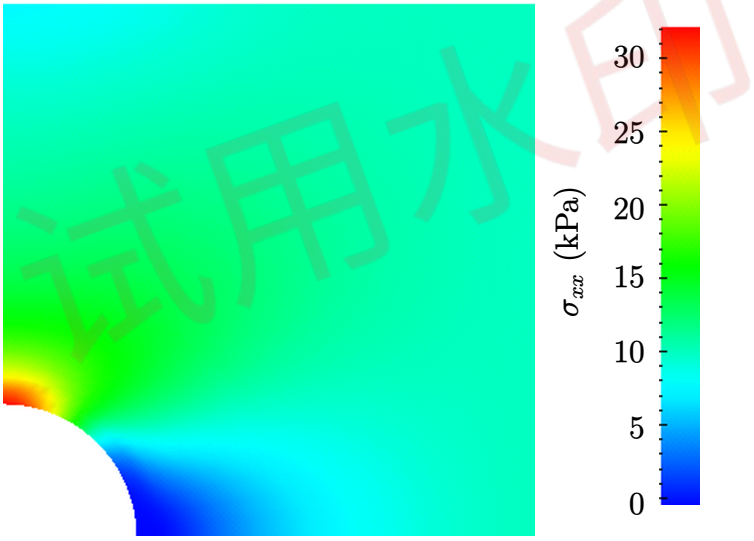


Figure 2.23: σ_{xx} stress field in the plate with hole.

In order to compare the solution to the analytical solution of Equation 2.7, data of σ_{xx} must be extracted along the left edge symmetry plane of our domain. The user may generate the required graph data using the foamPostProcess utility with the `graphUniform` function. Unlike earlier examples of foamPostProcess where no configuration is required, this example includes a *graphUniform* file pre-configured in the *system* directory. The sample line is set between $(0.0, 0.5, 0.25)$ and $(0.0, 2.0, 0.25)$, and the fields are specified in the `fields` list:

```
9       Writes graph data for specified fields along a line, specified by start and
10      end points. A specified number of graph points are used, distributed
11      uniformly along the line.
12
13  \*---------------------------------------------------------------------------*/
14
15  start           (0 0.5 0.25);
16  end             (0 2 0.25);
```

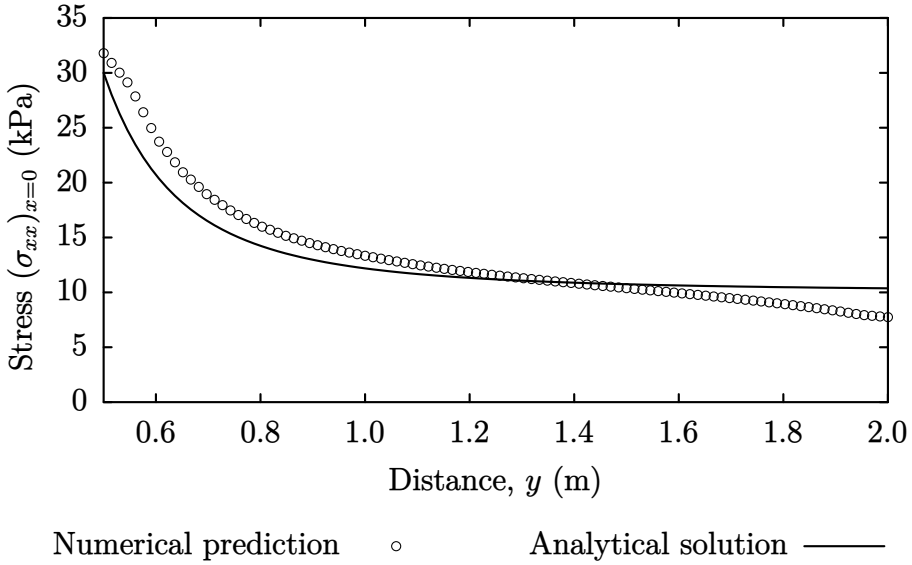


Figure 2.24: Normal stress along the vertical symmetry $(\sigma_{xx})_{x=0}$

```
17  nPoints         100;
18
19  fields          (sigmaxx);
20
21  axis            y;
22
23  #includeEtc "caseDicts/postProcessing/graphs/graphUniform.cfg"
24
25  // ************************************************************************* //
```

The user should execute postProcessing with the `graphUniform` function:

```
foamPostProcess -func graphUniform
```

Data is written in `raw` 2 column format into files within time subdirectories of a *postProcessing/graphUniform* directory, *e.g.* the data at $t = 100$ s is found within the file *graphUniform/100/line.xy*. If the user has GnuPlot installed they launch it (by typing `gnuplot`) and then plot both the numerical data and analytical solution as follows:

```
plot [0.5:2] [0:] "postProcessing/graphUniform/100/line.xy",
    1e4*(1+(0.125/(x**2))+(0.09375/(x**4)))
```

An example plot is shown in Figure 2.24.

Chapter 3

Applications and libraries

The examples in Chapter 2 show that OpenFOAM provides a range of software 'tools' that are run from a terminal command line. The tools include *applications* which are executable programs written in C++, the base programming language of OpenFOAM. Applications obtain most of the functionality from OpenFOAM's vast store of pre-compiled *libraries*, also written in C++. Since OpenFOAM is open source software, users have the freedom to create their own applications and libraries. Applications are generally split into two categories:

- *solvers*, *e.g.* foamRun, that perform CFD calculations involving fluid dynamics, energy, *etc.*;
- *utilities*, *e.g.* blockMesh and foamPostProcess, that perform other tasks in CFD like meshing and post-processing.

Prior to version 11 of OpenFOAM, there were many solvers, since separate ones were written for various different types of flow, *e.g.* simpleFoam, pimpleFoam, *etc.* However, most flow solvers are now written as modules, *e.g.* incompressibleFluid, incompressibleVoF and *e.g.*solid which are loaded by the general foamRun (or foamMultiRun) solvers. Rather than existing as an application, each solver module is compiled into a library of its own.

In addition to applications, the tools in OpenFOAM also include *shell scripts*, *e.g.* paraFoam, foamInfo and foamGet. Many of the scripts help with the configuration of cases.

This chapter gives an overview of applications and libraries, including their creation, modification, compilation and execution.

3.1 The programming language of OpenFOAM

This chapter provides some information to help understand how OpenFOAM applications and libraries are compiled. It provides some background knowledge of C++, the base language of OpenFOAM. Henry Weller chose C++ as the main programming language of OpenFOAM when he created it in the late 1980s.

The idea was to use object-oriented programming to express abstract concepts efficiently, just as verbal language and mathematics can do. For example, in fluid flow, we use the term "velocity field", which has meaning without any reference to the nature of the flow or any specific velocity data. The term encapsulates the idea of movement with direction and magnitude and relates to other physical properties. In mathematics,

"velocity field" can be replaced by a single symbol, *e.g.* **U**, and other symbols express operations and functions, *e.g.* "the field of velocity magnitude" by $|\mathbf{U}|$.

CFD deals with partial differential equations in 3 dimensions of space and time. The equations contain: fields of scalars, vectors and tensors; tensor algebra; tensor calculus; and, dimensional units. The solution to these equations involves discretisation procedures, matrices, solvers, and solution algorithms.

Rather than program CFD in terms of intrinsic entities known to a computer, *e.g.* bits, bytes, integers, floating point numbers, OpenFOAM provides *classes* that define the entities encountered in CFD. For example, a velocity field can be defined by a vectorField class, allowing a programmer to create an instance, or *object*, of that class. The object can be created with the name `U` to mimic the symbol used in mathematics. Associated functions can be created with names which also try to emulate the simplicity of mathematics, *e.g.* `mag(U)` can represent $|\mathbf{U}|$.

The class structure concentrates code development to contained regions of the code, *i.e.* the classes themselves, thereby making the code easier to manage. New classes can be derived or inherit properties from other classes, *e.g.* the vectorField can be derived from a vector class and a Field class. C++ provides the mechanism of *template classes* such that the template class Field<Type> can represent a field of any <Type>, *e.g.*scalar, vector, tensor. The general features of the template class are passed on to any class created from the template. Templating and inheritance reduce duplication of code and create class hierarchies that impose an overall structure on the code.

A theme of the OpenFOAM design is that it has a syntax that closely resembles the partial differential equations being solved. For example the equation

$$\frac{\partial \rho \mathbf{U}}{\partial t} + \nabla \bullet \phi \mathbf{U} - \nabla \bullet \mu \nabla \mathbf{U} = -\nabla p$$

is represented by the code

```
solve
(
    fvm::ddt(rho, U)
  + fvm::div(phi, U)
  - fvm::laplacian(mu, U)
    ==
  - fvc::grad(p)
);
```

The equation syntax is most evident in the code for solvers, both the modules and solver applications. Users do not need a deep knowledge of C++ programming to interpret the equations written in a solver. Instead, an understanding of the underlying equations, models and solution method and algorithms is perhaps more helpful, which can be found in ***Notes on Computational Fluid Dynamics: General Principles***.

It does help to have a rudimentary understanding of the the principles behind object-orientation and classes, and to have a basic knowledge of some C++ code syntax. To program in OpenFOAM, there is often little need for a user to immerse themselves in the code of any of the OpenFOAM classes. The essence of object-orientation is that the user should not have to go through the code of each class they use; merely the knowledge of the class' existence and its functionality are sufficient to use the class. A description of each class and its functions is supplied with the OpenFOAM distribution in HTML documentation generated with Doxygen at `https://cpp.openfoam.org`

3.2 Compiling applications and libraries

Compilation is an integral part of code development that requires careful management since every piece of code requires its own set instructions to access dependent components of the OpenFOAM library. In Linux systems there are various tools to help automate the management process, starting with the standard make utility. OpenFOAM uses its own wmake compilation script that is based on make. It is specifically designed for the large number of individual components that are compiled separately in OpenFOAM (approximately 150 applications and 150 libraries).

To understand the compilation process, we first need to explain certain aspects of C++ and its file structure, shown schematically in Figure 3.1. A class is defined through a set of instructions such as object construction, data storage and class member functions. The file that defines these functions — the class *definition* — takes a *.C* extension, *e.g.* a class nc would be written in the file *nc.C*. This file can be compiled independently of other code into a binary executable library file known as a shared object library with the *.so* file extension, *i.e.nc.so*. When compiling a piece of code, say *newApp.C*, that uses the nc class, *nc.C* need not be recompiled, rather *newApp.C* calls the *nc.so* library at runtime. This is known as *dynamic linking*.

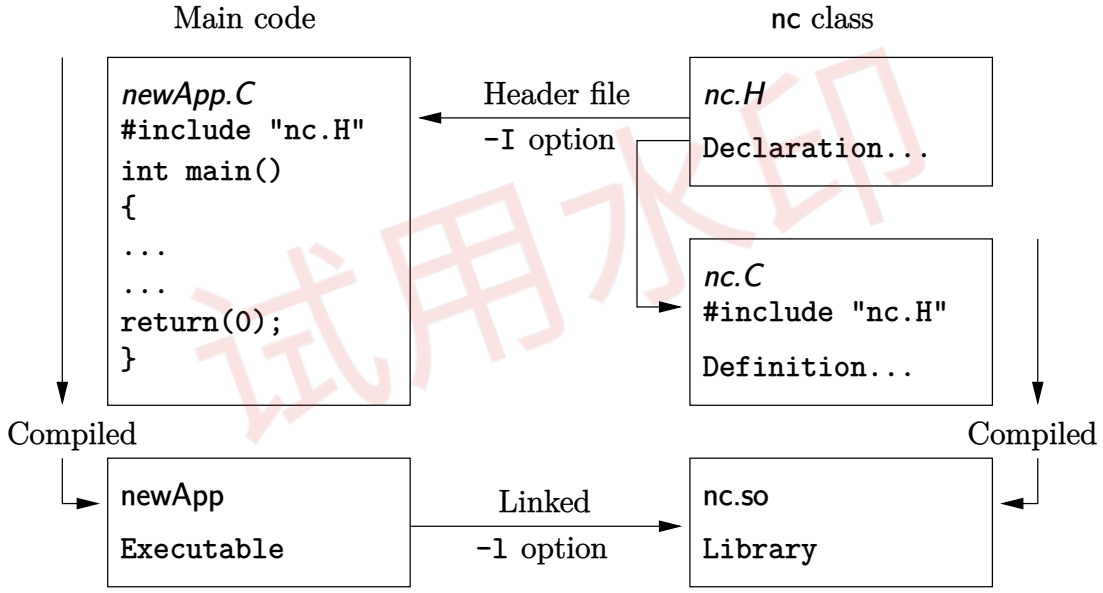


Figure 3.1: Header files, source files, compilation and linking

3.2.1 Header *.H* files

As a means of checking errors, the piece of code being compiled must know that the classes it uses and the operations they perform actually exist. Therefore each class requires a class *declaration*, contained in a header file with a *.H* file extension, *e.g. nc.H*, that includes the names of the class and its functions. This file is included at the beginning of any piece of code using the class, using the `#include` directive described below, including the class declaration code itself. Any piece of *.C* code can resource any number of classes and must begin by including all the *.H* files required to declare these classes. Those classes in turn can resource other classes and so also begin by including the relevant *.H* files. By searching recursively down the class hierarchy we can produce a complete list of header files for all the classes on which the top level *.C* code ultimately depends; these *.H* files are

known as the *dependencies*. With a dependency list, a compiler can check whether the source files have been updated since their last compilation and selectively compile only those that need to be.

Header files are included in the code using the `#include` directive, *e.g.*

```
#include "otherHeader.H";
```

This causes the compiler to suspend reading from the current file, to read the included file. This mechanism allows any self-contained piece of code to be put into a header file and included at the relevant location in the main code in order to improve code readability. For example, in most OpenFOAM applications the code for creating fields and reading field input data is included in a file *createFields.H* which is called at the beginning of the code. In this way, header files are not solely used as class declarations.

It is wmake that performs the task of maintaining file dependency lists amongst other functions listed below.

- Automatic generation and maintenance of file dependency lists, *i.e.* lists of files which are included in the source files and hence on which they depend.
- Multi-platform compilation and linkage, handled through appropriate directory structure.
- Multi-language compilation and linkage, *e.g.* C, C++, Java.
- Multi-option compilation and linkage, *e.g.* debug, optimised, parallel and profiling.
- Support for source code generation programs, *e.g.* lex, yacc, IDL, MOC.
- Simple syntax for source file lists.
- Automatic creation of source file lists for new codes.
- Simple handling of multiple shared or static libraries.

3.2.2 Compiling with wmake

OpenFOAM applications are organised using a standard convention that the source code of each application is placed in a directory whose name is that of the application. The top level source file then takes the application name with the *.C* extension. For example, the source code for an application called newApp would reside is a directory *newApp* and the top level file would be *newApp.C* as shown in Figure 3.2. wmake then requires the directory must contain a *Make* subdirectory containing 2 files, *options* and *files*, that are described in the following sections.

3.2.3 Including headers

The compiler searches for the included header files in the following order, specified with the `-I` option in wmake:

1. the *$WM_PROJECT_DIR/src/OpenFOAM/lnInclude* directory;
2. a local *lnInclude* directory, *i.e.newApp/lnInclude*;

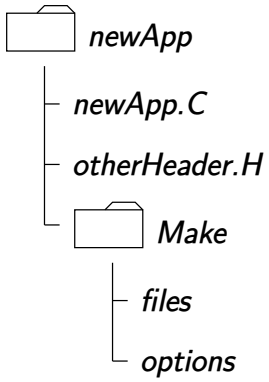


Figure 3.2: Directory structure for an application

3. the local directory, *i.e.newApp*;

4. platform dependent paths set in files in the *$WM_PROJECT_DIR/wmake/rules* directory, *e.g./usr/include/X11*;

5. other directories specified explicitly in the *Make/options* file with the `-I` option.

The *Make/options* file contains the full directory paths to locate header files using the syntax:

```
EXE_INC = \
    -I<directoryPath1> \
    -I<directoryPath2> \
    ...                \
    -I<directoryPathN>
```

Notice first that the directory names are preceded by the `-I` flag and that the syntax uses the `\` to continue the `EXE_INC` across several lines, with no `\` after the final entry.

3.2.4 Linking to libraries

The compiler links to shared object library files in the following directory **paths**, specified with the `-L` option in wmake:

1. the *$FOAM_LIBBIN* directory;

2. platform dependent paths set in files in the *$WM_DIR/rules* directory, *e.g.$(MPI_ARCH_PATH)/lib*;

3. other directories specified in the *Make/options* file.

The actual library **files** to be linked must be specified using the `-l` option and removing the `lib` prefix and `.so` extension from the library file name, *e.g.* libnew.so is included with the flag `-lnew`. By default, wmake loads the following libraries:

1. the libOpenFOAM.so library from the *$FOAM_LIBBIN* directory;

2. platform dependent libraries specified in set in files in the *$WM_DIR/rules* directory, *e.g.* libm.so and libdl.so;

3. other libraries specified in the *Make/options* file.

The *Make/options* file contains the full directory paths and library names using the syntax:

```
EXE_LIBS = \
    -L<libraryPath> \
    -l<library1>     \
    -l<library2>     \
    ...               \
    -l<libraryN>
```

To summarise: the directory paths are preceded by the `-L` flag, the library names are preceded by the `-l` flag.

3.2.5 Source files to be compiled

The compiler requires a list of *.C* source files that must be compiled. The list must contain the main *.C* file but also any other source files that are created for the specific application but are not included in a class library. For example, users may create a new class or some new functionality to an existing class for a particular application. The full list of *.C* source files must be included in the *Make/files* file. For many applications the list only includes the name of the main *.C* file, *e.g.* *newApp.C* in the case of our earlier example.

The *Make/files* file also includes a full path and name of the compiled executable, specified by the `EXE =` syntax. Standard convention stipulates the name is that of the application, *i.e.*newApp in our example. The OpenFOAM release offers two useful choices for path: standard release applications are stored in *$FOAM_APPBIN*; applications developed by the user are stored in *$FOAM_USER_APPBIN*.

If the user is developing their own applications, we recommend they create an *applications* subdirectory in their *$WM_PROJECT_USER_DIR* directory containing the source code for personal OpenFOAM applications. As with standard applications, the source code for each OpenFOAM application should be stored within its own directory. The only difference between a user application and one from the standard release is that the *Make/files* file should specify that the user's executables are written into their *$FOAM_USER_APPBIN* directory. The *Make/files* file for our example would appear as follows:

```
newApp.C

EXE = $(FOAM_USER_APPBIN)/newApp
```

3.2.6 Running wmake

The wmake script is generally executed by typing:

```
wmake <optionalDirectory>
```

The <`optionalDirectory`> is the directory path of the application that is being compiled. Typically, wmake is executed from within the directory of the application being compiled, in which case <`optionalDirectory`> can be omitted.

3.2.7 wmake environment variables

For information, the general environment variable settings used by wmake are listed below.

- $WM_PROJECT_INST_DIR: full path to the installation directory, *e.g.$HOME/-OpenFOAM*.
- $WM_PROJECT: name of the project being compiled, *i.e.* `OpenFOAM`.
- $WM_PROJECT_VERSION: version of the project being compiled, *i.e.* `11`.
- $WM_PROJECT_DIR: full path to the main directory of the OpenFOAM release, *e.g. $HOME/OpenFOAM/OpenFOAM-11*.
- $WM_PROJECT_USER_DIR: full path to the equivalent directory for customised developments in the user account, *e.g. $HOME/OpenFOAM/${USER}-11*.
- $WM_THIRD_PARTY_DIR: full path to the directory of *ThirdParty* software, *e.g. $HOME/OpenFOAM/ThirdParty-11*.

The environment variable settings for the compilation with wmake are listed below.

- $WM_ARCH: machine architecture, *e.g.* `linux`, `linux64`, `linuxArm64`, `linuxARM7`, `linuxPPC64`, `linuxPPC64le`.
- $WM_ARCH_OPTION: 32 or 64 bit architecture.
- $WM_DIR: full path of the *wmake* directory.
- $WM_LABEL_SIZE: 32 or 64 bit size for labels (integers).
- $WM_LABEL_OPTION: `Int32` or `Int64` compilation of labels.
- $WM_LINK_LANGUAGE: compiler used to link libraries and executables `c++`.
- $WM_MPLIB: parallel communications library, `SYSTEMOPENMPI` = system version of openMPI, alternatives include `OPENMPI`, `SYSTEMMPI`, `MPICH`, `MPICH-GM`, `HPMPI`, `MPI`, `QSMPI`, `INTELMPI` and `SGIMPI`.
- $WM_OPTIONS, *e.g.* `linux64GccDPInt32Opt`, formed by combining $WM_ARCH, $WM_COMPILER, $WM_PRECISION_OPTION, $WM_LABEL_OPTION, and $WM-_COMPILE_OPTION.
- $WM_PRECISION_OPTION: floating point precision of the compiled binares, `SP` = single precision, `DP` = double precision.

The environment variable settings relating to the choice of compiler and options withwmake are listed below.

- $WM_CC: choice of C compiler, `gcc`.
- $WM_CFLAGS: extra flags to the C compiler, *e.g.* `-m64 -fPIC`.
- $WM_CXX: choice of C++ compiler, `g++`.
- $WM_CXXFLAGS: extra flags to the C++ compiler, *e.g.* `-m64 -fPIC -std=c++0x`.

- $WM_COMPILER: compiler being used, *e.g.* `Gcc` = gcc, `Clang` = LLVM Clang
- $WM_COMPILE_OPTION: compilation option, `Debug` = debugging, `Opt` = optimised.
- $WM_COMPILER_LIB_ARCH: compiler library architecture, *e.g.* `64`.
- $WM_COMPILER_TYPE: choice of compiler, `system`, or `ThirdParty`, *i.e.* compiled in *ThirdParty* directory.
- $WM_LDFLAGS: extra flags for the linker, *e.g.* `-m64`.
- $WM_LINK_LANGUAGE: linker language, *e.g.* `c++`.
- $WM_OSTYPE: Operating system, `=POSIX c++`.

3.2.8 Removing dependency lists: wclean

When it is run, wmake builds a dependency list file with a *.dep* file extension, *e.g. newApp-.C.dep* in our example, in a *$WM_OPTIONS* sub-directory of the *Make* directory, *e.g. Make/linuxGccDPInt64Opt*. If the user wishes to remove these files, *e.g.* after making code changes, the user can run the wclean script by typing:

```
wclean <optionalDirectory>
```

Again, the <optionalDirectory> is a path to the directory of the application that is being compiled. Typically, wclean is executed from within the directory of the application, in which case the path can be omitted.

3.2.9 Compiling libraries

When compiling a library, there are 2 critical differences in the configuration of the file in the *Make* directory:

- in the *files* file, `EXE =` is replaced by `LIB =` and the target directory for the compiled entity changes from `$FOAM_APPBIN` to `$FOAM_LIBBIN` (and an equivalent `$FOAM_USER_LIBBIN` directory);
- in the *options* file, `EXE_LIBS =` is replaced by `LIB_LIBS =` to indicate libraries linked to library being compiled.

When wmake is executed it additionally creates a directory named *lnInclude* that contains soft links to all the files in the library. The *lnInclude* directory is deleted by the wclean script when cleaning library source code.

3.2.10 Compilation example: the foamRun application

The source code for application foamRun is in the *$FOAM_SOLVERS/foamRun* directory and the top level source file is named *foamRun.C*. The *foamRun.C* source code is:

```
/*---------------------------------------------------------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Copyright (C) 2022-2023 OpenFOAM Foundation
     \\/     M anipulation  |
-------------------------------------------------------------------------------
License
    This file is part of OpenFOAM.

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    (at your option) any later version.

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    ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or
    FITNESS FOR A PARTICULAR PURPOSE.  See the GNU General Public License
    for more details.

    You should have received a copy of the GNU General Public License
    along with OpenFOAM.  If not, see <http://www.gnu.org/licenses/>.

Application
    foamRun

Description
    Loads and executes an OpenFOAM solver module either specified by the
    optional \c solver entry in the \c controlDict or as a command-line
    argument.

    Uses the flexible PIMPLE (PISO-SIMPLE) solution for time-resolved and
    pseudo-transient and steady simulations.

Usage
    \b foamRun [OPTION]

      - \par -solver <name>
        Solver name

      - \par -libs '(\"lib1.so\" ... \"libN.so\")'
        Specify the additional libraries loaded

    Example usage:
      - To run a \c rhoPimpleFoam case by specifying the solver on the
        command line:
        \verbatim
            foamRun -solver fluid
        \endverbatim

      - To update and run a \c rhoPimpleFoam case add the following entries to
        the controlDict:
        \verbatim
            application     foamRun;

            solver          fluid;
        \endverbatim
        then execute \c foamRun

\*---------------------------------------------------------------------------*/

#include "argList.H"
#include "solver.H"
#include "pimpleSingleRegionControl.H"
#include "setDeltaT.H"

using namespace Foam;

// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

int main(int argc, char *argv[])
{
    argList::addOption
    (
        "solver",
        "name",
        "Solver name"
    );

    #include "setRootCase.H"
    #include "createTime.H"

    // Read the solverName from the optional solver entry in controlDict
```

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```
84      word solverName
85      (
86          runTime.controlDict().lookupOrDefault("solver", word::null)
87      );
88
89      // Optionally reset the solver name from the -solver command-line argument
90      args.optionReadIfPresent("solver", solverName);
91
92      // Check the solverName has been set
93      if (solverName == word::null)
94      {
95          args.printUsage();
96
97          FatalErrorIn(args.executable())
98              << "solver not specified in the controlDict or on the command-line"
99              << exit(FatalError);
100     }
101     else
102     {
103         // Load the solver library
104         solver::load(solverName);
105     }
106
107     // Create the default single region mesh
108     #include "createMesh.H"
109
110     // Instantiate the selected solver
111     autoPtr<solver> solverPtr(solver::New(solverName, mesh));
112     solver& solver = solverPtr();
113
114     // Create the outer PIMPLE loop and control structure
115     pimpleSingleRegionControl pimple(solver.pimple);
116
117     // Set the initial time-step
118     setDeltaT(runTime, solver);
119
120     // * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
121
122     Info<< nl << "Starting time loop\n" << endl;
123
124     while (pimple.run(runTime))
125     {
126         solver.preSolve();
127
128         // Adjust the time-step according to the solver maxDeltaT
129         adjustDeltaT(runTime, solver);
130
131         runTime++;
132
133         Info<< "Time = " << runTime.userTimeName() << nl << endl;
134
135         // PIMPLE corrector loop
136         while (pimple.loop())
137         {
138             solver.moveMesh();
139             solver.fvModels().correct();
140             solver.prePredictor();
141             solver.momentumPredictor();
142             solver.thermophysicalPredictor();
143             solver.pressureCorrector();
144             solver.postCorrector();
145         }
146
147         solver.postSolve();
148
149         runTime.write();
150
151         Info<< "ExecutionTime = " << runTime.elapsedCpuTime() << " s"
152             << "  ClockTime = " << runTime.elapsedClockTime() << " s"
153             << nl << endl;
154     }
155
156     Info<< "End\n" << endl;
157
158     return 0;
159 }
160
161
162 // ************************************************************************* //
```

The code begins with a description of the application contained within comments over

1 line (`//`) and multiple lines (`/*...*/`). Following that, the code contains several `#include` statements, *e.g.* `#include "argList.H"`, which causes the compiler to suspend reading from the current file, *foamRun.C* to read the *argList.H* file.

foamRun uses the finite volume numerics library and therefore requires the necessary header files, specified by the `EXE_INC = -I...` option, and links to the libraries with the `EXE_LIBS = -l...` option. The *Make/options* therefore contains the following:

```
EXE_INC = \
    -I$(LIB_SRC)/finiteVolume/lnInclude

EXE_LIBS = \
    -lfiniteVolume
```

foamRun contains the *foamRun.C* source and the executable is written to the *$FOAM_APPBIN* directory. The application uses functions to initialise and adjust the time step, defined in the *setDeltaT.C* file. The *Make/files* therefore contains:

```
setDeltaT.C
foamRun.C

EXE = $(FOAM_APPBIN)/foamRun
```

Following the recommendations of section 3.2.5, the user can compile a separate version of foamRun into their local *$FOAM_USER_DIR* directory as follows. First, the user should copy the foamRun source code to a local directory, *e.g.* *$FOAM_RUN*.

```
cd $FOAM_RUN
cp -r $FOAM_SOLVERS/foamRun .
```

They should then go into the *foamRun* directory.

```
cd foamRun
```

and edit the *Make/files* file as follows:

```
foamRun.C

EXE = $(FOAM_USER_APPBIN)/foamRun
```

Finally, they should run the wmake script.

```
wmake
```

The code should compile and produce a message similar to the following

```
Making dependency list for source file foamRun.C
g++ -std=c++14 -m64...
...
-o ... platforms/linux64GccDPInt32Opt/bin/foamRun
```

If the user tries recompiling without making any changes to the code file, nothing will happen. The user can compile the application from scratch by removing the dependency list with

```
wclean
```

and running wmake.

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3.2.11 Debug messaging and optimisation switches

OpenFOAM provides a system of messaging that is written during runtime, most of which are to help debugging problems encountered during running of a OpenFOAM case. The switches are listed in the *$WM_PROJECT_DIR/etc/controlDict* file; should the user wish to change the settings they should make a copy to their *$HOME* directory, *i.e.* *$HOME/.OpenFOAM/11/controlDict* file (see section 4.3 for more information). The list of possible switches is extensive, relating to a class or range of functionality, and can be switched on by their inclusion in the *controlDict* file, and by being set to `1`. For example, OpenFOAM can perform the checking of dimensional units in all calculations by setting the `dimensionSet` switch to `1`.

A small number of switches control messaging at three levels, `0`, `1` and `2`, most notably the overall `level` switch and `lduMatrix` which provides messaging for solver convergence during a run.

There are some switches that control certain operational and optimisation issues. Of particular importance is `fileModificationSkew`. OpenFOAM scans the write time of data files to check for modification. When running over a NFS with some disparity in the clock settings on different machines, field data files appear to be modified ahead of time. This can cause a problem if OpenFOAM views the files as newly modified and attempting to re-read this data. The `fileModificationSkew` keyword is the time in seconds that OpenFOAM will subtract from the file write time when assessing whether the file has been newly modified. The main optimisation switches are listed below:

- `fileModificationSkew`: a time in seconds that should be set higher than the maximum delay in NFS updates and clock difference for running OpenFOAM over a NFS.
- `fileModificationChecking`: method of checking whether files have been modified during a simulation, either reading the `timeStamp` or using `inotify`; versions that read only master-node data also exist, termed `timeStampMaster` and `inotifyMaster`.
- `commsType`: parallel communications type, `nonBlocking`, `scheduled` or `blocking`.
- `floatTransfer`: if `1`, will compact numbers to `float` precision before transfer; default is `0`.
- `nProcsSimpleSum`: optimises the global sum for parallel processing, by setting the number of processors above which a hierarchical sum is performed rather than a linear sum.

3.2.12 Dynamic linking at run-time

The situation may arise that a user creates a new library, say `new1`, and wishes the features within that library to be available across a range of applications. For example, the user may create a new boundary condition, compiled into `new1`, that would need to be recognised by a range of solver applications, pre- and post-processing utilities, mesh tools, *etc.* Under normal circumstances, the user would need to recompile every application with the `new1` linked to it.

Instead there is a simple mechanism to link one or more shared object libraries dynamically at run-time in OpenFOAM. The use can simply add the optional keyword entry `libs` to the *controlDict* file for a case and enter the full names of the libraries within a list

(as quoted string entries). For example, if a user wished to link the libraries new1 and new2 at run-time, they would simply need to add the following to the case *controlDict* file:

```
libs
(
    "libnew1.so"
    "libnew2.so"
);
```

3.3 Running applications

Each application is designed to be executed from a terminal command line, typically reading and writing a set of data files associated with a particular case. The data files for a case are stored in a directory named after the case as described in section 4.1; the directory name with full path is here given the generic name <*caseDir*>.

For any application, the form of the command line entry for any can be found by simply entering the application name at the command line with the `-help` option, *e.g.* typing

```
foamRun -help
```

returns the usage

```
Usage: foamRun [OPTIONS]
options:
  -case <dir>       specify alternate case directory, default is cwd
  -fileHandler <handler>
                    override the fileHandler
  -hostRoots <((host1 dir1) .. (hostN dirN))>
                    slave root directories for distributed running
  -libs '("lib1.so" ... "libN.so")'
                    pre-load libraries
  -noFunctionObjects
                    do not execute functionObjects
  -parallel         run in parallel
  -roots <(dir1 .. dirN)>
                    slave root directories for distributed running
  -solver <name>    Solver name
  -srcDoc           display source code in browser
  -doc              display application documentation in browser
  -help             print the usage
```

If the application is executed from within a case directory, it will operate on that case. Alternatively, the `-case` <`caseDir`> option allows the case to be specified directly so that the application can be executed from anywhere in the filing system.

Like any UNIX/Linux executable, applications can be run as a background process, *i.e.* one which does not have to be completed before the user can give the shell additional commands. If the user wished to run the foamRun example as a background process and output the case progress to a *log* file, they could enter:

```
foamRun > log &
```

3.4 Running applications in parallel

This section describes how to run OpenFOAM in parallel on distributed processors. The method of parallel computing used by OpenFOAM is known as domain decomposition, in which the geometry and associated fields are broken into pieces and allocated to separate processors for solution. The process of parallel computation involves: decomposition of mesh and fields; running the application in parallel; and, post-processing the decomposed case as described in the following sections. The parallel running uses the public domain openMPI implementation of the standard message passing interface (MPI) by default, although other libraries can be used.

3.4.1 Decomposition of mesh and initial field data

The mesh and fields are decomposed using the decomposePar utility. The underlying aim is to break up the domain with minimal effort but in such a way to guarantee an economic solution. The geometry and fields are broken up according to a set of parameters specified in a dictionary named *decomposeParDict* that must be located in the *system* directory of the case of interest. An example *decomposeParDict* dictionary can be copied into a case *system* directory using the foamGet script.

```
foamGet decomposeParDict
```

The dictionary entries within it are reproduced below.

```
16  numberOfSubdomains  8;
17
18  /*
19      Main methods are:
20      1) Geometric: "simple"; "hierarchical", with ordered sorting, e.g. xyz, yxz
21      2) Scotch: "scotch", when running in serial; "ptscotch", running in parallel
22  */
23
24  method              hierarchical;
25
26  simpleCoeffs
27  {
28      n               (4 2 1); // total must match numberOfSubdomains
29  }
30
31  hierarchicalCoeffs
32  {
33      n               (4 2 1); // total must match numberOfSubdomains
34      order           xyz;
35  }
36
37
38  // ************************************************************************* //
```

The user has a choice of four methods of decomposition, specified by the `method` keyword as described below.

`simple` Simple geometric decomposition in which the domain is split into pieces by direction, *e.g.* 2 pieces in the x direction, 1 in y *etc.*

`hierarchical` Hierarchical geometric decomposition which is the same as `simple` except the user specifies the order in which the directional split is done, *e.g.* first in the y-direction, then the x-direction *etc.*

`scotch` Scotch decomposition which requires no geometric input from the user and attempts to minimise the number of processor boundaries. The user can specify a weighting for the decomposition between processors, through an optional `processorWeights` keyword which can be useful on machines with differing performance between processors. There is also an optional keyword entry `strategy` that controls the decomposition strategy through a complex string supplied to Scotch. For more information, see the source code file: *$FOAM_SRC/parallel/decompose/scotchDecomp/scotchDecomp.C*

For each `method` there are a set of coefficients specified in a sub-dictionary of *decompositionDict*, named <*method*>*Coeffs* as shown in the dictionary listing. The full set of keyword entries in the *decomposeParDict* dictionary are explained below:

- `numberOfSubdomains`: total number of subdomains N.
- `method`: method of decomposition, `simple`, `hierarchical`, `scotch`.
- `n`: for `simple` and `hierarchical`, number of subdomains in x, y, z (`(`n_x n_y n_z`)`
- `order`: order of `hierarchical` decomposition, `xyz/xzy/yxz`...
- `processorWeights` option for `scotch`: list of weighting factors (`(`<`wt1`>`...`<`wtN`>`)`) for allocation of cells to processors; <`wt1`> is the weighting factor for processor 1, *etc.*; weights are normalised so can take any range of values.

The decomposePar utility is executed in the normal manner by typing

```
decomposePar
```

3.4.2 File input/output in parallel

Using standard file input/output completion, a set of subdirectories will have been created, one for each processor, in the case directory. The directories are named *processor*N where $N = 0, 1, \ldots$ represents a processor number and contains a time directory, containing the decomposed field descriptions, and a *constant/polyMesh* directory containing the decomposed mesh description.

While this file structure is well-organised, for large parallel cases, it generates a large number of files. In very large simulations, users can experience problems including hitting limits on the number of open files imposed by the operating system.

As an alternative, the **collated** file format was introduced in OpenFOAM in which the data for each decomposed field (and mesh) is collated into a single file that is written (and read) on the master processor. The files are stored in a single directory named *processors*.

The file writing can be threaded allowing the simulation to continue running while the data is being written to file — see below for details. NFS (Network File System) is not needed when using the collated format and, additionally, there is a `masterUncollated` option to write data with the original `uncollated` format without NFS.

The controls for the file handling are in the `OptimisationSwitches` of the global *etc/controlDict* file:

```
OptimisationSwitches
{
    ...
```

```
    //- Parallel IO file handler
    // uncollated (default), collated or masterUncollated
    fileHandler uncollated;

    //- collated: thread buffer size for queued file writes.
    // If set to 0 or not sufficient for the file size threading is not used.
    // Default: 2e9
    maxThreadFileBufferSize 2e9;

    //- masterUncollated: non-blocking buffer size.
    // If the file exceeds this buffer size scheduled transfer is used.
    // Default: 2e9
    maxMasterFileBufferSize 2e9;
}
```

The `fileHandler` can be set for a specific simulation by:

- over-riding the global `OptimisationSwitches {fileHandler ...;}` in the case *controlDict* file;
- using the `-fileHandler` command line argument to the solver;
- setting the $FOAM_FILEHANDLER environment variable.

A foamFormatConvert utility allows users to convert files between the collated and uncollated formats, e.g.

```
mpirun -np 2 foamFormatConvert -parallel -fileHandler uncollated
```

An example case demonstrating the file handling methods is provided in:
$FOAM_TUTORIALS/heatTransfer/buoyantFoam/iglooWithFridges

Collated file handling runs faster with threading, especially on large cases. But it requires threading support to be enabled in the underlying MPI. Without it, the simulation will "hang" or crash. For openMPI, threading support is not set by default prior to version 2, but is generally switched on from version 2 onwards. The user can check whether openMPI is compiled with threading support by the following command:

```
ompi_info -c | grep -oE "MPI_THREAD_MULTIPLE[^,]*"
```

When using the collated file handling, memory is allocated for the data in the thread. `maxThreadFileBufferSize` sets the maximum size of memory that is allocated in bytes. If the data exceeds this size, the write does not use threading.

Note: if threading is **not enabled** in the MPI, it must be disabled for collated file handling by setting in the global *etc/controlDict* file:

```
maxThreadFileBufferSize   0;
```

When using the `masterUncollated` file handling, non-blocking MPI communication requires a sufficiently large memory buffer on the master node. `maxMasterFileBufferSize` sets the maximum size of the buffer. If the data exceeds this size, the system uses scheduled communication.

3.4.3 Running a decomposed case

A decomposed OpenFOAM case is run in parallel using the openMPI implementation of MPI.

openMPI can be run on a local multiprocessor machine very simply but when running on machines across a network, a file must be created that contains the host names of the machines. The file can be given any name and located at any path. In the following description we shall refer to such a file by the generic name, including full path, <*machines*>.

The <*machines*> file contains the names of the machines listed one machine per line. The names must correspond to a fully resolved hostname in the */etc/hosts* file of the machine on which the openMPI is run. The list must contain the name of the machine running the openMPI. Where a machine node contains more than one processor, the node name may be followed by the entry `cpu=`n where n is the number of processors openMPI should run on that node.

For example, let us imagine a user wishes to run openMPI from machine `aaa` on the following machines: `aaa`; `bbb`, which has 2 processors; and `ccc`. The <*machines*> would contain:

```
aaa
bbb cpu=2
ccc
```

An application is run in parallel using mpirun.

```
mpirun --hostfile <machines> -np <nProcs>
    <foamExec> <otherArgs> -parallel > log &
```

where: <`nProcs`> is the number of processors; <`foamExec`> is the executable, *e.g.*foamRun; and, the output is redirected to a file named `log`. For example, if foamRun is run on 4 nodes, specified in a file named *machines*, then the following command should be executed:

```
mpirun --hostfile machines -np 4 foamRun -parallel > log &
```

3.4.4 Distributing data across several disks

Data files may need to be distributed if, for example, if only local disks are used in order to improve performance. In this case, the user may find that the root path to the case directory may differ between machines. The paths must then be specified in the *decomposeParDict* dictionary using `distributed` and `roots` keywords. The `distributed` entry should read

```
distributed  yes;
```

and the `roots` entry is a list of root paths, <`root0`>, <`root1`>, ..., for each node

```
roots
<nRoots>
(
   "<root0>"
```

```
    "<root1>"
    ...
);
```

where <nRoots> is the number of roots.

Each of the *processorN* directories should be placed in the case directory at each of the root paths specified in the *decomposeParDict* dictionary. The *system* directory and *files* within the *constant* directory must also be present in each case directory. Note: the files in the *constant* directory are needed, but the *polyMesh* directory is not.

3.4.5 Post-processing parallel processed cases

When post-processing cases that have been run in parallel the user has two options:

- reconstruction of the mesh and field data to recreate the complete domain and fields, which can be post-processed as normal;
- post-processing each segment of decomposed domain individually.

After a case has been run in parallel, it can be reconstructed for post-processing. The case is reconstructed by merging the sets of time directories from each *processorN* directory into a single set of time directories. The reconstructPar utility performs such a reconstruction by executing the command:

```
reconstructPar
```

The user may post-process decomposed cases using the paraFoam post-processor, described in section 7.1. The whole simulation can be post-processed by reconstructing the case or alternatively it is possible to post-process a segment of the decomposed domain individually by simply treating the individual processor directory as a case in its own right.

3.5 Solver modules

From OpenFOAM version 11, application solvers, *e.g.* simpleFoam have been largely replaced by the generic foamRun solver which loads a solver module, *e.g.* incompressibleFluid that defines the flow solution. Solver modules are located in the *$FOAM_APP/modules* directory. The current solver modules distributed with OpenFOAM are listed below.

3.5.1 Single-phase modules

fluid Solver module for steady or transient turbulent flow of compressible fluids with heat-transfer for HVAC and similar applications, with optional mesh motion and change.

incompressibleDenseParticleFluid Solver module for transient flow of incompressible isothermal fluids coupled with particle clouds including the effect of the volume fraction of particles on the continuous phase, with optional mesh motion and change.

incompressibleFluid Solver module for steady or transient turbulent flow of incompressible isothermal fluids with optional mesh motion and change.

multicomponentFluid Solver module for steady or transient turbulent flow of compressible multicomponent fluids with optional mesh motion and change.

shockFluid Solver module for density-based solution of compressible flow

XiFluid Solver module for compressible premixed/partially-premixed combustion with turbulence modelling.

3.5.2 Multiphase/VoF flow modules

compressibleMultiphaseVoF Solver module for the solution of multiple compressible, isothermal immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

compressibleVoF Solver module for for 2 compressible, non-isothermal immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

incompressibleDriftFlux Solver module for 2 incompressible fluids using the mixture approach with the drift-flux approximation for relative motion of the phases, with optional mesh motion and mesh topology changes including adaptive re-meshing.

incompressibleMultiphaseVoF Solver module for the solution of multiple incompressible, isothermal immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

incompressibleVoF Solver module for for 2 incompressible, isothermal immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

isothermalFluid Solver module for steady or transient turbulent flow of compressible isothermal fluids with optional mesh motion and change.

multiphaseVoFSolver Base solver module for the solution of multiple immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

3.5.3 Solid modules

solid Solver module for thermal transport in solid domains and regions for conjugate heat transfer, HVAC and similar applications, with optional mesh motion and mesh topology changes.

solidDisplacement Solver module for steady or transient segregated finite-volume solution of linear-elastic, small-strain deformation of a solid body, with optional thermal diffusion and thermal stresses.

3.5.4 Film modules

isothermalFilm Solver module for flow of compressible isothermal liquid films

film Solver module for flow of compressible liquid films

3.5.5 Utility modules

functions Solver module to execute the `functionObjects` for a specified

movingMesh Solver module to move the mesh.

3.5.6 Base classes for solver modules

fluidSolver Base solver module for fluid solvers.

twoPhaseSolver Solver module base-class for for 2 immiscible fluids, with optional mesh motion and mesh topology changes including adaptive re-meshing.

twoPhaseVoFSolver Solver module base-class for for 2 immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

VoFSolver Base solver module base-class for the solution of immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

3.6 Standard solvers

With the introduction of solver modules in OpenFOAM v11, the number of solver applications has much reduced. The applications which are relevant, including foamRun and foamMultiRun, are located in the *$FOAM_SOLVERS* directory, reached quickly by typing `sol` at the command line. These solver applications are listed in the following section.

There are also some legacy solver applications, which either have not been replaced yet by new solver modules or are included for teaching purposes. They are provided in the *$FOAM_APP/legacy* directory are listed in the subsequent section below.

3.6.1 Main solver applications

foamRun Loads and executes an OpenFOAM solver module either specified by the optional `solver` entry in the `controlDict` or as a command-line argument.

foamMultiRun Loads and executes an OpenFOAM solver modules for each region of a multiregion simulation e.g. for conjugate heat transfer.

boundaryFoam Steady-state solver for incompressible, 1D turbulent flow, typically to generate boundary layer conditions at an inlet, for use in a simulation.

chemFoam Solver for chemistry problems, designed for use on single cell cases to provide comparison against other chemistry solvers, that uses a single cell mesh, and fields created from the initial conditions.

potentialFoam Potential flow solver which solves for the velocity potential, to calculate the flux-field, from which the velocity field is obtained by reconstructing the flux.

3.6.2 Legacy solver applications

electrostaticFoam Solver for electrostatics.

magneticFoam Solver for the magnetic field generated by permanent magnets.

mhdFoam Solver for magnetohydrodynamics (MHD): incompressible, laminar flow of a conducting fluid under the influence of a magnetic field.

laplacianFoam Solves a simple Laplace equation, e.g. for thermal diffusion in a solid.

financialFoam Solves the Black-Scholes equation to price commodities.

dsmcFoam Direct simulation Monte Carlo (DSMC) solver for, transient, multi-species flows.

mdEquilibrationFoam Solver to equilibrate and/or precondition molecular dynamics systems.

mdFoam Molecular dynamics solver for fluid dynamics.

dnsFoam Direct numerical simulation solver for boxes of isotropic turbulence.

adjointShapeOptimizationFoam Steady-state solver for incompressible, turbulent flow of non-Newtonian fluids with optimisation of duct shape by applying "blockage" in regions causing pressure loss as estimated using an adjoint formulation.

icoFoam Transient solver for incompressible, laminar flow of Newtonian fluids.

shallowWaterFoam Transient solver for inviscid shallow-water equations with rotation.

porousSimpleFoam Steady-state solver for incompressible, turbulent flow with implicit or explicit porosity treatment and support for multiple reference frames (MRF).

rhoPorousSimpleFoam Steady-state solver for turbulent flow of compressible fluids, with implicit or explicit porosity treatment and optional sources.

PDRFoam Solver for compressible premixed/partially-premixed combustion with turbulence modelling.

3.7 Standard utilities

The utilities with the OpenFOAM distribution are in the *$FOAM_UTILITIES* directory. The names are reasonably descriptive, *e.g.* ideasToFoam converts mesh data from the format written by I-DEAS to the OpenFOAM format. The descriptions of current utilities distributed with OpenFOAM are given in the following Sections.

3.7.1 Pre-processing

applyBoundaryLayer Apply a simplified boundary-layer model to the velocity and turbulence fields based on the 1/7th power-law.

boxTurb Makes a box of turbulence which conforms to a given energy spectrum and is divergence free.

changeDictionary Utility to change dictionary entries, e.g. can be used to change the patch type in the field and `polyMesh`/boundary files.

createExternalCoupledPatchGeometry Application to generate the patch geometry (points and faces) for use with the `externalCoupled` boundary condition.

dsmcInitialise Initialise a case for `dsmcFoam` by reading the initialisation dictionary system/`dsmcInitialise`.

engineSwirl Generates a swirling flow for engine calculations.

faceAgglomerate Agglomerate boundary faces using the pairPatchAgglomeration algorithm. It writes a map from the fine to coarse grid.

foamSetupCHT Sets up a multi-region case using template files for material properties, field and system files.

mapFields Maps volume fields from one mesh to another, reading and interpolating all fields present in the time directory of both cases.

mapFieldsPar Maps volume fields from one mesh to another, reading and interpolating all fields present in the time directory of both cases. Parallel and non-parallel cases are handled without the need to reconstruct them first.

mdInitialise Initialises fields for a molecular dynamics (MD) simulation.

setAtmBoundaryLayer Applies atmospheric boundary layer models to the entire domain for case initialisation.

setFields Set values on a selected set of cells/patchfaces through a dictionary.

setWaves Applies wave models to the entire domain for case initialisation using level sets for second-order accuracy.

snappyHexMeshConfig Preconfigures *blockMeshDict*, *surfaceFeaturesDict* and *snappyHexMeshDict* files based on the case surface geometry files.

viewFactorsGen View factors are calculated based on a face agglomeration array (`finalAgglom` generated by `faceAgglomerate` utility).

3.7.2 Mesh generation

blockMesh A multi-block mesh generator.

extrudeMesh Extrude mesh from existing patch (by default outwards facing normals; optional flips faces) or from patch read from file.

extrude2DMesh Takes 2D mesh (all faces 2 points only, no front and back faces) and creates a 3D mesh by extruding with specified thickness.

extrudeToRegionMesh Extrude `faceZones` (internal or boundary faces) or `faceSets` (boundary faces only) into a separate mesh (as a different region).

snappyHexMesh Automatic split hex mesher. Refines and snaps to surface.

zeroDimensionalMesh Creates a zero-dimensional mesh.

3.7.3 Mesh conversion

ansysToFoam Converts an ANSYS input mesh file, exported from I-DEAS, to OpenFOAM format.

ccm26ToFoam Reads CCM files as written by Prostar/ccm using ccm 2.6

cfx4ToFoam Converts a CFX 4 mesh to OpenFOAM format.

datToFoam Reads in a `datToFoam` mesh file and outputs a points file. Used in conjunction with `blockMesh`.

fluent3DMeshToFoam Converts a Fluent mesh to OpenFOAM format.

fluentMeshToFoam Converts a Fluent mesh to OpenFOAM format including multiple region and region boundary handling.

foamMeshToFluent Writes out the OpenFOAM mesh in Fluent mesh format.

foamToStarMesh Reads an OpenFOAM mesh and writes a pro-STAR (v4) bnd/cel/vrt format.

foamToSurface Reads an OpenFOAM mesh and writes the boundaries in a surface format.

gambitToFoam Converts a GAMBIT mesh to OpenFOAM format.

gmshToFoam Reads .msh file as written by Gmsh.

ideasUnvToFoam I-Deas unv format mesh conversion.

kivaToFoam Converts a KIVA3v grid to OpenFOAM format.

mshToFoam Converts .msh file generated by the Adventure system.

netgenNeutralToFoam Converts neutral file format as written by Netgen v4.4.

plot3dToFoam Plot3d mesh (ascii/formatted format) converter.

sammToFoam Converts a Star-CD (v3) SAMM mesh to OpenFOAM format.

star3ToFoam Converts a Star-CD (v3) pro-STAR mesh into OpenFOAM format.

star4ToFoam Converts a Star-CD (v4) pro-STAR mesh into OpenFOAM format.

tetgenToFoam Converts .ele and .node and .face files, written by tetgen.

vtkUnstructuredToFoam Converts ascii .vtk (legacy format) file generated by vtk/paraview.

writeMeshObj For mesh debugging, writes mesh as three separate OBJ files for visualisation.

3.7.4 Mesh manipulation

attachMesh Attach topologically detached mesh using prescribed mesh modifiers.

autoPatch Divides external faces into patches based on (user supplied) feature angle.

checkMesh Checks validity of a mesh.

createBaffles Makes internal faces into boundary faces. Does not duplicate points, unlike `mergeOrSplitBaffles`.

createNonConformalCouples Utility to create non-conformal couples between non-coupled patches.

createPatch Utility to create patches out of selected boundary faces. Faces come either from existing patches or from a `faceSet`.

deformedGeom Deforms a `polyMesh` using a displacement field U and a scaling factor supplied as an argument.

flattenMesh Flattens the front and back planes of a 2D cartesian mesh.

insideCells Picks up cells with cell centre 'inside' of surface. Requires surface to be closed and singly connected.

mergeBaffles Detects faces that share points (baffles) and merges them into internal faces.

mergeMeshes Merges two meshes.

mirrorMesh Mirrors a mesh around a given plane.

moveMesh Mesh motion and topological mesh changes utility.

objToVTK Read obj line (not surface!) file and convert into vtk.

orientFaceZone Corrects orientation of `faceZone`.

polyDualMesh Calculates the dual of a `polyMesh`. Adheres to all the feature and patch edges.

refineMesh Utility to refine cells in multiple directions.

renumberMesh Renumbers the cell list in order to reduce the bandwidth, reading and renumbering all fields from all the time directories.

rotateMesh Rotates the mesh and fields from the direction n1 to direction n2.

setsToZones Add `pointZones`, `faceZones` or `cellZones` to the mesh from similar named `pointSets`, `faceSets` or `cellSets`.

singleCellMesh Reads all fields and maps them to a mesh with all internal faces removed (`singleCellFvMesh`) which gets written to region `singleCell`.

splitBaffles Detects faces that share points (baffles) and duplicates the points to separate them.

splitMesh Splits mesh by making internal faces external. Uses `attachDetach`.

splitMeshRegions Splits mesh into multiple regions.

stitchMesh Stitches a mesh.

subsetMesh Selects a section of mesh based on a `cellSet`.

topoSet Operates on `cellSets`/`faceSets`/`pointSets` through a dictionary.

transformPoints Transforms the mesh points in the `polyMesh` directory according to the translate, rotate and scale options.

zipUpMesh Reads in a mesh with hanging vertices and zips up the cells to guarantee that all polyhedral cells of valid shape are closed.

3.7.5 Other mesh tools

autoRefineMesh Utility to refine cells near to a surface.

collapseEdges Collapses short edges and combines edges that are in line.

combinePatchFaces Checks for multiple patch faces on same cell and combines them. Multiple patch faces can result from e.g. removal of refined neighbouring cells, leaving 4 exposed faces with same owner.

modifyMesh Manipulates mesh elements.

PDRMesh Mesh and field preparation utility for PDR type simulations.

refineHexMesh Refines a hex mesh by 2x2x2 cell splitting.

refinementLevel Tries to figure out what the refinement level is on refined Cartesian meshes. Run BEFORE snapping.

refineWallLayer Utility to refine cells next to patches.

removeFaces Utility to remove faces (combines cells on both sides).

selectCells Select cells in relation to surface.

splitCells Utility to split cells with flat faces.

3.7.6 Post-processing

engineCompRatio Calculate the geometric compression ratio. Note that if you have valves and/or extra volumes it will not work, since it calculates the volume at BDC and TCD.

foamPostProcess Execute the set of `functionObjects` specified in the selected dictionary (which defaults to *system/controlDict*) or on the command-line for the selected set of times on the selected set of fields.

noise Utility to perform noise analysis of pressure data using the `noiseFFT` library.

pdfPlot Generates a graph of a probability distribution function.

temporalInterpolate Interpolate fields between time-steps e.g. for animation.

3.7.7 Post-processing data converters

foamDataToFluent Translates OpenFOAM data to Fluent format.

foamToEnsight Translates OpenFOAM data to EnSight format.

foamToEnsightParts Translates OpenFOAM data to Ensight format. An Ensight part is created for each `cellZone` and patch.

foamToGMV Translates foam output to GMV readable files.

foamToTetDualMesh Converts `polyMesh` results to `tetDualMesh`.

foamToVTK Legacy VTK file format writer.

smapToFoam Translates a STAR-CD SMAP data file into OpenFOAM field format.

3.7.8 Surface mesh (e.g. OBJ/STL) tools

surfaceAdd Add two surfaces. Does geometric merge on points. Does not check for overlapping/intersecting triangles.

surfaceAutoPatch Patches surface according to feature angle. Like `autoPatch`.

surfaceBooleanFeatures Generates the *extendedFeatureEdgeMesh* for the interface between a boolean operation on two surfaces. Assumes that the orientation of the surfaces is correct.

surfaceCheck Checks geometric and topological quality of a surface.

surfaceClean Removes baffles - collapses small edges, removing triangles. - converts sliver triangles into split edges by projecting point onto base of triangle.

surfaceCoarsen Surface coarsening using bunnylod

surfaceConvert Converts from one surface mesh format to another.

surfaceFeatureConvert Convert between `edgeMesh` formats.

surfaceFeatures Identifies features in a surface geometry and writes them to file, based on control parameters specified by the user.

surfaceFind Finds nearest face and vertex.

surfaceHookUp Find close open edges and stitches the surface along them

surfaceInertia Calculates the inertia tensor, principal axes and moments of a command line specified `triSurface`. Inertia can either be of the solid body or of a thin shell.

surfaceLambdaMuSmooth Smooths a surface using lambda/mu smoothing.

surfaceMeshConvert Converts between surface formats with optional scaling or transformations (rotate/translate) on a `coordinateSystem`.

surfaceMeshExport Export from `surfMesh` to various third-party surface formats with optional scaling or transformations (rotate/translate) on a `coordinateSystem`.

surfaceMeshImport Import from various third-party surface formats into `surfMesh` with optional scaling or transformations (rotate/translate) on a `coordinateSystem`.

surfaceMeshInfo Miscellaneous information about surface meshes.

surfaceMeshTriangulate Extracts surface from a `polyMesh`. Depending on output surface format triangulates faces.

surfaceOrient Set normal consistent with respect to a user provided 'outside' point. If the `-inside` option is used the point is considered inside.

surfacePointMerge Merges points on surface if they are within absolute distance. Since absolute distance use with care!

surfaceRedistributePar (Re)distribution of `triSurface`. Either takes an non-decomposed surface or an already decomposed surface and redistributes it so that each processor has all triangles that overlap its mesh.

surfaceRefineRedGreen Refine by splitting all three edges of triangle ('red' refinement). Neighbouring triangles which are not marked for refinement get split in half ('green' refinement).

surfaceSplitByPatch Writes regions of `triSurface` to separate files.

surfaceSplitByTopology Strips any baffle parts of a surface. A baffle region is one which is reached by walking from an open edge, and stopping when a multiply connected edge is reached.

surfaceSplitNonManifolds Takes multiply connected surface and tries to split surface at multiply connected edges by duplicating points. Introduces concept of - `borderEdge`. Edge with 4 faces connected to it. - `borderPoint`. Point connected to exactly 2 `borderEdges`. - `borderLine`. Connected list of `borderEdges`.

surfaceSubset A surface analysis tool which sub-sets the `triSurface` to choose only a part of interest. Based on `subsetMesh`.

surfaceToPatch Reads surface and applies surface regioning to a mesh. Uses `boundaryMesh` to do the hard work.

surfaceTransformPoints Transform (scale/rotate) a surface. Like `transformPoints` but for surfaces.

3.7.9 Parallel processing

decomposePar Automatically decomposes a mesh and fields of a case for parallel execution of OpenFOAM.

reconstructPar Reconstructs fields of a case that is decomposed for parallel execution of OpenFOAM.

redistributePar Redistributes existing decomposed mesh and fields according to the current settings in the `decomposeParDict` file.

3.7.10 Thermophysical-related utilities

adiabaticFlameT Calculates the adiabatic flame temperature for a given fuel over a range of unburnt temperatures and equivalence ratios.

chemkinToFoam Converts CHEMKINIII thermodynamics and reaction data files into OpenFOAM format.

equilibriumCO Calculates the equilibrium level of carbon monoxide.

equilibriumFlameT Calculates the equilibrium flame temperature for a given fuel and pressure for a range of unburnt gas temperatures and equivalence ratios; the effects of dissociation on O2, H2O and CO2 are included.

mixtureAdiabaticFlameT Calculates the adiabatic flame temperature for a given mixture at a given temperature.

3.7.11 Miscellaneous utilities

foamDictionary Interrogates and manipulates dictionaries.

foamFormatConvert Converts all IOobjects associated with a case into the format specified in the `controlDict`.

foamListTimes List times using `timeSelector`.

foamToC Run-time selection table of contents printing and interrogation.

patchSummary Writes fields and boundary condition info for each patch at each requested time instance.

试用水印

Chapter 4

OpenFOAM cases

This chapter deals with the file structure and organisation of OpenFOAM cases. Normally, a user would assign a name to a case, *e.g.* the tutorial case of aerodynamics of a motorbike is simply named motorBike. This name becomes the name of a directory in which all the case files and sub-directories are stored.

When running a simulation, a case directory can be located anywhere on a user's filing system. However, we recommend putting cases within a *run* subdirectory of the user's filing system, *i.e.$HOME/OpenFOAM/${USER}-11* as described at the beginning of chapter 2. The $FOAM_RUN environment variable is set to *$HOME/OpenFOAM/-${USER}-11/run* by default and the user can quickly move to that directory by executing a preset alias, `run`, at the command line.

The tutorial cases that accompany the OpenFOAM distribution provide useful examples of the case directory structures. The tutorials are located in the *$FOAM_TUTORIALS* directory, reached quickly by executing the `tut` alias at the command line. Users can view tutorial examples at their leisure while reading this chapter.

4.1 File structure of OpenFOAM cases

The basic directory structure of an OpenFOAM case, containing the minimum set of files required to run an application, is shown in Figure 4.1 and described as follows:

constant **directory** that contains a full description of the case mesh in a subdirectory *polyMesh* and files specifying properties and models for the application concerned, *e.g.* *physicalProperties* and *momentumTransport*.

system **directory** for setting parameters associated with the solution procedure itself. It contains *at least* the following three files: *controlDict* where run control parameters are set including start/end time, time step and parameters for data output; *fvSchemes* where discretisation schemes used in the solution are selected; and, *fvSolution* where the equation solvers, tolerances and other algorithm controls are set for the run.

'time' directories containing individual files of data for particular fields, *e.g.* velocity and pressure. The data can be: either, initial values and boundary conditions that the user must specify to define the problem; or, results written to file by OpenFOAM. Fields must always be initialised, even when the solution does not strictly require it, as in steady-state problems. The name of each time directory is based on the simulated time at which the data is written and is described fully in section 4.4.

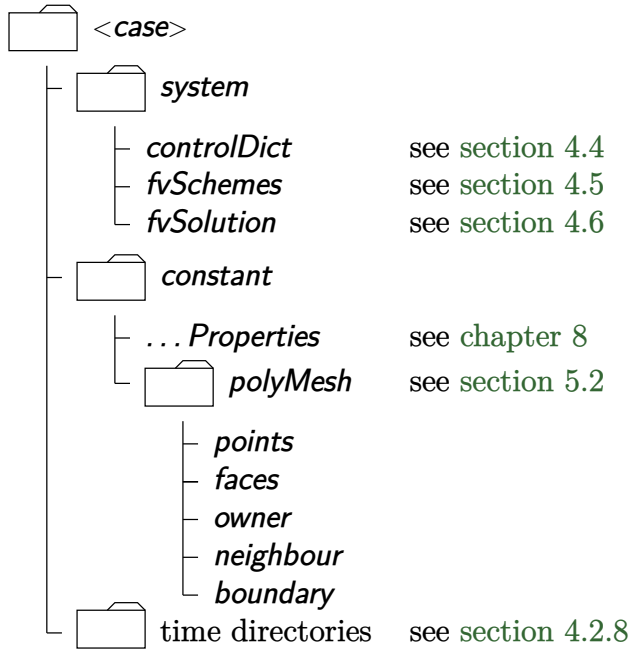


Figure 4.1: Case directory structure

Since we usually start our simulations at time $t = 0$, the initial conditions are usually stored in a directory named *0*. For example, in the motorBike tutorial, the velocity field **U** and pressure field p are initialised from files *0/U* and *0/p* respectively.

4.2 Basic input/output file format

OpenFOAM needs to read a range of data structures such as strings, words, scalars, vectors, tensors, lists and fields. The input/output (I/O) format of files is extremely flexible, following a consistent set of rules that make the files easy to interpret. The OpenFOAM file format is described in the following sections.

4.2.1 General syntax rules

The format resembles C++ code, following the general principles below.

- Files have free form, with no particular meaning assigned to any column and no need to indicate continuation across lines.
- Lines have no particular meaning except to a `//` comment delimiter which makes OpenFOAM ignore any text that follows it until the end of line.
- A comment over multiple lines is done by enclosing the text between `/*` and `*/` delimiters.

4.2.2 Dictionaries

OpenFOAM mainly uses *dictionaries* to specify data, in which data entries are retrieved by means of *keywords*. Each keyword entry follows the general format, beginning with the keyword and ending in semi-colon (`;`).

```
<keyword>  <dataEntry1> ... <dataEntryN>;
```

Many entries include only a single data entry as shown below.

```
<keyword>  <dataEntry>;
```

Most data files, *e.g.* *controlDict*, are themselves dictionaries since they contain a series of keyword entries. Any dictionary can contain one or more sub-dictionaries, usually denoted by a dictionary *name* and its keyword entries contained within curly braces `{}` as follows.

```
<dictionaryName>
{
    ... keyword entries ...
}
```

(Sub-)dictionaries can be nested within others, as shown in the following example. The extract, from an *fvSolution* dictionary file, containing two dictionaries, *solvers* and *PIMPLE*. The *solvers* dictionary contains nested sub-dictionary for different matrix equations based on different solution variables, *e.g.* `p`, `U` and `k` (with some entries using regular expressions described in section 4.2.12).

```
16
17  solvers
18  {
19      p
20      {
21          solver          GAMG;
22          tolerance       1e-7;
23          relTol          0.01;
24
25          smoother        DICGaussSeidel;
26
27      }
28
29      pFinal
30      {
31          $p;
32          relTol          0;
33      }
34
35      "(U|k|epsilon)"
36      {
37          solver          smoothSolver;
38          smoother        symGaussSeidel;
39          tolerance       1e-05;
40          relTol          0.1;
41      }
42
43      "(U|k|epsilon)Final"
44      {
45          $U;
46          relTol          0;
47      }
48  }
49
50  PIMPLE
51  {
52      nNonOrthogonalCorrectors 0;
53      nCorrectors         2;
54  }
55
56
57  // ************************************************************************* //
```

4.2.3 The data file header

All data files that are read and written by OpenFOAM begin with a dictionary named `FoamFile` containing a standard set of keyword entries, listed below:

- `version`: I/O format version, optional, defaults to `2.0`
- `format`: data format, `ascii` or `binary`
- `class`: class relating to the data, either `dictionary` or a field, *e.g.* `volVectorField`
- `object`: filename, *e.g.* `controlDict` (mandatory, but not used)
- `location`: path to the file (optional)

A example header for a *controlDict* file is shown below.

```
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "system";
    object      controlDict;
}
```

4.2.4 Lists

OpenFOAM applications contain lists, *e.g.* a list of vertex coordinates for a mesh description. Lists are commonly found in I/O and have a format of their own in which the entries are contained within round braces `( )`. When a user specifies a list in an input file, *e.g.* the `vertices` list in a *blockMeshDict* file, it just includes the `vertices` keyword and the data in `( )`, *e.g.*

```
vertices
(
    ... entries ...
);
```

When OpenFOAM writes out a list, it invariably prefixes it with the number of elements in the list. For example the *points* file for the mesh in the pizDailySteady case contains the following (abbreviated) list, where `25012` denotes the number of vertex points in the mesh.

```
25012
(
    (-0.0206 0 -0.0005)
    (-0.01901716308 0 -0.0005)
    ... entries ...
);
```

In some cases, when OpenFOAM writes out a list, it further prefixes it with the class name of the list. For example, the `inGroups` entry in a *boundary* file of a mesh contains a list where each group name is a word. The entry for the `lowerWall` patch from the pizDailySteady case is shown below, indicating the List<word> class with a single (1) element.

```
lowerWall
{
    type            wall;
    inGroups        List<word> 1(wall); // Note!
    nFaces          250;
    startFace       24480;
}
```

4.2.5 Scalars, vectors and tensors

A scalar is a single number represented as such in a data file. A vector contains three values, expressed using the simple List format so that the vector $(1.0, 1.1, 1.2)$ is written:

```
(1.0 1.1 1.2)
```

In OpenFOAM, a tensor contains nine elements, such that the identity tensor can be written:

```
( 1 0 0 0 1 0 0 0 1 )
```

The user can write the entry over multiple lines to give the "look" of a tensor as a 3×3 entity.

```
(
    1 0 0
    0 1 0
    0 0 1
)
```

4.2.6 Dimensional units

In continuum mechanics, properties are represented in some chosen units, *e.g.* mass in kilograms (kg), volume in cubic metres (m^3), pressure in Pascals ($kg\,m^{-1}\,s^{-2}$). Algebraic operations must be performed on these properties using consistent units of measurement; in particular, addition, subtraction and equality are only physically meaningful for properties of the same dimensional units. As a safeguard against implementing a meaningless operation, OpenFOAM attaches dimensions to field data and physical properties and performs dimension checking on any operation.

Dimensions are described by the dimensionSet class which has its own unique I/O syntax using square brackets, *e.g.*

```
[0 2 -1 0 0 0 0]
```

where each of the values corresponds to the power of each of the base units of measurement listed in sequence below:

1. mass, *e.g.* kilogram (kg), pound-mass (lbm);
2. length, *e.g.* metre (m), foot (ft);
3. time, *e.g.* second (s);
4. temperature, *e.g.* Kelvin (K), degree Rankine (°R);
5. quantity, *e.g.* mole (mol);
6. current, *e.g.* ampere (A);
7. luminous intensity, *e.g.* candela (cd).

The list includes base units for the Système International (SI) and the United States Customary System (USCS) but OpenFOAM can be used with any system of units. All that is required is that the *input data is correct for the chosen set of units*. The input data may include physical constants, *e.g.* the Universal Gas Constant R, which are specified in a *DimensionedConstant* sub-dictionary of main *controlDict* file of the OpenFOAM installation (*$WM_PROJECT_DIR/etc/controlDict*). By default these constants are set in SI units. Those wishing to use the USCS or any other system of units should modify these constants to their chosen set of units accordingly, as described in section 4.3.

4.2.7 Dimensioned types

Physical properties are typically specified with their associated dimensions. They are often described by the dimensioned class which includes three components: a word name; a dimensionSet and a value (scalar, vector, *etc.*).

The I/O for a dimensioned entry can include all three components, *e.g.*

```
nu    nu  [0 2 -1 0 0 0 0]  1e-5;
```

Note that the first `nu` is the keyword; the second `nu` is the word name stored in class word; the next entry is the dimensionSet and the final entry is the scalar value.

Very often, however, the word and dimensionSet are specified in the code with default values, so can be omitted from the I/O as shown below.

```
nu    1e-5;
```

4.2.8 Fields

Field files, *e.g.* *U* and *p*, that are read from and written into the time directories, possess their own customised I/O with the following three key entries.

- `dimensions`: the dimensions of the field, *e.g.* `[1 1 -2 0 0 0 0]`.
- `internalField`: values within the internal field, *e.g.* within each cell of a mesh.
- `boundaryField`: condition (`type`) and data for each patch of the mesh boundary.

The `internalField` can be specified in two ways. First, when the user edits a field file to initialise it, they generally specify a single value across all elements, *i.e.* the cells (or faces, points, depending on the type of field) of the mesh. A single value of `0` is denoted by the `uniform` keyword as shown below.

```
internalField uniform 0;
```

When results are written out, fields cannot generally be represented by a single value. The output uses the `nonuniform` keyword, followed by a suitable list of values. The abbreviated example below is from an output *p* file for a mesh of 12225 cells.

```
internalField  nonuniform  List<scalar>
12225
(
-4.92806
-5.42676
...
);
```

The `boundaryField` is a dictionary containing a set of entries corresponding to each patch listed in the *boundary* file in the *polyMesh* directory. Each entry is a sub-dictionary containing a list of keyword entries. The mandatory entry, `type`, describes the patch field condition specified for the field. The remaining entries correspond to the type of patch field condition selected and can typically include field data specifying initial conditions on patch faces. A selection of patch field conditions available in OpenFOAM are listed in section 6.2, section 6.3 and section 6.4, with a description and the data that must be specified with it. Example field dictionary entries for velocity `U` are shown below:

```
16  dimensions      [0 1 -1 0 0 0 0];
17
18  internalField   uniform (0 0 0);
19
20  boundaryField
21  {
22      inlet
23      {
24          type            fixedValue;
25          value           uniform (10 0 0);
26      }
27
28      outlet
29      {
30          type            zeroGradient;
31      }
32
33      upperWall
34      {
35          type            noSlip;
36      }
37
38      lowerWall
39      {
40          type            noSlip;
41      }
42
43      frontAndBack
44      {
45          type            empty;
46      }
47  }
48
49  // ************************************************************************* //
```

4.2.9 Macro expansion

The configuration of case files can benefit from a macro syntax which uses the dollar (`$`) symbol in front of a keyword to expand the data associated with the keyword. For example the value set for keyword `a` below, 10, is expanded in the following line, so that the value of `b` is also 10.

```
a 10;
b $a;
```

Variables can be accessed within different levels of sub-dictionaries, or scope. Scoping is performed using a '`/`' (slash) syntax, illustrated by the following example, where `b` is set to the value of `a`, specified in a sub-dictionary called `subdict`.

```
subdictA
{
    a 20;
}
b $subdictA/a;
```

There are further syntax rules for macro expansions:

- to traverse up one level of sub-dictionary, use the '`..`' (double-dot) prefix, see below;
- to traverse up two levels use '`../..`' prefix, *etc.*;
- to traverse to the top level dictionary use the '`!`' (exclamation mark) prefix (most useful), see below;
- to traverse into a separate file named `otherFile`, use '`otherFile!`', see below;
- for multiple levels of macro substitution, each specified with the '`$`' dollar syntax, '`{}`' brackets are required to protect the expansion, see below.

When accessing parameters from another file, the $FOAM_CASE environment variable is useful to specify the path to the file as described in Section 4.2.11 and illustrated below.

```
a 10;
b a;
c ${$b}; // returns 10, since $b returns "a", and $a returns 10

subdictA
{
    a 20;
}

subdictB
{
    // double-dot takes scope up 1 level, then into "subdictA" => 20
    b $../subdictA/a;

    subsubdict
```

```
    {
        // exclamation mark takes scope to top level => 10
        b $!a;

        // "a" from another file named "otherFile"
        c $otherFile!a;

        // "a" from another file "otherFile" in the case directory
        d ${${FOAM_CASE}/otherFile!a};
    }
}
```

4.2.10 Including files

Directives are commands that begin with the hash (`#`) symbol which provide further flexibility when configuring case files. There is a set of directive commands for reading a data file from within another data file. If a case requires a single value of pressure of 100 kPa, used in different input files, we could create a file, *e.g.* named *initialConditions*, which contains the following entry:

```
pressure 1e+05;
```

In order to use this pressure for internal and initial boundary fields, the user could simply include the *initialConditions* file using the `#include` directive, then use a macro expansion on the `pressure` keyword, as follows.

```
#include "initialConditions"
internalField uniform $pressure;
boundaryField
{
    patch1
    {
        type fixedValue;
        value $internalField;
    }
}
```

This example works if the included file is in the same directory as the file that includes it. Otherwise, more generally the path to the file is required, *e.g.* if *initialConditions* is in the *constant* directory:

```
#include "$FOAM_CASE/constant/initialConditions"
```

Here `$FOAM_CASE` represents is the path of the case directory as described in the following section. The following special forms of the `#include` directive also exist.

- `#includeIfPresent`: reads a file if it exists.
- `#includeEtc`: reads a file with the *$FOAM_ETC* directory as the starting path.

- `#includeFunc`: reads file containing a single functionObject configuration, first searching the case *system* directory, followed by the *$FOAM_ETC* directory.
- `#includeModel`: reads a file containing a single fvModel configuration, first searching the case *constant* directory, followed by the *$FOAM_ETC* directory.
- `#includeConstraint`: reads a file containing a single fvConstraint configuration, first searching the case *system* directory, followed by the *$FOAM_ETC* directory.

Keyword entries can also be removed with the directive:

```
#remove <keywordEntry>
```

where <`keywordEntry`> can be either a single keyword or a regular expression.

4.2.11 Environment variables

Environment variables can be used in input files. For example, the $FOAM_RUN environment variable can be used to identify the *run* directory, as described in the introduction to Chapter 2. This could be used to include a file, *e.g.* by

```
#include "$FOAM_RUN/pitzDailySteady/0/U"
```

In addition to environment variables like $FOAM_RUN, set within the operating system, a number of "internal" environment variables are recognised, including the following.

- $FOAM_CASE: the path and directory of the running case.
- $FOAM_CASENAME: the directory *name* of the running case.
- $FOAM_APPLICATION: the name of the running application.

4.2.12 Regular expressions

As discussed, data is looked up from files using keywords. If a particular keyword does not exist, the I/O system will try to match the keyword with any POSIX regular expression, specified inside double-quotations (`"..."`) in the input file.

In some cases, when the I/O system searches for a keyword in a case file, a can be used to match the keyword

When running an application, data is initialised by looking up keywords from dictionaries. The user can either provide an entry with a keyword that directly matches the one being looked up, or can provide a that matches the keyword, specified inside double-quotations (`"..."`).

Regular expressions have an extensive syntax for various matches of text patterns but in OpenFOAM input files there are only two expressions that are generally used. Firstly, '`.`' denoting "any character", and '`*`' denoting "repeated any number of times, including 0 times" is often used in combination to match "any characters". For example, to specify a `noSlip` boundary condition for any patch whose name ends `Wall...`, the user could specify in the `boundaryField` for *U*:

```
".*Wall"
{
    type   noSlip;
}
```

The other common regular expression uses `()` to group expressions. For example, to a `noSlip` boundary condition on two wall patches named `upper` and `lower`, the user could specify:

```
"(upper|lower)"
{
    type   noSlip;
}
```

4.2.13 Keyword ordering

The order in which keywords are listed does not matter, except when *the same keyword is specified multiple times*. Where the same keyword is duplicated, the last instance is used. The most common example of a duplicate keyword occurs when a keyword is included from the file or expanded from a macro, and then overridden. The example below demonstrates this, where `pFinal` adopts all the keyword entries, including `relTol 0.05` in the `p` sub-dictionary by the macro expansion `$p`, then overrides the `relTol` entry.

```
p
{
    solver          PCG;
    preconditioner  DIC;
    tolerance       1e-6;
    relTol          0.05;
}
pFinal
{
    $p;
    relTol          0;
}
```

Where a data lookup matches both a keyword and a regular expression, the keyword match takes precedence irrespective of the order of the entries.

4.2.14 Inline calculations

There are two further directives that enable calculations from within input files: `#calc`, described here, for simple calculations; and `#codeStream`, for more complex calculations, described in section 4.2.14.

The pipeCyclic tutorial in *$FOAM_TUTORIALS/incompressibleFluid* demonstrates the `#calc` directive through its blockMesh configuration in *blockMeshDict*:

```
//- Half angle of wedge in degrees
halfAngle 45.0;

//- Radius of pipe [m]
radius 0.5;

radHalfAngle    #calc "degToRad($halfAngle)";
y               #calc "$radius*sin($radHalfAngle)";
z               #calc "$radius*cos($radHalfAngle)";
```

The file contains several calculations that calculate vertex ordinates, *e.g.* `y`, `z`, *etc.*, from geometry dimensions, *e.g.* `radius`.

Calculations include standard C++ functions including unit conversions, *e.g.* `degToRad`, and trigonometric functions, *e.g.* `sin`. They can also include OpenFOAM mathematical functions if the relevant header files are included for those functions. The `#calcInclude` directive enables header files to be included for use with `#calc`.

The aerofoilNACA0012Steady example, using the fluid solver module, sets the inlet velocity using an angle of attack using the code below. The `transform` function is provided by the *transform.H* header file, to rotate unit vectors by the angle of attack to set the lift and drag directions.

```
angleOfAttack   5; // degs

angle           #calc "-degToRad($angleOfAttack)";

#calcInclude    "transform.H"
liftDir         #calc "transform(Ry($angle), vector(0, 0, 1))";
dragDir         #calc "transform(Ry($angle), vector(1, 0, 0))";
```

Dictionary entries constructed with `#calc` or `#codeStream` (see below) can use variables that represent OpenFOAM classes, or *types*, such as vector, tensor, List, Field, string *etc.*. To create a typed variable, the type is specified inside angled brackets `<>`, immediately after the `$` symbol, *e.g.* `$<vector>var` or `$<vector>{var}` substitutes a variable named `var` as a vector. The following example shows a calculation $c = \mathbf{a} \bullet \mathbf{b}$ using `#calc`.

```
a       (1 2 3);
b       (1 1 0);
c       #calc "$<vector>a & $<vector>b";
```

Care is required with calculations involving a division because the `/` character is otherwise used to identify keywords in sub-dictionaries, `"$a/b"` looks for a keyword `b` within a sub-dictionary named `a`. Where a division is required, the user can put spaces around the `/`, *e.g.*

```
c     #calc "$a / $b";
```

or they can apply brackets around the first variable, *e.g.*

```
    c     #calc "$(a)/$b";
```

The code string can also be delimited by `#{...#}` instead of quotation marks `"..."`. The former delimiter supports code strings across multiple lines and avoids problems with `string` typed variables that may contain quotation marks, as shown in the following example.

```
    s "field";
    fieldName #calc
    #{
        $<string>s + "Name"
    #};
```

Further examples can be found in files in the *test/dictionary* directory in the OpenFOAM installation.

4.2.15 Inline code

The `#codeStream` directive takes C++ code which is compiled and executed to deliver the dictionary entry. The code and compilation instructions are specified through the following keywords.

- `code`: specifies the code using arguments `OStream& os` and `const dictionary& dict` which can be used in the code, *e.g.* to lookup keyword entries from within the current case file.
- `codeInclude` (optional): specifies additional C++ `#include` statements to include code files.
- `codeOptions` (optional): specifies any extra compilation flags to be added to `EXE_INC` in *Make/options*.
- `codeLibs` (optional): specifies any extra compilation flags to be added to `LIB_LIBS` in *Make/options*.

Code, like any string, can be written across multiple lines by enclosing it within hash-bracket delimiters, *i.e.* `#{...#}`. Anything in between these two delimiters becomes a string with all newlines, quotes, *etc.* preserved.

An example of `#codeStream` is given below, where the code in the calculates moment of inertia of a box shaped geometry.

```
momentOfInertia #codeStream
{
    codeInclude
    #{
        #include "diagTensor.H"
    #};

    code
    #{
        scalar sqrLx = sqr($Lx);
        scalar sqrLy = sqr($Ly);
        scalar sqrLz = sqr($Lz);
        os  <<
            $mass
           *diagTensor(sqrLy + sqrLz, sqrLx + sqrLz, sqrLx + sqrLy)/12.0;
    #};
};
```

4.2.16 Conditionals

Input files support two conditional directives: `#if...#else...#endif`; and, `#ifEq... #else... #endif`. The `#if` conditional reads a switch that can be generated by a `#calc` directive, *e.g.*:

```
angle 65;

laplacianSchemes
{
#if #calc "${angle} < 75"
    default  Gauss linear corrected;
#else
    default  Gauss linear limited corrected 0.5;
#endif
}
```

The `#ifEq` compares a word or string, and executes based on a match, *e.g.*:

```
rotating
{
    timeScheme      ${${FOAM_CASE}/system/fvSchemes!ddtSchemes/default};
#ifeq $timeScheme steadyState
    type            MRFnoSlip;
#else
    type            movingWallVelocity;
#endif
    value           uniform (0 0 0);
}
```

4.3 Global controls

OpenFOAM includes a large number of global parameters that are configured by default in a file named *controlDict*. This is the so-called "global" *controlDict* file, as opposed to a case *controlDict* file that is described in the following section.

The global *controlDict* file can be found in the installation within a directory named *etc*, represented by the environment variable $FOAM_ETC. The file contains sub-dictionaries for the following items.

- `Documentation`: for opening documentation in a web browser.
- `InfoSwitches`: controls information printed to standard output, *i.e.* the terminal.
- `OptimisationSwitches`: for parallel communication and I/O, see section 3.4.2.
- `DebugSwitches`: messaging switches to help debug code failures, as described in section 3.2.11.
- `DimensionedConstants`: defines fundamental physical constants, *e.g.* Boltzmann's Constant.
- `DimensionSets`: defines a notation for dimensional units, *e.g.* `kg`.

4.3.1 Overriding global controls

The *values* of the `DimensionedConstants` depend on the unit system being adopted, *i.e.* the International System of Units (SI units), or US Customary system (USCS), based on English units (pounds, feet, *etc.*). The default system is naturally SI, but some users may wish to override this with USCS units, either globally or for a specific case. The system is set through the `unitSet` keyword, *i.e.*

```
DimensionedConstants
{
    unitSet  SI; // USCS
}
```

While a user could modify this setting in the *etc/controlDict* file in the installation, it is better practice to use a file in their user directory. OpenFOAM provides a set of directory locations, where global configuration files can be included, which it looks up in an order of precedence. To list the locations, simply run the following command.

```
foamEtcFile -list
```

The listed locations include a local *$HOME/.OpenFOAM* directory and follow a descending order of precedence, *i.e.* the last location listed (*etc*) is lowest precedence.

If a user therefore wished to work permanently in USCS units, they could maintain a *controlDict* file in their *$HOME/.OpenFOAM* directory that includes the following entry.

```
DimensionedConstants
{
    unitSet  USCS;
}
```

OpenFOAM would read the `unitSet` entry from this file, but read all other *controlDict* keyword entries from the global *controlDict* file.

Alternatively, if a user wished to work on a *single case* in USCS units, they could add the same entry into the *controlDict* file in the *system* directory for their *case*. This file is discussed in the next section.

4.4 Time and data input/output control

The OpenFOAM solvers begin all runs by setting up a database. The database controls I/O and, since output of data is usually requested at intervals of time during the run, time is an inextricable part of the database. The *controlDict* dictionary sets input parameters *essential* for the creation of the database. The keyword entries in *controlDict* are listed in the following sections. Only the time control and `writeInterval` entries are mandatory, with the database using default values for any of the optional entries that are omitted. Example entries from a *controlDict* dictionary are given below:

```
16
17  application     foamRun;
18
19  solver          incompressibleFluid;
20
21  startFrom       latestTime;
22
23  startTime       0;
24
25  stopAt          endTime;
26
27  endTime         0.3;
28
29  deltaT          0.0001;
30
31  writeControl    adjustableRunTime;
32
33  writeInterval   0.01;
34
35  purgeWrite      0;
36
37  writeFormat     ascii;
38
39  writePrecision  6;
40
```

```
41  writeCompression off;
42
43  timeFormat      general;
44
45  timePrecision   6;
46
47  runTimeModifiable yes;
48
49  adjustTimeStep  yes;
50
51  maxCo           5;
52
53  functions
54  {
55      #includeFunc patchAverage(patch=inlet, fields=(p U))
56  }
57
58  // ************************************************************************* //
```

4.4.1 Modules

`solver` Choice of solver module for the simulation, *e.g.* `incompressibleFluid`

`regionSolvers` Dictionary of solvers for different domain regions, *e.g.* heat transfer of water flowing over a plate might use the fluid and solid modules as follows:

```
regionSolvers
{
    water fluid;
    plate solid;
}
```

`libs` List of additional libraries (existing on $LD_LIBRARY_PATH) to be loaded at run-time, *e.g.* `("libNew1.so" "libNew2.so")`

4.4.2 Time control

`startFrom` Controls the start time of the simulation.

- `firstTime`: Earliest time step from the set of time directories.
- `startTime`: Time specified by the `startTime` keyword entry.
- `latestTime`: Most recent time step from the set of time directories.

`startTime` Start time for the simulation with `startFrom startTime;`

`stopAt` Controls the end time of the simulation.

- `endTime`: Time specified by the `endTime` keyword entry.
- `writeNow`: Stops simulation on completion of current time step and writes data.
- `noWriteNow`: Stops simulation on completion of current time step and does not write out data.
- `nextWrite`: Stops simulation on completion of next scheduled write time, specified by `writeControl`.

`endTime` End time for the simulation when `stopAt endTime;` is specified.

`deltaT` Time step of the simulation.

4.4.3 Data writing

`writeControl` Controls the timing of write output to file.

- `timeStep`: Writes data every `writeInterval` time steps.
- `runTime`: Writes data every `writeInterval` seconds of simulated time.
- `adjustableRunTime`: Writes data every `writeInterval` seconds of simulated time, adjusting the time steps to coincide with the `writeInterval` if necessary — used in cases with automatic time step adjustment.
- `cpuTime`: Writes data every `writeInterval` seconds of CPU time.
- `clockTime`: Writes data out every `writeInterval` seconds of real time.

`writeInterval` Scalar used in conjunction with `writeControl` described above.

`purgeWrite` Integer representing a limit on the number of time directories that are stored by overwriting time directories on a cyclic basis. For example, if the simulations starts at $t = 5$s and $\Delta t = 1$s, then with `purgeWrite 2;`, data is first written into 2 directories, *6* and *7*, then when *8* is written, *6* is deleted, and so on so that only 2 new results directories exists at any time. *To disable the purging, specify* `purgeWrite 0;` (default).

`writeFormat` Specifies the format of the data files.

- `ascii` (default): ASCII format, written to `writePrecision` significant figures.
- `binary`: binary format.

`writePrecision` Integer used in conjunction with `writeFormat` described above, 6 by default.

`writeCompression` Switch to specify whether files are compressed with `gzip` when written: `on/off` (`yes/no`, `true/false`)

`timeFormat` Choice of format of the naming of the time directories.

- `fixed`: $\pm m.dddddd$ where the number of ds is set by `timePrecision`.
- `scientific`: $\pm m.dddddde\pm xx$ where the number of ds is set by `timePrecision`.
- `general` (default): Specifies `scientific` format if the exponent is less than -4 or greater than or equal to that specified by `timePrecision`.

`timePrecision` Integer used in conjunction with `timeFormat` described above, 6 by default.

`graphFormat` Format for graph data written by an application.

- `raw` (default): Raw ASCII format in columns.
- `gnuplot`: Data in `gnuplot` format.
- `csv`: Comma-separated values.
- `vtk`: Visualisation Toolkit (VTK) format.
- `ensight`: Ensight format.

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4.4.4 Other settings

`beginTime` Optional entry to for cases with an unusual start time that causes inconvenient write times. With `beginTime`, the write times are multiples of `writeInterval`, starting at the `beginTime`. For example, if the start time of `1.52` and a `writeInterval` of `1`, results would be written at `2.52`, `3.52`, ... If `beginTime` is set to `0` (or `1`), the write times would be `2`, `3`, *etc.*

Switch used by some solvers to adjust the time step during the simulation, usually according to `maxCo`.

`adjustTimeStep` Switch used by some solvers to adjust the time step during the simulation, usually according to `maxCo`.

`maxCo` Maximum Courant number, *e.g.* `0.5`

`runTimeModifiable` Switch for whether dictionaries, *e.g.controlDict*, are re-read during a simulation at the beginning of each time step, allowing the user to modify parameters during a simulation.

`functions` Dictionary of functions, *e.g.* `probes` to be loaded at run-time; see examples in *$FOAM_TUTORIALS*

4.5 Numerical schemes

The *fvSchemes* dictionary in the *system* directory sets the numerical schemes for terms, such as derivatives in equations, that are calculated during a simulation. This section describes how to specify the schemes in the *fvSchemes* dictionary. Details of the schemes are described in Chapter 3 of ***Notes on Computational Fluid Dynamics: General Principles***.

The aim for *fvSchemes* is to provide an unrestricted choice of schemes to the user for everything from derivatives, *e.g.* gradient ∇, to interpolations of values from one set of points to another. OpenFOAM uses the finite volume method so spatial derivatives are based on Gaussian integration which sums values on cell faces, which must be interpolated from cell centres. The user has a wide range of options for interpolation schemes, with certain schemes being specifically designed for particular derivative terms, especially the advection divergence $\nabla \bullet$ terms.

The set of terms, for which numerical schemes must be specified, are subdivided within the *fvSchemes* dictionary into the categories below, using Ψ as an example field variable.

- `timeScheme`: first and second time derivatives, *e.g.* $\partial\Psi/\partial t, \partial^2\Psi/\partial^2 t$
- `gradSchemes`: gradient $\nabla\Psi$
- `divSchemes`: divergence $\nabla \bullet \Psi$
- `laplacianSchemes`: Laplacian $\nabla \bullet \Gamma\nabla\Psi$, with diffusivity Γ
- `interpolationSchemes`: cell to face interpolations of values.
- `snGradSchemes`: component of gradient normal to a cell face.
- `wallDist`: distance to wall calculation, where required.

Each keyword is the name of a sub-dictionary which contains terms of a particular type, *e.g.* `gradSchemes` contains all the gradient derivative terms such as `grad(p)` (which represents ∇p). Further examples can be seen in the extract from an *fvSchemes* dictionary below:

```
16
17  ddtSchemes
18  {
19      default         Euler;
20  }
21
22  gradSchemes
23  {
24      default         Gauss linear;
25  }
26
27  divSchemes
28  {
29      default             none;
30
31      div(phi,U)          Gauss linearUpwind grad(U);
32      div(phi,k)          Gauss upwind;
33      div(phi,epsilon)    Gauss upwind;
34      div(phi,R)          Gauss upwind;
35      div(R)              Gauss linear;
36      div(phi,nuTilda)    Gauss upwind;
37
38      div((nuEff*dev2(T(grad(U))))) Gauss linear;
39  }
40
41  laplacianSchemes
42  {
43      default         Gauss linear corrected;
44  }
45
46  interpolationSchemes
47  {
48      default         linear;
49  }
50
51  snGradSchemes
52  {
53      default         corrected;
54  }
55
56
57  // ************************************************************************* //
```

The example shows *fvSchemes* with 6 *...Schemes* subdictionaries, each containing keyword entries including: a `default` entry; other entries for the particular term specified, *e.g.* `div(phi, k)` for $\nabla \bullet (\mathbf{U}k)$. If a `default` scheme is specified in a particular *...Schemes* sub-dictionary, it is assigned to all of the terms to which the sub-dictionary refers, *e.g.* specifying a `default` in *gradSchemes* sets the scheme for all gradient terms in the application, *e.g.* ∇p, $\nabla \mathbf{U}$. With a `default` specified, the specific terms are not required in that sub-dictionary, *i.e.* the entries for `grad(p)`, `grad(U)` are omitted in this example. Specifying a particular will however override the `default` scheme.

The user can specify no `default` scheme by the `none` entry, as in the `divSchemes` in the example above. The user is then obliged to specify all terms in that sub-dictionary individually. Setting `default` to `none` ensures the user specifies all terms individually which is common for `divSchemes` which requires precise configuration.

OpenFOAM includes a vast number of discretisation schemes, from which only a few are typically recommended for real-world, engineering applications. The user can get help with scheme selection by interrogating the tutorial cases for example scheme settings. They should look at the schemes used in relevant cases, *e.g.* for running a large-eddy simulation (LES), look at schemes used in tutorials running LES. Additionally, foamSearch is a useful tool to list the schemes used in all the tutorials. For example,

to print all the `default` entries for `ddtSchemes` for cases in the *$FOAM_TUTORIALS* directory, the user can type:

```
foamSearch $FOAM_TUTORIALS fvSchemes ddtSchemes/default
```

which returns:

```
default         backward;
default         CrankNicolson 0.9;
default         Euler;
default         localEuler;
default         none;
default         steadyState;
```

The schemes listed using foamSearch are described in the following sections.

4.5.1 Time schemes

The first time derivative ($\partial/\partial t$) terms are specified in the *ddtSchemes* sub-dictionary. The discretisation schemes for each term can be selected from those listed below.

- `steadyState`: sets time derivatives to zero.
- `Euler`: transient, first order implicit, bounded.
- `backward`: transient, second order implicit, potentially unbounded.
- `CrankNicolson`: transient, second order implicit, bounded; requires an off-centering coefficient ψ where:

$$\psi = \begin{cases} 1 & \text{corresponds to pure } \texttt{CrankNicolson}, \\ 0 & \text{corresponds to } \texttt{Euler}; \end{cases}$$

  generally $\psi = 0.9$ is used to stabilise the scheme for practical engineering problems.

- `localEuler`: pseudo transient for accelerating a solution to steady-state using local-time stepping; first order implicit.

Any second time derivative ($\partial^2/\partial t^2$) terms are specified in the *d2dt2Schemes* sub-dictionary. Only the `Euler` scheme is available for *d2dt2Schemes*.

4.5.2 Gradient schemes

The *gradSchemes* sub-dictionary contains gradient terms. The `default` discretisation scheme that is primarily used for gradient terms is:

```
default         Gauss linear;
```

The `Gauss` entry specifies the standard finite volume discretisation with Gaussian integration which requires the interpolation of values from cell centres to face centres. The interpolation scheme is then given by the `linear` entry, meaning linear interpolation or central differencing.

In some tutorials cases, particular involving poorer quality meshes, the discretisation of specific gradient terms is then overridden to improve boundedness and stability. The terms that are overridden in those cases are the velocity gradient

```
grad(U)          cellLimited Gauss linear 1;
```

and, less frequently, the gradient of turbulence fields, *e.g.*

```
grad(k)          cellLimited Gauss linear 1;
grad(epsilon)    cellLimited Gauss linear 1;
```

They use the `cellLimited` scheme which limits the gradient such that when cell values are extrapolated to faces using the calculated gradient, the face values do not fall outside the bounds of values in surrounding cells. A limiting coefficient is specified after the underlying scheme for which 1 guarantees boundedness and 0 applies no limiting; 1 is invariably used.

Other schemes that are rarely used are as follows.

- `leastSquares`: a second-order, least squares distance calculation using all neighbour cells.
- `Gauss cubic`: third-order scheme that appears in solidDisplacement and dnsFoam examples.

4.5.3 Divergence schemes

The *divSchemes* sub-dictionary contains divergence terms, *i.e.* terms of the form $\nabla\bullet\ldots$, excluding Laplacian terms (of the form $\nabla\bullet\Gamma\nabla\Psi$). This includes both advection terms, *e.g.* $\nabla\bullet(\mathbf{U}k)$, where velocity $\mathbf{U}$ provides the advective flux, and other terms, that are often diffusive in nature, *e.g.* $\nabla\bullet\nu(\nabla\mathbf{U})^{\mathrm{T}}$.

The fact that terms that are fundamentally different reside in one sub-dictionary means that the `default` scheme in generally set to `none` in *divSchemes*. The non-advective terms then generally use the `Gauss` integration with `linear` interpolation, *e.g.*

```
div(U)          Gauss linear;
```

The treatment of advective terms is one of the major challenges in CFD numerics and so the options are more extensive. The keyword identifier for the advective terms are usually of the form `div(phi,...)`, where `phi` denotes the (volumetric) flux of velocity on the cell faces for constant-density flows and the mass flux for compressible flows, *e.g.* `div(phi,U)` for the advection of velocity, `div(phi,e)` for the advection of internal energy, *etc.* For advection of velocity, the user can run the foamSearch script to extract the `div(phi,U)` keyword from all tutorials.

```
foamSearch $FOAM_TUTORIALS fvSchemes "divSchemes/div(phi,U)"
```

The schemes are all based on `Gauss` integration, using the flux `phi` and the advected field being interpolated to the cell faces by one of a selection of schemes, *e.g.* `linear`, `linearUpwind`, *etc.* There is a `bounded` variant of the discretisation, discussed later.

Ignoring 'V'-schemes (with keywords ending "`V`"), and rarely-used schemes such as `Gauss cubic` and `vanLeerV`, the interpolation schemes used in the tutorials are as follows.

- `linear`: second order, unbounded.
- `linearUpwind`: second order, upwind-biased, unbounded (but much less so than `linear`), that requires discretisation of the velocity gradient to be specified.
- `LUST`: blended 75% `linear`/ 25%`linearUpwind` scheme, that requires discretisation of the velocity gradient to be specified.
- `limitedLinear`: `linear` scheme that limits towards `upwind` in regions of rapidly changing gradient; requires a coefficient, where 1 is strongest limiting, tending towards `linear` as the coefficient tends to 0.
- `upwind`: first-order bounded, generally too inaccurate for velocity but more often used for transport of scalar fields.

Example syntax for these schemes is as follows.

```
div(phi,U)      Gauss linear;
div(phi,U)      Gauss linearUpwind grad(U);
div(phi,U)      Gauss LUST grad(U);
div(phi,U)      Gauss LUST unlimitedGrad(U);
div(phi,U)      Gauss limitedLinear 1;
div(phi,U)      Gauss upwind;
```

'V'-schemes are specialised versions of schemes designed for vector fields. They differ from conventional schemes by calculating a single limiter which is applied to all components of the vectors, rather than calculating separate limiters for each component of the vector. The 'V'-schemes' single limiter is calculated based on the direction of most rapidly changing gradient, resulting in the strongest limiter being calculated which is most stable but arguably less accurate. Example syntax is as follows.

```
div(phi,U)      Gauss limitedLinearV 1;
div(phi,U)      Gauss linearUpwindV grad(U);
```

The **`bounded`** variants of schemes relate to the treatment of the material time derivative which can be expressed in terms of a spatial time derivative and convection, *e.g.* for field e in incompressible flow

$$\frac{De}{Dt} = \frac{\partial e}{\partial t} + \mathbf{U} \bullet \nabla e = \frac{\partial e}{\partial t} + \nabla \bullet (\mathbf{U}e) - (\nabla \bullet \mathbf{U})e \tag{4.1}$$

For numerical solution of incompressible flows, $\nabla \bullet \mathbf{U} = 0$ at convergence, at which point the third term on the right hand side is zero. Before convergence is reached, however, $\nabla \bullet \mathbf{U} \neq 0$ and in some circumstances, particularly steady-state simulations, it is better to include the third term within a numerical solution to help maintain boundedness of the solution variable and promote better convergence. The `bounded` variant of the `Gauss` scheme provides this, automatically including the discretisation of the third-term with the advection term. Example syntax is as follows, as seen in *fvSchemes* files for steady-state cases.

```
div(phi,U)      bounded Gauss limitedLinearV 1;
div(phi,U)      bounded Gauss linearUpwindV grad(U);
```

The schemes used for advection of scalar fields are similar to those for advection of velocity, although in general there is greater emphasis placed on boundedness than accuracy when selecting the schemes. For example, a search for schemes for advection of internal energy (`e`) reveals the following.

```
foamSearch $FOAM_TUTORIALS fvSchemes "divSchemes/div(phi,e)"

div(phi,e)      bounded Gauss upwind;
div(phi,e)      Gauss limitedLinear 1;
div(phi,e)      Gauss linearUpwind limited;
div(phi,e)      Gauss LUST grad(e);
div(phi,e)      Gauss upwind;
div(phi,e)      Gauss vanAlbada;
```

In comparison with advection of velocity, there are no cases set up to use `linear`. The `limitedLinear` and `upwind` schemes are commonly used, with the additional appearance of `vanLeer`, another limited scheme, with less strong limiting than `limitedLinear`.

There are specialised versions of the limited schemes for scalar fields that are commonly bounded between 0 and 1, *e.g.* the laminar flame speed regress variable b. A search for the discretisation used for advection in the laminar flame transport equation yields:

```
div(phiSt,b)    Gauss limitedLinear01 1;
```

The underlying scheme is `limitedLinear`, specialised for stronger bounding between 0 and 1 by adding `01` to the name of the scheme.

The `multivariateSelection` mechanism also exists for grouping multiple equation terms together, and applying the same limiters on all terms, using the strongest limiter calculated for all terms. A good example of this is in a set of mass transport equations for fluid species, where it is good practice to apply the same discretisation to all equations for consistency. The example below comes from the smallPoolFire3D tutorial in *$FOAM_TUTORIALS/multicompon/entFluid*, in which the equation for enthalpy h is included with the specie mass transport equations in the calculation of a single limiter.

```
div(phi,Yi_h)   Gauss multivariateSelection
{
    O2 limitedLinear01 1;
    CH4 limitedLinear01 1;
    N2 limitedLinear01 1;
    H2O limitedLinear01 1;
    CO2 limitedLinear01 1;
    h limitedLinear 1 ;
}
```

4.5.4 Surface normal gradient schemes

It is worth explaining the *snGradSchemes* sub-dictionary that contains surface normal gradient terms, before discussion of *laplacianSchemes*, because they are required to evaluate a Laplacian term using Gaussian integration. A surface normal gradient is evaluated at a cell face; it is the component, normal to the face, of the gradient of values at the centres of the 2 cells that the face connects.

A search for the `default` scheme for *snGradSchemes* reveals the following entries.

```
default         corrected;
default         limited corrected 0.33;
default         limited corrected 0.5;
default         orthogonal;
default         uncorrected;
```

The basis of the gradient calculation at a face is to subtract the value at the cell centre on one side of the face from the value in the centre on the other side and divide by the distance. The calculation is second-order accurate for the gradient *normal to the face* if the vector connecting the cell centres is orthogonal to the face, *i.e.* they are at right-angles. This is the `orthogonal` scheme.

Orthogonality requires a regular mesh, typically aligned with the Cartesian co-ordinate system, which does not generally occur with real world, engineering geometries. Therefore, to maintain second-order accuracy, an explicit non-orthogonal correction can be added to the orthogonal component, known as the `corrected` scheme. The correction increases in size as the non-orthogonality, *i.e.* the angle α between the cell-cell vector and face normal vector, increases.

As α tends towards 90°, typically beyond 75°, the explicit correction can be so large to cause a solution to go unstable. The solution can be stabilised by applying the `limited` scheme to the correction which requires a coefficient $\psi, 0 \leq \psi \leq 1$ where

$$\psi = \begin{cases} 0 & \text{corresponds to } \texttt{uncorrected}, \\ 0.333 & \text{non-orthogonal correction} \leq 0.5 \times \text{orthogonal part}, \\ 0.5 & \text{non-orthogonal correction} \leq \text{orthogonal part}, \\ 1 & \text{corresponds to } \texttt{corrected}. \end{cases} \tag{4.2}$$

Typically, *psi* is chosen to be 0.33 or 0.5, where 0.33 offers greater stability and 0.5 greater accuracy.

The corrected scheme applies under-relaxation in which the implicit orthogonal calculation is increased by $cos^{-1}\alpha$, with an equivalent boost within the non-orthogonal correction. The `uncorrected` scheme is equivalent to the `corrected` scheme, without the non-orthogonal correction, so is like `orthogonal` but with the additional $cos^{-1}\alpha$ under-relaxation.

Generally the `uncorrected` and `orthogonal` schemes are only recommended for meshes with very low non-orthogonality (*e.g.* maximum 5°). The `corrected` scheme is generally recommended, but for maximum non-orthogonality above 75°, `limited` may be required. At non-orthogonality above 85°, convergence is generally hard to achieve.

4.5.5 Laplacian schemes

The *laplacianSchemes* sub-dictionary contains Laplacian terms. A typical Laplacian term is $\nabla \bullet (\nu \nabla \mathbf{U})$, the diffusion term in the momentum equations, which corresponds to the

keyword `laplacian(nu,U)` in *laplacianSchemes*. The `Gauss` scheme is the only choice of discretisation and requires a selection of both an interpolation scheme for the diffusion coefficient, *i.e.* ν in our example, and a surface normal gradient scheme, *i.e.* $\nabla\mathbf{U}$. To summarise, the entries required are:

```
Gauss <interpolationScheme> <snGradScheme>
```

The user can search for the `default` scheme for *laplacianSchemes* in all the cases in the *$FOAM_TUTORIALS* directory.

```
foamSearch $FOAM_TUTORIALS fvSchemes laplacianSchemes/default
```

It reveals the following entries.

```
default         Gauss linear corrected;
default         Gauss linear limited corrected 0.33;
default         Gauss linear limited corrected 0.5;
default         Gauss linear orthogonal;
default         Gauss linear uncorrected;
```

In all cases, the `linear` interpolation scheme is used for interpolation of the diffusivity. The cases uses the same array of `snGradSchemes` based on the maximum non-orthogonality in the mesh, as described in section 4.5.4.

4.5.6 Interpolation schemes

The *interpolationSchemes* sub-dictionary contains terms that are interpolations of values typically from cell centres to face centres, primarily used in the interpolation of velocity to face centres for the calculation of flux `phi`. There are numerous interpolation schemes in OpenFOAM, but a search for the `default` scheme in all the tutorial cases reveals that `linear` interpolation is used in almost every case, except for one stress analysis example which uses `cubic` interpolation.

4.6 Solution and algorithm control

Once a matrix equation is constructed according to the schemes in the previous section, a linear solver is applied. A set of equations is also framed within an algorithm containing loops and controls. For information about the main algorithms and solvers in OpenFOAM, refer to Chapter 5 of ***Notes on Computational Fluid Dynamics: General Principles***.

The controls for algorithms and solvers are found in the *fvSolution* dictionary in the *system* directory. Below is an example set of entries from the *fvSolution* dictionary for a case using the incompressibleFluid modular solver.

```
16
17  solvers
18  {
19      p
20      {
21          solver          GAMG;
22          tolerance       1e-7;
23          relTol          0.01;
24
25          smoother        DICGaussSeidel;
26
```

```
27      }
28
29      pFinal
30      {
31          $p;
32          relTol          0;
33      }
34
35      "(U|k|epsilon)"
36      {
37          solver          smoothSolver;
38          smoother        symGaussSeidel;
39          tolerance       1e-05;
40          relTol          0.1;
41      }
42
43      "(U|k|epsilon)Final"
44      {
45          $U;
46          relTol          0;
47      }
48  }
49
50  PIMPLE
51  {
52      nNonOrthogonalCorrectors 0;
53      nCorrectors         2;
54  }
55
56
57  // ************************************************************************* //
```

fvSolution contains a set of subdictionaries, described in the remainder of this section that includes: *solvers*; *relaxationFactors*; and, *SIMPLE* for steady-state cases or *PIMPLE* for transient or pseudo-transient cases.

4.6.1 Linear solver control

The first sub-dictionary in our example is `solvers`. It specifies each linear solver that is used for each discretised equation; here, the term *linear* solver refers to the method of number-crunching to solve a matrix equation, as opposed to an *modular* solver, such as incompressibleFluid which describes the entire set of equations and algorithms to solve a particular problem. The term 'linear solver' is abbreviated to 'solver' in much of what follows; hopefully the context of the term avoids any ambiguity.

The syntax for each entry within *solvers* starts with a keyword that for the variable being solved in the particular equation. For example, incompressibleFluid solves equations for pressure p, velocity **U** and often turbulence fields, hence the entries for `p`, `U`, `k` and `epsilon`. The keyword introduces a sub-dictionary containing the type of solver and the parameters that the solver uses. The solver is selected through the `solver` keyword from the options listed below. The parameters, including `tolerance`, `relTol`, `preconditioner`, *etc.* are described in following sections.

- `PCG/PBiCGStab`: Stabilised preconditioned (bi-)conjugate gradient, for both symmetric and asymmetric matrices.
- `PCG/PBiCG`: preconditioned (bi-)conjugate gradient, with `PCG` for symmetric matrices, `PBiCG` for asymmetric matrices.
- `smoothSolver`: solver that uses a smoother.
- `GAMG`: generalised geometric-algebraic multi-grid.
- `diagonal`: diagonal solver for explicit systems.

The solvers distinguish between symmetric matrices and asymmetric matrices. The symmetry of the matrix depends on the terms of the equation being solved, *e.g.* time derivatives and Laplacian terms form coefficients of a symmetric matrix, whereas an advective derivative introduces asymmetry. If the user specifies a symmetric solver for an asymmetric matrix, or vice versa, an error message will be written to advise the user accordingly, *e.g.*

```
--> FOAM FATAL IO ERROR : Unknown asymmetric matrix solver PCG
Valid asymmetric matrix solvers are :
4
(
GAMG
PBiCG
PBiCGStab
smoothSolver
)
```

4.6.2 Solution tolerances

The finite volume method generally solves equations in a segregated, decoupled manner, meaning that each matrix equation solves for only one variable. The matrices are consequently sparse, meaning they predominately include coefficients of 0. The solvers are generally iterative, *i.e.* they are based on reducing the equation residual over successive solutions. The residual is ostensibly a measure of the error in the solution so that the smaller it is, the more accurate the solution. More precisely, the residual is evaluated by substituting the current solution into the equation and taking the magnitude of the difference between the left and right hand sides; it is also normalised to make it independent of the scale of the problem being analysed.

Before solving an equation for a particular field, the initial residual is evaluated based on the current values of the field. After each solver iteration the residual is re-evaluated. The solver stops if *any one* of the following conditions are reached:

- the residual falls below the *solver tolerance*, `tolerance`;
- the ratio of current to initial residuals falls below the *solver relative tolerance*, `relTol`;
- the number of iterations exceeds a *maximum number of iterations*, `maxIter`;

The solver tolerance should represent the level at which the residual is small enough that the solution can be deemed sufficiently accurate. The solver relative tolerance limits the relative improvement from initial to final solution. In transient simulations, it is usual to set the solver relative tolerance to 0 to force the solution to converge to the solver tolerance in each time step. The tolerances, `tolerance` and `relTol` must be specified in the dictionaries for all solvers; `maxIter` is optional and defaults to a value of 1000.

Equations are very often solved multiple times within one solution step, or time step. For example, when using the PIMPLE algorithm, a pressure equation is solved according to the number specified by `nCorrectors`, as described in section 4.6.7. Where this occurs, the solver is very often set up to use different settings when solving the particular equation for the final time, specified by a keyword that adds `Final` to the field name. For example,

in the transient pitzDaily example for the incompressibleFluid solver, the solver settings for pressure are as follows.

```
p
{
    solver          GAMG;
    tolerance       1e-07;
    relTol          0.01;
    smoother        DICGaussSeidel;
}

pFinal
{
    $p;
    relTol          0;
}
```

If the case is specified to solve pressure 4 times within one time step, then the first 3 solutions would use the settings for `p` with `relTol` of 0.01, so that the cost of solving each equation is relatively low. Only when the equation is solved the final (4th) time, it solves to a residual level specified by `tolerance` (since `relTol` is 0, effectively deactivating it) for greater accuracy, but at greater cost.

4.6.3 Preconditioned conjugate gradient solvers

There are a range of options for preconditioning of matrices in the conjugate gradient solvers, represented by the `preconditioner` keyword in the solver dictionary, listed below. Note that the `DIC/DILU` preconditioners are exclusively specified in the tutorials in OpenFOAM.

- `DIC/DILU`: diagonal incomplete-Cholesky (symmetric) and incomplete-LU (asymmetric)
- `FDIC`: slightly faster diagonal incomplete-Cholesky (`DIC` with caching, symmetric)
- `GAMG`: geometric-algebraic multi-grid.
- `diagonal`: diagonal preconditioning, not generally used.
- `none`: no preconditioning.

4.6.4 Smooth solvers

The solvers that use a smoother require the choice of smoother to be specified. The smoother options are listed below. The `symGaussSeidel` and `GaussSeidel` smoothers are preferred in the tutorials.

- `GaussSeidel`: Gauss-Seidel.
- `symGaussSeidel`: symmetric Gauss-Seidel.
- `DIC/DILU`: diagonal incomplete-Cholesky (symmetric), incomplete-LU (asymmetric).

- `DICGaussSeidel`: diagonal incomplete-Cholesky/LU with Gauss-Seidel (symmetric/-asymmetric).

When using the smooth solvers, the user can optionally specify the number of sweeps, by the `nSweeps` keyword, before the residual is recalculated. Without setting it, it reverts to a default value of 1.

4.6.5 Geometric-algebraic multi-grid solvers

The geometric-algebraic multi-grid (GAMG) method uses the principle of: generating a quick solution on a mesh with a small number of cells; mapping this solution onto a finer mesh; using it as an initial guess to obtain an accurate solution on the fine mesh. GAMG is faster than standard methods when the increase in speed by solving first on coarser meshes outweighs the additional costs of mesh refinement and mapping of field data. In practice, GAMG starts with the original mesh and coarsens/refines the mesh in stages. The coarsening is limited by a specified minimum number of cells at the most coarse level.

The agglomeration of cells is performed by the method specified by the `agglomerator` keyword. The tutorials all use the default `faceAreaPair` method. The agglomeration can be controlled using the following optional entries, most of which default in the tutorials.

- `cacheAgglomeration`: switch specifying caching of the agglomeration strategy (default `true`).
- `nCellsInCoarsestLevel`: approximate mesh size at the most coarse level in terms of the number of cells (default 10).
- `directSolveCoarset`: use a direct solver at the coarsest level (default `false`).
- `mergeLevels`: keyword controls the speed at which coarsening or refinement is performed; the default is 1, which is safest, but for simple meshes, the solution speed can be increased by coarsening/refining 2 levels at a time, *i.e.* setting `mergeLevels 2`.

Smoothing is specified by the `smoother` as described in section 4.6.4. The number of sweeps used by the smoother at different levels of mesh density are specified by the following optional entries.

- `nPreSweeps`: number of sweeps as the algorithm is coarsening (default 0).
- `preSweepsLevelMultiplier`: multiplier for the number of sweeps between each coarsening level (default 1).
- `maxPreSweeps`: maximum number of sweeps as the algorithm is coarsening (default 4).
- `nPostSweeps`: number of sweeps as the algorithm is refining (default 2).
- `postSweepsLevelMultiplier`: multiplier for the number of sweeps between each refinement level (default 1).
- `maxPostSweeps`: maximum number of sweeps as the algorithm is refining (default 4).
- `nFinestSweeps`: number of sweeps at finest level (default 2).

4.6.6 Solution under-relaxation

The *fvSolution* file usually includes a *relaxationFactors* sub-dictionary which controls under-relaxation. Under-relaxation is used to improve stability of a computation, particularly for steady-state problems. It works by limiting the amount which a variable changes from one iteration to the next, either by manipulating the matrix equation prior to solving for a field, or by modifying the field afterwards. An under-relaxation factor $\alpha, 0 < \alpha \leq 1$ specifies the amount of under-relaxation, as described below.

- No specified α: no under-relaxation.
- $\alpha = 1$: guaranteed matrix diagonal equality/dominance.
- α decreases, under-relaxation increases.
- $\alpha = 0$: solution does not change with successive iterations.

An optimum choice of α is one that is small enough to ensure stable computation but large enough to move the iterative process forward quickly; values of α as high as 0.9 can ensure stability in some cases and anything much below, say, 0.2 can be prohibitively restrictive in slowing the iterative process.

Relaxation factors for under-relaxation of fields are specified within a *field* sub-dictionary; relaxation factors for equation under-relaxation are within a *equations* sub-dictionary. An example is shown below from a case with the incompressibleFluid solver module running in steady-state mode. The factors are specified for pressure p, pressure U, and turbulent fields grouped using a regular expression.

```
54  relaxationFactors
55  {
56      fields
57      {
58          p               0.3;
59      }
60      equations
61      {
62          U               0.7;
63          "(k|omega|epsilon).*" 0.7;
64      }
65  }
66
67  // ************************************************************************* //
```

For a transient case with the incompressibleFluid module, under-relaxation would simply slow convergence of the solution within each time step. Instead, the following setting is generally adopted to ensure diagonal equality which is generally required for convergence of a matrix equation.

```
60  relaxationFactors
61  {
62      equations
63      {
64          ".*"            1;
65      }
66  }
67
68
69  // ************************************************************************* //
```

4.6.7 SIMPLE and PIMPLE algorithms

Most fluid solver modules use algorithms to couple equations for mass and momentum conservation known as:

- SIMPLE (semi-implicit method for pressure-linked equations);
- PIMPLE, a combination of PISO (pressure-implicit split-operator) and SIMPLE.

Within in time, or solution, step, the algorithms solve a pressure equation, to enforce mass conservation, with an explicit correction to velocity to satisfy momentum conservation. They optionally begin each step by solving the momentum equation — the so-called momentum predictor.

While all the algorithms solve the same governing equations (albeit in different forms), the algorithms principally differ in how they loop over the equations. The looping is controlled by input parameters that are listed below. They are set in a dictionary named after the algorithm, *i.e.* *SIMPLE*, or *PIMPLE*.

- `nCorrectors`: used by PIMPLE, sets the number of times the algorithm solves the pressure equation and momentum corrector in each step; typically set to 2 or 3.
- `nNonOrthogonalCcorrectors`: used by all algorithms, specifies repeated solutions of the pressure equation, used to update the explicit non-orthogonal correction, described in section 4.5.4, of the Laplacian term $\nabla \bullet ((1/A)\nabla p)$; typically set to 0 for steady-state and 1 for transient cases.
- `nOuterCorrectors`: used by PIMPLE, it enables looping over the entire system of equations within on time step, representing the total number of times the system is solved; must be ≥ 1 and is typically set to 1.
- `momentumPredictor`: switch that controls solving of the momentum predictor; typically set to `off` for some flows, including low Reynolds number and multiphase.

4.6.8 Pressure referencing

In a closed incompressible system, pressure is relative: it is the pressure range that matters not the absolute values. In these cases, the solver sets a reference level of `pRefValue` in cell `pRefCell`. These entries are generally stored in the *SIMPLE* or *PIMPLE* sub-dictionary.

4.7 Case management tools

There are a set of applications and scripts that help with managing case files and help the user find and set keyword data entries in case files. The tools are described in the following sections.

4.7.1 General file management

There are some tools for general management of case files, including foamListTimes, foamCleanCase and foamCloneCase. A case includes configuration files for various processes such as meshing, case initialisation, simulation and post-processing. Each process generates new data files in various directories, *e.g.* mesh data is stored in *constant/polyMesh*, CFD results in time directories, and further post-processing in a *postProcessing* directory.

The foamListTimes utility lists the time directories for a case, omitting the *0* directory by default. Prior to re-running a CFD simulation, it can be useful to delete the results from the previous simulation. The foamListTimes utility provides this function through the `-rm` option which deletes the listed time directories, executed by the following command.

```
foamListTimes -rm
```

The foamCleanCase script aims to reset the case files to their original state, removing all files generated during a workflow including the meshing, post-processing. It deletes directories including: *postProcessing* and *VTK*; the *constant/polyMesh* directory; *processor* directories from parallel decomposition; and, *dynamicCode* for run-time compiled code. The script is simply run as follows.

```
foamCleanCase
```

The foamCloneCase script creates a new case, by copying the *0*, *system* and *constant* directories from an existing case. The copied case is ready to run since the mesh is copied through the *constant* directory. If the original case is set up to run in parallel, the *processor* directories can also be copied using the `-processor` option. The basic command is executed simply as follows, where *oldCase* refers to an existing case directory.

```
foamCloneCase oldCase newCase
```

4.7.2 The foamDictionary script

The foamDictionary utility offers several options for printing, editing and adding keyword entries in case files. The utility is executed with a case dictionary file as an argument, *e.g.* from within a case directory on the *fvSchemes* file.

```
foamDictionary system/fvSchemes
```

Without options, the utility prints the entries from the file, removing comments, *e.g.* as follows for the *fvSchemes* file in the pitzDailySteady tutorial case.

```
FoamFile
{
    format          ascii;
    class           dictionary;
    location        "system";
    object          fvSchemes;
}

ddtSchemes
{
    default         steadyState;
}

gradSchemes
{
    default         Gauss linear;
    grad(U)         cellLimited Gauss linear 1;
}

divSchemes
{
    default         none;
    div(phi,U)      bounded Gauss linearUpwind grad(U);
    turbulence      bounded Gauss limitedLinear 1;
    div(phi,k)      $turbulence;
    div(phi,epsilon) $turbulence;
    div(phi,omega)  $turbulence;
    div(phi,v2)     $turbulence;
    div((nuEff*dev2(T(grad(U))))) Gauss linear;
    div(nonlinearStress) Gauss linear;
}
```

```
laplacianSchemes
{
    default         Gauss linear corrected;
}

interpolationSchemes
{
    default         linear;
}

snGradSchemes
{
    default         corrected;
}
```

The output includes the macros before expansion, indicated by the `$` symbol, *e.g.* `$turbulence`. The macros can be expanded by the `-expand` option as shown below

```
foamDictionary -expand system/fvSchemes
```

The `-entry` option prints the entry for a particular keyword, expanding the macros by default, *e.g.* `divSchemes` in the example below

```
foamDictionary -entry divSchemes system/fvSchemes
```

The example clearly extracts the `divSchemes` dictionary.

```
divSchemes
{
    default         none;
    div(phi,U)      bounded Gauss linearUpwind grad(U);
    turbulence      bounded Gauss limitedLinear 1;
    div(phi,k)      $turbulence;
    div(phi,epsilon) $turbulence;
    div(phi,omega)  $turbulence;
    div(phi,v2)     $turbulence;
    div((nuEff*dev2(T(grad(U))))) Gauss linear;
    div(nonlinearStress) Gauss linear;
}
```

The "`/`" syntax allows access to keywords with levels of sub-dictionary. For example, the `div(phi,U)` keyword can be accessed within the `divSchemes` sub-dictionary by the following command.

```
foamDictionary -entry "divSchemes/div(phi,U)" system/fvSchemes
```

The example returns the single `divSchemes/div(phi,U)` entry.

```
div(phi,U)      bounded Gauss linearUpwind grad(U);
```

The `-value` option prints only the entry.

```
foamDictionary -entry "divSchemes/div(phi,U)" -value system/fvSchemes
```

The example removes the keyword and terminating semicolon, leaving just the data.

```
bounded Gauss linearUpwind grad(U)
```

The `-keywords` option prints only the keywords.

```
foamDictionary -entry divSchemes -keywords system/fvSchemes
```

The example produces a list of keywords inside the `divSchemes` dictionary.

```
default
div(phi,U)
div(phi,k)
div(phi,epsilon)
div(phi,omega)
div(phi,v2)
div((nuEff*dev2(T(grad(U)))))
div(nonlinearStress)
```

foamDictionary can set entries with the `-set` option. If the user wishes to change the `div(phi,U)` to the upwind scheme, they can enter the following.

```
foamDictionary -entry "divSchemes/div(phi,U)" \
    -set "bounded Gauss upwind" system/fvSchemes
```

An alternative "=" syntax can be used with the `-set` option which is particularly useful when modifying multiple entries:

```
foamDictionary -set "startFrom=startTime, startTime=0" \
    system/controlDict
```

foamDictionary can add entries with the `-add` option. If the user wishes to add an entry named `turbulence` to `divSchemes` with the upwind scheme, they can enter the following.

```
foamDictionary -entry "divSchemes/turbulence" \
   -add "bounded Gauss upwind" system/fvSchemes
```

4.7.3 The foamSearch script

The foamSearch script, demonstrated extensively in section 4.5, uses foamDictionary to extract and sort keyword entries from all files of a specified name in a specified dictionary. The `-c` option counts the number of entries of each type, *e.g.* the user could searche for the choice of `solver` for the `p` equation in all the *fvSolution* files in the tutorials.

```
foamSearch -c $FOAM_TUTORIALS fvSolution solvers/p/solver
```

The search shows `GAMG` to be the most common choice in all the tutorials.

```
 64 solver           GAMG;
  2 solver           PBiCGStab;
 26 solver           PCG;
  5 solver           smoothSolver;
```

4.7.4 The foamGet script

The foamGet script copies configuration files into a case quickly and conveniently. The user must be inside a case directory to run the script or identify the case directory with the `-case` option. Its operation can be described using the pitzDailySteady case described in section 2.1. The example begins by copying the case directory as follows:

```
run
cp -r $FOAM_TUTORIALS/modules/incompressibleFluid/pitzDailySteady .
```

The mesh is generated for the case by going into the case directory and running blockMesh:

```
cd pitzDailySteady
blockMesh
```

The user might decide before running the simulation to configure some automatic post-processing as described in section 7.2. The user can list the pre-configured function objects by the following command.

```
foamPostProcess -list
```

From the output, the user could select the `patchFlowRate` function to monitor the flow rate at the outlet patch. The *patchFlowRate* configuration file can be copied into the *system* directory using foamGet:

```
foamGet patchFlowRate
```

In order to monitor the flow through the `outlet` patch, the `patch` entry in *patchFlowRate* file should be set as follows:

```
patch    outlet;
```

The *patchFlowRate* configuration is then included in the case by adding to the `functions` sub-dictionary in the *controlDict* file:

```
functions
{
    ...
    //#includeFunc writeObjects(kEpsilon:G) // existing entry
    #includeFunc patchFlowRate
}
```

4.7.5 The foamInfo script

The foamInfo script provides quick information and examples relating to thing in OpenFOAM that the user can "select". The selections include models (including boundary conditions and packaged function objects), solver modules, applications and scripts. For example, foamInfo prints information about the incompressibleFluid solver module by typing the following:

```
foamInfo incompressibleFluid
```

Information for the flowRateInletVelocity boundary condition can similarly be obtained by typing the following command.

```
foamInfo flowRateInletVelocity
```

The output includes: the location of the source code header file for this boundary condition; the description and usage details from the header file; a list of other models of the same type, *i.e.* other boundary conditions; and, a list of example cases that use the boundary condition. This example is demonstrated in section 2.1.17.

When the user requests information about a model with foamInfo, it attempts to provide a list of other models in the "family". For example, if the user requests information about the `kEpsilon` turbulence model by

```
foamInfo kEpsilon
```

the output includes the following

```
Model
 This appears to be the 'kEpsilon' model of the 'RAS' family.
 The models in the 'RAS' family are:
 + kEpsilon
 + kOmega
 + kOmega2006
 + kOmegaSST
 + kOmegaSSTLM
 + kOmegaSSTSAS
 + LaunderSharmaKE
 + LRR
 + RAS
 + realizableKE
 + RNGkEpsilon
 + SpalartAllmaras
 + SSG
 + v2f
```

It provides fairly complete lists for fvModels and fvConstraints, which can be demonstrated by typing the following commands.

```
foamInfo clouds
foamInfo limitTemperature
```

It also lists options used in input files, *e.g.* for Function1 entries in boundary conditions, searchableSurface entries in the configuration of snappyHexMesh. Users could try searching for specific models within those families or the families themselves, *e.g.*

```
foamInfo linearRamp
foamInfo triSurfaceMesh
foamInfo Function1
foamInfo searchableSurfaces
```

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4.7.6 The foamToC utility

The foamToC utility lists all the options in OpenFOAM which the user can select through input files. The functionality overlaps with foamInfo to some extent but foamToC produces more definitive reporting since it is an OpenFOAM application which directly interrogates the run-time selection tables in the compiled libraries. The "ToC" in the name is an abbreviation for "Table of Contents."

As well as providing general options to interrogate anything in OpenFOAM, foamToC includes specific options that replicate most of the "`-list...`" options included in application solvers prior to v11. These options included: `-listScalarBCs` and `-listVectorBCs` to list boundary conditions; `-listFunctionObjects` to list functionObjects; `-listFvModels` to list fvModels; and, `-listFvConstraints` to list fvConstraints. The equivalent options in foamToC are listed below, with an additional `-solvers` option:

- `-scalarBCs` and `-vectorBCs` to list boundary conditions;
- `-functionObjects` to list functionObjects;
- `-fvModels` to list fvModels;
- `-fvConstraints` to list fvConstraints; and,
- `-solvers` to list the solver modules.

For example, with the last option, foamToC prints the following output

```
>> foamToC -fvConstraints

fvConstraints:
Contents of table fvConstraint:
    bound                                libfvConstraints.so
    fixedTemperatureConstraint           libfvConstraints.so
    fixedValueConstraint                 libfvConstraints.so
    limitMag                             libfvConstraints.so
    limitPressure                        libfvConstraints.so
    limitTemperature                     libfvConstraints.so
    meanVelocityForce                    libfvConstraints.so
    patchMeanVelocityForce               libfvConstraints.so
    zeroDimensionalFixedPressure         libfvConstraints.so
```

Each fvConstraint is listed in the left column with the library to which it belongs in the right column. The options listed above essentially invoke the more general `-table` option that lists the contents of a run-time selection table. The `-fvConstraints` option is equivalent to the following command which lists the items in the `fvConstraint` table.

```
foamToC -table fvConstraint
```

All the selection tables in OpenFOAM are listed by running foamToC with the `-tables` option as shown below.

```
foamToC -tables
```

This is also the default response of foamToC without options, *i.e.*

```
foamToC
```

By default foamToC loads **all** the libraries in OpenFOAM to produce a complete list of tables. The user can control the libraries that are loaded with the following options.

- `-noLibs`: only loads the core libOpenFOAM.so library by default.
- `-solver` <`solver`>: only loads libraries associated with the specified solver module <`solver`>.
- `-libs '("lib1.so" ... "libN.so")'`: *additionally* pre-loads specified libraries, *e.g.* customised libraries of the user.

An important use of foamToC is to enable users to find alternative models to the one currently configured for their case. The challenge is to find the table that contains the models they wish to list. The `-search` option helps with this, since it takes an entry, *e.g.* a model, and reports the table in which it belongs. For example, if the user was familiar with the `BirdCarreau` model for viscosity and wished to list alternative non-Newtonian models, they could first issue the following command.

```
    >> foamToC -search BirdCarreau

BirdCarreau is in table
    generalisedNewtonianViscosityModel  libmomentumTransportModels.so
```

Having identified the table, the user can then list its contents using the `-table` option.

```
    >> foamToC -table generalisedNewtonianViscosityModel

Contents of table generalisedNewtonianViscosityModel:
    BirdCarreau             libmomentumTransportModels.so
    Casson                  libmomentumTransportModels.so
    CrossPowerLaw           libmomentumTransportModels.so
    HerschelBulkley         libmomentumTransportModels.so
    Newtonian               libmomentumTransportModels.so
    powerLaw                libmomentumTransportModels.so
    strainRateFunction      libmomentumTransportModels.so
```

If the user wished to list the solver modules, they can run

```
    foamToC -solvers
```

which is equivalent to running

```
    foamToC -table solver
```

With turbulence models, a search for `kEpsilon` lists the following set of tables (the corresponding libraries are not shown here).

```
kEpsilon is in table
    RAS
        RAScompressibleMomentumTransportModel
        RASincompressibleMomentumTransportModel
        RASphaseCompressibleMomentumTransportModel
        RASphaseIncompressibleMomentumTransportModel
```

The output indicates the model exists in different tables corresponding to both incompressible and compressible flows, and both single phase and multi-phase flows (`phase` indicates multi-phase in the table name). For single phase, incompressible flows, the available RAS turbulence models can be listed as follows.

```
foamToC -table RASincompressibleMomentumTransportModel
```

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Chapter 5

Mesh generation and conversion

This chapter describes all topics relating to the creation of meshes in OpenFOAM: section 5.1 gives an overview of the way a mesh is be described in OpenFOAM; section 5.2 lists the basic data files that describe a mesh; section 5.3 discusses mesh boundaries and introduces boundary conditions; section 5.4 covers the blockMesh utility for generating simple meshes of blocks of hexahedral cells; section 5.5 covers the snappyHexMesh utility for generating complex meshes of hexahedral and split-hexahedral cells automatically from triangulated surface geometries; section 5.6 describes the options for conversion of a mesh that has been generated by a third-party product into a format that OpenFOAM can read.

5.1 Mesh description

This section provides a specification of the way OpenFOAM describes a mesh. The mesh is an integral part of the numerical solution and must satisfy certain criteria to ensure a valid, and hence accurate, solution. OpenFOAM defines a mesh of arbitrary polyhedral cells in 3-D, bounded by arbitrary polygonal faces, *i.e.* the cells can have an unlimited number of faces where, for each face, there is no limit on the number of edges nor any restriction on its alignment. This flexible description of a mesh offers great freedom in mesh generation and manipulation when the geometry of the domain is complex.

An OpenFOAM mesh begins with **points** (or **vertices**). Each point is a location in 3D space, defined by a vector. The set of points forms a list where each point can be indexed by its position in the list, starting from zero. The list does not contain points that are unused.

The points are used to from mesh **faces**, where each face is defined as an ordered list of points, described by their where a point is referred to by its label. The ordering of point labels in a face is such that each two neighbouring points are connected by an edge, *i.e.* you follow points as you travel around the circumference of the face. The set of faces forms a list where each face is referred to by its label, representing its position in the list.

Each face can be characterised by a vector whose direction is normal to the face. The normal vector follows the right-hand rule, *i.e.* looking towards a face, if the numbering of the points follows a clockwise path, the normal vector points away from you, as shown in Figure 5.1. Note that faces can be warped, i.e. the points of the face may not necessarily lie on a plane. There are two types of face, described below.

- *Internal faces*, which connect two cells (and it can never be more than two). For each internal face, the ordering of the point labels is such that the face normal

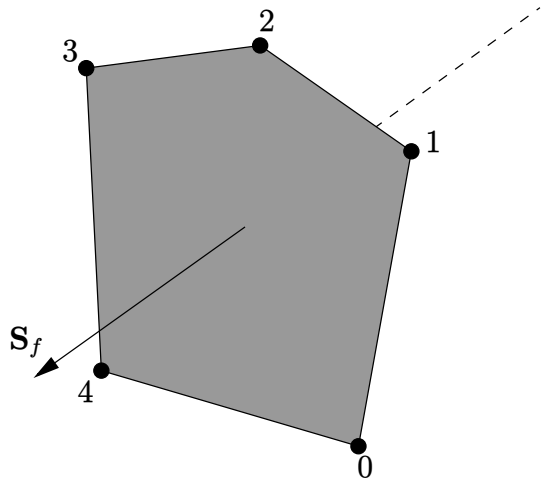


Figure 5.1: Face area vector from point numbering on the face

points into the cell with the larger label, *i.e.* for cells labelled '2' and '5', the normal points into '5'.

- *Boundary faces*, which belong to one cell since they coincide with the boundary of the domain. A boundary face is therefore addressed by one cell(only) and a boundary patch. The ordering of the point labels is such that the face normal points outside of the computational domain.

A **cell** is a list of faces in arbitrary order. Under normal circumstances, cells must have the properties listed below.

- The cells must be *contiguous*,*i.e.* completely cover the computational domain and must not overlap one another.
- Every cell must be *closed geometrically*, such that when all face area vectors are oriented to point outwards of the cell, their sum should equal the zero vector to machine accuracy;
- Every cell must be *closed topologically* such that all the edges in a cell are used by exactly two faces of the cell in question.

The **boundary** is formed by the boundary faces. It should be closed, *i.e.* the sum all boundary face area vectors equates to zero to machine tolerance. It is split into regions known as **patches** so that different boundary conditions can be applied to different parts of the boundary. A patch is defined by the labels of the faces it contains.

5.2 Mesh files

When a mesh is written out by OpenFOAM, the data files go into a *polyMesh* sub-directory. Usually the *polyMesh* directory is written to the the *constant* directory, but simulations with *dynamic* meshes (*e.g.* mesh motion, refinement, *etc.*) write the modified meshes into time directories along with the field data files.

The data files are based around faces rather than cells. Each face is therefore assigned an 'owner' cell and 'neighbour' cell so that the connectivity across a given face can simply be described by the owner and neighbour cell labels. In the case of boundaries, there is no neighbour cell. With this in mind, the I/O specification consists of the following files:

points a list of vectors describing the cell vertices, where the first vector in the list represents vertex 0, the second vector represents vertex 1, *etc.*;

faces a list of faces, each face being a list of indices to vertices in the points list, where again, the first entry in the list represents face 0, *etc.*;

owner a list of owner cell labels, starting with the owner cell of face 0, then 1, 2, ...

neighbour a list of neighbour cell labels;

boundary a list of patches, containing a dictionary entry for each patch, declared using the patch name.

Critically, the *faces* list is ordered so that all internal faces are listed first, followed by the boundary faces. The boundary faces are themselves ordered so that they begin with the faces in the first patch, followed by the second, *etc.* As a consequence the patch entries in the *boundary* file are very compact, *e.g.*

```
inlet
{
    type            patch;
    nFaces          30;
    startFace       24170;
}
```

Due to the face ordering, the patch faces are simply described by: `startFace`, the index into the face list of the first face in the patch; and, `nFaces`, the number of faces in the patch.

5.3 Mesh boundary

As we saw in section 5.2, the domain boundary is defined by patches within the mesh, listed within the *boundary* mesh file. Each patch includes a `type` entry which can apply a *geometric constraint* to the patch. These geometric constraints include conditions that represent a geometric approximation, *e.g.* a symmetry plane, and conditions which form numerical connections between patches, *e.g.* a *cyclic* (or *periodic*) boundary. An example *boundary* file is shown below which includes some patches with geometric constraints.

```
16
17  5
18  (
19      top
20      {
21          type            wall;
22          inGroups        List<word> 1(wall);
23          nFaces          60;
24          startFace       3510;
25      }
26      inlet
27      {
28          type            patch;
29          nFaces          30;
30          startFace       3570;
31      }
32      outlet
33      {
34          type            patch;
35          nFaces          30;
36          startFace       3600;
37      }
```

```
38      bottom
39      {
40          type            symmetryPlane;
41          inGroups        List<word> 1(symmetryPlane);
42          nFaces          60;
43          startFace       3630;
44      }
45      frontAndBack
46      {
47          type            empty;
48          inGroups        List<word> 1(empty);
49          nFaces          3600;
50          startFace       3690;
51      }
52  )
53
54  // ************************************************************************* //
```

A `type` entry is specified for every patch (`inlet`, `outlet`, *etc.*), with types assigned that include `patch`, `wall`, `symmetryPlane` and `empty`. Some patches also include an `inGroups` entry which is discussed in section 5.3.6.

5.3.1 Generic patch and wall

The patch types specified in the *boundary* file, which are **not** associated with a geometric constraint are the generic patch and wall. The patch type is assigned to open boundaries such as an inlet or outlet which does not involve any special handling of geometric approximation or numerical connections.

The wall type also provides no special geometric or numerical handling, but is used for patches which coincide with a solid wall. The wall 'tag' is required by some models, *e.g.* wall functions in turbulence modelling which require the distance to nearest wall.

5.3.2 1D/2D and axi-symmetric problems

OpenFOAM is designed as a code for 3D space and defines all meshes as such. However, 1D and 2D and axi-symmetric problems can be simulated in OpenFOAM by generating a mesh in 3 dimensions and applying special boundary conditions on any patch in the plane(s) normal to the direction(s) of interest. 1D and 2D problems apply the empty patch type to the relevant patches. Often the two regions of the boundary, on the 'front' and 'back' of the domain, are combined into a single patch, as in the `frontAndBack` patch in the quoted example above.

For axi-symmetric cases, the geometry, *e.g.* a cylinder, is approximated by a wedge-shaped mesh of small angle (*e.g.* 1°) and 1 cell thick, running along the centre line, straddling one of the coordinate planes, as shown in Figure 5.2. The axi-symmetric wedge planes must be specified as separate patches of wedge type. The generation of wedge geometries for axi-symmetric problems is discussed in section 5.4.10.

5.3.3 Symmetry condition

A symmetry plane is a boundary condition that imagines the solution within the domain is 'mirrored' across the boundary. It can therefore be applied reliably to a domain with a plane of symmetry where the flow is believed to be symmetric across the plane. When the flow involves something like vortex shedding that breaks symmetry, the condition is less applicable.

There are two patch types relating to symmetry. Firstly, the symmetryPlane condition is a pure symmetry plane which can only be applied to a patch which is perfectly planar.

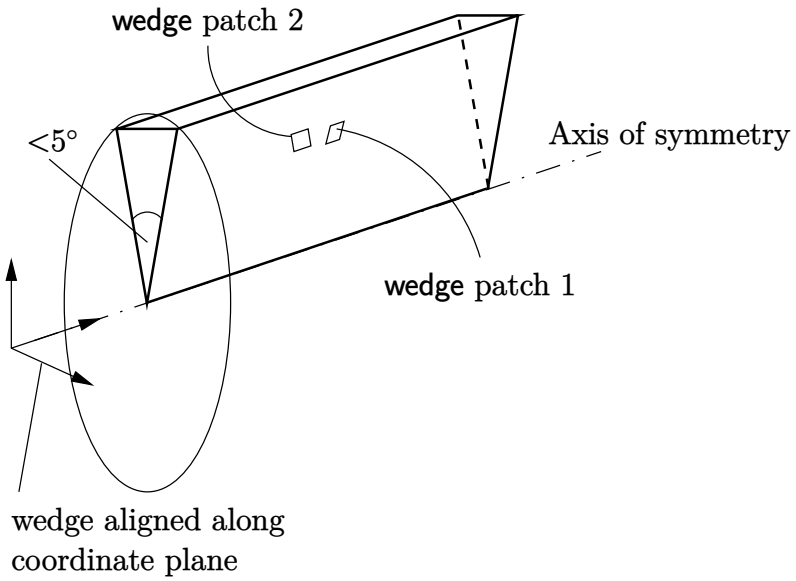


Figure 5.2: Axi-symmetric geometry using the wedge patch type.

There is then a symmetry condition, which can be applied to any patch, including those that are non-planar.

5.3.4 Cyclic conditions

The cyclic boundary conditions form a numerical connections between patches that are physically disconnected. The cyclic condition connects patches which have the same area to within a tolerance given by the `matchTolerance` keyword. Each patch specifies the name of the patch to which it connects through the `neighbourPatch` keyword. The condition can transform the field between patches, *e.g.* by a rotation, so the patches do not require the same orientation.

OpenFOAM also includes non-conformal coupling (NCC) which connects regions of a domain with independent meshes. It is and is used particularly for cases when one or more regions are moving, *e.g.* to simulate rotating geometry. Non-conformal coupling uses the nonConformalCyclic condition which are usually generated with the createNonConformalCouples utility. NCC examples can be located by searching for the createNonConformalCouples utility in Allrun scripts in the *tutorials* directory, *e.g.* by running

```
find $FOAM_TUTORIALS -name Allrun | \
    xargs grep -l createNonConformalCouples
```

5.3.5 Processor patches

Running applications in parallel is described in section 3.4. It involves decomposition of the mesh using decomposePar as described in section 3.4.1. Decomposition splits the domain which creates new patches at the exposed faces. Those patches are assigned the processor type which forms a numerical connection between sub-domains. Each processor patch entry in the *boundary* file includes a `myProcNo` entry for the processor (sub-domain) index and a `neighbProcNo` entry for the index of the matching patch on the sub-domain it connects with.

5.3.6 Patch groups

The *boundary* file example shows some patches include an `inGroups` entry, *e.g.* the `top` patch:

```
top
{
    type          wall;
    inGroups      List<word> 1(wall);
    nFaces        60;
    startFace     3510;
}
```

The `inGroups` entry is optional. It specifies one or more patch groups to which a patch can belong. A patch group is specified by a name which the user can choose. Group names can be used to specify boundary conditions in field files, simplifying the configuration. For example, if all inlet patches can be included in an `inlet` group, then one `inlet` entry can specify a boundary condition for all the patches.

Every non-generic patch, *i.e.* one which is not patch type, is *included in a patch group of the same name as its type.* For example, a patch of type wall is included in a `wall` group, one of type symmetry is included in a `symmetry` group, *etc.* The use of group names to specify boundary conditions in described further in chapter 6.

5.3.7 Constraint type examples

The user can scan the tutorials for mesh generation configuration files, *e.g. blockMeshDict* for blockMesh (see section 5.4) and *snappyHexMeshDict* for snappyHexMesh (see section 5.5, for examples of different types being used. The following example provides documentation and lists cases that use the symmetryPlane condition.

```
foamInfo -a symmetryPlane
```

The next example searches for *snappyHexMeshDict* files that specify the wall condition.

```
find $FOAM_TUTORIALS -name snappyHexMeshDict | \
    xargs grep -El "type[\t ]*wall"
```

5.4 Mesh generation with the blockMesh utility

This section describes the mesh generation utility, blockMesh, supplied with OpenFOAM. The blockMesh utility creates parametric meshes with grading and curved edges. The mesh is generated from a dictionary file named *blockMeshDict* located in the *system* directory of a case. blockMesh reads this dictionary, generates the mesh and writes out the mesh data to *points*, *faces*, *cells* and *boundary* files in the *polyMesh* directory.

The principle behind blockMesh is to decompose the domain geometry into a set of 1 or more 3D hexahedral blocks. Edges of the blocks can be straight lines, arcs or splines. The mesh is ostensibly specified as a number of cells in each direction of the block, sufficient information for blockMesh to generate the mesh data.

5.4.1 Overview of a blockMeshDict file

The *blockMeshDict* file is a dictionary including keywords described below.

- `convertToMeters`: scaling factor for the vertex coordinates, *e.g.* `0.001` scales to mm.
- `vertices`: list of vertex coordinates, see section 5.4.2.
- `edges`: optional entry to describe curved geometry, see section 5.4.3.
- `blocks`: ordered list of vertex labels and mesh size, see section 5.4.4.
- `boundary`: sub-dictionary of boundary patches, see section 5.4.6.
- `defaultPatch`: optional entry describing a default patch, see section 5.4.6.
- `mergePatchPairs`: optional list of patches to be merged, see section 5.4.7.

5.4.2 The vertices

The vertices of the blocks of the mesh are given next as a standard list named `vertices`. An example set of vertices, corresponding to a block in Figure 5.3, is provided below.

```
vertices
(
    ( 0    0    0  )    // vertex number 0
    ( 1    0    0.1)    // vertex number 1
    ( 1.1  1    0.1)    // vertex number 2
    ( 0    1    0.1)    // vertex number 3
    (-0.1 -0.1  1  )    // vertex number 4
    ( 1.3  0    1.2)    // vertex number 5
    ( 1.4  1.1  1.3)    // vertex number 6
    ( 0    1    1.1)    // vertex number 7
);
```

The `convertToMeters` keyword specifies a scaling factor by which all vertex coordinates in the mesh description are multiplied. For example,

```
convertToMeters     0.001;
```

means that all coordinates are multiplied by 0.001, *i.e.* the values quoted in the *blockMeshDict* file are in mm.

5.4.3 The edges

Each edge joining 2 vertex points is assumed to be straight by default. However any edge may be specified to be curved by entries in a list named `edges`. The list is optional; if the geometry contains no curved edges, it may be omitted.

Each entry for a curved edge begins with a keyword specifying the type of curve from those listed below.

- `arc`: a circular arc with a single interpolation point or angle + axis (see below).

- `spline`: spline curve using a list of interpolation points
- `polyLine`: a set of lines with list of interpolation points
- `BSpline`: a B-spline curve with list of interpolation points
- `line`: a straight line, the default which requires no edge specification.

The keyword is then followed by the labels of the 2 vertices that the edge connects. Following that, interpolation points must be specified through which the edge passes. For an `arc`, either of the following is required: a single interpolation point, which the circular arc will intersect; or an angle and rotation axis for the arc. For `spline`, `polyLine` and `BSpline`, a list of interpolation points is required. For our example block in Figure 5.3 we specify an `arc` edge connecting vertices 1 and 5 as follows through the interpolation point $(1.1, 0.0, 0.5)$:

```
edges
(
    arc 1 5 (1.1 0.0 0.5)
);
```

For the angle and axis specification of an arc, the syntax is of the form:

```
edges
(
    arc 1 5 25 (0 1 0) // 25 degrees, y-normal
);
```

5.4.4 The blocks

The block definitions are contained in a list named `blocks`. Each block of the geometry is defined by 8 vertices, one at each corner of a hexahedron. An example block is shown in Figure 5.3 with each vertex numbered according to the list in section 5.4.2. The edge connecting vertices 1 and 5 is curved as a reminder that edges can be curved in blockMesh.

Each block has a local coordinate system (x_1, x_2, x_3) that must be right-handed. A right-handed set of axes is defined such that to an observer looking down the Oz axis, with O nearest them, the arc from a point on the Ox axis to a point on the Oy axis is in a clockwise sense.

The local coordinate system is defined by the order in which the vertices are presented in the block definition according to:

- the axis origin is the first entry in the block definition, vertex 0 in our example;
- the x_1 direction is described by moving from vertex 0 to vertex 1;
- the x_2 direction is described by moving from vertex 1 to vertex 2;
- vertices 0, 1, 2, 3 define the plane $x_3 = 0$;
- vertex 4 is found by moving from vertex 0 in the x_3 direction;
- vertices 5,6 and 7 are similarly found by moving in the x_3 direction from vertices 1,2 and 3 respectively.

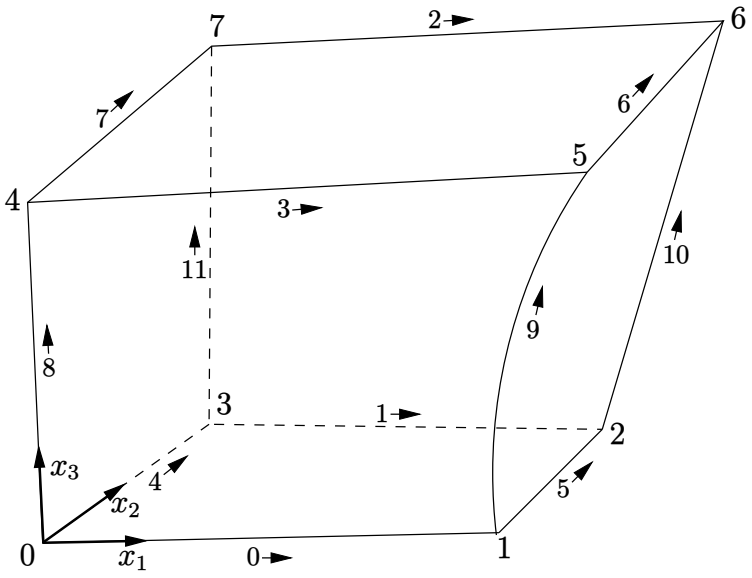


Figure 5.3: A single block

An example block specification is shown below.

```
blocks
(
    hex (0 1 2 3 4 5 6 7)    // vertex numbers
    (10 10 10)               // numbers of cells in each direction
    simpleGrading (1 2 3)    // cell expansion ratios
);
```

It begins with the shape identifier of the block (defined in the *$FOAM_ETC/cellModels* file). The shape is always `hex` since the blocks are always hexahedra. The list of vertex numbers follows, ordered in the manner described above.

The second entry gives the number of cells in each of the x_1 x_2 and x_3 directions for that block. The third entry gives the cell expansion ratios for each direction in the block. The expansion ratio enables the mesh to be graded, or refined, in specified directions. The ratio is that of the width of the end cell δ_e along one edge of a block to the width of the start cell δ_s along that edge, as shown in Figure 5.4.

There are two types of grading specification available in blockMesh. The most common one is `simpleGrading` which specifies uniform expansions in the local x_1, x_2 and x_3 directions respectively with only 3 expansion ratios, *e.g.*

```
simpleGrading (1 2 3)
```

The more complex alternative is `edgeGrading`. This full cell expansion description gives a ratio for each edge of the block, numbered according to the scheme shown in Figure 5.3 with the arrows representing the direction from first cell...to last cell *e.g.*

```
edgeGrading (1 1 1 1 2 2 2 2 3 3 3 3)
```

This example is directly equivalent to the `simpleGrading` example given above because it uses a ratio of cell widths of 1 along edges 0-3, 2 along edges 4-7 and 3 along 8-11. Note that it is possible to generate blocks with fewer than 8 vertices by collapsing one or more pairs of vertices on top of each other, as described in section 5.4.10.

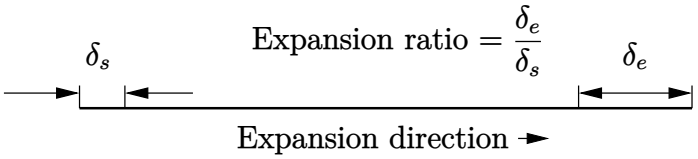


Figure 5.4: Mesh grading along a block edge

5.4.5 Multi-grading of a block

Using a single expansion ratio to describe mesh grading permits only "one-way" grading within a mesh block. In some cases, it reduces complexity and effort to be able to control grading within separate divisions of a single block, rather than have to define several blocks with one grading per block. For example, to mesh a channel with two opposing walls and grade the mesh towards the walls requires three regions: two with grading to the wall with one in the middle without grading.

OpenFOAM v2.4+ includes multi-grading functionality that can divide a block in an given direction and apply different grading within each division. This multi-grading is specified by replacing any single value expansion ratio in the grading specification of the block, *e.g.* "1", "2", "3" in

```
blocks
(
    hex (0 1 2 3 4 5 6 7) (100 300 100)
    simpleGrading (1 2 3);
);
```

We will present multi-grading for the following example:

- split the block into 3 divisions in the y-direction, representing 20%, 60% and 20% of the block length;
- include 30% of the total cells in the y-direction (300) in *each* divisions 1 and 3 and the remaining 40% in division 2;
- apply 1:4 expansion in divisions 1 and 3, and zero expansion in division 2.

We can specify this by replacing the y-direction expansion ratio "2" in the example above with the following:

```
blocks
(
    hex (0 1 2 3 4 5 6 7) (100 300 100)
    simpleGrading
    (
        1                   // x-direction expansion ratio
        (
            (0.2 0.3 4)     // 20% y-dir, 30% cells, expansion = 4
            (0.6 0.4 1)     // 60% y-dir, 40% cells, expansion = 1
            (0.2 0.3 0.25) // 20% y-dir, 30% cells, expansion = 0.25 (1/4)
        )
        3                   // z-direction expansion ratio
    )
);
```

Both the fraction of the block and the fraction of the cells are normalized automatically. They can be specified as percentages, fractions, absolute lengths, *etc.* and do not need to sum to 100, 1, *etc.* The example above can be specified using percentages, *e.g.*

```
blocks
(
    hex (0 1 2 3 4 5 6 7) (100 300 100)
    simpleGrading
    (
        1
        (
            (20 30 4)   // 20%, 30%...
            (60 40 1)
            (20 30 0.25)
        )
        3
    )
);
```

5.4.6 The boundary

The boundary of the mesh is given in a list named `boundary`. The boundary is broken into patches (regions), where each patch in the list has its name as the keyword, which is the choice of the user, although we recommend something that conveniently identifies the patch, *e.g.*`inlet`; the name is used as an identifier for setting boundary conditions in the field data files. The patch information is then contained in sub-dictionary with:

- `type`: the patch type, either a generic `patch` on which some boundary conditions are applied or a particular geometric condition, as listed in section 5.3;
- `faces`: a list of block faces that make up the patch and whose name is the choice of the user, although we recommend something that conveniently identifies the patch, *e.g.*`inlet`; the name is used as an identifier for setting boundary conditions in the field data files.

blockMesh collects *block* faces that are omitted from the patches in the `boundary` list and assigns them to a default patch. The default patch can be configured through a `defaultPatch` sub-dictionary, including `type` and `name`, *e.g.*

```
    defaultPatch
    {
        name    frontAndBack;
        type    empty;
    }
```

In absence of any of these entries a default patch uses the name `defaultFaces` and type `empty` by default. This means that for a 2D, the user has the option to omit block faces lying in the 2D plane, knowing that they will be collected into an `empty` patch as required.

Returning to the example block in Figure 5.3, if it has an inlet on the left face, an output on the right face and the four other faces are walls then the patches could be defined as follows:

```
boundary                // keyword
(
    inlet               // patch name
    {
        type patch;     // patch type for patch 0
        faces
        (
            (0 4 7 3)   // block face in this patch
        );
    }                   // end of 0th patch definition

    outlet              // patch name
    {
        type patch;     // patch type for patch 1
        faces
        (
            (1 2 6 5)
        );
    }

    walls
    {
        type wall;
        faces
        (
            (0 1 5 4)
            (0 3 2 1)
            (3 7 6 2)
            (4 5 6 7)
        );
    }
);
```

Each block face is defined by a list of 4 vertex numbers. The list can begin with any vertex but needs to follow a sequence through connecting edges, with no restriction on the direction.

Where a patch type requires additional data in the resulting *boundary* file, the data is simply added in the patch entry in *blockMeshDict*. For example, with the cyclic patch, the user must specify the name of the related patch through the `neighbourPatch` keyword, *e.g.*

```
left
{
    type            cyclic;
    neighbourPatch  right;
    faces           ((0 4 7 3));
}
right
```

试用水印

```
{
    type            cyclic;
    neighbourPatch  left;
    faces           ((1 5 6 2));
}
```

5.4.7 Multiple blocks

A mesh can be created using more than 1 block. In such circumstances, the mesh is created as described in the preceeding text. The only additional issue is the connection between blocks. Firstly, if a face of one block also belongs to another block, the block face will not form an external patch but instead a set of internal faces of the cells in the resulting mesh.

Alternatively if the user wishes to combine block faces which do not exactly match one another, *i.e.* through shared vertices, they can first include the block faces within the `patches` list. Each pair of patches whose faces are to be merged can then be included in an optional list named `mergePatchPairs`. The format of `mergePatchPairs` is:

```
mergePatchPairs
(
    ( <masterPatch> <slavePatch> ) // merge patch pair 0
    ( <masterPatch> <slavePatch> ) // merge patch pair 1
    ...
)
```

See for example $FOAM_TUTORIALS/multiphaseEuler/LBend. The pairs of patches are interpreted such that the first patch becomes the *master* and the second becomes the *slave*. The rules for merging are as follows:

- the faces of the master patch remain as originally defined, with all vertices in their original location;
- the faces of the slave patch are projected onto the master patch where there is some separation between slave and master patch;
- the location of any vertex of a slave face might be adjusted by blockMesh to eliminate any face edge that is shorter than a minimum tolerance;
- if patches overlap as shown in Figure 5.5, each face that does not merge remains as an external face of the original patch, on which boundary conditions must then be applied;
- if all the faces of a patch are merged, then the patch itself will contain no faces and is removed.

The consequence is that the original geometry of the slave patch will not necessarily be completely preserved during merging. Therefore in a case, say, where a cylindrical block is being connected to a larger block, it would be wise to the assign the master patch to the cylinder, so that its cylindrical shape is correctly preserved. There are some additional recommendations to ensure successful merge procedures:

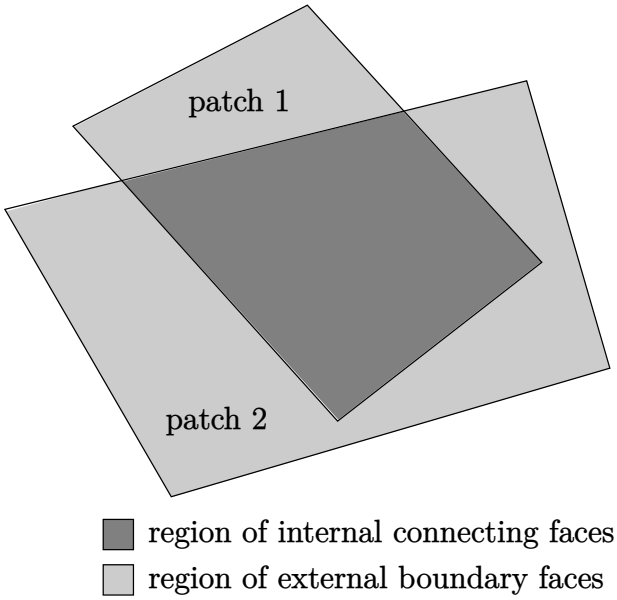


Figure 5.5: Merging overlapping patches

- in 2 dimensional geometries, the size of the cells in the third dimension, *i.e.* out of the 2D plane, should be similar to the width/height of cells in the 2D plane;
- it is inadvisable to merge a patch twice, *i.e.* include it twice in `mergePatchPairs`;
- where a patch to be merged shares a common edge with another patch to be merged, both should be declared as a master patch.

5.4.8 Projection of vertices, edges and faces

blockMesh can be configured to create body fitted meshes using projection of vertices, edges and/or faces onto specified geometry. The functionality can be used to mesh spherical and cylindrical geometries such as pipes and vessels conveniently. The user can specify within the *blockMeshDict* file within an optional `geometry` dictionary with the same format as used in the *snappyHexMeshDict* file. For example to specify a cylinder using the built in geometric type the user could configure with the following:

```
geometry
{
    cylinder
    {
        type searchableCylinder;
        point1 (0 -4 0);
        point2 (0 4 0);
        radius 0.7;
    }
};
```

The user can then project vertices, edges and/or faces onto the cylinder surface with the `project` keyword using example syntax shown below:

```
vertices
(
    project (-1 -0.1 -1) (cylinder)
    project ( 1 -0.1 -1) (cylinder)
    ...
);

edges
```

```
(
    project 0 1 (cylinder)
    ...
);

faces
(
    project (0 4 7 3) cylinder
    ...
);
```

The use of this functionality is demonstrated in tutorials which can be located by searching for the `project` keyword in all the *blockMeshDict* files by:

```
find $FOAM_TUTORIALS -name blockMeshDict | xargs grep -l project
```

5.4.9 Naming vertices, edges, faces and blocks

Vertices, edges, faces and blocks can be named in the configuration of a *blockMeshDict* file, which can make it easier to manage more complex examples. It is done simply using the `name` keyword. The following syntax shows naming using the example for projection in the previous subsection:

```
vertices
(
    name v0 project (-1 -0.1 -1) (cylinder)
    name v1 project ( 1 -0.1 -1) (cylinder)
    ...
);

edges
(
    project v0 v1 (cylinder)
    ...
);
```

When a name is provided for a given entity, it can be used to replace the index. In the example about, rather than specify the edge using vertex indices `0` and `1`, the names `v0` and `v1` are used.

5.4.10 Blocks with fewer than 8 vertices

It is possible to collapse one or more pair(s) of vertices onto each other in order to create a block with fewer than 8 vertices. The most common example of collapsing vertices is when creating a 6-sided wedge shaped block for 2-dimensional axi-symmetric cases that use the **wedge** patch type described in section 5.3.2. The process is best illustrated by using a simplified version of our example block shown in Figure 5.6. Let us say we wished to create a wedge shaped block by collapsing vertex 7 onto 4 and 6 onto 5. This is simply done by exchanging the vertex number 7 by 4 and 6 by 5 respectively so that the block numbering would become:

```
hex (0 1 2 3 4 5 5 4)
```

The same applies to the patches with the main consideration that the block face containing the collapsed vertices, previously `(4 5 6 7)` now becomes `(4 5 5 4)`. This is a block face of zero area which creates a patch with no faces in the **polyMesh**, as the user can see in a *boundary* file for such a case. The patch should be specified as **empty** in the *blockMeshDict* and the boundary condition for any fields should consequently be **empty** also.

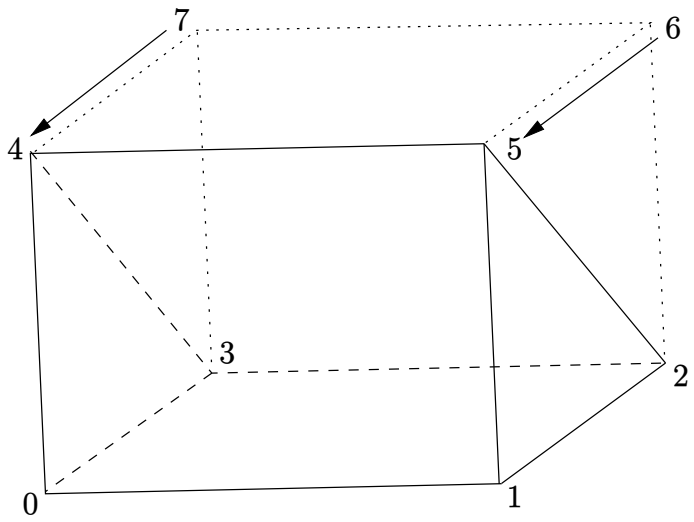


Figure 5.6: Creating a wedge shaped block with 6 vertices

5.4.11 Running blockMesh

As described in section 3.3, blockMesh can be run from within the case directory by:

```
blockMesh
```

Like many utilities, it can also be run using a configuration file named differently from *blockMeshDict*. Several examples in the *tutorials* directory for example use the pitzDaily geometry. They use a common blockMesh configuration file named *pitzDaily* in the *$FOAM_TUTORIALS/resources/blockMesh*. The meshes for these cases are generated using the `-dict` option by

```
blockMesh -dict $FOAM_TUTORIALS/resources/blockMesh/pitzDaily
```

5.5 Mesh generation with the snappyHexMesh utility

This section describes the mesh generation utility, snappyHexMesh, supplied with OpenFOAM. The snappyHexMesh utility generates 3-dimensional meshes containing hexahedra (hex) and split-hexahedra (split-hex) automatically from triangulated surface geometries, or tri-surfaces, in Stereolithography (STL) or Wavefront Object (OBJ) format. The mesh approximately conforms to the surface by iteratively refining a starting mesh and morphing the resulting split-hex mesh to the surface. An optional phase will shrink back the resulting mesh and insert cell layers. The specification of mesh refinement level is very flexible and the surface handling is robust with a pre-specified final mesh quality. It runs in parallel with a load balancing step every iteration.

5.5.1 The mesh generation process of snappyHexMesh

The process of generating a mesh using snappyHexMesh will be described using the schematic in Figure 5.7. The objective is to mesh a rectangular shaped region (shaded grey in the figure) surrounding an object described by a tri-surface, *e.g.* typical for an external aerodynamics simulation. Note that the schematic is 2-dimensional to make it easier to understand, even though the snappyHexMesh is a 3D meshing tool.

In order to run snappyHexMesh, the user requires the following:

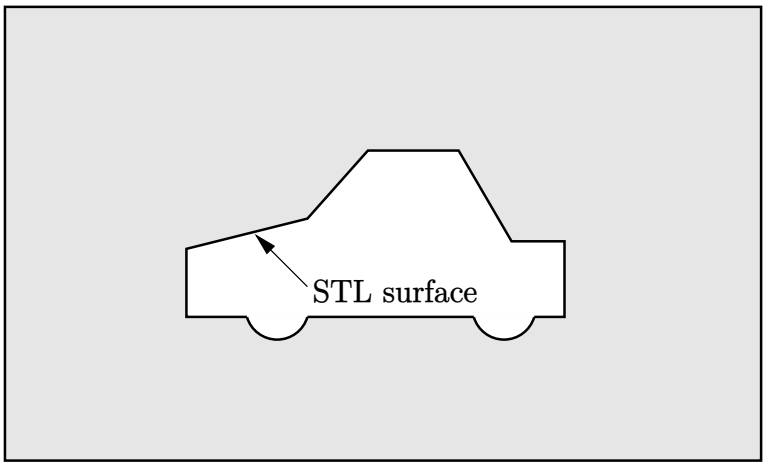


Figure 5.7: Schematic 2D meshing problem for snappyHexMesh

- one or more tri-surface files located in a *constant/geometry* sub-directory of the case directory;
- a background hex mesh which defines the extent of the computational domain and a base level mesh density; typically generated using blockMesh, discussed in section 5.5.2.
- a *snappyHexMeshDict* dictionary, with appropriate entries, located in the *system* sub-directory of the case.

The *snappyHexMeshDict* dictionary includes: switches at the top level that control the various stages of the meshing process; and, individual sub-directories for each process. The entries are listed below.

- `castellatedMesh`: to switch on creation of the castellated mesh.
- `snap`: to switch on surface snapping stage.
- `addLayers`: to switch on surface layer insertion.
- `mergeTolerance`: merge tolerance as fraction of bounding box of initial mesh.
- `geometry`: sub-dictionary of all surface geometry used.
- `castellatedMeshControls`: sub-dictionary of controls for castellated mesh.
- `snapControls`: sub-dictionary of controls for surface snapping.
- `addLayersControls`: sub-dictionary of controls for layer addition.
- `meshQualityControls`: sub-dictionary of controls for mesh quality.

All the geometry used by snappyHexMesh is specified in a *geometry* sub-dictionary in the *snappyHexMeshDict* dictionary. The geometry can be specified through a tri-surface or bounding geometry entities in OpenFOAM. An example is given below:

```
geometry
{
    duct        // User defined region name
    {
        type    triSurfaceMesh;
```

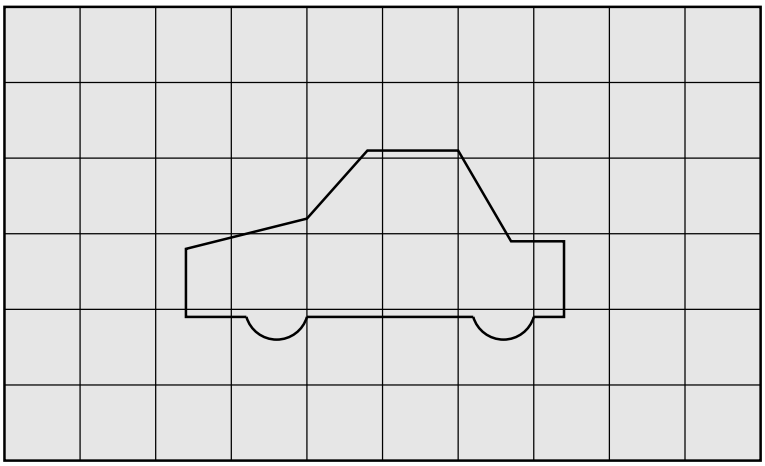


Figure 5.8: Initial mesh generation in snappyHexMesh meshing process

```
        file    "duct.obj";       // surface geometry OBJ file
        regions
        {
            leftOpening           // Named region in the OBJ file
            {
                name inlet;       // User-defined patch name
            }                     // otherwise given sphere1_secondSolid
        }
    }

    box         // User defined region name
    {
        type    searchableBox;    // region defined by bounding box
        min     (1.5 1 -0.5);
        max     (3.5 2 0.5);
    }

    sphere      // User defined region name
    {
        type    searchableSphere; // region defined by bounding sphere
        centre  (1.5 1.5 1.5);
        radius  1.03;
    }
};
```

5.5.2 Creating the background hex mesh

Before snappyHexMesh is executed the user must create a background mesh of hexahedral cells that fills the entire region within by the external boundary as shown in Figure 5.8. This can be done simply using blockMesh. The following criteria must be observed when creating the background mesh:

- the mesh must consist purely of hexes;
- the cell aspect ratio should be approximately 1, at least near surfaces at which the subsequent snapping procedure is applied, otherwise the convergence of the snapping procedure is slow, possibly to the point of failure;
- there must be at least one intersection of a cell edge with the tri-surface, *i.e.* a mesh of one cell will not work.

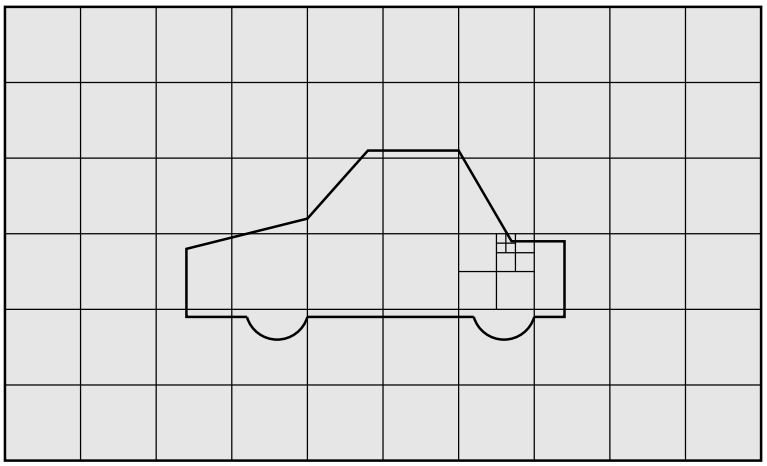


Figure 5.9: Cell splitting by feature edge in snappyHexMesh meshing process

5.5.3 Cell splitting at feature edges and surfaces

Cell splitting is performed according to the specification supplied by the user in the *castellatedMeshControls* sub-dictionary in the *snappyHexMeshDict*. The entries for *castellatedMeshControls* are presented below.

- `insidePoint`: location vector inside the region to be meshed; vector must not coincide with a cell face either before or during refinement.
- `maxLocalCells`: max number of cells per processor during refinement.
- `maxGlobalCells`: overall cell limit during refinement (*i.e.* before removal).
- `minRefinementCells`: if `minRefinementCells` $\geq$ number of cells to be refined, surface refinement stops.
- `nCellsBetweenLevels`: number of buffer layers of cells between successive levels of refinement (typically set to 3).
- `resolveFeatureAngle`: applies maximum level of refinement to cells that can see intersections whose angle exceeds `resolveFeatureAngle` (typically set to 30).
- `features`: list of features for refinement.
- `refinementSurfaces`: dictionary of surfaces for refinement.
- `refinementRegions`: dictionary of regions for refinement.

The splitting process begins with cells being selected according to specified edge features first within the domain as illustrated in Figure 5.9. The `features` list in the *castellatedMeshControls* sub-dictionary permits dictionary entries containing a name of an edgeMesh file and the `level` of refinement, *e.g.*:

```
features
(
    {
        file "features.eMesh"; // file containing edge mesh
        level 2;               // level of refinement
    }
);
```

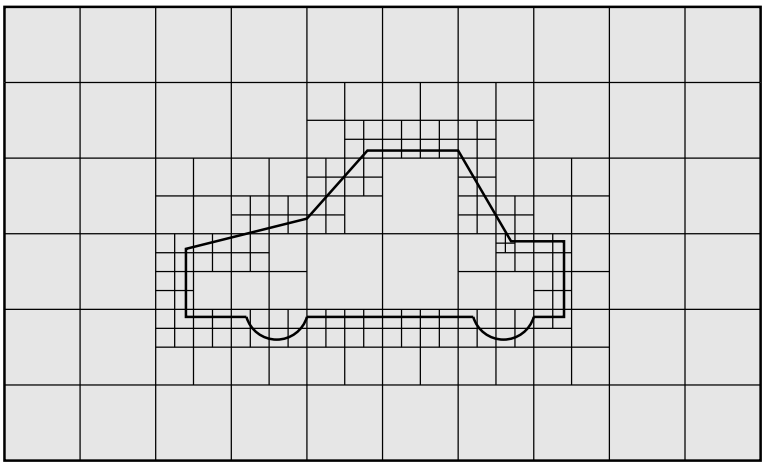


Figure 5.10: Cell splitting by surface in snappyHexMesh meshing process

The edgeMesh containing the features can be extracted from the tri-surface file using the surfaceFeatures utility which specifies the tri-surface and controls such as included angle through a *surfaceFeaturesDict* configuration file, examples of which can be found in several tutorials and at *$FOAM_ETC/caseDicts/surface/surfaceFeaturesDict*. The utility is simply run by executing the following in a terminal

```
surfaceFeatures
```

Following feature refinement, cells are selected for splitting in the locality of specified surfaces as illustrated in Figure 5.10. The `refinementSurfaces` dictionary in *castellatedMeshControls* requires dictionary entries for each STL surface and a default `level` specification of the minimum and maximum refinement in the form `(<min> <max>)`. The minimum level is applied generally across the surface; the maximum level is applied to cells that can see intersections that form an angle in excess of that specified by `resolveFeatureAngle`.

The refinement can optionally be overridden on one or more specific region of an STL surface. The region entries are collected in a `regions` sub-dictionary. The keyword for each region entry is the name of the region itself and the refinement level is contained within a further sub-dictionary. An example is given below:

```
refinementSurfaces
{
    duct
    {
        level (2 2); // default (min max) refinement for whole surface
        regions
        {
            leftOpening
            {
                level (3 3); // optional refinement for secondSolid region
            }
        }
    }
}
```

5.5.4 Cell removal

Once the feature and surface splitting is complete a process of cell removal begins. Cell removal requires one or more regions enclosed entirely by a bounding surface within the domain. The region in which cells are retained are simply identified by a location vector

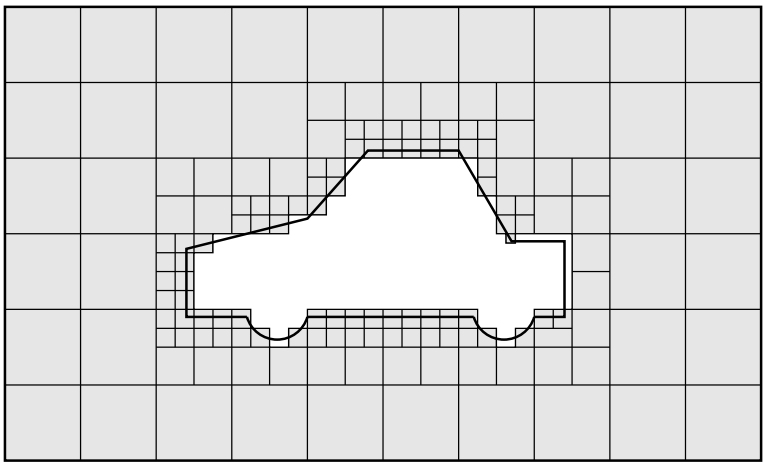


Figure 5.11: Cell removal in snappyHexMesh meshing process

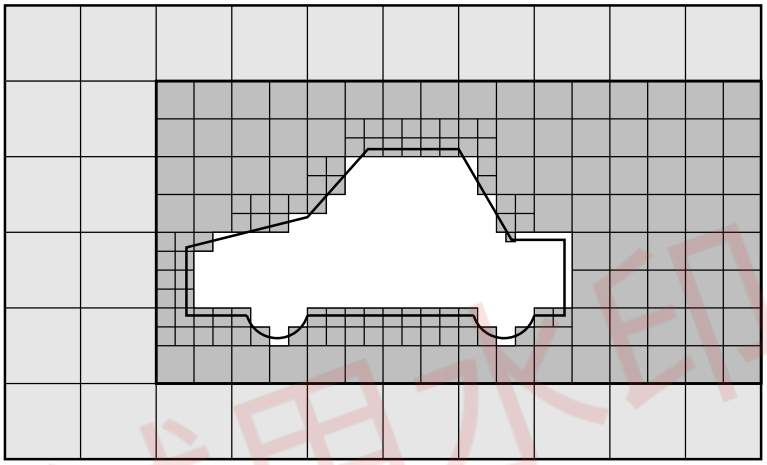


Figure 5.12: Cell splitting by region in snappyHexMesh meshing process

within that region, specified by the `insidePoint` keyword in *castellatedMeshControls*. Cells are retained if, approximately speaking, 50% or more of their volume lies within the region. The remaining cells are removed accordingly as illustrated in Figure 5.11.

5.5.5 Cell splitting in specified regions

Those cells that lie within one or more specified volume regions can be further split as illustrated in Figure 5.12 by a rectangular region shown by dark shading. The `refinementRegions` sub-dictionary in *castellatedMeshControls* contains entries for refinement of the volume regions specified in the *geometry* sub-dictionary. A refinement `mode` is applied to each region which can be:

- `inside` refines inside the volume region;
- `outside` refines outside the volume region
- `distance` refines according to distance to the surface; and can accommodate different levels at multiple distances with the `levels` keyword.

For the `refinementRegions`, the refinement level is specified by the `level` keyword for `inside` and `outside` refinement. For `distance` refinement, the keyword is `levels`

(**plural!**) which contains list of entries with the format (<distance> <level>). Examples are shown below:

```
refinementRegions
{
    box
    {
        mode  inside;
        level 4;                  // refinement level 4
    }

    sphere
    {                             // refinement level 5 within 1.0 m
        mode  distance;           // refinement level 3 within 2.0 m
        levels ((1.0 5) (2.0 3)); // levels must be ordered nearest first
    }
}
```

5.5.6 Cell splitting based on local span

Refinement of cells can also be specified to guarantee a specified number of cells across the span between opposing surfaces. This refinement option can ensure that there are sufficient cells to resolve the flow in a region of the domain, e.g. across a narrow pipe. The method requires closeness data to be provided on the surface geometry. The closeness can be calculated by the surfaceFeatures utility with the following entry in the *surfaceFeaturesDict* file:

```
surfaces
(
    "pipeWall.obj"
);

closeness
{
    pointCloseness          yes;
}
```

This writes closeness data to a file named *pipeWall.closeness.internalPointCloseness* into the *constant/geometry* directory. The closeness is then be used for span-based refinement by the addition of an entry in the `refinementRegions` sub-dictionary in *snappyHexMeshDict*, *e.g.*:

```
refinementRegions
{
    pipeWall
    {
        mode    insideSpan;
        level   (1000 2);
        cellsAcrossSpan 40;
    }
}
```

The example shows a refinement region inside the `pipeWall` surface in which a maximum 2 levels of refinement is guaranteed within a specified distance of 1000 from the wall. The span-based refinement, specified by the `insideSpan` mode, enables the user to guarantee at least 40 `cellsAcrossSpan`, *i.e.* across the pipe diameter.

5.5.7 Snapping to surfaces

The next stage of the meshing process involves moving cell vertex points onto surface geometry to remove the jagged castellated surface from the mesh. The process is:

1. displace the vertices in the castellated boundary onto the STL surface;

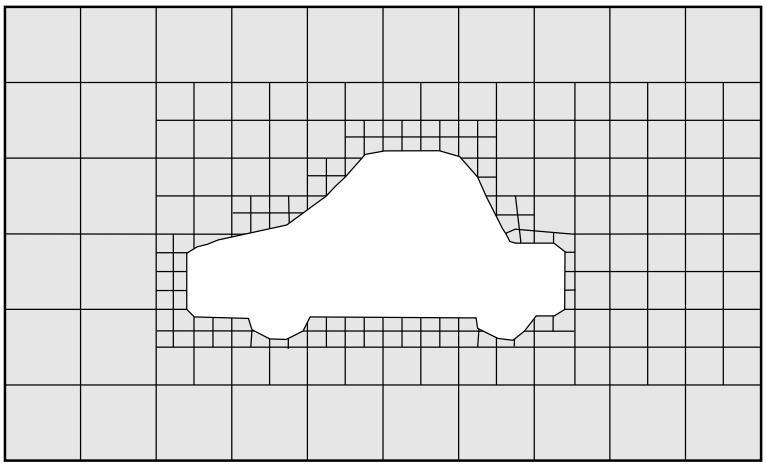


Figure 5.13: Surface snapping in snappyHexMesh meshing process

2. solve for relaxation of the internal mesh with the latest displaced boundary vertices;
3. find the vertices that cause mesh quality parameters to be violated;
4. reduce the displacement of those vertices from their initial value (at 1) and repeat from 2 until mesh quality is satisfied.

The method uses the settings in the *snapControls* sub-dictionary in *snappyHexMeshDict*, listed below.

- `nSmoothPatch`: number of patch smoothing iterations before finding correspondence to surface (typically 3).
- `tolerance`: ratio of distance for points to be attracted by surface feature point or edge, to local maximum edge length (typically 2.0).
- `nSolveIter`: number of mesh displacement relaxation iterations (typically 30-100).
- `nRelaxIter`: maximum number of snapping relaxation iterations (typically 5).

An example is illustrated in the schematic in Figure 5.13 (albeit with mesh motion that looks slightly unrealistic).

5.5.8 Mesh layers

The mesh output from the snapping stage may be suitable for the purpose, although it can produce some irregular cells along boundary surfaces. There is an optional stage of the meshing process which introduces additional layers of hexahedral cells aligned to the boundary surface as illustrated by the dark shaded cells in Figure 5.14.

The process of mesh layer addition involves shrinking the existing mesh from the boundary and inserting layers of cells, broadly as follows:

1. the mesh is projected back from the surface by a specified thickness in the direction normal to the surface;
2. solve for relaxation of the internal mesh with the latest projected boundary vertices;
3. check if validation criteria are satisfied otherwise reduce the projected thickness and return to 2; if validation cannot be satisfied for any thickness, do not insert layers;

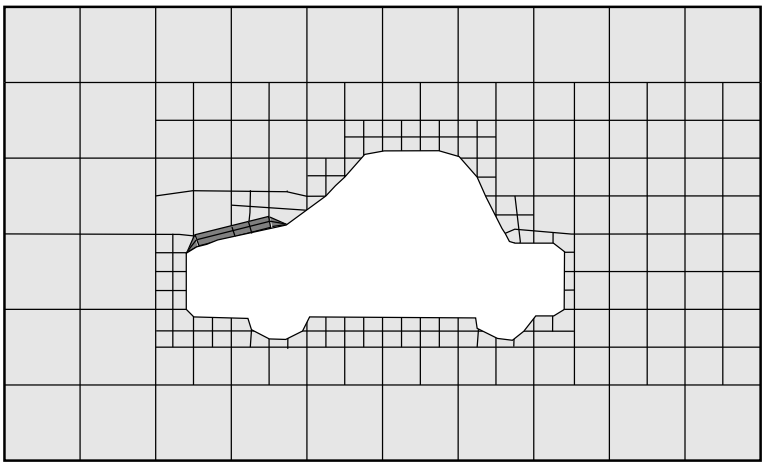


Figure 5.14: Layer addition in snappyHexMesh meshing process

4. if the validation criteria can be satisfied, insert mesh layers;

5. the mesh is checked again; if the checks fail, layers are removed and we return to 2.

The layer addition procedure uses the settings in the *addLayersControls* sub-dictionary in *snappyHexMeshDict*; entries are listed below. The user has the option of 4 different layer thickness parameters — `expansionRatio`, `finalLayerThickness`, `firstLayerThickness`, `thickness` — *from which they must specify 2 only*; more than 2, and the problem is over-specified.

- `layers`: dictionary specifying layers to be inserted.
- `relativeSizes`: switch that sets whether the specified layer thicknesses are relative to undistorted cell size outside layer or absolute.
- `expansionRatio`: expansion factor for layer mesh, increase in size from one layer to the next.
- `finalLayerThickness`: thickness of layer furthest from the wall, usually in combination with relative sizes according to the `relativeSizes` entry.
- `firstLayerThickness`: thickness of layer nearest the wall, usually in combination with absolute sizes according to the `relativeSizes` entry.
- `thickness`: total thickness of all layers of cells, usually in combination with absolute sizes according to the
- `relativeSizes` entry.
- `minThickness`: minimum thickness of cell layer, either relative or absolute (as above).
- `nGrow`: number of layers of connected faces that are not grown if points do not get extruded; helps convergence of layer addition close to features.
- `featureAngle`: angle above which surface is not extruded.
- `nRelaxIter`: maximum number of snapping relaxation iterations (typcially 5).

- `nSmoothSurfaceNormals`: number of smoothing iterations of surface normals (typically 1).
- `nSmoothNormals`: number of smoothing iterations of interior mesh movement direction (typically 3).
- `nSmoothThickness`: smooth layer thickness over surface patches (typically 10).
- `maxFaceThicknessRatio`: stop layer growth on highly warped cells (typically 0.5).
- `maxThicknessToMedialRatio`: reduce layer growth where ratio thickness to medial distance is large (typically 0.3)
- `minMedianAxisAngle`: angle used to pick up medial axis points (typically 90).
- `nBufferCellsNoExtrude`: create buffer region for new layer terminations (typically 0).
- `nLayerIter`: overall max number of layer addition iterations (typically 50).
- `nRelaxedIter`: max number of iterations after which the controls in the *relaxed* sub dictionary of `meshQuality` are used (typically 20).

The `layers` sub-dictionary contains entries for each *patch* on which the layers are to be applied and the number of surface layers required. The patch name is used because the layers addition relates to the existing mesh, not the surface geometry; hence applied to a patch, not a surface region. An example `layers` entry is as follows:

```
layers
{
    sphere1_firstSolid
    {
        nSurfaceLayers 1;
    }
    maxY
    {
        nSurfaceLayers 1;
    }
}
```

5.5.9 Mesh quality controls

The mesh quality is controlled by the entries in the *meshQualityControls* sub-dictionary in *snappyHexMeshDict*; entries are listed below.

- `maxNonOrtho`: maximum non-orthogonality allowed (degrees, typically 65).
- `maxBoundarySkewness`: max boundary face skewness allowed (typically 20).
- `maxInternalSkewness`: max internal face skewness allowed (typically 4).
- `maxConcave`: max concaveness allowed (typically 80).
- `minFlatness`: ratio of minimum projected area to actual area (typically 0.5)
- `minTetQuality`: minimum quality of tetrahedral cells from cell decomposition; generally deactivated by setting large negative number since v5.0 when new barycentric tracking was introduced, which could handle negative tets.

- `minVol`: minimum cell pyramid volume (typically 1e-13, large negative number disables).
- `minArea`: minimum face area (typically -1).
- `minTwist`: minimum face twist (typically 0.05).
- `minDeterminant`: minimum normalised cell determinant; 1 = hex; ≤ 0 = illegal cell (typically 0.001).
- `minFaceWeight`: 0→0.5 (typically 0.05).
- `minVolRatio`: 0→1.0 (typically 0.01).
- `minTriangleTwist`: > 0 for Fluent compatibility (typically -1).
- `nSmoothScale`: number of error distribution iterations (typically 4).
- `errorReduction`: amount to scale back displacement at error points (typically 0.75).
- `relaxed`: sub-dictionary that can include modified values for the above keyword entries to be used when `nRelaxedIter` is exceeded in the layer addition process.

5.6 Mesh conversion

The user can generate meshes using other packages and convert them into the format that OpenFOAM uses. There are numerous mesh conversion utilities listed in section 3.7.3. Some of the more popular mesh converters are listed below and their use is presented in this section.

fluentMeshToFoam reads a Fluent `.msh` mesh file, working for both 2-D and 3-D cases;

starToFoam reads STAR-CD/PROSTAR mesh files.

gambitToFoam reads a GAMBIT `.neu` neutral file;

ideasToFoam reads an I-DEAS mesh written in ANSYS `.ans` format;

cfx4ToFoam reads a CFX mesh written in `.geo` format;

5.6.1 fluentMeshToFoam

Fluent writes mesh data to a single file with a *.msh* extension. The file must be written in ASCII format, which is not the default option in Fluent. It is possible to convert single-stream Fluent meshes, including the 2 dimensional geometries. In OpenFOAM, 2 dimensional geometries are currently treated by defining a mesh in 3 dimensions, where the front and back plane are defined as the empty boundary patch type. When reading a 2 dimensional Fluent mesh, the converter automatically extrudes the mesh in the third direction and adds the empty patch, naming it `frontAndBackPlanes`.

The following features should also be observed.

- The OpenFOAM converter will attempt to capture the Fluent boundary condition definition as much as possible; however, since there is no clear, direct correspondence between the OpenFOAM and Fluent boundary conditions, the user should check the boundary conditions before running a case.

- Creation of axi-symmetric meshes from a 2 dimensional mesh is currently not supported but can be implemented on request.
- Multiple material meshes are not permitted. If multiple fluid materials exist, they will be converted into a single OpenFOAM mesh; if a solid region is detected, the converter will attempt to filter it out.
- Fluent allows the user to define a patch which is internal to the mesh, *i.e.* consists of the faces with cells on both sides. Such patches are not allowed in OpenFOAM and the converter will attempt to filter them out.
- There is currently no support for embedded interfaces and refinement trees.

The procedure of converting a Fluent.*msh* file is first to create a new OpenFOAM case by creating the necessary directories/files: the case directory containing a *controlDict* file in a *system* subdirectory. Then at a command prompt the user should execute:

```
fluentMeshToFoam <meshFile>
```

where <`meshFile`> is the name of the *.msh* file, including the full or relative path.

5.6.2 starToFoam

This section describes how to convert a mesh generated on the STAR-CD code into a form that can be read by OpenFOAM mesh classes. The mesh can be generated by any of the packages supplied with STAR-CD, *i.e.*PROSTAR, SAMM, ProAM and their derivatives. The converter accepts any single-stream mesh including integral and arbitrary couple matching and all cell types are supported. The features that the converter does not support are:

- multi-stream mesh specification;
- baffles, *i.e.* zero-thickness walls inserted into the domain;
- partial boundaries, where an uncovered part of a couple match is considered to be a boundary face;
- sliding interfaces.

For multi-stream meshes, mesh conversion can be achieved by writing each individual stream as a separate mesh and reassemble them in OpenFOAM.

OpenFOAM adopts a policy of only accepting input meshes that conform to the fairly stringent validity criteria specified in section 5.1. It will simply not run using invalid meshes and cannot convert a mesh that is itself invalid. The following sections describe steps that must be taken when generating a mesh using a mesh generating package supplied with STAR-CD to ensure that it can be converted to OpenFOAM format. To avoid repetition in the remainder of the section, the mesh generation tools supplied with STAR-CD will be referred to by the collective name STAR-CD.

We strongly recommend that the user run the STAR-CD mesh checking tools before attempting a starToFoam conversion and, after conversion, the checkMesh utility should be run on the newly converted mesh. Alternatively, starToFoam may itself issue warnings containing PROSTAR commands that will enable the user to take a closer look at cells with

problems. Problematic cells and matches should be checked and fixed before attempting to use the mesh with OpenFOAM. Remember that an invalid mesh will not run with OpenFOAM, but it may run in another environment that does not impose the validity criteria.

Some problems of tolerance matching can be overcome by the use of a matching tolerance in the converter. However, there is a limit to its effectiveness and an apparent need to increase the matching tolerance from its default level indicates that the original mesh suffers from inaccuracies.

When mesh generation in is completed, remove any extraneous vertices and compress the cells boundary and vertex numbering, assuming that fluid cells have been created and all other cells are discarded. This is done with the following PROSTAR commands:

```
CSET NEWS FLUID
CSET INVE
```

The `CSET` should be empty. If this is not the case, examine the cells in `CSET` and adjust the model. If the cells are genuinely not desired, they can be removed using the PROSTAR command:

```
CDEL CSET
```

Similarly, vertices will need to be discarded as well:

```
CSET NEWS FLUID
VSET NEWS CSET
VSET INVE
```

Before discarding these unwanted vertices, the unwanted boundary faces have to be collected before purging:

```
CSET NEWS FLUID
VSET NEWS CSET
BSET NEWS VSET ALL
BSET INVE
```

If the `BSET` is not empty, the unwanted boundary faces can be deleted using:

```
BDEL BSET
```

At this time, the model should contain only the fluid cells and the supporting vertices, as well as the defined boundary faces. All boundary faces should be fully supported by the vertices of the cells, if this is not the case, carry on cleaning the geometry until everything is clean.

By default, STAR-CD assigns wall boundaries to any boundary faces not explicitly associated with a boundary region. The remaining boundary faces are collected into a `default` boundary region, with the assigned boundary type `0`. OpenFOAM deliberately does not have a concept of a `default` boundary condition for undefined boundary faces since it invites human error, *e.g.* there is no means of checking that we meant to give all the unassociated faces the default condition.

Therefore **all** boundaries for each OpenFOAM mesh must be specified for a mesh to be successfully converted. The `default` boundary needs to be transformed into a real one using the procedure described below:

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1. Plot the geometry with `Wire Surface` option.

2. Define an extra boundary region with the same parameters as the `default` region 0 and add all visible faces into the new region, say 10, by selecting a zone option in the boundary tool and drawing a polygon around the entire screen draw of the model. This can be done by issuing the following commands in PROSTAR:

   ```
   RDEF 10 WALL
   BZON 10 ALL
   ```

3. We shall remove all previously defined boundary types from the set. Go through the boundary regions:

   ```
   BSET NEWS REGI 1
   BSET NEWS REGI 2
   ... 3, 4, ...
   ```

   Collect the vertices associated with the boundary set and then the boundary faces associated with the vertices (there will be twice as many of them as in the original set).

   ```
   BSET NEWS REGI 1
   VSET NEWS BSET
   BSET NEWS VSET ALL
   BSET DELE REGI 1
   REPL
   ```

   This should give the faces of boundary Region 10 which have been defined on top of boundary Region 1. Delete them with `BDEL BSET`. Repeat these for all regions.

Renumber and check the model using the commands:

```
CSET NEW FLUID
CCOM CSET

VSET NEWS CSET
VSET INVE (Should be empty!)
VSET INVE
VCOM VSET

BSET NEWS VSET ALL
BSET INVE (Should be empty also!)
BSET INVE
BCOM BSET

CHECK ALL
GEOM
```

Internal PROSTAR checking is performed by the last two commands, which may reveal some other unforeseeable error(s). Also, take note of the scaling factor because PROSTAR only applies the factor for STAR-CD and not the geometry. If the factor is not 1, use the scalePoints utility in OpenFOAM.

Once the mesh is completed, place all the integral matches of the model into the couple type 1. All other types will be used to indicate arbitrary matches.

```
CPSET NEWS TYPE INTEGRAL
CPMOD CPSET 1
```

The components of the computational grid must then be written to their own files. This is done using PROSTAR for boundaries by issuing the command

```
BWRITE
```

by default, this writes to a *.23* file (versions prior to 3.0) or a *.bnd* file (versions 3.0 and higher). For cells, the command

```
CWRITE
```

outputs the cells to a *.14* or *.cel* file and for vertices, the command

```
VWRITE
```

outputs to file a *.15* or *.vrt* file. The current default setting writes the files in ASCII format. If couples are present, an additional couple file with the extension *.cpl* needs to be written out by typing:

```
CPWRITE
```

After outputting to the three files, exit PROSTAR or close the files. Look through the panels and take note of all STAR-CD sub-models, material and fluid properties used – the material properties and mathematical model will need to be set up by creating and editing OpenFOAM dictionary files.

The procedure of converting the PROSTAR files is first to create a new OpenFOAM case by creating the necessary directories. The PROSTAR files must be stored within the same directory and the user must change the file extensions: from *.23*, *.14* and *.15* (below STAR-CD version 3.0), or *.pcs*, *.cls* and *.vtx* (STAR-CD version 3.0 and above); to *.bnd*, *.cel* and *.vrt* respectively.

The *.vrt* file is written in columns of data of specified width, rather than free format. A typical line of data might be as follows, giving a vertex number followed by the coordinates:

```
19422       -0.105988957     -0.413711881E-02 0.000000000E+00
```

If the ordinates are written in scientific notation and are negative, there may be no space between values, *e.g.*:

```
19423       -0.953953117E-01-0.338810333E-02 0.000000000E+00
```

The starToFoam converter reads the data using spaces to delimit the ordinate values and will therefore object when reading the previous example. Therefore, OpenFOAM includes a simple script, foamCorrectVrt to insert a space between values where necessary, *i.e.* it would convert the previous example to:

```
19423       -0.953953117E-01 -0.338810333E-02 0.000000000E+00
```

The foamCorrectVrt script should therefore be executed if necessary before running the starToFoam converter, by typing:

```
foamCorrectVrt <file>.vrt
```

The translator utility starToFoam can now be run to create the boundaries, cells and points files necessary for a OpenFOAM run:

```
starToFoam <meshFilePrefix>
```

where <meshFilePrefix> is the name of the prefix of the mesh files, including the full or relative path. After the utility has finished running, OpenFOAM boundary types should be specified by editing the *boundary* file by hand.

5.6.3 gambitToFoam

GAMBIT writes mesh data to a single file with a *.neu* extension. The procedure of converting a GAMBIT*.neu* file is first to create a new OpenFOAM case, then at a command prompt, the user should execute:

```
gambitToFoam <meshFile>
```

where <**meshFile**> is the name of the *.neu* file, including the full or relative path.

The GAMBIT file format does not provide information about type of the boundary patch, *e.g.* wall, symmetry plane, cyclic. Therefore all the patches have been created as type patch. Please reset after mesh conversion as necessary.

5.6.4 ideasToFoam

OpenFOAM can convert a mesh generated by I-DEAS but written out in ANSYS format as a *.ans* file. The procedure of converting the *.ans* file is first to create a new OpenFOAM case, then at a command prompt, the user should execute:

```
ideasToFoam <meshFile>
```

where <**meshFile**> is the name of the *.ans* file, including the full or relative path.

5.6.5 cfx4ToFoam

CFX writes mesh data to a single file with a *.geo* extension. The mesh format in CFX is block-structured, *i.e.* the mesh is specified as a set of blocks with glueing information and the vertex locations. OpenFOAM will convert the mesh and capture the CFX boundary condition as best as possible. The 3 dimensional 'patch' definition in CFX, containing information about the porous, solid regions *etc.* is ignored with all regions being converted into a single OpenFOAM mesh. CFX supports the concept of a 'default' patch, where each external face without a defined boundary condition is treated as a wall. These faces are collected by the converter and put into a **defaultFaces** patch in the OpenFOAM mesh and given the type wall; of course, the patch type can be subsequently changed.

Like, OpenFOAM 2 dimensional geometries in CFX are created as 3 dimensional meshes of 1 cell thickness. If a user wishes to run a 2 dimensional case on a mesh created

by CFX, the boundary condition on the front and back planes should be set to empty; the user should ensure that the boundary conditions on all other faces in the plane of the calculation are set correctly. Currently there is no facility for creating an axi-symmetric geometry from a 2 dimensional CFX mesh.

The procedure of converting a CFX.*geo* file is first to create a new OpenFOAM case, then at a command prompt, the user should execute:

```
cfx4ToFoam <meshFile>
```

where <meshFile> is the name of the *.geo* file, including the full or relative path.

5.7 Mapping fields between different geometries

The mapFields utility maps one or more fields relating to a given geometry onto the corresponding fields for another geometry. It is completely generalised in so much as there does not need to be any similarity between the geometries to which the fields relate. However, for cases where the geometries are consistent, mapFields can be executed with a special option that simplifies the mapping process.

For our discussion of mapFields we need to define a few terms. First, we say that the data is mapped from the *source* to the *target*. The fields are deemed *consistent* if the geometry *and* boundary types, or conditions, of both source and target fields are identical. The field data that mapFields maps are those fields within the time directory specified by `startFrom`/`startTime` in the *controlDict* of the target case. The data is read from the equivalent time directory of the source case and mapped onto the equivalent time directory of the target case.

5.7.1 Mapping consistent fields

A mapping of consistent fields is simply performed by executing mapFields on the (target) case using the `-consistent` command line option as follows:

```
mapFields <source dir> -consistent
```

5.7.2 Mapping inconsistent fields

When the fields are not consistent, as shown in Figure 5.15, mapFields requires a *mapFieldsDict* dictionary in the *system* directory of the target case. The following rules apply to the mapping:

- the field data is mapped from source to target wherever possible, *i.e.* in our example all the field data within the target geometry is mapped from the source, except those in the shaded region which remain unaltered;
- the patch field data is left unaltered unless specified otherwise in the *mapFieldsDict* dictionary.

The *mapFieldsDict* dictionary contain two lists that specify mapping of patch data. The first list is `patchMap` that specifies mapping of data between pairs of target and source patches that are geometrically coincident, as shown in Figure 5.15. The list contains each pair of names of target patch (first) and source patch (second). The second list is

`cuttingPatches` that contains names of target patches whose values are to be mapped from the source internal field through which the target patch cuts. In the situation where the target patch only cuts through part of the source internal field, *e.g.* bottom left target patch in our example, those values within the internal field are mapped and those outside remain unchanged. An example *mapFieldsDict* dictionary is shown below:

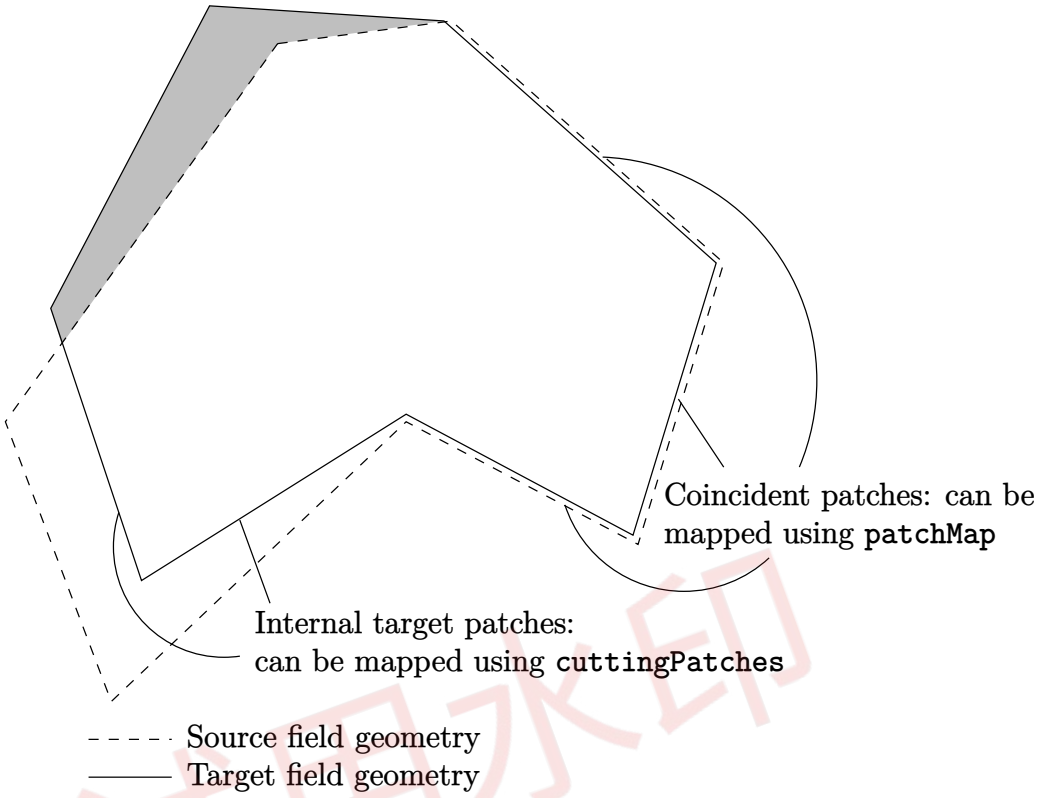


Figure 5.15: Mapping inconsistent fields

```
16
17  patchMap        (lid movingWall);
18
19  cuttingPatches  ();
20
21
22  // ************************************************************************* //
```

`mapFields <source dir>`

5.7.3 Mapping parallel cases

If either or both of the source and target cases are decomposed for running in parallel, additional options must be supplied when executing mapFields:

`-parallelSource` if the source case is decomposed for parallel running;

`-parallelTarget` if the target case is decomposed for parallel running.

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Chapter 6

Boundary conditions

Boundary conditions are specified in field files, *e.g.* p, U, in time directories. The structure of these files is introduced in sections 2.1.4 and 4.2.8. They include three entries: `dimensions` for the dimensional units; `internalField` for the initial internal field values; and, `boundaryField` where the boundary conditions are specified. The `boundaryField` requires an entry for each patch in the mesh. The patches are specified in the *boundary* file; below is a sample file from a 2D incompressibleFluid example in OpenFOAM.

```
5
(
    outlet
    {
        type            patch;
        nFaces          320;
        startFace       198740;
    }
    up
    {
        type            symmetry;
        inGroups        List<word> 1(symmetry);
        nFaces          760;
        startFace       199060;
    }
    hole
    {
        type            wall;
        inGroups        List<word> 1(wall);
        nFaces          1120;
        startFace       199820;
    }
    frontAndBack
    {
        type            empty;
        inGroups        List<word> 1(empty);
        nFaces          200000;
        startFace       200940;
    }
    inlet
    {
        type            patch;
        nFaces          320;
        startFace       400940;
    }
)
```

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The corresponding pressure field file, *p*, is shown below.

```
16  dimensions      [0 2 -2 0 0 0 0];
17
18  internalField   uniform 0;
19
20  boundaryField
21  {
22      inlet
23      {
24          type            zeroGradient;
25      }
26      outlet
27      {
28          type            fixedValue;
29          value           uniform 0;
30      }
31      up
32      {
33          type            symmetry;
34      }
35      hole
36      {
37          type            zeroGradient;
38      }
39      frontAndBack
40      {
41          type            empty;
42      }
43  }
44
45  // ************************************************************************* //
```

The `boundaryField` is a sub-dictionary containing an entry for every patch in the mesh. Each entry begins with the patch name and configures the boundary condition through entries in a sub-dictionary. A `type` entry is required for every patch which specifies the type of boundary condition. The examples above include zeroGradient and fixedValue conditions corresponding to generic patches defined in the *boundary* file. They also include symmetry and empty types corresponding to equivalent constraint patches, *e.g.* the up patch is defined as symmetry in the mesh and uses a symmetry condition in the field file.

For details about the main boundary conditions used in OpenFOAM, refer to Chapter 4 of ***Notes on Computational Fluid Dynamics: General Principles***.

6.1 Patch selection in field files

There are three different ways an entry can be specified for a patch in the `boundaryField` of a field file: 1) by patch name; 2) by group name; 3) matching a patch name with a regular expression. They are listed here in order of precedence which is obeyed if multiple entries are valid of a particular patch. The different specifications can be illustrated by imagining a mesh with the following patches.

- `inlet`: a generic patch.
- `lowerWall` and `upperWall`: two wall patches.
- `outletSmall`, `outletMedium` and `outletLarge`: three outlet patches of generic type, all in a patch group named `outlet`.

Then imagine the following `boundaryField` for a field, *e.g.* *p*, corresponding to the patches above.

```
boundaryField
{
    inlet
    {
       type              zeroGradient;
    }
    ".*Wall"
    {
        type              zeroGradient;
    }
    outletSmall
    {
        type              fixedValue;
        value             uniform 1;
    }
    outlet
    {
        type              fixedValue;
        value             uniform 0;
    }
}
```

In this example, the `inlet` field entry is read for the `inlet` patch, following rule 1 above (matching patch name). Similarly, the `outletSmall` entry will be read for the patch of the same name.

The `outletMedium` and `outletLarge` patches do not have matching entries in the field file, so they instead the `outlet` entry will be applied (rule 2), since it matches the group name to which the patches belong. Note that the `outletSmall` patch does not use the `outlet` entry because a matching patch entry takes precedence over a matching group entry.

Finally, the `lowerWall` and `upperWall` match the regular expression `".*Wall"`. Regular expressions are described in section 4.2.12; they must be included in double quotations `"..."`. The `".*"` component matches any expression (including nothing), so matches the wall patch names here. The regular expression could use word grouping to provide a more precise match to the patch names, *e.g.*

```
    "(lower|upper)Wall"
    {
        type              zeroGradient;
    }
```

Alternatively a patch entry could cover the wall patches taking advantage of the fact that every non-generic patch is automatically placed in a group of the same name as its type, as discussed in section 5.3.6. In this case, all wall patches are placed in a group named wall, so the following entry would be read for both patches.

```
    wall
    {
        type              zeroGradient;
    }
```

6.2 Geometric constraints

Section 5.3 describes the mesh boundary, which is split into patches and written in the mesh *boundary* file. Each patch includes a `type` entry which can be specified as a generic `patch`, a `wall` or a geometric constraint, *e.g.* `empty`, `symmetry`, `cyclic` *etc.*

For each geometric constraint type for a patch in the mesh, there is an equivalent boundary condition type that must be applied to the same patch in the `boundaryField` of a field file. The type names in the mesh and `boundaryField` are the same, *e.g.* the symmetry boundary condition must be applied to a symmetry patch.

To simplify the configuration of field files, OpenFOAM includes a file named *setConstraintTypes* in the *$FOAM_ETC/caseDicts* of the installation. The *setConstraintTypes* file contains the following entries.

```
cyclic
{
    type  cyclic;
}

cyclicSlip
{
    type  cyclicSlip;
}

nonConformalCyclic
{
    type  nonConformalCyclic;
    value $internalField;
}

nonConformalError
{
    type  nonConformalError;
}

empty
{
    type  empty;
}

processor
{
    type  processor;
    value $internalField;
}

processorCyclic
{
    type  processorCyclic;
    value $internalField;
}

nonConformalProcessorCyclic
{
    type  nonConformalProcessorCyclic;
    value $internalField;
}

symmetryPlane
{
    type  symmetryPlane;
}

symmetry
{
    type  symmetry;
}

wedge
{
    type  wedge;
}

internal
```

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```
69  {
70      type  internal;
71  }
72
73
74  // ************************************************************************* //
```

The file exploits the fact that a patch which is a geometric constraint is automatically included in a group of the constraint name, *e.g.* a `symmetry` patch is in a group named `symmetry`. The entries therefore set a boundary type for each constraint group (to the name of the group). All constraint conditions are covered by an entry for each condition.

The user can then include this file inside the `boundaryField` of their field files. Since the file is in the *$FOAM_ETC* directory it can be included using the special `#includeEtc` directive, *e.g.* in the `boundaryField` entry below.

```
boundaryField
{
    inlet
    {
        type            zeroGradient;
    }

    outlet
    {
        type            fixedValue;
        value           uniform 0;
    }

    wall
    {
        type            zeroGradient;
    }

    #includeEtc "caseDicts/setConstraintTypes"
}
```

With the *setContraintTypes* file included in the field files, the only patches that generally need to be configured are: the generic patches, corresponding to open boundaries; and, wall patches.

6.3 Basic boundary conditions

The main basic boundary condition types available in OpenFOAM are summarised below using a patch field named Ψ. This is not a complete list; for all types see *$FOAM_SRC/finiteVolume/fields/fvPatchFields/basic*.

- fixedValue: value of Ψ is specified by `value`.
- fixedGradient: normal gradient of Ψ ($\partial\Psi/\partial n$) is specified by `gradient`.
- zeroGradient: normal gradient of Ψ is zero.
- calculated: patch field Ψ calculated from other patch fields.

- mixed: mixed fixedValue/ fixedGradient condition depending on `valueFraction` ($0 \leq$ `valueFraction` ≤ 1) where

$$\texttt{valueFraction} = \begin{cases} 1 & \text{corresponds to } \Psi = \texttt{refValue}, \\ 0 & \text{corresponds to } \partial\Psi/\partial n = \texttt{refGradient}. \end{cases} \tag{6.1}$$

- directionMixed: mixed condition with tensorial `valueFraction`, to allow different conditions in normal and tangential directions of a vector patch field, *e.g.* fixedValue in the tangential direction, zeroGradient in the normal direction.

6.4 Derived boundary conditions

There are numerous more complex boundary conditions derived from the basic conditions. For example, many complex conditions are derived from fixedValue, where the value is calculated by a function of other patch fields, time, geometric information, *etc.* Some other conditions derived from mixed/directionMixed switch between fixedValue and fixedGradient (usually with a zero gradient).

The available boundary conditions can be listed with foamToC using the -scalarBCs and -vectorBCs options, corresponding to boundary conditions for scalar fields and vector fields, respectively. For example, for scalar fields, boundary conditions are listed by

```
foamToC -scalarBCs
```

These produce long lists which the user can scan through. If the user wants more information of a particular condition, they can run the foamInfo script which provides a description of the boundary condition and lists example cases where it is used. For example, for the totalPressure boundary condition, run the following.

```
foamInfo totalPressure
```

In the following sections we will highlight some particular important, commonly used boundary conditions.

6.4.1 The inlet/outlet condition

The inletOutlet condition is one derived from mixed, which switches between zeroGradient when the fluid flows out of the domain at a patch face, and fixedValue, when the fluid is flowing into the domain. For inflow, the inlet value is specified by an `inletValue` entry. A good example of its use can be seen in the damBreakLaminar tutorial, where it is applied to the phase fraction on the upper `atmosphere` boundary. Where there is outflow, the condition is well posed, where there is inflow, the phase fraction is fixed with a value of 0, corresponding to 100% air.

```
16  dimensions      [0 0 0 0 0 0 0];
17
18  internalField   uniform 0;
19
20  boundaryField
21  {
22      leftWall
23      {
24          type            zeroGradient;
25      }
```

```
26
27      rightWall
28      {
29          type            zeroGradient;
30      }
31
32      lowerWall
33      {
34          type            zeroGradient;
35      }
36
37      atmosphere
38      {
39          type            inletOutlet;
40          inletValue      uniform 0;
41          value           uniform 0;
42      }
43
44      defaultFaces
45      {
46          type            empty;
47      }
48  }
49
50  // ************************************************************************* //
```

6.4.2 Entrainment boundary conditions

The combination of the totalPressure condition on pressure and pressureInletOutletVelocity on velocity is extremely common for patches where some inflow occurs and the inlet flow velocity is not known. The conditions are used on the `atmosphere` boundary in the damBreak tutorial, inlet conditions where only pressure is known, outlets where flow reversal may occur, and where flow in entrained, *e.g.* on boundaries surrounding a jet through a nozzle.

The idea behind this combination is that the condition is a standard combination in the case of outflow, but for inflow the normal velocity is allowed to find its own value. Under these circumstances, a rapid rise in velocity presents a risk of instability, but the rise is moderated by the reduction of inlet pressure, and hence driving pressure gradient, as the inflow velocity increases.

The totalPressure condition specifies:

$$p = \begin{cases} p_0 & \text{for outflow} \\ p_0 - \frac{1}{2}\rho|\mathbf{U}^2| & \text{for inflow (dynamic pressure, subsonic)} \end{cases} \tag{6.2}$$

where the user specifies p_0 through the p0 keyword. Solver applications which include buoyancy effects, though a gravitational force $\rho\mathbf{g}$ (per unit volume) source term, tend to solve for a pressure field $p_{\rho gh} = p - \rho|\mathbf{g}|\Delta h$, where the hydrostatic component is subtracted based on a height Δh above some reference. For such solvers, *e.g.* interFoam, an equivalent prghTotalPressure condition is applied which specifies:

$$p_{\rho gh} = \begin{cases} p_0 & \text{for outflow} \\ p_0 - \rho|\mathbf{g}|\Delta h - \frac{1}{2}\rho|\mathbf{U}^2| & \text{for inflow (dynamic pressure, subsonic)} \end{cases} \tag{6.3}$$

The pressureInletOutletVelocity condition specifies zeroGradient at all times, except on the tangential component which is set to fixedValue for inflow, with the tangentialVelocity defaulting to 0.

The specification of these boundary conditions in the *U* and *p_rgh* files, in the damBreak case, are shown below.

```
16
17  dimensions      [0 1 -1 0 0 0 0];
18
19  internalField   uniform (0 0 0);
20
21  boundaryField
22  {
23      leftWall
24      {
25          type            noSlip;
26      }
27      rightWall
28      {
29          type            noSlip;
30      }
31      lowerWall
32      {
33          type            noSlip;
34      }
35      atmosphere
36      {
37          type            pressureInletOutletVelocity;
38          value           uniform (0 0 0);
39      }
40      defaultFaces
41      {
42          type            empty;
43      }
44  }
45
46
47  // ************************************************************************* //
```

```
16  dimensions      [1 -1 -2 0 0 0 0];
17
18  internalField   uniform 0;
19
20  boundaryField
21  {
22      leftWall
23      {
24          type            fixedFluxPressure;
25          value           uniform 0;
26      }
27
28      rightWall
29      {
30          type            fixedFluxPressure;
31          value           uniform 0;
32      }
33
34      lowerWall
35      {
36          type            fixedFluxPressure;
37          value           uniform 0;
38      }
39
40      atmosphere
41      {
42          type            prghTotalPressure;
43          p0              uniform 0;
44      }
45
46      defaultFaces
47      {
48          type            empty;
49      }
50  }
51
52  // ************************************************************************* //
```

6.4.3 Fixed flux pressure

In the above example, it can be seen that all the wall boundaries use a boundary condition named fixedFluxPressure. This boundary condition is used for pressure in situations where zeroGradient is generally used, but where body forces such as gravity and surface tension are present in the solution equations. The condition adjusts the gradient accordingly.

6.4.4 Time-varying boundary conditions

There are several boundary conditions for which some input parameters are specified by a function of time (using Function1 functionality) class. They can be searched by the following command.

```
find $FOAM_SRC/finiteVolume/fields/fvPatchFields -type f -name "*.H" |\
    xargs grep -l Function1 | xargs dirname | sort
```

They include conditions such as uniformFixedValue, which is a fixedValue condition which applies a single value which is a function of time through a `uniformValue` keyword entry.

The Function1 is specified by a keyword following the `uniformValue` entry, followed by parameters that relate to the particular function. The Function1 options are list below.

- `constant`: constant value.
- `table`: inline list of `(time value)` pairs; interpolates values linearly between times.
- `tableFile`: as above, but with data supplied in a separate file.
- `square`: square-wave function.
- `squarePulse`: single square pulse.
- `sine`: sine function.
- `one` and `zero`: constant one and zero values.
- `polynomial`: polynomial function using a list `(coeff exponent)` pairs.
- `coded`: function specified by user coding.
- `scale`: scales a given `value` function by a scalar `scale` function; both entries can be themselves Function1; `scale` function is often a ramp function (below), with `value` controlling the ramp value.
- `linearRamp`, `quadraticRamp`, `exponentialSqrRamp`, `halfCosineRamp`, `quarterCosineRamp` and `quarterSineRamp`: monotonic ramp functions which ramp from 0 to 1 over specified duration.
- `reverseRamp`: reverses the values of a ramp function, e.g. from 1 to 0.

Examples or a time-varying inlet for a scalar are shown below.

```
inlet
{
    type         uniformFixedValue;
    uniformValue constant 2;
}

inlet
{
    type         uniformFixedValue;
    uniformValue table ((0 0) (10 2));
}

inlet
{
    type         uniformFixedValue;
    uniformValue polynomial ((1 0) (2 2)); // = 1*t^0 + 2*t^2
```

```
}

inlet
{
    type         uniformFixedValue;
    uniformValue
    {
        type             tableFile;
        format           csv;
        nHeaderLine      4;              // number of header lines
        refColumn        0;              // time column index
        componentColumns (1);            // data column index
        separator        ",";            // optional (defaults to ",")
        mergeSeparators  no;             // merge multiple separators
        file             "dataTable.csv";
   }
}

inlet
{
    type         uniformFixedValue;
    uniformValue
    {
        type             square;
        frequency        10;
        amplitude        1;
        scale            2;  // Scale factor for wave
        level            1;  // Offset
    }
}

inlet
{
    type         uniformFixedValue;
    uniformValue
    {
        type             sine;
        frequency        10;
        amplitude        1;
        scale            2;  // Scale factor for wave
        level            1;  // Offset
    }
}

input  // ramp from 0 -> 2, from t = 0 -> 0.4
{
    type         uniformFixedValue;
    uniformValue
    {
        type             scale;
        scale            linearRamp;
        start            0;
        duration         0.4;
        value            2;
    }
}

input  // ramp from 2 -> 0, from t = 0 -> 0.4
{
    type         uniformFixedValue;
    uniformValue
    {
        type             scale;
        scale            reverseRamp;
        ramp             linearRamp;
        start            0;
        duration         0.4;
        value            2;
    }
}

inlet  // pulse with value 2, from t = 0 -> 0.4
{
    type         uniformFixedValue;
```

试用水印

```
    uniformValue
    {
        type            scale;
        scale           squarePulse
        start           0;
        duration        0.4;
        value           2;
    }
}

inlet
{
    type            uniformFixedValue;
    uniformValue    coded;
    name            pulse;
    codeInclude
    #{
        #include "mathematicalConstants.H"
    #};

    code
    #{
        return scalar
        (
            0.5*(1 - cos(constant::mathematical::twoPi*min(x/0.3, 1)))
        );
    #};
}
```

试用水印

试用水印

Chapter 7

Post-processing

This chapter describes options for post-processing with OpenFOAM. Post-processing in its most general sense involves data processing (processing results) and visualisation. The functionality for data processing is described in sections 7.2, 7.3 and 7.4. For visualisation, OpenFOAM relies on ParaView, a third-party open source application described in the example cases in chapter 2, with some additional information provided in the following section 7.1. Other methods of visualisation using third party software are described in section 7.5.

7.1 ParaView/paraFoam graphical user interface (GUI)

OpenFOAM includes a native reader module to visualise data with ParaView, an open-source, visualisation application. The module comprises of the PVFoamReader and vtkPVFoam libraries, which currently supports version 5.10.1 of ParaView. It is recommended that this version of ParaView is used, although it is possible that the latest binary release of the software will run adequately. Further details about ParaView can be found at `http://www.paraview.org`.

ParaView uses the Visualisation Toolkit (VTK) as its data processing and rendering engine and can therefore read any data in VTK format. OpenFOAM includes a variety of tools which can write data in VTK and other supported formats, which can be read directly by ParaView. Entire case data can be converted to VTK using the foamToVTK utility if the user wishes to process their results without the OpenFOAM reader.

In summary, we recommend the reader module for ParaView as the primary visualisation option for OpenFOAM. Alternatively OpenFOAM data can be converted into VTK format to be read by ParaView or any other VTK-based graphics tools.

7.1.1 Overview of ParaView/paraFoam

paraFoam is a script that launches ParaView using the reader module supplied with OpenFOAM. It is executed like any of the OpenFOAM utilities either by the single command from within the case directory or with the `-case` option with the case path as an argument, *e.g.*:

```
paraFoam -case <caseDir>
```

ParaView is launched and opens the window shown in Figure 7.1. The case is controlled from the left panel, which contains the following:

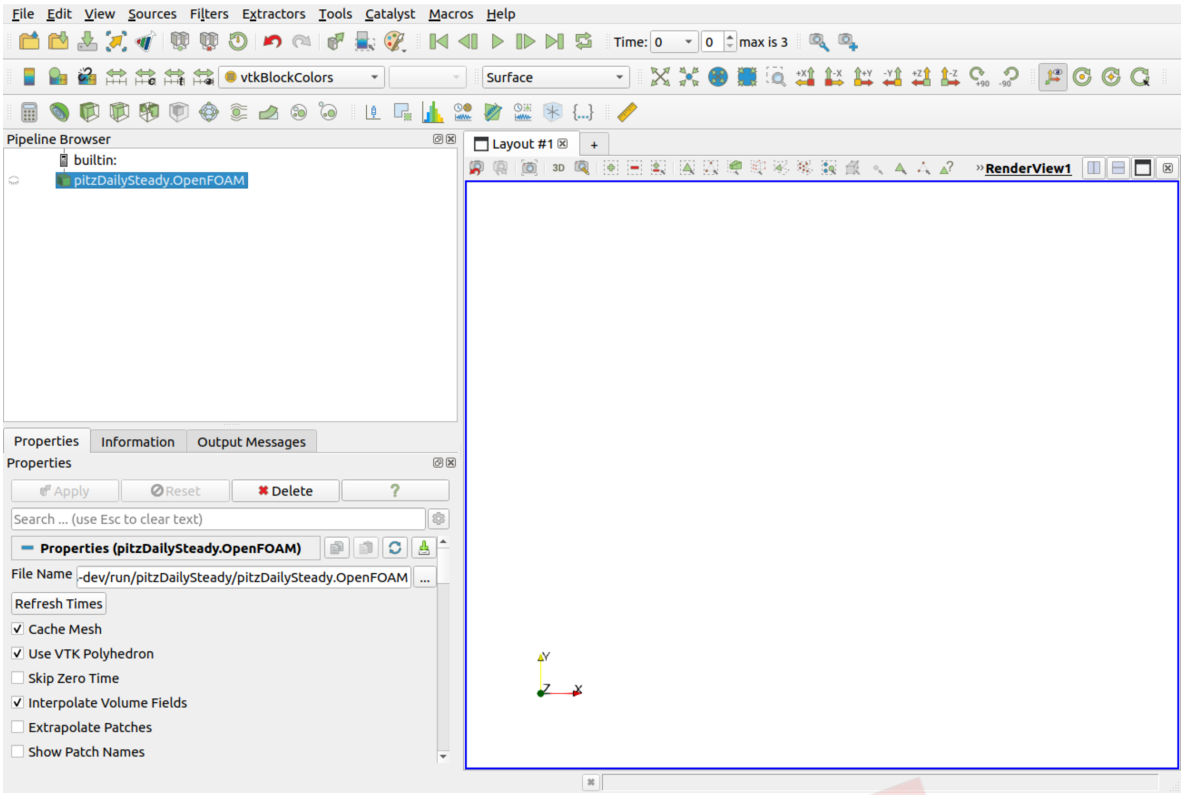


Figure 7.1: The ParaView window

- The Pipeline Browser lists the *modules* opened in ParaView, where the selected modules are highlighted in blue and the graphics for the given module can be enabled/disabled by clicking the eye button alongside;

- The Properties panel contains the input selections for the case, such as times, regions and fields; it includes the Display panel that controls the visual representation of the selected module, *e.g.* colours;

- Other panels can be selected from the `View` menu, including the Information panel which gives case statistics such as mesh geometry and size.

ParaView operates a tree-based structure in which data can be filtered from the top-level case module to create sets of sub-modules. For example, a contour plot of, say, pressure could be a sub-module of the case module which contains all the pressure data. The strength of ParaView is that the user can create a number of sub-modules and display whichever ones they need to create the desired image or animation. For example, they may add some solid geometry, mesh and velocity vectors, to a contour plot of pressure, switching any of the items on and off as necessary.

The general operation of the system is based on the user making a selection and then clicking the green Apply button in the Properties panel. The additional buttons are: the Reset button which can be used to reset the settings if necessary; and, the Delete button that will delete the active module.

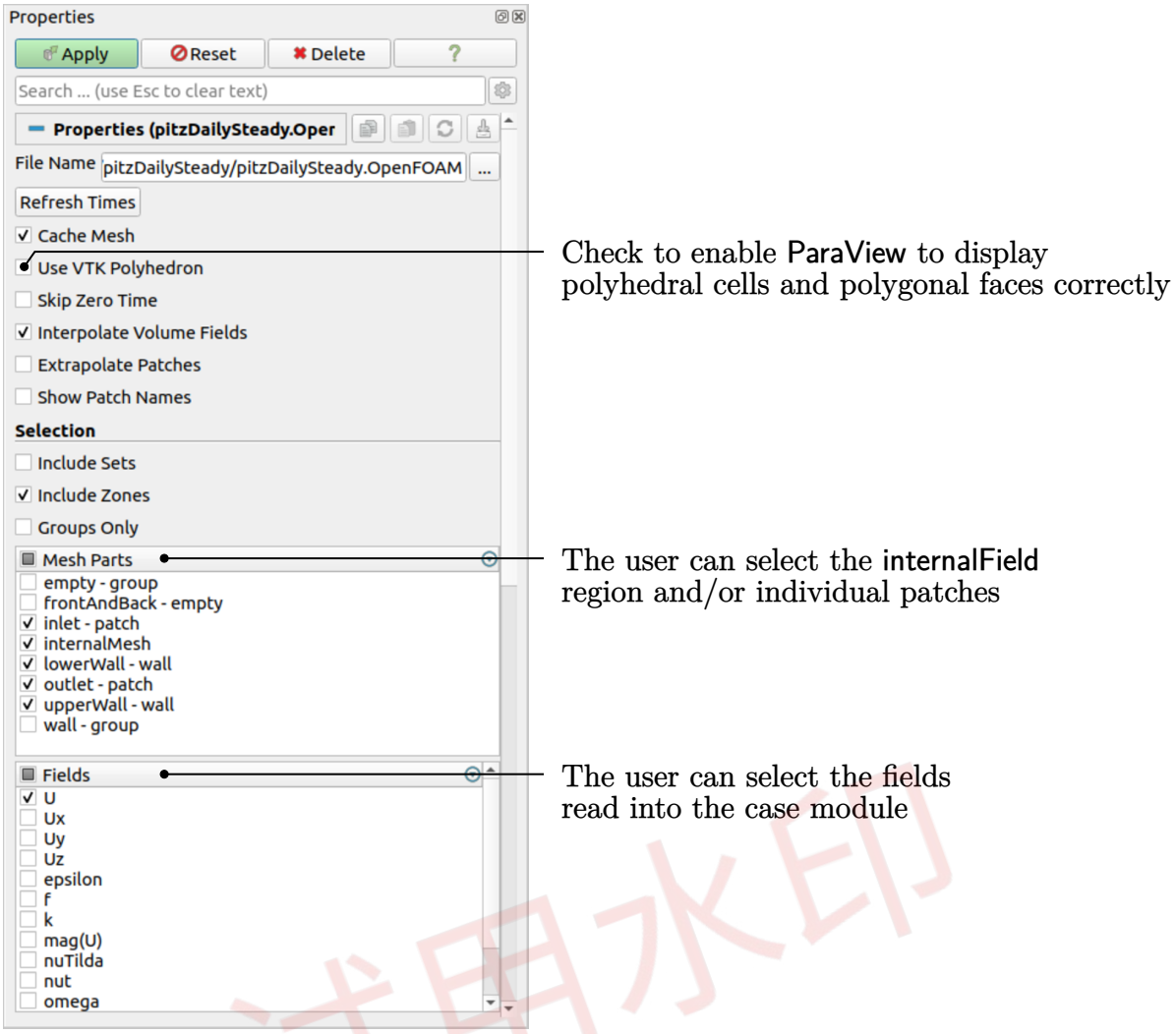


Figure 7.2: The Properties panel for the case module

7.1.2 The Parameters panel

The Properties window for the case module includes the Parameters panel that contains the settings for mesh, fields and global controls. The controls are described in Figure 7.2. The user can select mesh and field data which is loaded for all time directories into ParaView. The buttons in the `Current Time Controls` and `VCR Controls` toolbars then select the time data to be displayed, as shown is section 7.1.4.

As with any operation in ParaView, the user must click Apply after making any changes to any selections. The Apply button is highlighted in green to alert the user if changes have been made but not accepted. This method of operation has the advantage of allowing the user to make a number of selections before accepting them, which is particularly useful in large cases where data processing is best kept to a minimum.

If new data is written to time directories while the user is running ParaView, the user must load the additional time directories by checking the Refresh Times button. Where there are occasions when the case data changes on file and ParaView needs to load the changes, the user can also toggle the Cache Mesh button in the Parameters panel and apply the changes.

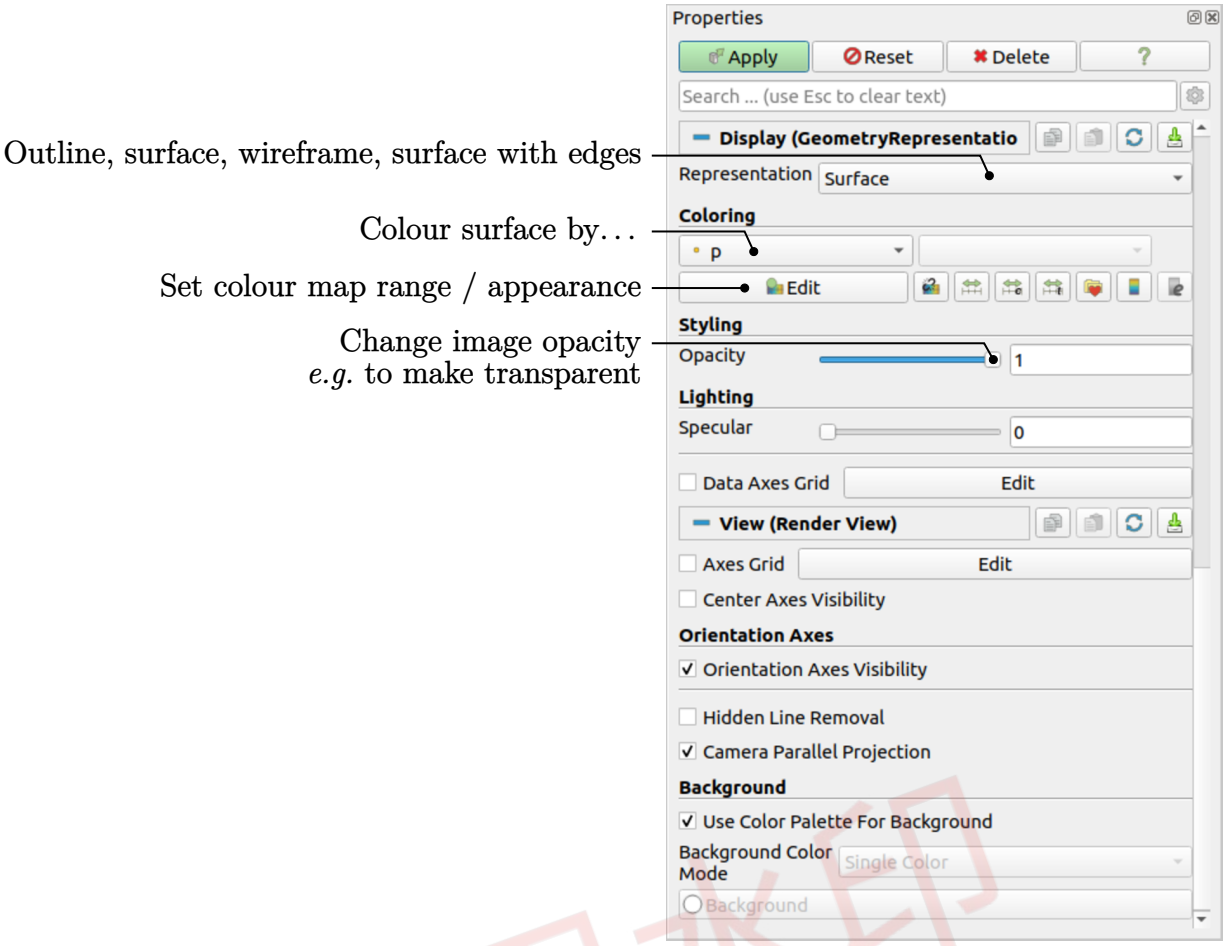


Figure 7.3: The Display panel

7.1.3 The Display panel

The Properties window contains the Display panel that includes the settings for visualising the data for a given case module. The following points are particularly important:

- the data range may not be automatically updated to the max/min limits of a field, so the user should take care to select Rescale at appropriate intervals, in particular after loading the initial case module;
- clicking the Edit Color Map button, brings up a window in which there are two panels:
  1. The Color Scale panel in which the colours within the scale can be chosen. The standard blue to red colour scale for CFD can be selected by clicking Choose Preset and searching for `Blue to Red Rainbow` and selecting.
  2. The Color Legend panel has a toggle switch for a colour bar legend and contains settings for the layout of the legend, *e.g.* font.
- the underlying mesh can be represented by selecting `Wireframe` in the `Representation` menu of the Style panel;
- the geometry, *e.g.* a mesh (if `Wireframe` is selected), can be visualised as a single colour by selecting `Solid Color` from the `Color By` menu and specifying the colour in the Set Ambient Color window;

- the image can be made translucent by editing the value in the Opacity text box (1 = solid, 0 = invisible) in the Style panel.

7.1.4 The button toolbars

ParaView duplicates functionality from pull-down menus at the top of the main window and the major panels, within the toolbars below the main pull-down menus. The displayed toolbars can be selected from `Toolbars` in the main `View` menu. The default layout with all toolbars is shown in Figure 7.4 with each toolbar labelled. The function of many of the buttons is clear from their icon and, with tooltips enabled in the `Help` menu, the user is given a concise description of the function of any button.

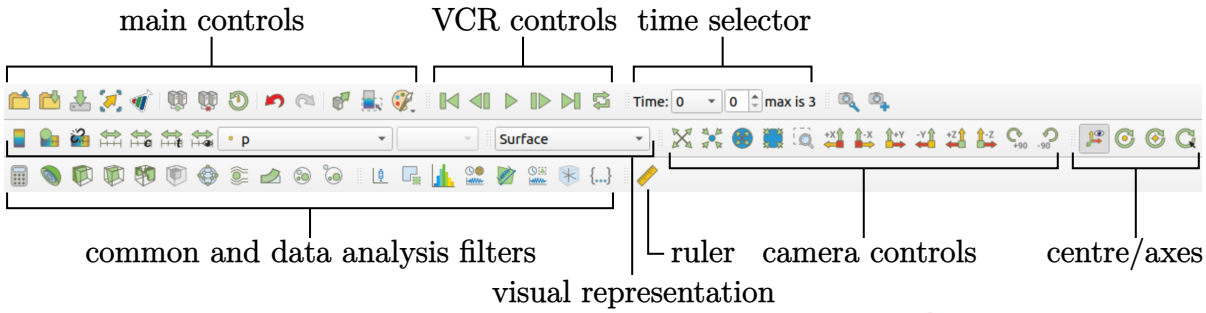


Figure 7.4: Toolbars in ParaView

7.1.5 Manipulating the view

This section describes operations for setting and manipulating the view in ParaView. Firstly, the `View Settings` are available in the Render View panel below the Display panel in the Properties window. Settings that are generally important only appear when the user checks the gearwheel button at the top of the Properties window, next to the search bar. These *advanced properties* include setting the background colour, where white is often a preferred choice for creating images for printed and website material.

The Lights button opens detailed lighting controls within the Light Kit panel. A separate Headlight panel controls the direct lighting of the image. Checking the Headlight button with white light colour of strength 1 seems to help produce images with strong bright colours, e.g. with an isosurface.

The Camera Parallel Projection is the usual choice for CFD, especially for 2D cases, and so should generally be checked. Other settings include Cube Axes which displays axes on the selected object to show its orientation and geometric dimensions.

The general `Settings` are selected from the `Edit` menu, which opens a general Options window with General, Camera, Render View Color Arrays and Color Palette menu items.

The General panel controls some default behaviour of ParaView. In particular, there is an Auto Apply button that enables ParaView to accept changes automatically without clicking the green Apply button in the Properties window. For larger cases, this option is generally not recommended: the user does not generally want the image to be re-rendered between each of a number of changes he/she selects, but be able to apply a number of changes to be re-rendered in their entirety once.

The Render View panel contains level of detail (LOD) which controls the rendering of the image while it is being manipulated, *e.g.* translated, resized, rotated; lowering the

levels set by the sliders, allows cases with large numbers of cells to be re-rendered quickly during manipulation.

The Camera panel includes control settings for 3D and 2D movements. This presents the user with a map of rotation, translate and zoom controls using the mouse in combination with Shift- and Control-keys. The map can be edited to suit by the user.

7.1.6 Contour plots

A contour plot is created by selecting `Contour` from the `Filter` menu at the top menu bar. The filter acts on a given module so that, if the module is the 3D case module itself, the contours will be a set of 2D surfaces that represent a constant value, *i.e.* isosurfaces. The Properties panel for contours contains an Isosurfaces list that the user can edit, most conveniently by the New Range window. The chosen scalar field is selected from a pull down menu.

Very often a user will wish to create a contour plot across a plane rather than producing isosurfaces. To do so, the user must first use the `Slice` filter to create the cutting plane, on which the contours can be plotted. The `Slice` filter allows the user to specify a cutting `Plane`, `Box` or `Sphere` in the `Slice Type` menu by a center and normal/radius respectively. The user can manipulate the cutting plane like any other using the mouse.

The user can then run the `Contour` filter on the cut plane to generate contour lines.

7.1.7 Vector plots

Vector plots are created using the `Glyph` filter. The filter reads the field selected in `Vectors` and offers a range of `Glyph Types` for which the `Arrow` provides a clear vector plot images. Each glyph has a selection of graphical controls in a panel which the user can manipulate to best effect.

The remainder of the Properties panel contains mainly the `Scale Mode` menu for the glyphs. The most common options for `Scale Mode` are: `Vector`, where the glyph length is proportional to the vector magnitude; and, `Off` where each glyph is the same length. The Set Scale Factor parameter controls the base length of the glyphs.

Vectors are by default plotted on cell vertices but, very often, we wish to plot data at cell centres. This is done by first applying the `Cell Centers` filter to the case module, and then applying the `Glyph` filter to the resulting cell centre data.

7.1.8 Streamlines

Streamlines are created by first creating tracer lines using the `Stream Tracer` filter. The tracer Seed panel specifies a distribution of tracer points over a `Line Source` or `Point Cloud`. The user can view the tracer source, *e.g.* the line, but it is displayed in white, so they may need to change the background colour in order to see it.

The distance the tracer travels and the length of steps the tracer takes are specified in the text boxes in the main Stream Tracer panel. The process of achieving desired tracer lines is largely one of trial and error in which the tracer lines obviously appear smoother as the step length is reduced but with the penalty of a longer calculation time.

Once the tracer lines have been created, the `Tubes` filter can be applied to the *Tracer* module to produce high quality images. The tubes follow each tracer line and are not strictly cylindrical but have a fixed number of sides and given radius. When the number of sides is set above, say, 10, the tubes do however appear cylindrical, but again this adds a computational cost.

7.1.9 Image output

The simplest way to output an image to file from ParaView is to select `Save Screenshot` from the `File` menu. On selection, a window appears in which the user can select the resolution for the image to save. There is a button that, when clicked, locks the aspect ratio, so if the user changes the resolution in one direction, the resolution is adjusted in the other direction automatically. After selecting the pixel resolution, the image can be saved. To achieve high quality output, the user might try setting the pixel resolution to 1000 or more in the x-direction so that when the image is scaled to a typical size of a figure in an A4 or US letter document, perhaps in a PDF document, the resolution is sharp.

7.1.10 Animation output

To create an animation, the user should first select `Save Animation` from the `File` menu. A dialogue window appears in which the user can specify a number of things including the image resolution. The user should specify the resolution as required. The other noteworthy setting is number of frames per timestep. While this would intuitively be set to 1, it can be set to a larger number in order to introduce more frames into the animation artificially. This technique can be particularly useful to produce a slower animation because some movie players have limited speed control, particularly over `mpeg` movies.

On clicking the Save Animation button, another window appears in which the user specifies a file name *root* and file format for a set of images. On clicking OK, the set of files will be saved according to the naming convention "<`fileRoot`>_<`imageNo`>.<`fileExt`>", *e.g.* the third image of a series with the file root "`animation`", saved in `jpg` format would be named "`animation_0002.jpg`" (<`imageNo`> starts at `0000`).

Once the set of images are saved the user can convert them into a movie using their software of choice. One option is to use the built in foamCreateVideo script from the command line whose usage is shown with the `-help` option.

7.2 Post-processing command line interface (CLI)

Post-processing is provided directly within OpenFOAM through the command line including data processing, sampling (*e.g.* probes, graph plotting) visualisation, case control and run-time I/O. Functionality can be executed by:

- conventional *post-processing*, a data processing activity that occurs *after* a simulation has run;
- *run-time processing*, data processing that is performed *during* the running of a simulation.

Both approaches have advantages. Conventional post-processing allows the user to choose how to analyse data after the results are obtained. Run-time processing offers greater flexibility because it has access to *all* the data in the database of the run at all times, rather than just the data written during the simulation. It also allows the user to monitor processed data during a simulation and provides a greater level of convenience because the processed results can be available immediately to the user when the simulation ends.

There are 3 methods of post-processing that cover the options described above.

- The case can be configured to include run-time processing during the simulation.
- The foamPostProcess utility provides conventional post-processing of data after a simulation is completed.
- The foamPostProcessutility is run with a `-solver` which provides additional access to data available on the database for the particular solver.

All modes of post-processing access the same functionality implemented in OpenFOAM in the *function object* framework. Function objects can be listed using foamToC by the following command.

```
foamToC -functionObjects
```

The list represents the underlying post-processing functionality. Almost all the functionality is packaged into a set of configured tools that are conveniently integrated within the post-processing CLI. Those tools are located in *$FOAM_ETC/caseDicts/postProcessing* and are listed by running foamPostProcesswith the `-list` option.

```
foamPostProcess -list
```

This produces a list of tools catalogued in section 7.3.

7.2.1 Run-time data processing

When a user wishes to process data during a simulation, they need to configure the case accordingly. The configuration process is as follows, using an example of monitoring flow rate at an outlet patch named `outlet`.

Firstly, the user should include the patchFlowRate function in `functions` sub-dictionary in the case *controlDict* file, using the `#includeFunc` directive.

```
functions
{
    #includeFunc  patchFlowRate
    ... other function objects here ...
}
```

That will include the functionality in the *patchFlowRate* configuration file, located in the directory hierarchy beginning with *$FOAM_ETC/caseDicts/postProcessing*.

The configuration of *patchFlowRate* requires the `name` of the patch to be supplied. **Option 1** for doing this is that the user copies the *patchFlowRate* file into their case *system* directory. The foamGet script copies the file conveniently, *e.g.*

```
foamGet patchFlowRate
```

The patch `name` can be edited in the copied file to be `outlet`. When the solver is run, it will pick up an included function in the local case *system* directory, in precedence over *$FOAM_ETC/caseDicts/postProcessing*. The flow rate through the patch will be calculated and written out into a file within a directory named *postProcessing*.

Option 2 for specifying the patch name is to provide the name as an argument to the patchFlowRate in the `#includeFunc` directive, using the syntax `keyword=entry`.

```
functions
{
    #includeFunc  patchFlowRate(patch=outlet)
    ... other function objects here ...
}
```

In the case where the keyword is `field` or `fields`, only the entry is needed when specifying an argument to a function. For example, if the user wanted to calculate and write out the magnitude of velocity into time directories during a simulation they could simply add the following to the `functions` sub-dictionary in *controlDict*.

```
functions
{
    #includeFunc  mag(U)
    ... other function objects here ...
}
```

This works because the function's argument `U` is represented by the keyword `field`, see *$FOAM_ETC/caseDicts/postProcessing/fields/mag*.

Some functions require the setting of many parameters, *e.g.* to calculate forces and generate elements for visualisation, *etc.* For those functions, it is more reliable and convenient to copy and configure the function using option 1 (above) rather than through arguments.

7.2.2 The foamPostProcess utility

The user can execute post-processing functions after the simulation is complete using the foamPostProcessutility. We can us illustrate the use of foamPostProcessusing the pitzDailySteady case from section 2.1. The tutorial does not need to be run to use the case, it can instead be copied into the user's *run* directory and run using its accompanying Allrun script as follows.

```
run
cp -r $FOAM_TUTORIALS/incompressibleFluid/pitzDailySteady .
cd pitzDaily
./Allrun
```

Now the user can run execute post-processing functions with foamPostProcess. The `-help` option provides a summary of its use.

```
foamPostProcess -help
```

Simple functions like mag can be executed using the `-func` option; text on the command line generally needs to be quoted (`"..."`) if it contains punctuation characters.

```
foamPostProcess -func "mag(U)"
```

This operation calculates and writes the field of magnitude of velocity into a file named *mag(U)* in each time directory. Similarly, the patchFlowRate example can be executed using foamPostProcess.

```
foamPostProcess -func "patchFlowRate(name=outlet)"
```

Let us say the user now wants to calculate total pressure $= p + |\mathbf{U}|^2/2$ for incompressible flow with kinematic pressure, p. The function is available, named totalPressureIncompressible, which requires a `rhoInf` parameter to be specified. The user could attempt first to run as follows.

```
foamPostProcess -func "totalPressureIncompressible(rhoInf=1.2)"
```

This returns the following error message.

```
--> FOAM FATAL IO ERROR:
request for volVectorField U from objectRegistry region0 failed
```

The error message is telling the user that the velocity field *U* is not loaded. For the function to work, both the field needs to be loaded using the `-field` option as follows.

```
foamPostProcess -func "totalPressureIncompressible(rhoInf=1.2)" -field U
```

A more complex example is calculating wall shear stress using the wallShearStress function.

```
foamPostProcess -fields "(p U)" -func wallShearStress
```

Even loading relevant fields, the post-processing fails with the following message.

```
--> FOAM FATAL ERROR:
Unable to find turbulence model in the database
```

The message is telling us that the foamPostProcessutility has not constructed the necessary models, *i.e.* a turbulence model, that the incompressibleFluid solver module used when running the simulation. This is a situation where we need to post-process (as opposed to run-time process) using the `-solver` option modelling will be available that the post-processing function needs.

```
foamPostProcess -solver incompressibleFluid -func wallShearStress
```

Note that no fields need to be supplied, *e.g.* using "`-field U`", because incompressibleFluid module constructs and stores the required fields. Functions can also be selected by the `#includeFunc` directive in `functions` in the *controlDict* file, instead of the `-func` option.

7.3 Post-processing functionality

The packaged function objects are catalogued in this section. Each packaged function object is a configuration file stored in *$FOAM_ETC/caseDicts/postProcessing*. As a reminder, they can be listed by the following command.

```
foamPostProcess -list
```

7.3.1 Field calculation

age Calculates and writes out the time taken for a particle to travel from an inlet to the location.

components Writes the component scalar fields (*e.g.* Ux, Uy, Uz) of a field (*e.g.* U).

CourantNo Calculates the Courant Number field from the flux field.

ddt Calculates the Eulerian time derivative of a field.

div Calculates the divergence of a field.

enstrophy Calculates the enstrophy of the velocity field.

fieldAverage Calculates and writes the time averages of a given list of fields.

flowType Calculates and writes the `flowType` of velocity field where: -1 = rotational flow; 0 = simple shear flow; +1 = planar extensional flow.

grad Calculates the gradient of a field.

Lambda2 Calculates and writes the second largest eigenvalue of the sum of the square of the symmetrical and anti-symmetrical parts of the velocity gradient tensor.

log Calculates the natural logarithm of the specified scalar field.

MachNo Calculates the Mach Number field from the velocity field.

mag Calculates the magnitude of a field.

magSqr Calculates the magnitude-squared of a field.

massFractions Calculates mass-fraction fields from mole-fraction fields, or moles fields, and a multi-component thermophysical model.

moleFractions Calculates mole-fraction fields from the mass-fraction fields of a multi-component thermophysical model.

PecletNo Calculates the Peclet Number field from the flux field.

Q Calculates the second invariant of the velocity gradient tensor.

randomise Adds a random component to a field, with a specified perturbation magnitude.

reconstruct Calculates the reconstruction of a field; *e.g.* to construct a cell-centred velocity `U` from the face-centred flux `phi`.

scale Multiplies a field by a scale factor

shearStress Calculates the shear stress, outputting the data as a `volSymmTensorField`.

streamFunction Writes the stream-function `pointScalarField`, calculated from the specified flux `surfaceScalarField`.

surfaceInterpolate Calculates the surface interpolation of a field.

totalEnthalpy Calculates and writes the total enthalpy $h_a + K$ as the `volScalarField` Ha.

turbulenceFields Calculates specified turbulence fields and stores it on the database.

turbulenceIntensity Calculates and writes the turbulence intensity field I.

vorticity Calculates the vorticity field, i.e. the curl of the velocity field.

wallHeatFlux Calculates the heat flux at wall patches, outputting the data as a `volVectorField`.

wallHeatTransferCoeff Calculates the estimated incompressible flow heat transfer coefficient at wall patches, outputting the data as a `volScalarField`.

wallShearStress Calculates the shear stress at wall patches, outputting the data as a `volVectorField`.

writeCellCentres Writes the cell-centres volVectorField and the three component fields as `volScalarFields`; useful for post-processing thresholding.

writeCellVolumes Writes the cell-volumes `volScalarField`

writeVTK Writes out specified objects in VTK format, *e.g.* fields, stored on the case database.

yPlus Calculates the turbulence y+, outputting the data as a `yPlus` field.

7.3.2 Field operations

add Add a list of fields.

divide From the first field, divide the remaining fields in the list.

multiply Multiply a list of fields.

subtract From the first field, subtracts the remaining fields in the list.

uniform Create a uniform field.

7.3.3 Forces and force coefficients

forceCoeffsCompressible Calculates lift, drag and moment coefficients by summing forces on specified patches for a case where the solver is compressible (pressure is in units M/(LT^2), *e.g.* Pa).

forceCoeffsIncompressible Calculates lift, drag and moment coefficients by summing forces on specified patches for a case where the solver is incompressible (pressure is kinematic, *e.g.* m^2/s^2).

forcesCompressible Calculates pressure and viscous forces over specified patches for a case where the solver is compressible (pressure is in units M/(LT^2), *e.g.* Pa).

forcesIncompressible Calculates pressure and viscous forces over specified patches for a case where the solver is incompressible (pressure is kinematic, *e.g.* m^2/s^2).

7.3.4 Sampling for graph plotting

graphCell Writes graph data for specified fields along a line, specified by start and end points. One graph point is generated in each cell that the line intersects.

graphUniform Writes graph data for specified fields along a line, specified by start and end points. A specified number of graph points are used, distributed uniformly along the line.

graphCellFace Writes graph data for specified fields along a line, specified by start and end points. One graph point is generated on each face and in each cell that the line intersects.

graphFace Writes graph data for specified fields along a line, specified by start and end points. One graph point is generated on each face that the line intersects.

graphLayerAverage Generates plots of fields averaged over the layers in the mesh.

graphPatchCutLayerAverage Writes graphs of patch face values, area-averaged in planes perpendicular to a given direction. It adaptively grades the distribution of graph points to match the resolution of the mesh

7.3.5 Lagrangian data

dsmcFields Calculate intensive fields `UMean`, `translationalT`, `internalT`, `overallT` from averaged extensive fields from a DSMC calculation.

stopAtEmptyClouds Stops the run when all clouds are empty, *i.e.* have no particles.

7.3.6 Volume fields

cellMax Writes out the maximum cell value for one or more fields.

cellMaxMag Writes out the maximum cell value magnitude for one or more fields.

cellMin Writes out the minimum cell value for one or more fields.

cellMinMag Writes out the maximum cell value magnitude for one or more fields.

volAverage Writes out the volume-weighted average of one or more fields.

volIntegrate Writes out the volume integral of one or more fields.

7.3.7 Numerical data

residuals For specified fields, writes out the initial residuals for the first solution of each time step; for non-scalar fields (*e.g.* vectors), writes the largest of the residuals for each component (*e.g.* x, y, z).

7.3.8 Control

adjustTimeStepToChemistry Adjusts the time step to a chemistry model's bulk chemical time scales

adjustTimeStepToCombustion Adjusts the time step to a combustion model's bulk reaction time scales

stopAtClockTime Stops the run when the specified clock time in second has been reached and optionally write results before stopping.

stopAtFile Stops the run when the file *stop* is created in the case directory.

stopAtTimeStep Stops the run if the time-step drops below the specified value in seconds and optionally write results before stopping.

time Writes run time, CPU time and clock time and optionally the CPU and clock times per time step.

timeStep Writes the time step to a file for monitoring.

writeObjects Writes out specified objects, *e.g.* fields, stored on the case database.

7.3.9 Pressure tools

staticPressureIncompressible Calculates the pressure field in normal units, *i.e.* Pa in SI, from kinematic pressure by scaling by a specified density.

totalPressureCompressible Calculates the total pressure field in normal units, *i.e.* Pa in SI, for a case where the solver is compressible.

totalPressureIncompressible Calculates the total pressure field for a case where the solver is incompressible, in kinematic units, *i.e.* m^2/s^2 in SI.

7.3.10 Combustion

Qdot Calculates and outputs the heat release rate for the current combustion model.

XiReactionRate Writes the turbulent flame-speed and reaction-rate `volScalarFields` for the Xi-based combustion models.

7.3.11 Multiphase

populationBalanceMoments Calculates and writes out integral (integer moments) or mean properties (mean, variance, standard deviation) of a size distribution computed with multiphaseEulerFoam. Requires solver post-processing.

phaseForces Calculates the blended interfacial forces acting on a given phase, *i.e.* drag, virtual mass, lift, wall-lubrication and turbulent dispersion. Note that it works only in solver post-processing mode and in combination with multiphaseEulerFoam. For a simulation involving more than two phases, the accumulated force is calculated by looping over all `phasePairs` the phase is a part of.

phaseMap Writes the phase-fraction map field `alpha.map` with incremental value ranges for each phase e.g., with values 0 for water, 1 for air, 2 for oil, *etc.*

populationBalanceSizeDistribution Writes out the size distribution computed with multiphaseEulerFoam for the entire domain or a volume region. Requires solver post-processing.

wallBoilingProperties Looks up wall boiling wall functions and collects and writes out out fields of bubble departure diameter, bubble departure frequency, nucleation site density, effective liquid fraction at the wall, quenching heat flux, and evaporative heat flux.

7.3.12 Probes

boundaryProbes Writes out values of fields at a cloud of points, interpolated to specified boundary patches.

interfaceHeight Reports the height of the interface above a set of locations. For each location, it writes the vertical distance of the interface above both the location and the lowest boundary. It also writes the point on the interface from which these heights are computed.

internalProbes Writes out values of fields interpolated to a specified cloud of points.

probes Writes out values of fields from cells nearest to specified locations.

7.3.13 Surface fields

faceZoneAverage Calculates the average value of one or more fields on a `faceZone`.

faceZoneFlowRate Calculates the flow rate through a specified face zone by summing the flux on patch faces. For solvers where the flux is volumetric, the flow rate is volumetric; where flux is mass flux, the flow rate is mass flow rate.

patchAverage Calculates the average value of one or more fields on a patch.

patchDifference Calculates the difference between the average values of fields on two specified patches. Calculates the average value of one or more fields on a patch.

patchFlowRate Calculates the flow rate through a specified patch by summing the flux on patch faces. For solvers where the flux is volumetric, the flow rate is volumetric; where flux is mass flux, the flow rate is mass flow rate.

patchIntegrate Calculates the surface integral of one or more fields on a patch.

triSurfaceAverage Calculates the average on a specified triangulated surface by interpolating onto the triangles and integrating over the surface area. Triangles need to be small (<= cell size) for an accurate result.

triSurfaceDifference Calculates the difference between the average values of fields on two specified triangulated surfaces.

triSurfaceVolumetricFlowRate Calculates volumetric flow rate through a specified triangulated surface by interpolating velocity onto the triangles and integrating over the surface area. Triangles need to be small (<= cell size) for an accurate result.

7.3.14 Meshing

checkMesh Executes `primitiveMesh::checkMesh` to check the distortion of moving meshes.

7.3.15 'Pluggable' solvers

particles Tracks a cloud of parcels driven by the flow of the continuous phase.

phaseScalarTransport Solves a transport equation for a scalar field within one phase of a multiphase simulation.

scalarTransport Solves a transport equation for a scalar field.

7.3.16 Visualisation tools

cutPlaneSurface Writes out cut-plane surface files with interpolated field data in VTK format.

isoSurface Writes out iso-surface files with interpolated field data in VTK format.

patchSurface Writes out patch surface files with interpolated field data in VTK format.

streamlinesLine Writes out files of stream lines with interpolated field data in VTK format, with initial points uniformly distributed along a line.

streamlinesPatch Writes out files of stream lines with interpolated field data in VTK format, with initial points randomly selected within a patch.

streamlinesPoints Writes out files of stream lines with interpolated field data in VTK format, with specified initial points.

streamlinesSphere Writes out files of stream lines with interpolated field data in VTK format, with initial points randomly selected within a sphere.

7.4 Sampling and monitoring data

There are a set of general post-processing functions for sampling data across the domain for graphs and visualisation. Several functions also provide data in a single file, in the form of time versus values, that can be plotted onto graphs. This time-value data can be monitored during a simulation with the foamMonitor script.

7.4.1 Probing data

The functions for probing data are boundaryProbes, internalProbes and probes as listed in section 7.3.12. All functions work on the basis that the user provides some point locations and a list of fields, and the function writes out values of the fields are those locations. The differences between the functions are as follows.

- probes identifies the nearest cells to the probe locations and writes out the cell values; data is written into a single file in time-value format, suitable for plotting a graph.

- boundaryProbes and internalProbes interpolate field data to the probe locations, with the locations being snapped onto boundaries for boundaryProbes; data sets are written to separate files at scheduled write times (like fields). data.

Generally probes is more suitable for monitoring values at smaller numbers of locations, whereas the other functions are typically for sampling at large numbers of locations.

As an example, the user could use the pitzDailySteady case set up in section 2.1. The probes function is best configured by copying the file to the local system directory using foamGet.

```
foamGet probes
```

The user can modify the `probeLocations` in the *probes* file as follows.

```
12
13  #includeEtc "caseDicts/postProcessing/probes/probes.cfg"
14
15  fields (p U);
16  probeLocations
17  (
18      (0.01 0 0)
19  );
20
21  // ************************************************************************* //
```

The configuration is completed by adding the `#includeFunc` directive to `functions` in the *controlDict* file.

```
functions
{
    #includeFunc  probes
    ... other function objects here ...
}
```

When the simulation runs, time-value data is written into *p* and *U* files in *postProcessing/probes/0*.

7.4.2 Sampling for graphs

The graphUniform function samples data for graph plotting. To use it, the *graphUniform* file can be copied into the *system* directory to be configured. We will configure it here using the pitzDaily case as before. The file is simply copied using foamGet.

```
foamGet graphUniform
```

The start and end points of the line, along which data is sampled, should be edited; the entries below provide a vertical line across the full height of the geometry 0.01 m beyond the back step.

```
14
15  start           (0.01 -0.025 0);
16  end             (0.01  0.025 0);
17  nPoints         100;
18
19  fields          (U p);
20
21  axis            distance; // The independent variable of the graph. Can be "x",
22                            // "y", "z", "xyz" (all coordinates written out), or
23                            // "distance" (from the start point).
24
25  #includeEtc "caseDicts/postProcessing/graphs/graphUniform.cfg"
26
27  // ************************************************************************* //
```

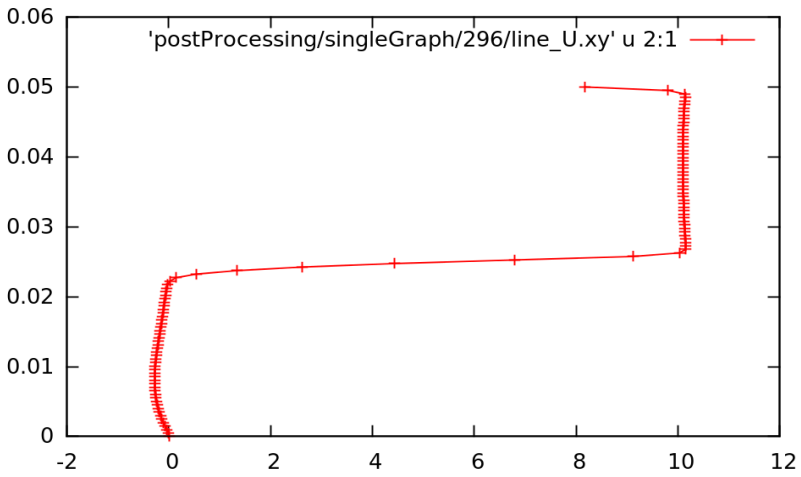


Figure 7.5: Graph of U_x at $x = 0.01$, uniform sampling

The configuration is completed by adding the `#includeFunc` directive to `functions` in the *controlDict* file.

```
functions
{
    #includeFunc  graphUniform
    ... other function objects here ...
}
```

The simulation can be then re-run or the user could run the post-processing with the following command.

```
foamPostProcess -solver incompressibleFluid
```

Either way, distance-value data is written into files in time directories within *postProcessing/graphUniform*. The user can quickly display the data for x-component of velocity, U_x in the last time *e.g. 285*, by running gnuplot and plotting values.

```
gnuplot
gnuplot> set style data linespoints
gnuplot> plot "postProcessing/graphUniform/285/line_U.xy" u 2:1
```

This produces the graph shown in Figure 7.5. This graph corresponds to the velocity inlet with a uniform profile, rather than a boundary layer profile. The formatting of the graph is specified in configuration files in *$FOAM_ETC/caseDicts/postProcessing/graphs*. The *graphUniform.cfg* file in that directory includes the configuration as follows.

```
 8
 9  #includeEtc "caseDicts/postProcessing/graphs/graph.cfg"
10
11  sets
12  (
13      line
14      {
15          type            lineUniform;
16          axis            $axis;
17          start           $start;
18          end             $end;
19          nPoints         $nPoints;
20      }
21  );
22
23  // ************************************************************************* //
```

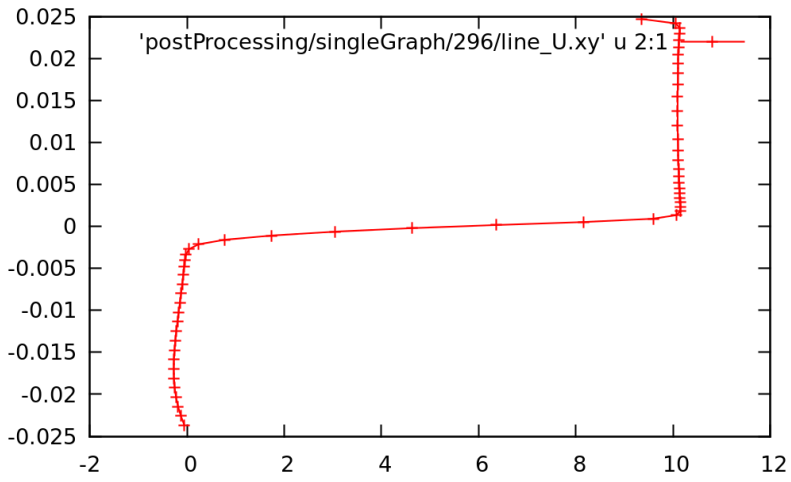


Figure 7.6: Graph of U_x at $x = 0.01$, mid-point sampling

It shows that the sampling type is `lineUniform`, meaning the sampling uses a uniform distribution of points along a line. The other parameters are included by macro expansion from the main file and specify the line start and end, the number of points and the distance parameter specified on the horizontal axis of the graph.

An alternative graph function object, *graphCell*, samples the data at locations nearest to the cell centres. The user can copy that function object file and configure it as shown below.

```
13
14  start    (0.01 -0.025 0);
15  end      (0.01  0.025 0);
16  fields   (U p);
17
18  axis     distance; // The independent variable of the graph. Can be "x",
19                     // "y", "z", "xyz" (all coordinates written out), or
20                     // "distance" (from the start point).
21
22  #includeEtc "caseDicts/postProcessing/graphs/graphCell.cfg"
23
24  // ************************************************************************* //
```

Running the post-processing produces the graph in Figure 7.6.

7.4.3 Sampling for visualisation

There are several surfaces and streamlines functions, listed in Section 7.3.16, that can be used to generate files for visualisation. The use of streamlinesLine is already configured in the pitzDailySteady case.

To generate a cutting plane, the cutPlaneSurface function can be configured by copying the *cutPlaneSurface* file to the *system* directory using foamGet.

```
foamGet cutPlaneSurface
```

The file is configured by setting the origin and normal of the plane and the field data to be sampled. We can edit the file to produce a cutting plane along the pitzDaily geometry, normal to the z-direction.

```
16
17  fields      (p U);
18
19  interpolate true; // If false, write cell data to the surface triangles.
20                    // If true, write interpolated data at the surface points.
```

```
21
22  #includeEtc "caseDicts/postProcessing/surface/cutPlaneSurface.cfg"
23
24  // ************************************************************************* //
```

The function can be included as normal by adding the `#includeFunc` directive to `functions` in the *controlDict* file. Alternatively, the user could test running the function using the solver post-processing by the following command.

```
foamPostProcess -solver incompressibleFluid -func cutPlaneSurface
```

This produces VTK format files of the cutting plane with pressure and velocity data in time directories in the *postProcessing/cutPlaneSurface* directory. The user can display the cutting plane by opening ParaView (type `paraview`), then doing `File->Open` and selecting one of the files, *e.g. postProcessing/cutPlaneSurface/285/U_zNormal.vtk* as shown in Figure 7.7.

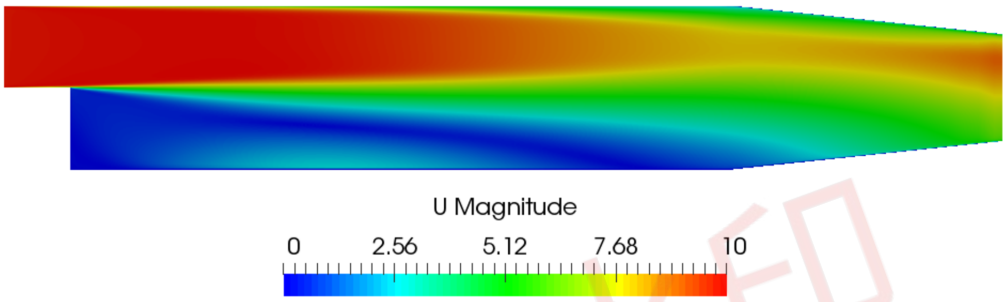


Figure 7.7: Cutting plane with velocity

7.4.4 Live monitoring of data

Functions like probes produce a single file of time-value data, suitable for graph plotting. When the function is executed during a simulation, the user may wish to monitor the data live on screen. The foamMonitor script enables this; to discover its functionality, the user run it with the `-help` option. The help option includes an example of monitoring residuals that we can demonstrate in this section.

Firstly, include the residuals function in the *controlDict* file.

```
functions
{
#includeFunc  residuals
... other function objects here ...
}
```

The default fields whose residuals are captured are *p* and *U*. Should the user wish to configure other fields, they should make copy the *residuals* file in their *system* and edit the `fields` entry accordingly. All functions files are within the *$FOAM_ETC/caseDicts* directory. The *residuals* file can be located using foamInfo:

```
foamInfo residuals
```

It can then be copied into the *system* directory conveniently using foamGet:

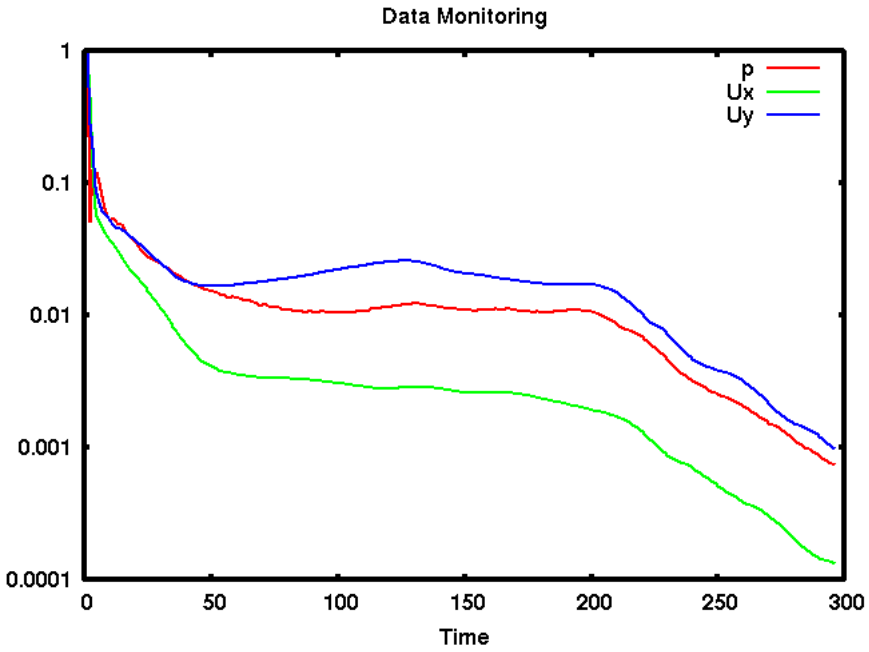


Figure 7.8: Live plot of residuals with foamMonitor

```
foamGet residuals
```

The user can then run the case in the background.

```
foamRun > log &
```

The user should then run foamMonitor using the `-l` option for a log scale y-axis on the *residuals* file as follows. If the command is executed before the simulation is complete, they can see the graph being updated live.

```
foamMonitor -l postProcessing/residuals/0/residuals.dat &
```

It produces the graph of residuals for pressure and velocity in Figure 7.8.

7.5 Third-Party post-processing

OpenFOAM includes the following applications for converting data to formats for post-processing with several third-party tools. For EnSight, it additionally includes a reader module, described in the next section.

foamDataToFluent Translates OpenFOAM data to Fluent format.

foamToEnsight Translates OpenFOAM data to EnSight format.

foamToEnsightParts Translates OpenFOAM data to Ensight format. An Ensight part is created for each `cellZone` and patch.

foamToGMV Translates foam output to GMV readable files.

foamToTetDualMesh Converts `polyMesh` results to `tetDualMesh`.

foamToVTK Legacy VTK file format writer.

smapToFoam Translates a STAR-CD SMAP data file into OpenFOAM field format.

7.5.1 Post-processing with Ensight

OpenFOAM offers the capability for post-processing OpenFOAM cases with EnSight, with a choice of 2 options:

- converting the OpenFOAM data to EnSight format with the foamToEnsight utility;
- reading the OpenFOAM data directly into EnSight using the ensight74FoamExec module.

The foamToEnsight utility converts data from OpenFOAM to EnSight file format. For a given case, foamToEnsight is executed like any normal application. foamToEnsight creates a directory named *Ensight* in the case directory, *deleting any existing Ensight directory in the process*. The converter reads the data in all time directories and writes into a case file and a set of data files. The case file is named *EnSight_Case* and contains details of the data file names. Each data file has a name of the form *EnSight_nn.ext*, where *nn* is an incremental counter starting from 1 for the first time directory, 2 for the second and so on and *ext* is a file extension of the name of the field that the data refers to, as described in the case file, *e.g.*`T` for temperature, `mesh` for the mesh. Once converted, the data can be read into EnSight by the normal means:

1. from the EnSight GUI, the user should select `Data (Reader)` from the `File` menu;
2. the appropriate *EnSight_Case* file should be highlighted in the Files box;
3. the `Format` selector should be set to `Case`, the EnSight default setting;
4. the user should click (Set) Case and Okay.

EnSight provides the capability of using a user-defined module to read data from a format other than the standard EnSight format. OpenFOAM includes its own reader module ensightFoamReader that is compiled into a library named libuserd-foam. It is this library that EnSight needs to use which means that it must be able to locate it on the filing system as described in the following section.

In order to run the EnSight reader, it is necessary to set some environment variables correctly. The settings are made in the *bashrc* (or *cshrc*) file in the *$WM_PROJECT_DIR/-etc/apps/ensightFoam* directory. The environment variables associated with EnSight are prefixed by $CEI_ or $ENSIGHT7_ and listed in Table 7.1. With a standard user setup, only $CEI_HOME may need to be set manually, to the path of the EnSight installation.

The principal difficulty in using the EnSight reader lies in the fact that EnSight expects that a case to be defined by the contents of a particular file, rather than a directory as it is in OpenFOAM. Therefore in following the instructions for the using the reader below, the user should pay particular attention to the details of case selection, since EnSight does not permit selection of a directory name.

1. from the EnSight GUI, the user should select `Data (Reader)` from the `File` menu;
2. The user should now be able to select the `OpenFOAM` from the `Format` menu; if not, there is a problem with the configuration described above.

Environment variable	Description and options
$CEI_HOME	Path where EnSight is installed, eg */usr/local/ensight*, added to the system path by default
$CEI_ARCH	Machine architecture, from a choice of names corresponding to the machine directory names in *$CEI_HOME/ensight74/machines*; default settings include `linux_2.4` and `sgi_6.5_n32`
$ENSIGHT7_READER	Path that EnSight searches for the user defined libuserd-foam reader library, set by default to `$FOAM_LIBBIN`
$ENSIGHT7_INPUT	Set by default to `dummy`

Table 7.1: Environment variable settings for EnSight.

3. The user should find their case directory from the File Selection window, highlight one of top 2 entries in the Directories box ending in `/.` or `/..` and click (Set) Geometry.

4. The path field should now contain an entry for the case. The (Set) Geometry text box should contain a '/'.

5. The user may now click Okay and EnSight will begin reading the data.

6. When the data is read, a new Data Part Loader window will appear, asking which part(s) are to be read. The user should select Load all.

7. When the mesh is displayed in the EnSight window the user should close the Data Part Loader window, since some features of EnSight will not work with this window open.

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Chapter 8

Models and physical properties

OpenFOAM includes a large range of solvers, each designed for a specific class of flow, as described in section 3.6. Each solver uses a particular set of models which calculate physical properties and simulate phenomena like transport, turbulence, thermal radiation, *etc.*

From OpenFOAM v10 onwards, a distinction is made between material *properties* and models for phenomena such as those mentioned above. **Properties are specified in *physicalProperties* file in the *constant* directory.** In the case of fluids, properties in *physicalProperties* relate to a fluid at rest. They are the properties you might look up from a table in a book, so can be dependent on temperature T, based on some function.

Properties described in *physicalProperties* do not include any dependency on the flow itself. For example, turbulence, visco-elasticity and the variation of viscosity ν with strain-rate, are all specified in a *momentumTransport* file in the *constant* directory. This chapter includes a description of models for viscosity which are dependent on strain-rate in section 8.3 and turbulence models in section 8.2. Thermophysical models, which are specified in the *physicalProperties* file (since they represent temperature dependency of properties) are described in section 8.1.

8.1 Thermophysical models

Thermophysical models are concerned with: thermodynamics, *e.g.* relating internal energy e to temperature T; transport, *e.g.* the dependence of properties such as ν on temperature; and state, *e.g.* dependence of density ρ on T and pressure p. Thermophysical models are specified in the *physicalProperties* dictionary.

A thermophysical model required an entry named `thermoType` which specifies the package of thermophysical modelling that is used in the simulation. OpenFOAM includes a large set of pre-compiled combinations of modelling, built within the code using C++ templates. It can also compile on-demand a combination which is not pre-compiled during a simulation.

Thermophysical modelling packages begin with the equation of state and then adding more layers of thermophysical modelling that derive properties from the previous layer(s). The keyword entries in `thermoType` reflects the multiple layers of modelling and the underlying framework in which they combined. Below is an example entry for `thermoType`:

```
thermoType
{
    type            hePsiThermo;
```

```
    mixture         pureMixture;
    transport       const;
    thermo          hConst;
    equationOfState perfectGas;
    specie          specie;
    energy          sensibleEnthalpy;
}
```

The keyword entries specify the choice of thermophysical models, *e.g.* `transport constant` (constant viscosity, thermal diffusion), `equationOfState perfectGas` , *etc.* In addition there is a keyword entry named `energy` that allows the user to specify the form of energy to be used in the solution and thermodynamics. The following sections explains the entries and options in the `thermoType` package.

8.1.1 Thermophysical and mixture models

Each solver that uses thermophysical modelling constructs an object of a specific thermophysical model class. The model classes are listed below.

fluidThermo Thermophysical model for a general fluid with fixed composition used by the isoThermalFluid and fluid solver modules.

rhoThermo Thermophysical model for liquids and solids, used by the isothermalFilm and film solver module.

psiThermo Thermophysical model for gases only, with fixed composition, used by the shockFluid solver module.

fluidMulticomponentThermo Thermophysical model for fluid of varying composition used by the multicomponentFluid solver module.

psiuMulticomponentThermo Thermophysical model for combustion that modelled by a laminar flame speed and regress variable used by the XiFluid solver module.

compressibleMultiphaseVoFMixtureThermo Thermophysical models for multiple phases used by the compressibleMultiphaseVoF solver module.

solidThermo **and** solidDisplacementThermo Thermophysical models for solids used by by the solid and solidDisplacement solver modules, respectively.

The `type` keyword (in the *thermoType* sub-dictionary) specifies the underlying thermophysical model used by the solver. The user can select from the following.

- `hePsiThermo`: available for solvers that construct fluidThermo, psiThermo, fluidMulticomponentThermo and .
- `heRhoThermo`: available for solvers that construct fluidThermo, rhoThermo, fluidMulticomponentThermo, compressibleMultiphaseVoFMixtureThermo.
- `heheuPsiThermo`: for solvers that construct psiuMulticomponentThermo.
- `heSolidThermo`: for solvers that construct solidThermo or solidDisplacementThermo.

The `mixture` specifies the mixture composition. The options available are listed below.

- `pureMixture`: mixture with fixed composition, which reads properties from a a sub-dictionary called `mixture`.
- `multicomponentMixture`: mixture with variable composition, with species, *e.g.* `O2`, `N2`, listed by the `species` keyword, and properties specified for each specie within sub-dictionaries named after each specie.
- `coefficientWilkeMulticomponentMixture`: as `multicomponentMixture`, but applies Wilke's equation to calculate transport properties for the mixture.
- `valueMulticomponentMixture`: as `multicomponentMixture`, but applies mole-fraction weighting to calculate transport properties for the mixture.
- `homogeneousMixture`, `inhomogeneousMixture` and `veryInhomogeneousMixture`: for combustion based on laminar flame speed and regress variables, constituents are a set of mixtures, such as `fuel`, `oxidant` and `burntProducts`.

8.1.2 Transport model

The transport modelling concerns evaluating dynamic viscosity μ, thermal conductivity κ and thermal diffusivity α (for internal energy and enthalpy equations). The current `transport` models are as follows:

`const` assumes a constant μ and Prandtl number $Pr = c_p\mu/\kappa$ which is simply specified by a two keywords, `mu` and `Pr`, respectively.

`sutherland` calculates μ as a function of temperature T from a Sutherland coefficient A_s and Sutherland temperature T_s, specified by keywords `As` and `Ts`; μ is calculated according to:

$$\mu = \frac{A_s\sqrt{T}}{1 + T_s/T}. \tag{8.1}$$

`polynomial` calculates μ and κ as a function of temperature T from a polynomial of any order N, *e.g.*:

$$\mu = \sum_{i=0}^{N-1} a_i T^i. \tag{8.2}$$

`logPolynomial` calculates $\ln(\mu)$ and $\ln(\kappa)$ as a function of $\ln(T)$ from a polynomial of any order N; from which μ, κ are calculated by taking the exponential, *e.g.*:

$$\ln(\mu) = \sum_{i=0}^{N-1} a_i [\ln(T)]^i. \tag{8.3}$$

`Andrade` calculates $\ln(\mu)$ and $\ln(\kappa)$ as a polynomial function of T, *e.g.* for μ:

$$\ln(\mu) = a_0 + a_1 T + a_2 T^2 + \frac{a_3}{a_4 + T}. \tag{8.4}$$

`tabulated` uses uniform tabulated data for viscosity and thermal conductivity as a function of pressure and temperature.

icoTabulated uses non-uniform tabulated data for viscosity and thermal conductivity as a function of temperature.

WLF (Williams-Landel-Ferry) calculates μ as a function of temperature from coefficients C_1 and C_2 and reference temperature T_r specified by keywords `C1`, `C2` and `Tr`; μ is calculated according to:

$$\mu = \mu_0 \exp\left(\frac{-C_1(T - T_r)}{C_2 + T - T_r}\right) \tag{8.5}$$

8.1.3 Thermodynamic models

The thermodynamic models are concerned with evaluating the specific heat c_p from which other properties are derived. The current thermo models are as follows:

eConst assumes a constant c_v and a heat of fusion H_f which is simply specified by a two values c_v H_f, given by keywords `Cv` and `Hf`.

eIcoTabulated calculates c_v by interpolating non-uniform tabulated data of (T, c_p) value pairs, *e.g.*:

```
( (200 1005) (400 1020) );
```

ePolynomial calculates c_v as a function of temperature by a polynomial of any order N:

$$c_v = \sum_{i=0}^{N-1} a_i T^i. \tag{8.6}$$

ePower calculates c_v as a power of temperature according to:

$$c_v = c_0 \left(\frac{T}{T_{\text{ref}}}\right)^{n_0}. \tag{8.7}$$

eTabulated calculates c_v by interpolating uniform tabulated data of (T, c_p) value pairs, *e.g.*:

```
( (200 1005) (400 1020) );
```

hConst assumes a constant c_p and a heat of fusion H_f which is simply specified by a two values c_p H_f, given by keywords `Cp` and `Hf`.

hIcoTabulated calculates c_p by interpolating non-uniform tabulated data of (T, c_p) value pairs, *e.g.*:

```
( (200 1005) (400 1020) );
```

hPolynomial calculates c_p as a function of temperature by a polynomial of any order N:

$$c_p = \sum_{i=0}^{N-1} a_i T^i. \tag{8.8}$$

hPower calculates c_p as a power of temperature according to:

$$c_p = c_0 \left(\frac{T}{T_{\text{ref}}}\right)^{n_0}. \tag{8.9}$$

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hTabulated calculates c_p by interpolating uniform tabulated data of (T, c_p) value pairs, *e.g.*:
```
( (200 1005) (400 1020) );
```

janaf calculates c_p as a function of temperature T from a set of coefficients taken from JANAF tables of thermodynamics. The ordered list of coefficients is given in Table 8.1. The function is valid between a lower and upper limit in temperature T_l and T_h respectively. Two sets of coefficients are specified, the first set for temperatures above a common temperature T_c (and below T_h), the second for temperatures below T_c (and above T_l). The function relating c_p to temperature is:

$$c_p = R((((a_4T + a_3)T + a_2)T + a_1)T + a_0). \tag{8.10}$$

In addition, there are constants of integration, a_5 and a_6, both at high and low temperature, used to evaluating h and s respectively.

Description	Entry	Keyword
Lower temperature limit	T_l (K)	`Tlow`
Upper temperature limit	T_h (K)	`Thigh`
Common temperature	T_c (K)	`Tcommon`
High temperature coefficients	$a_0 \dots a_4$	`highCpCoeffs (a0 a1 a2 a3 a4...`
High temperature enthalpy offset	a_5	`a5...`
High temperature entropy offset	a_6	`a6)`
Low temperature coefficients	$a_0 \dots a_4$	`lowCpCoeffs (a0 a1 a2 a3 a4...`
Low temperature enthalpy offset	a_5	`a5...`
Low temperature entropy offset	a_6	`a6)`

Table 8.1: JANAF thermodynamics coefficients.

8.1.4 Composition of each constituent

There is currently only one option for the `specie` model which specifies the composition of each constituent. That model is itself named `specie`, which is specified by the following entries.

- `nMoles`: number of moles of component. This entry is only used for combustion modelling based on regress variable with a homogeneous mixture of reactants; otherwise it is set to 1.
- `molWeight` in grams per mole of specie.

8.1.5 Equation of state

The following equations of state are available in the thermophysical modelling library.

adiabaticPerfectFluid Adiabatic perfect fluid:

$$\rho = \rho_0 \left(\frac{p + B}{p_0 + B} \right)^{1/\gamma}, \tag{8.11}$$

where ρ_0, p_0 are reference density and pressure respectively, and B is a model constant.

Boussinesq Boussinesq approximation

$$\rho = \rho_0 \left[1 - \beta \left(T - T_0\right)\right] \tag{8.12}$$

where β is the coeffient of volumetric expansion and ρ_0 is the reference density at reference temperature T_0.

icoPolynomial Incompressible, polynomial equation of state:

$$\rho = \sum_{i=0}^{N-1} a_i T^i, \tag{8.13}$$

where a_i are polynomial coefficients of any order N.

icoTabulated Tabulated data for an incompressible fluid using (T, ρ) value pairs, *e.g.*

```
rho ( (200 1010) (400 980) );
```

incompressiblePerfectGas Perfect gas for an incompressible fluid:

$$\rho = \frac{1}{RT} p_{\rm ref}, \tag{8.14}$$

where $p_{\rm ref}$ is a reference pressure.

linear Linear equation of state:

$$\rho = \psi p + \rho_0, \tag{8.15}$$

where ψ is compressibility (not necessarily $(RT)^{-1}$).

PengRobinsonGas Peng Robinson equation of state:

$$\rho = \frac{1}{zRT} p, \tag{8.16}$$

where the complex function $z = z(p, T)$ can be referenced in the source code in *PengRobinsonGasI.H*, in the *$FOAM_SRC/thermophysicalModels/specie/equationOfState/* directory.

perfectFluid Perfect fluid:

$$\rho = \frac{1}{RT} p + \rho_0, \tag{8.17}$$

where ρ_0 is the density at $T = 0$.

perfectGas Perfect gas:

$$\rho = \frac{1}{RT} p. \tag{8.18}$$

rhoConst Constant density:

$$\rho = \text{constant}. \tag{8.19}$$

rhoTabulated Uniform tabulated data for a compressible fluid, calculating ρ as a function of T and p.

rPolynomial Reciprocal polynomial equation of state for liquids and solids:

$$\frac{1}{\rho} = C_0 + C_1 T + C_2 T^2 - C_3 p - C_4 p T \tag{8.20}$$

where C_i are coefficients.

8.1.6 Selection of energy variable

The user must specify the form of energy to be used in the solution, either internal energy e and enthalpy h, and in forms that include the heat of formation Δh_f or not. This choice is specified through the `energy` keyword.

We refer to *absolute* energy where heat of formation is included, and *sensible* energy where it is not. For example absolute enthalpy h is related to sensible enthalpy h_s by

$$h = h_s + \sum_i c_i \Delta h_f^i \tag{8.21}$$

where c_i and h_f^i are the molar fraction and heat of formation, respectively, of specie i. In most cases, we use the sensible form of energy, for which it is easier to account for energy change due to reactions. Keyword entries for `energy` therefore include *e.g.* `sensibleEnthalpy`, `sensibleInternalEnergy` and `absoluteEnthalpy`.

8.1.7 Thermophysical property data

The basic thermophysical properties are specified for each species from input data. Data entries must contain the name of the specie as the keyword, *e.g.* `O2`, `H2O`, `mixture`, followed by sub-dictionaries of coefficients, including:

`specie` containing *i.e.* number of moles, `nMoles`, of the specie, and molecular weight, `molWeight` in units of g/mol;

`thermodynamics` containing coefficients for the chosen thermodynamic model (see below);

`transport` containing coefficients for the chosen tranpsort model (see below).

The following is an example entry for a specie named `fuel` modelled using `sutherland` transport and `janaf` thermodynamics:

```
fuel
{
    specie
    {
        nMoles       1;
        molWeight    16.0428;
    }
    thermodynamics
    {
        Tlow         200;
        Thigh        6000;
        Tcommon      1000;
        highCpCoeffs (1.63543 0.0100844 -3.36924e-06 5.34973e-10
                      -3.15528e-14 -10005.6 9.9937);
        lowCpCoeffs  (5.14988 -0.013671 4.91801e-05 -4.84744e-08
                      1.66694e-11 -10246.6 -4.64132);
    }
    transport
    {
        As           1.67212e-06;
```

```
        Ts              170.672;
    }
}
```

The following is an example entry for a specie named `air` modelled using const transport and hConst thermodynamics:

```
air
{
    specie
    {
        nMoles          1;
        molWeight       28.96;
    }
    thermodynamics
    {
        Cp              1004.5;
        Hf              2.544e+06;
    }
    transport
    {
        mu              1.8e-05;
        Pr              0.7;
    }
}
```

8.2 Turbulence models

Turbulence modelling is part of general **momentum transport** which is concerned with models for the viscous stress in a fluid. Momentum transport is configured through the *momentumTransport* file in the *constant* directory of a case. The file includes the mandatory `simulationType` keyword that specifies how turbulence is modelled, which includes the following options:

.

The file then includes a sub-dictionary of the same name as the chosen `simulationType` which contains the model selections. A typical example is shown below that uses the k–ε (k-epsilon) turbulence model.

```
16
17  simulationType RAS;
18
19  RAS
20  {
21      model           kEpsilon;
22
23      turbulence      on;
24
25      printCoeffs     on;
26  }
27
28  // ************************************************************************* //
```

The file shows the selected `RAS` simulation followed by the `RAS` sub-dictionary containing the model selections, in particular the `model` which is set to `kEpsilon`. The choice of RAS models is described in section 8.2.1 and more information can be found in Chapter 7 of ***Notes on Computational Fluid Dynamics: General Principles***. The LES models are listed in section 8.2.3.

Where the `laminar` option is selected, the sub-dictionary is optional and will default to a Newtonian model, using the viscosity specified in the *physicalProperties* file. Other models, including non-Newtonian and visco-elastic models, are described in section 8.3. Non-Newtonian models can also be combined with turbulence models (whereas visco-elastic models cannot).

For a general introduction to turbulence for CFD, the reader may also wish to consult Chapter 6 of ***Notes on Computational Fluid Dynamics: General Principles***.

8.2.1 Reynolds-averaged simulation (RAS) modelling

If `RAS` is selected, the choice of RAS modelling is specified in a `RAS` sub-dictionary which requires the following entries.

- `model`: name of RAS turbulence model.
- `turbulence`: switch to turn the solving of turbulence modelling on/off.
- `printCoeffs`: optional switch to print model coeffs to terminal at simulation start up, defaults to `false`.
- <`model`>`Coeffs`: optional dictionary of coefficients for the respective `model`, defaults to standard coefficients.

Turbulence models can be listed by running foamToC with a relevant table listed from the `RAS` or `LES` tables. For example, the `RAS` tables are listed by running the following command.

This returns several sub-tables. The user can then list the models within one of those tables, *e.g.* the incompressible models.

The RAS models used in the tutorials can be listed using foamSearch with the following command. The lists of available models are given in the following sections.

```
foamSearch $FOAM_TUTORIALS momentumTransport RAS/model
```

Users can locate tutorials using a particular model, *e.g.* buoyantKEpsilon, using foamInfo.

8.2.2 RAS turbulence models

For **incompressible flows**, the RAS `model` can be chosen from the list below.

LRR Launder, Reece and Rodi Reynolds-stress turbulence model for incompressible flows.

LamBremhorstKE Lam and Bremhorst low-Reynolds number k-epsilon turbulence model for incompressible flows.

LaunderSharmaKE Launder and Sharma low-Reynolds k-epsilon turbulence model for incompressible flows.

LienCubicKE Lien cubic non-linear low-Reynolds k-epsilon turbulence models for incompressible flows.

LienLeschziner Lien and Leschziner low-Reynolds number k-epsilon turbulence model for incompressible flows.

RNGkEpsilon Renormalization group k-epsilon turbulence model for incompressible flows.

SSG Speziale, Sarkar and Gatski Reynolds-stress turbulence model for incompressible flows.

ShihQuadraticKE Shih's quadratic algebraic Reynolds stress k-epsilon turbulence model for incompressible flows

SpalartAllmaras Spalart-Allmaras one-eqn mixing-length model for incompressible external flows.

kEpsilon Standard k-epsilon turbulence model for incompressible flows.

kEpsilonLopesdaCosta Variant of the standard k-epsilon turbulence model with additional source terms to handle the changes in turbulence in porous regions for atmospheric flows over forested terrain.

kOmega Standard high Reynolds-number k-omega turbulence model for incompressible flows.

kOmega2006 Standard (2006) high Reynolds-number k-omega turbulence model for incompressible flows.

kOmegaSST Implementation of the k-omega-SST turbulence model for incompressible flows.

kOmegaSSTLM Langtry-Menter 4-equation transitional SST model based on the k-omega-SST RAS model.

kOmegaSSTSAS Scale-adaptive URAS model based on the k-omega-SST RAS model.

kkLOmega Low Reynolds-number k-kl-omega turbulence model for incompressible flows.

qZeta Gibson and Dafa'Alla's q-zeta two-equation low-Re turbulence model for incompressible flows

realizableKE Realizable k-epsilon turbulence model for incompressible flows.

v2f Lien and Kalitzin's v2-f turbulence model for incompressible flows, with a limit imposed on the turbulent viscosity given by Davidson et al.

For **compressible flows**, the RAS `model` can be chosen from the list below.

LRR Launder, Reece and Rodi Reynolds-stress turbulence model for compressible flows.

LaunderSharmaKE Launder and Sharma low-Reynolds k-epsilon turbulence model for compressible and combusting flows including rapid distortion theory (RDT) based compression term.

RNGkEpsilon Renormalization group k-epsilon turbulence model for compressible flows.

SSG Speziale, Sarkar and Gatski Reynolds-stress turbulence model for compressible flows.

SpalartAllmaras Spalart-Allmaras one-eqn mixing-length model for compressible external flows.

buoyantKEpsilon Additional buoyancy generation/dissipation term applied to the k and epsilon equations of the standard k-epsilon model.

kEpsilon Standard k-epsilon turbulence model for compressible flows including rapid distortion theory (RDT) based compression term.

kOmega Standard high Reynolds-number k-omega turbulence model for compressible flows.

kOmega2006 Standard (2006) high Reynolds-number k-omega turbulence model for compressible flows.

kOmegaSST Implementation of the k-omega-SST turbulence model for compressible flows.

kOmegaSSTLM Langtry-Menter 4-equation transitional SST model based on the k-omega-SST RAS model.

kOmegaSSTSAS Scale-adaptive URAS model based on the k-omega-SST RAS model.

realizableKE Realizable k-epsilon turbulence model for compressible flows.

v2f Lien and Kalitzin's v2-f turbulence model for compressible flows, with a limit imposed on the turbulent viscosity given by Davidson et al.

8.2.3 Large eddy simulation (LES) modelling

If `LES` is selected, the choice of LES modelling is specified in a `LES` sub-dictionary which requires the following entries.

- `model`: name of LES turbulence model.
- `turbulence`: switch to turn the solving of turbulence modelling on/off.
- `delta`: name of delta δ model.
- `printCoeffs`: optional switch to print model coeffs to terminal at simulation start up, defaults to `false`.

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- <`model`>`Coeffs`:
- <`model`>`Coeffs`: optional dictionary of coefficients for the respective `model`, to override the default coefficients.
- <`delta`>`Coeffs`: dictionary of coefficients for the `delta` model.

The LES models used in the tutorials can be listed using foamSearch with the following command. The lists of available models are given in the following sections.

```
foamSearch $FOAM_TUTORIALS momentumTransport LES/model
```

8.2.4 LES turbulence models

For **incompressible and compressible flows**, the LES `model` can be chosen from the list below.

DeardorffDiffStress Differential SGS Stress Equation Model for incompressible flows

Smagorinsky The Smagorinsky SGS model.

SpalartAllmarasDDES SpalartAllmaras DDES turbulence model for incompressible flows

SpalartAllmarasDES SpalartAllmarasDES DES turbulence model for incompressible flows

SpalartAllmarasIDDES SpalartAllmaras IDDES turbulence model for incompressible flows

WALE The Wall-adapting local eddy-viscosity (WALE) SGS model.

dynamicKEqn Dynamic one equation eddy-viscosity model

dynamicLagrangian Dynamic SGS model with Lagrangian averaging

kEqn One equation eddy-viscosity model

kOmegaSSTDES Implementation of the k-omega-SST-DES turbulence model for incompressible flows.

8.2.5 Model coefficients

The coefficients for the RAS turbulence models are given default values in their respective source code. If the user wishes to override these default values, then they can do so by adding a sub-dictionary entry to the `RAS` sub-dictionary file, whose keyword name is that of the model with `Coeffs` appended, *e.g.* `kEpsilonCoeffs` for the `kEpsilon` model. If the `printCoeffs` switch is `on` in the `RAS` sub-dictionary, an example of the relevant `...Coeffs` dictionary is printed to standard output when the model is created at the beginning of a run. The user can simply copy this into the `RAS` sub-dictionary file and edit the entries as required.

8.2.6 Wall functions

A range of wall function models is available in OpenFOAM that are applied as boundary conditions on individual patches. This enables different wall function models to be applied to different wall regions. The choice of wall function model is specified through the turbulent viscosity field ν_t in the *0/nut* file. For example, a *0/nut* file:

```
16
17  dimensions      [0 2 -1 0 0 0 0];
18
19  internalField   uniform 0;
20
21  boundaryField
22  {
23      inlet
24      {
25          type            calculated;
26          value           uniform 0;
27      }
28      outlet
29      {
30          type            calculated;
31          value           uniform 0;
32      }
33      upperWall
34      {
35          type            nutkWallFunction;
36          value           uniform 0;
37      }
38      lowerWall
39      {
40          type            nutkWallFunction;
41          value           uniform 0;
42      }
43      frontAndBack
44      {
45          type            empty;
46      }
47  }
48
49
50  // ************************************************************************* //
```

There are a number of wall function models available in the release, *e.g.* `nutWallFunction`, `nutRoughWallFunction`, `nutUSpaldingWallFunction`, `nutkWallFunction` and `nutkAtmWallFunction`. The user can get the full list of wall function models using foamInfo:

```
foamToC -scalarBCs | grep nut
```

Within each wall function boundary condition the user can over-ride default settings for E, κ and C_μ through optional `E`, `kappa` and `Cmu` keyword entries.

Having selected the particular wall functions on various wall patches in the *nut* file, the user should select the following boundary conditions at wall patches for other turbulence fields.

- *epsilon* field: apply the epsilonWallFunction to corresponding patches.
- *omega* field: apply the omegaWallFunction to corresponding patches.
- *k*, *q* or *R* field: apply kqRwallFunction to corresponding patches.

8.3 Transport/rheology models

The *momentumTransport* file includes any model for the viscous stress in a fluid. That includes turbulence models, described in the previous section 8.2, but also non-Newtonian

and visco-elastic models described in this section. These models are described as `laminar`, located in *$FOAM_SRC/MomentumTransportModels/momentumTransportModels/laminar*, including:

- a family of generalisedNewtonian models for a non-uniform viscosity which is a function of strain rate $\dot{\gamma} = \sqrt{2}|\,\mathrm{symm}(\nabla \mathbf{U})|$, described in sections 8.3.1, 8.3.2, 8.3.3, 8.3.4, 8.3.5 and 8.3.6;
- a set of visco-elastic models, including Maxwell, Giesekus and PTT (Phan-Thien & Tanner), described in sections 8.3.7, 8.3.8 and 8.3.9, respectively;
- the lambdaThixotropic model, described in section 8.3.10.

When turbulence modelling is selected in the *momentumTransport* file, the generalisedNewtonian model is used by default to calculate the molecular viscosity. The choice of generalisedNewtonian model, specified by the `viscosityModel` keyword, is set to `Newtonian` by default, which simply uses the viscosity `nu` specified in the *physicalProperties* file. The following example exposes the default settings used with turbulence modelling.

```
simulationType RAS

RAS
{
    model           kEpsilon;  // RAS model
    turbulence      on;
    printCoeffs     on;

    // "laminar" model generalisedNewtonian is used by default
    viscosityModel  Newtonian; // default
}
```

While the `viscosityModel` entry is generally omitted when turbulence models are used, it can be included to set any of the non-Newtonian generalisedNewtonian models.

When turbulence modelling is not selected, by setting the `laminar` simulation type, the user can select any of the laminar models through the `model` keyword entry in the `laminar` sub-dictionary, including the visco-elastic models. The laminar models are listed by the following command.

```
foamToC -table laminarincompressibleMomentumTransportModel
```

If the `generalisedNewtonian` model is selected, the user must then specify the viscosity model through the `viscosityModel` keyword as mentioned above. The viscosity models are listed by the following command.

```
foamToC -table generalisedNewtonianViscosityModel
```

The example below shows how the the Bird-Carreau viscosity model is selected in a configuration without turbulence modelling.

```
simulationType laminar

laminar
{
    model           generalisedNewtonian;
    viscosityModel  BirdCarreau;
    // ... followed by the BirdCarreau parameters
}
```

The laminar models still use the viscosity *property* ν (`nu`) specified in the *physicalProperties* file, *e.g.*

```
viscosityModel constant;

nu              1.5e-05;
```

This viscosity is a single value which is **constant** in time and **uniform** over the solution domain. The non-Newtonian models adopt ν as the zero strain-rate viscosity ν_0. The visco-elastic models incorporate a linear viscous stress using ν, in addition to stress calculated by the respective models. The details of the models are provided in the following sections.

8.3.1 Bird-Carreau model

The Bird-Carreau generalisedNewtonian model is

$$\nu = \nu_\infty + (\nu_0 - \nu_\infty)\left[1 + (k\dot{\gamma})^a\right]^{(n-1)/a} \tag{8.22}$$

where the coefficient a has a default value of 2. An example specification of the model in *momentumTransport* is:

```
viscosityModel BirdCarreau;

nuInf   1e-05;
k       1;
n       0.5;
```

The constant, uniform viscosity at zero strain-rate, ν_0, is specified by `nu` in the *physicalProperties* file.

8.3.2 Cross Power Law model

The Cross Power Law generalisedNewtonian model is:

$$\nu = \nu_\infty + \frac{\nu_0 - \nu_\infty}{1 + (m\dot{\gamma})^n} \tag{8.23}$$

An example specification of the model in *momentumTransport* is:

```
viscosityModel CrossPowerLaw;

nuInf   1e-05;
m       1;
n       0.5;
```

The constant, uniform viscosity at zero strain-rate, ν_0, is specified by `nu` in the *physicalProperties* file.

8.3.3 Power Law model

The Power Law generalisedNewtonian model provides a function for viscosity, limited by minimum and maximum values, $\nu_{\min}$ and $\nu_{\max}$ respectively. The function is:

$$\nu = k\dot{\gamma}^{n-1} \quad \nu_{\min} \leq \nu \leq \nu_{\max} \tag{8.24}$$

An example specification of the model in *momentumTransport* is:

```
viscosityModel powerLaw;

nuMax    1e-03;
nuMin    1e-05;
k        1e-05;
n        0.5;
```

8.3.4 Herschel-Bulkley model

The Herschel-Bulkley generalisedNewtonian model combines the effects of Bingham plastic and power-law behaviour in a fluid. For low strain rates, the material is modelled as a very viscous fluid with viscosity ν_0. Beyond a threshold in strain-rate corresponding to threshold stress τ_0, the viscosity is described by a power law. The model is:

$$\nu = \min\left(\nu_0, \tau_0/\dot{\gamma} + k\dot{\gamma}^{n-1}\right) \tag{8.25}$$

An example specification of the model in *momentumTransport* is:

```
viscosityModel HerschelBulkley;

tau0     0.01;
k        0.001;
n        0.5;
```

The constant, uniform viscosity at zero strain-rate, ν_0, is specified in the *physicalProperties* file.

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8.3.5 Casson model

The Casson generalisedNewtonian model is a basic model used in blood rheology that specifies minimum and maximum viscosities, $\nu_{\min}$ and $\nu_{\max}$ respectively. Beyond a threshold in strain-rate corresponding to threshold stress τ_0, the viscosity is described by a "square-root" relationship. The model is:

$$\nu = \left(\sqrt{\tau_0/\dot{\gamma}} + \sqrt{m}\right)^2 \quad \nu_{\min} \leq \nu \leq \nu_{\max} \tag{8.26}$$

An example specification of model parameters for blood is:

```
viscosityModel Casson;

m         3.934986e-6;
tau0      2.9032e-6;
nuMax     13.3333e-6;
nuMin     3.9047e-6;
```

8.3.6 General strain-rate function

A `strainRateFunction` generalisedNewtonian model exists that allows a user to specify viscosity as a function of strain rate at run-time. It uses the same Function1 functionality to specify the function of strain-rate, used by time varying properties in boundary conditions described in section 6.4.4. An example specification of the model in *momentumTransport* is shown below using the `polynomial` function:

```
viscosityModel  strainRateFunction;

function polynomial ((0 0.1) (1 1.3));
```

8.3.7 Maxwell model

The Maxwell laminar visco-elastic model solves an equation for the fluid stress tensor τ:

$$\frac{\partial \tau}{\partial t} + \nabla \cdot (\mathbf{U}\tau) = 2\,\mathrm{symm}\left[\tau \cdot \nabla \mathbf{U}\right] - 2\frac{\nu_{\mathrm{M}}}{\lambda}\,\mathrm{symm}(\nabla \mathbf{U}) - \frac{1}{\lambda}\tau \tag{8.27}$$

where ν_{M} (`nuM`) is the "Maxwell" viscosity and λ (`lambda`) is the relaxation time. An example specification of model parameters is shown below:

```
simulationType laminar;

laminar
{
    model               Maxwell;

    MaxwellCoeffs
    {
        nuM             0.002;
        lambda          0.03;
    }
}
```

If an additional constant, uniform viscosity at zero strain-rate, ν_0, is specified in the *physicalProperties* file, the model becomes equivalent to an Oldroyd-B visco-elastic model. The Maxwell model includes a multi-mode option where τ is a sum of stresses, each with an associated relaxation time λ.

8.3.8 Giesekus model

The Giesekus laminar visco-elastic model is similar to the Maxwell model but includes an additional "mobility" term in the equation for τ:

$$\frac{\partial \tau}{\partial t} + \nabla \bullet (\mathbf{U}\tau) = 2\,\text{symm}\left[\tau \bullet \nabla \mathbf{U}\right] - 2\frac{\nu_\text{M}}{\lambda}\,\text{symm}(\nabla \mathbf{U}) - \frac{1}{\lambda}\tau - \frac{\alpha_\text{G}}{\nu_\text{M}}\left[\tau_i \bullet \tau_i\right] \tag{8.28}$$

where α_G (`alphaG`) is the mobility parameter. An example specification of model parameters is shown below:

```
simulationType laminar;

laminar
{
    model               Giesekus;

    GiesekusCoeffs
    {
        nuM             0.002;
        lambda          0.03;
        alphaG          0.1;
    }
}
```

The Giesekus model includes a multi-mode option where τ is a sum of stresses, each with an associated relaxation time λ and mobility coefficient α_G.

8.3.9 Phan-Thien-Tanner (PTT) model

The Phan-Thien-Tanner (PTT) laminar visco-elastic model is also similar to the Maxwell model but includes an additional "extensibility" term in the equation for τ, suitable for polymeric liquids:

$$\frac{\partial \tau}{\partial t} + \nabla \bullet (\mathbf{U}\tau) = 2\,\text{symm}\left[\tau \bullet \nabla \mathbf{U}\right] - 2\frac{\nu_\text{M}}{\lambda}\,\text{symm}(\nabla \mathbf{U}) - \frac{1}{\lambda}\exp\left(-\frac{\varepsilon\lambda}{\nu_\text{M}}\,\text{tr}(\tau)\right)\tau \tag{8.29}$$

where ε (`epsilon`) is the extensibility parameter. An example specification of model parameters is shown below:

```
simulationType laminar;

laminar
{
    model               PTT;
```

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```
    PTTCoeffs
    {
        nuM             0.002;
        lambda          0.03;
        epsilon         0.25;
    }
}
```

The PTT model includes a multi-mode option where τ is a sum of stresses, each with an associated relaxation time λ and extensibility coefficient ε.

8.3.10 Lambda thixotropic model

The Lambda Thixotropic laminar model calculates the evolution of a structural parameter λ (`lambda`) according to:

$$\frac{\partial \lambda}{\partial t} + \nabla \bullet (\mathbf{U}\lambda) = a(1-\lambda)^b - c\dot{\gamma}^d\lambda \tag{8.30}$$

with model coefficients a, b, c and d. The viscosity ν is then calculated according to:

$$\nu = \frac{\nu_\infty}{1 - K\lambda^2} \tag{8.31}$$

where the parameter $K = \sqrt{\nu_\infty/\nu_0}$. The viscosities ν_0 and ν_∞ are limiting values corresponding to $\lambda = 1$ and $\lambda = 0$.

An example specification of the model in *momentumTransport* is:

```
simulationType laminar;

laminar
{
    model               lambdaThixotropic;

    lambdaThixotropicCoeffs
    {
        a       1;
        b       2;
        c       1e-3;
        d       3;
        nu0     0.1;
        nuInf   1e-4;
    }
}
```

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